AN AI-NATIVE TEXTBOOK

Advanced Mathematical Decision Making

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Advanced Mathematical Decision Making

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This book is the student textbook of the AMDM (Advanced Mathematical Decision Making) series. A student workbook and a teacher guide are published separately.

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CHAPTER 1

Number, Operations, and Applications

How to Learn with AI in This Book

This is not an ordinary math textbook. Every section of this book contains an **AI Learning Lab**: copy-and-paste prompts that turn any AI chatbot into your personal tutor. You will not just read about mathematics — you will *discuss* it, *argue* with it, and *verify* it, with an AI as your sparring partner.

What you need

Any AI chatbot works: ChatGPT, Claude, Gemini, Copilot, or any other you have access to. The free versions are fine for everything in this book. The prompts are written generically — if your AI phrases things differently, that is normal and even useful: comparing two AIs' answers is itself a learning activity.

The four levels

Each AI Learning Lab has four levels. Follow them in order:

- **Level 1 — Understand.** Ask the AI to explain the concept, then explain it back *in your own words*. If you cannot rephrase it, you do not understand it yet.
- **Level 2 — Apply.** Ask the AI to generate practice problems and check your work. You do the solving; the AI does the grading.
- **Level 3 — Matter.** Ask why the section's real-life story matters: what does it mean for you, and what are its implications? This is where mathematics meets judgment.
- **Level 4 — Decide.** Put yourself in the situation: what decision would you make? Work a connected decision problem with the AI as your coach — you decide, the AI advises.

The golden rule: AI makes mistakes — verify everything

AI chatbots are confident, fluent, and sometimes wrong — especially at arithmetic. **Never trust an AI's number without checking it.** In §1.2 you will learn *percentage error*, and from that point on, here is your standing assignment: whenever an AI gives you a numerical answer, compute it yourself and find the percentage error between the two. If it is 0%, excellent. If it is not, one of you is wrong — and finding out *who* is the real lesson.

CAUTION

An AI is a sparring partner, not a ghostwriter. Pasting your homework

into an AI and copying the answer teaches you nothing and violates academic honesty. In this book, the AI asks *you* questions as often as you ask it.

How to use the [PROMPT BOX]es

Each prompt is in a box marked [PROMPT BOX]. Copy it, paste it into your AI, and follow the conversation. Feel free to adapt the wording — good prompting is itself a skill you will build chapter by chapter.

About This Chapter

Every decision you make — from comparing prices at the grocery store to estimating how long a road trip will take — runs on numbers. This chapter builds the number skills you will use all year in AMDM: comparing quantities with ratios and rates, working with percents, taming very large and very small numbers, making smart estimates, weighting averages fairly, and even checking whether the numbers printed on a product are real.

By the end of this chapter, you will be able to:

- Use ratios, rates, and unit rates to compare quantities and solve problems.
- Convert between fractions, decimals, and percents, and compute percent change and percentage error.
- Use exponents and scientific notation for very large and very small numbers.
- Solve Fermi problems by breaking big questions into smaller estimates.
- Compute simple and weighted averages and explain when each is appropriate.
- Compute and verify check digits (UPC, ISBN-13, and credit card numbers).
- Use AI prompts at four levels (understand, apply, matter, decide) to learn mathematics — and verify AI-generated math using percentage error.

1.1 Ratios and Rates

THE DECISION

The "bulk is cheaper" trap — olive oil.

A 1-liter bottle of olive oil costs $12.99. A 3-liter tin costs $33.99. Unit rates: $12.99/L vs $33.99 ÷ 3 = $11.33/L. The tin wins on unit price.

But your family uses about 0.4 liters per month, and olive oil goes stale about 6 months after opening. The tin lasts 3 ÷ 0.4 = 7.5 months — so the last 1.5 months' worth goes stale before you use it.

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Start with how much oil your family can actually use before it goes stale — 0.4 liters a month for 6 months.
- **Hint 2:** This is a unit-rate problem, but the "unit" that counts is usable liters, not the liters printed on the tin.
- **Hint 3:** Divide each price by its usable liters and compare — the sticker unit price is not the real one.

What Is a Ratio?

A **ratio** compares two quantities. If a recipe calls for 2 cups of syrup for every 5 cups of ice, we write the ratio as **2 : 5** or as the fraction **2/5**. The order matters: "2 cups of syrup to 5 cups of ice" is not the same as "5 cups of ice to 2 cups of syrup." A ratio is a *relationship*, and flipping it tells a different story.

Ratios can be **simplified** just like fractions. The ratio 18 : 24 simplifies to 3 : 4 (divide both sides by 6). Simplifying does not change the relationship — 18 : 24 and 3 : 4 describe the same mixture, the same deal, the same shape. Two ratios are **equivalent** when they simplify to the same ratio: 4/6 and 10/15 are equivalent because both reduce to 2/3.

A **proportion** is simply a statement that two ratios are equal, and it is the main tool for solving ratio problems: find the multiplier that turns one side into the other, then apply it to both sides. If 2/5 = x/10, the multiplier is 2 (5 × 2 = 10), so x = 2 × 2 = 4.

STORY

The Slushie Contest.

Maya, Jordan, and Rio entered a slushie-making contest with one rule: scale Maya's original recipe — 2 cups of syrup for every 5 cups of ice — up to ten times the size, and the judges will crown whoever reproduces the original taste most faithfully. Jordan used 20 cups of syrup but only 40 cups of ice — too sweet, too far from the original. Rio used 20 cups of syrup but poured in 60 cups of ice — watery and weak. Maya kept the ratio exactly: 20 cups of syrup to 50 cups of ice, and the judges crowned her the winner. There is no single "best" ratio — taste is personal — but once a recipe is set, only equivalent ratios reproduce it. That is how recipes scale, how maps work, and how models are built.

Rates and Unit Rates

A **rate** is a ratio that compares quantities with *different* units — miles per hour, dollars per pound, boxes per hour. Prices in a grocery store are rates. Gasoline prices, hourly wages, and streaming subscriptions are all rates.

A **unit rate** is a rate with a denominator of 1: dollars *per one* pound, miles *per one* hour. To find a unit rate, divide: \$4.99 for 12 ounces becomes \$4.99 ÷ 12 ≈ \$0.416 per ounce. Unit rates make comparisons easy — but read the fine print in the AI Learning Lab below, because the cheapest *per unit* is not always the smartest buy.

STORY

The Pizza Box Challenge.

Ava folds 144 pizza boxes in 3 hours. Her rate is 144 ÷ 3 = **48 boxes per hour**. The manager needs 720 boxes ready before the dinner rush. At Ava's rate, that takes 720 ÷ 48 = 15 hours for Ava alone — so the manager had better recruit Ben and Chris to help!

TRY IT

Open your kitchen cabinet. Find two different sizes of the same

product (cereal, rice, detergent — anything). Compute each unit price. Which is the better buy per unit? Now ask yourself: is there any reason you would still buy the worse one?

Worked Example 1 — Scaling a ratio. According to Greg, perfect cherry pies have a ratio of 240 cherries to 3 pies. How many cherries does Greg need for 9 perfect cherry pies? What does the answer mean?

Solution. First find the cherries per pie: 240 ÷ 3 = 80 cherries per pie. For 9 pies: 9 × 80 = **720 cherries**. The answer means that to keep the same cherry-to-pie ratio, 9 pies require 720 cherries in total.

Worked Example 2 — A rate over time. Jaime's dog-walking service walks 40 dogs in 8 hours. How many dogs can he walk in 12 hours, assuming the same pace? What does the answer mean?

Solution. Jaime's rate is 40 ÷ 8 = 5 dogs per hour. In 12 hours: 12 × 5 = **60 dogs**. The answer means that at a steady pace of 5 dogs per hour, 12 hours of work walks 60 dogs.

Worked Example 3 — Comparing with unit rates. A 12-ounce bag of coffee costs $4.99 and a 20-ounce bag costs $7.49. Which is the better buy?

Solution. Compute the unit price of each: 12-oz bag: $4.99 ÷ 12 ≈ $0.416 per ounce. 20-oz bag: $7.49 ÷ 20 = $0.3745 per ounce. The 20-ounce bag costs less per ounce, so it is the better buy — even though its sticker price is higher.

DECIDE — THE VERDICT

The "bulk is cheaper" trap — olive oil.

Back to the question that opened this section — now with the math to settle it.

Solution. Usable oil from the tin: 0.4 L/month × 6 months = 2.4 L. Wasted: 3 − 2.4 = 0.6 L. True cost per *usable* liter: $33.99 ÷ 2.4 ≈ **$14.16/L** — worse than the small bottle's $12.99/L.

The decision: the unit rate told you to buy the tin; the *waste-adjusted* unit rate tells you to buy the bottle. The math didn't change — the question did. "Cheapest per unit" is only the right question when you will actually use every unit.

AI AI Learning Lab 1.1

AI Level 1 — Understand.

Explain what a unit rate is as if I'm 15 years old. Use a grocery store example with real numbers. Then give me 3 practice problems, one at a time, and check each of my answers before moving on to the next.

AI Level 2 — Apply.

Give me 4 real-world unit-rate comparison problems (grocery prices, gas mileage, hourly pay). Hide the answers. I'll solve each one and show my work step by step; then check my work, point out any mistake, and explain what I did wrong.

AI Level 3 — Matter.

The olive-oil decision showed that "cheapest per unit" can be the wrong question when food goes stale before you finish it. But this story is bigger than olive oil. Discuss with me: WHY does this matter to ordinary families — who benefits when shoppers always grab the biggest size? What are the implications for a family's yearly grocery budget, for food waste, and for the environment? Then flip it: describe a situation where buying bulk is genuinely the smart move, and tell me what makes it different. End by asking me for one product in MY kitchen where I'd now rethink the size I buy.

Level 4 — Decide. *If you were the one buying, what would you choose?*

> **Example decision problem.** Your family drinks 5 oz of coffee per week, and ground coffee goes stale 3 weeks after opening. A 12-oz bag costs $4.99; a 20-oz bag costs $7.49. Which bag should you buy?

AI Level 4 — Decide.

I need to make a real decision: my family drinks 5 oz of coffee per week, coffee goes stale 3 weeks after opening, a 12-oz bag is $4.99 and a 20-oz bag is $7.49. Act as my decision coach. Don't give me the answer — ask me questions one at a time until I compute the waste-adjusted price per ounce for each bag myself. Then make me state my decision with reasons. Finally, challenge me: what would have to change — the price? our drinking habits? — for the big bag to win?

Check your verdict: usable coffee from the big bag is $5 \times 3 = 15$ oz, so $\$7.49 \div 15 \approx \$0.50$/oz usable; the small bag is $\$4.99 \div 12 \approx \$0.42$/oz. The "better unit price" loses once waste is counted.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 1.1.

Ask your AI: "A 5-oz candy bar costs $1.00 and a 12-oz bag costs $2.20. The bag has the better unit price — but give me three reasons I might still buy the bar." Read its reasons, then write a one-paragraph rule for *when to ignore the unit rate*. Mathematics tells you the price; judgment tells you the purchase.

Practice Problems 1.1

1. A paint mixture uses 3 cups of blue paint for every 2 cups of white paint. How much white paint is needed for 12 cups of blue paint?
2. A car travels 270 miles on 9 gallons of gas. What is the unit rate (miles per gallon)? How far can it travel on 14 gallons?
3. Simplify the ratio 18 : 24. Are the ratios 4/6 and 10/15 equivalent? Explain.
4. Ben folds 175 pizza boxes in 5 hours. What is his rate in boxes per hour? How many boxes can he fold in 8 hours at that pace?
5. A store sells 6 apples for $2.40 and 10 apples for $3.80. Which is the better buy? Show your work with unit rates.

1.2 Percent and Percentage Error

THE DECISION

Minimum payment vs. real payment — a $2,400 credit card balance.

You owe $2,400 on a card charging 24% APR (2% per month). The minimum payment is $50/month. Two plans:

- **Plan A:** pay the $50 minimum.
- **Plan B:** pay $200/month.

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Each month, 2% interest is added to the remaining balance first, then your payment is subtracted.
- **Hint 2:** In month one, what fraction of the $50 minimum just pays interest? This is a percent-of-a-changing-amount problem.
- **Hint 3:** Estimate how many months each plan lasts and the total paid — the minimum payment barely covers the interest each month.

Percent Basics

STORY

The Two Sales.

Daniel wants sneakers that cost $80. Store A offers 25% off; Store B offers $18 off. "Percent off" and "dollars off" speak different languages — but percent translates everything onto the same scale: 25% of $80 is **$20**, which beats $18. *Which store should Daniel choose, and what single number made the comparison possible?*

Percent means "per hundred." 25% is 25 out of 100, or 25/100 = 0.25. To convert a fraction to a percent, divide and multiply by 100: 5/6 ≈ 0.8333, so 5/6 ≈ **83.3%**. To go the other way, divide by 100: 6.5% = 0.065.

Percents are powerful because they *standardize* comparisons. Saying "we raised $200 of our $500 goal" means little to an outsider; saying "we reached 40% of our goal" is instantly meaningful. But that same power makes percents easy to abuse — as you will see in the percent trap below, and as advertisers know very well.

Percent Increase and Decrease

Percent change measures how much a quantity grew or shrank *relative to where it started*:

Percent change = (new value − old value) ÷ old value × 100%

The denominator is the key: percent change is always relative to the *original* value. Forget that, and you walk straight into the trap.

KEY IDEA

The percent trap.

A number is increased by 40% and then

decreased by 40%. The net change is *not* zero. Start with 100: increase by 40% → 100 × 1.40 = 140. Now decrease *140* by 40% → 140 × 0.60 = 84. The net result is 84, a **16% decrease** from 100. The 40% decrease applies to the *larger* number, so it removes more than the increase added. Whenever someone says "up X%, then down X%, so we're even" — do the math.

STORY

The School Fundraiser.

Emma, Leo, and Chris want to raise $500 for

new library books. After week one they have $125; by mid-week two they add $75 more, for $200 total. As a percentage of the goal, that's 200 ÷ 500 = **40%**. They still need $300, which is 60% of the goal. Percents turn raw dollar amounts into progress you can feel.

CAUTION

Percent vs. percentage points.

If support for a policy rises

from 40% to 50%, that is a 10-*percentage-point* increase — but a 25% *percent* increase (10 ÷ 40 = 0.25). News headlines mix these up constantly. When you see a percent in the wild, always ask: *percent of what?*

Percentage Error

STORY

The Delivery Estimate.

Jamal's pizza app estimated 25 minutes; the pizza arrived in 32. Seven minutes late — but is that a big error or a small one? Percentage error puts the true value on the bottom: $|25 - 32| \div 32 \approx$ **21.9%**. *What percentage error would make you stop trusting an app's estimates?*

When you estimate a quantity, **percentage error** tells you how far off your estimate was, relative to the true value:

Percentage error = |estimated value − actual value| ÷ actual value × 100%

The absolute value bars mean the error is always reported as a positive number — being 10% too high and 10% too low are both "10% off." Note the denominator: error is relative to the *actual* value, not your estimate. A $1 error on a $10 item (10%) matters; a $1 error on a $1,000 item (0.1%) does not.

Why this formula is your AI lie detector. From this section on, every time your AI gives you a number, compute it yourself and plug both into this formula. An AI that says a fence is 70 m when it is really 78.3 m has a percentage error of about 10.6% — big enough to notice, small enough that you would never catch it without doing the math.

Worked Example 1 — The percent trap, worked out. A number is increased by 40% and then decreased by 40%. What is the net change?

Solution. Start with 100. Increase by 40%: 100 × 1.40 = 140. Now decrease *140* by 40%: 140 × 0.60 = 84. The net result is 84, which is a **16% decrease** from 100 — not zero! The trap: the 40% decrease applies to the *larger* number, so it removes more than the increase added.

Worked Example 2 — Measuring a fence. You estimate a fence's length at 70 m, but measuring shows its true length is 78.3 m. Find (a) the error and (b) the absolute percentage error, to one decimal place.

Solution. (a) Error = $|70 - 78.3|$ = **8.3 m**. (b) Percentage error = 8.3 ÷ 78.3 × 100% ≈ **10.6%**.

DECIDE — THE VERDICT

Minimum payment vs. real payment — a $2,400 credit card balance.

Back to the question that opened this section — now with the math to settle it.

Solution. (Using the loan payoff formula — or a spreadsheet:)

- Plan A: 0.02 × $2,400 = $48 of the $50 goes to interest in month one. The balance barely shrinks. Payoff takes ≈ **163 months (13.5 years)**; total paid ≈ **$8,130**; interest ≈ **$5,730**.
- Plan B: payoff takes ≈ **14 months**; total paid ≈ **$2,772**; interest ≈ **$372**.

The decision: Plan A repays 339% of what you borrowed ($8,130 ÷ $2,400); Plan B repays 116%. The minimum payment is not a plan — it is a trap priced in percent. An extra $150/month for a little over a year saves about **$5,360** in interest. Percent change isn't abstract: it is the price tag on your decisions.

AI AI Learning Lab 1.2

AI Level 1 — Understand.

Explain the difference between "percent change" and "percentage points" as if I'm 15 years old, with one store example and one news example. Then quiz me with 3 questions, one at a time, checking each answer before moving on.

AI Level 2 — Apply.

Create 3 percent-change word problems about sale prices and test scores. Hide the answers. I'll solve each one and show my steps; then check my work and explain any error I made.

AI Level 3 — Matter.

The credit-card decision put a price tag on the minimum payment: about $5,360 in extra interest on a $2,400 balance. This story matters because it will be YOUR story the day you get your first card. Discuss with me: WHY does this matter to teenagers specifically — who profits when borrowers don't understand percent? What are the implications for someone's twenties if they learn this at 25 instead of 15? Then ask me: if you had 60 seconds to warn a friend who is about to pay only the minimum, what would you say — and what one number would you show them?

Level 4 — Decide. *If you were at the register, what would you choose?*

Example decision problem. A store offers 10% off a $600 TV today if you open their credit card at 29.99% APR (about 2.5% per month). You would owe $540 and plan to pay the $25 minimum each month. Is the "discount" worth it?

AI Level 4 — Decide.

I'm at the register deciding whether to open a store credit card for 10% off a $600 TV. The card charges 29.99% APR (about 2.5% per month), I'd owe $540, and I'd pay $25 a month. Be my decision coach: first ask me what I THINK the total cost will be, then walk me through the month-by-month math one step at a time until I find the real total myself. Then make me decide: discount or trap? End by asking me what monthly payment would flip your verdict.

Check your verdict: at 2.5%/month, $25 barely covers the interest — payoff takes about 32 months and the total is about $786, versus $600 cash. The $60 "discount" becomes roughly $186 extra.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 1.2.

Ask your AI: "Find me a real news headline that misuses percent change — something like 'crime up 200%!' — and explain what the headline hides." Then do the math yourself: reconstruct the original numbers, compute the honest percent change, and write the headline the way it *should* have been written. Bring both versions to class.

TRY IT

Find a "SALE — 30% OFF!" tag (in a store or online). Compute the

sale price yourself, then ask your AI to compute it too. Compute the percentage error between its answer and yours.

Practice Problems 1.2

1. Convert 3/8 to a percent.
2. A $60 shirt is on sale for 25% off. What is the sale price?
3. Your phone estimates 6 hours of battery life, but it actually lasts 5.2 hours. Find the absolute percentage error.
4. A town's population grows by 10% one year, then shrinks by 10% the next. What is the net percent change? Explain.
5. A fundraiser's goal is $800. The team raises $200 in week one and $350 in week two. What percent of the goal have they reached?

1.3 Exponents (Indices) and Scientific Notation

THE DECISION

How long should your password be?

A hacker can try about 1 billion (10^9) passwords per second. Your password uses 62 possible characters (a–z, A–Z, 0–9). Compare an 8-character password with a 12-character one.

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** First count the total passwords for each length — 62 multiplied by itself how many times?
- **Hint 2:** Divide each count by 10^9 (tries per second) to get seconds, then convert to days or years.
- **Hint 3:** Compare the two cracking times with a ratio — the exponent laws let you simplify $62^{12} \div 62^{8}$ to a single power of 62.

Exponents: Repeated Multiplication

STORY

The Doubling Penny.

Luis's uncle Marco offered him a deal: take $1,000 in cash right now, or take a single penny that doubles every day for 30 days. Luis almost grabbed the $1,000 — then did the math. After 10 doublings: 2^{10} = 1,024 pennies ≈ $10. After 20 doublings: 2^{20} = 1,048,576 pennies ≈ $10,486. After 30 doublings: 2^{30} = 1,073,741,824 pennies ≈ **$10.7 million**. Luis took the penny. *Would you?*

An **exponent** (also called an **index**) tells you how many times to multiply a number by itself. In 2^4, the **base** is 2 and the exponent is 4: $2^4 = 2 \times 2 \times 2 \times 2 = 16$. The small raised number is doing enormous work: 2^{10} is only a little more ink than 2^4, but it equals 1,024 — and 2^{100} has 31 digits.

Worked Example 1 — Exponent form. Write in exponent form and evaluate: $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3$.

Solution. $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 = 2^4 \times 3^4 = 16 \times 81 =$ **1,296**.

CAUTION

Common trap.

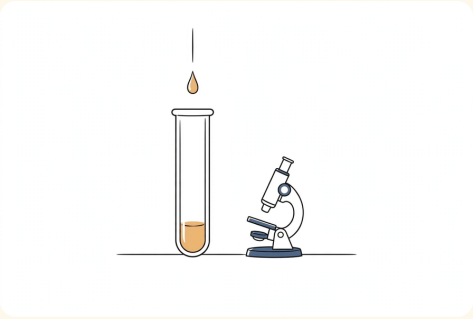
Repeated *addition* is not exponentiation.

$2 + 2 + 2 + 2 + 3 + 3 + 3 + 3 = 20$, which is very different from $2^4 \times 3^4 = 1{,}296$. Exponents mean repeated *multiplication*.

Zero and Negative Exponents

STORY

The Dilution.



A biology lab dilutes a pond-water sample by a factor of 10, three times in a row, before it goes under the microscope. The original held 2,000,000 bacteria per milliliter; three dilutions means multiplying by 10^{-3}, leaving **2,000** per milliliter. *A negative exponent didn't make anything negative — so what did it do?*

Consider the pattern of powers of 3: $3^3 = 27$, $3^2 = 9$, $3^1 = 3$. Each step down *divides* by 3. Continuing the pattern: $3^0 = 3 \div 3 = \mathbf{1}$, $3^{-1} = 1 \div 3 = \mathbf{1/3}$, $3^{-2} = \mathbf{1/9}$. So 3^0 is 1, not zero — and here is the cleanest proof: $3^0 = 3^3 \div 3^3 = 27 \div 27 = 1$. Anything divided by itself is 1, so any nonzero base to the zero power is 1.

A negative exponent means "take the reciprocal": $a^{-n} = 1/a^n$. So $4^{-2} = 1/4^2 = 1/16$, and $10^{-3} = 1/1{,}000 = 0.001$. Negative exponents are how we write very small numbers compactly — which is exactly what scientific notation needs.

Three **laws of exponents** (for the same base) handle everything else:

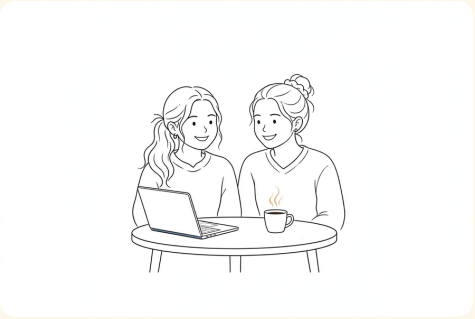
- **Product:** $a^m \times a^n = a^{m+n}$ (add exponents when multiplying)
- **Quotient:** $a^m \div a^n = a^{m-n}$ (subtract exponents when dividing)
- **Power of a power:** $(a^m)^n = a^{mn}$ (multiply exponents)

Example: $10^3 \times 10^4 = 10^7$ (not 10^{12} — a classic mistake), and $(2^2)^3 = 2^6 = 64$.

Scientific Notation

STORY

The Download.



A movie file is 4.2 GB — that's 4.2×10^9 bytes. The café Wi-Fi manages 25 MB per second, or 2.5×10^7 bytes per second. Divide: $4.2 \times 10^9 \div 2.5 \times 10^7 = 1.68 \times 10^2 =$ **168 seconds**, under three minutes. Scientific notation turned a ten-digit division into subtracting two small exponents. *How long would the same movie take on a connection half as fast?*

Scientific notation writes a number as $a \times 10^n$, where $1 \le a < 10$ and n is an integer. It keeps very large and very small numbers readable: Georgia's population of about 11,180,000 people is 1.118×10^7, and 0.00045 is 4.5×10^{-4}. The exponent tells you the *scale* at a glance — and in §1.4, scale is everything.

Multiplying in scientific notation is easy because of the product law: multiply the a-parts, add the exponents. $(2 \times 10^3)(3 \times 10^4) = 6 \times 10^7$.

Worked Example 2 — Scientific notation meets percentage error. Suppose Georgia has exactly 4,523,417 housing units. Write this in scientific notation, round it to one digit (4.5×10^6), and find the percentage error of the rounded form.

Solution. $4{,}523{,}417 = 4.523417 \times 10^6 \approx \mathbf{4.5 \times 10^6}$. Error = $|4{,}500{,}000 - 4{,}523{,}417| = 23{,}417$. Percentage error = $23{,}417 \div 4{,}523{,}417 \times 100\% \approx$ **0.52%**. The rounded form is off by about half a percent — perfectly acceptable for a quick estimate, which is exactly what scientific notation is for.

DECIDE — THE VERDICT

How long should your password be?

Back to the question that opened this section — now with the math to settle it.

Solution. Possible passwords: 62^8 vs 62^{12}. The ratio is $62^{12} \div 62^8 = 62^4 = 14{,}776{,}336$ — the 12-character password has about $\mathbf{1.5 \times 10^7}$ **times** more combinations.

- 8 characters: $62^8 \approx 2.18 \times 10^{14}$ combos → $2.18 \times 10^{14} \div 10^9 \approx 2.18 \times 10^5$ seconds ≈ **2.5 days** to crack.
- 12 characters: $2.18 \times 10^{14} \times 1.48 \times 10^7 \approx 3.23 \times 10^{21}$ combos → 3.23×10^{12} seconds ≈ **100,000 years**.

The decision: four extra characters move the crack time from a long weekend to longer than human civilization. Exponents don't add security — they *multiply* it. This is why a *long* passphrase beats a *short* password in practice, and why "password123!" is a decision you can now price in years.

AI AI Learning Lab 1.3

AI Level 1 — Understand.

Explain why any nonzero number to the power of 0 equals 1, using the dividing pattern 3^3, 3^2, 3^1, 3^0. Explain it as if I'm 15. Then quiz me with 3 questions, one at a time, checking each answer.

AI Level 2 — Apply.

Give me 5 problems mixing the laws of exponents and scientific notation — for example, multiplying $(4 \times 10^5)(2 \times 10^{-3})$ or simplifying $(x^3)^4$. Hide the answers. I'll show every step; then check my work and flag the exact step where I go wrong, if I do.

AI Level 3 — Matter.

The password decision turned four extra characters into 100,000 years of safety. This story matters because every account you own — bank, email, school — is protected by exactly this math. Discuss with me: WHY does exponent intuition matter to ordinary people, not just hackers? What are the implications if millions of people all choose short, clever passwords — who pays the price? Then broaden it: where else does exponential growth quietly run the world — viruses? rumors? investments? — and what goes wrong when people think linearly about it? Ask me for one place in my own life where I'm underestimating an exponent.

Level 4 — Decide. *If this were your bank account, what would you choose?*

> **Example decision problem.** For a new bank password you consider (A) a 4-word passphrase from a 2,000-word list, or (B) a 10-character password from 94 possible characters. A hacker tries a billion (10^9) guesses per second. Which is stronger — and which would you actually choose?

AI Level 4 — Decide.

I need to choose a new bank password: (A) 4 random words from a 2,000-word list, or (B) 10 characters drawn from 94 possible characters. A hacker tries a billion guesses per second. Walk me through the cracking-time math for EACH option — make me do the exponent arithmetic myself. Then interview me: which one would I actually remember, never reuse, and type correctly? Give me a verdict that weighs BOTH the math and the human, and tell me which factor mattered more.

Check your verdict: (A) $2000^4 = 1.6 \times 10^{13}$ combos → about 4.4 hours to crack. (B) $94^{10} \approx 5.4 \times 10^{19}$ combos → about 1,700 years. The math favors (B) by ~3.4 million times — but if you'd reuse (B) everywhere and remember (A) uniquely, the human factor can overrule the math.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 1.3.

Ask your AI: "If you fold a paper 42 times, it would reach the Moon. How many folds would it take to reach the Sun? Show the exponents." Then **verify its arithmetic yourself** — redo the calculation by hand and compute the percentage error of the AI's answer. (The Moon is about 3.84×10^8 m away; the Sun about 1.496×10^{11} m. A folded paper doubles in thickness each time: thickness after n folds = 0.0001×2^n meters.)

TRY IT

Write your height in meters in scientific notation. Now write the

distance to the Moon in scientific notation. How many "yous" stacked up would reach the Moon? Subtract the exponents (the quotient law) to estimate the order of magnitude — then check with your AI.

Practice Problems 1.3

1. Write $5 \times 5 \times 5 \times 5 \times 5$ in exponent form and evaluate it.
2. Evaluate 4^{-2}. Explain in words what a negative exponent means.
3. Explain why $7^0 = 1$ instead of 0.
4. Write 0.00045 in scientific notation.
5. Multiply $(2 \times 10^3)(3 \times 10^4)$. Give the answer in scientific notation.
6. A city has 2,340,000 people. Write this in scientific notation, round it to 2×10^6, and find the percentage error of the rounded value.

1.4 Fermi Problems and Estimation

THE DECISION

Is it worth driving to the cheaper gas station?

Station A (2 minutes away) charges $3.29/gallon. Station B (12 miles away) charges $3.09/gallon — $0.20 cheaper. Your tank holds 13 gallons and your car gets 28 miles per gallon. Should you drive to B?

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Break it into two pieces: money saved at the pump versus money spent on the detour.
- **Hint 2:** The detour is 24 miles round trip — how many gallons is that at 28 mpg, and what does that cost?
- **Hint 3:** Compare $0.20 × 13 gallons against the fuel cost of the detour — and don't forget the drive back.

Thinking in Powers of Ten

STORY

The Cafeteria Fry Count.

At lunch, Sofia bet Priya a cookie that their cafeteria serves more than a million fries a year. Priya laughed — until they estimated it. About 1,200 students buy lunch; roughly half order fries; each serving is about 20 fries; there are 180 school days. That's 1,200 × 0.5 × 20 × 180 ≈ **2.16 million** fries. Sofia collected her cookie. *What would you bet on in your school — and how would you estimate it?*

A **Fermi problem** asks for a quick estimate of a quantity that seems impossible to know exactly — how many piano tuners work in Chicago, or how many ping-pong balls would fill your classroom. Named after physicist Enrico Fermi, who famously estimated the energy of the first atomic bomb by dropping torn paper strips and watching how far the blast wave carried them, the method is:

1. **Break** the big question into smaller pieces you *can* estimate.
2. **Estimate** each piece to the nearest power of ten (don't chase digits).
3. **Multiply** the pieces together.
4. **Sanity-check** the answer — is it in a believable ballpark?

The goal isn't the exact number; it's the right **order of magnitude**. Being off by a factor of 2 is a triumph; being off by a factor of 1,000 means one of your pieces is wrong — and finding *which* piece is the real skill.

KEY IDEA

Fermi estimation works because errors tend to cancel. If you overestimate one piece by 3× and underestimate another by 3×, the product is still about right. That is why rough pieces can build a solid answer — as long as your pieces aren't all biased the same way — and why the sanity check in step 4 is non-negotiable.

CAUTION

Fermi estimates fail when the pieces are not independent. Estimating

"microwaves sold per year" from "households × replacement rate" works; but if your pieces secretly depend on each other (or on the very thing you are estimating), the errors multiply instead of canceling.

Worked Example 1 — Microwaves in Georgia. About how many microwaves are sold in Georgia each year?

Solution. Break it down:

- Georgia's population $\approx 1.1 \times 10^7$ people.
- Average household ≈ 2.6 people → households $\approx 4 \times 10^6$.
- About 90% of households have a microwave → $\approx 3.6 \times 10^6$ microwaves in use.
- A microwave lasts ~10 years → yearly sales $\approx 3.6 \times 10^6 \div 10 \approx \mathbf{4 \times 10^5}$. Answer: on the order of a few hundred thousand per year.

Worked Example 2 — Filling the classroom. How much would it cost to fill an empty math classroom with ping-pong balls?

Solution. Break it down: - Classroom ≈ 8 m × 7 m × 3 m = 168 m³. - A ping-pong ball is ~4 cm across; treat it as a 4-cm cube: volume $\approx (0.04 \text{ m})^3 = 6.4 \times 10^{-5}$ m³. - Number of balls $\approx 168 \div (6.4 \times 10^{-5}) \approx 2.6 \times 10^6$ — a few million balls (packing gaps knock this down a bit, still millions). - At ~$0.25 per ball in bulk: $\approx$ **$650,000**. Answer: on the order of half a million dollars. A fun number to be wrong about — the point is we got there in five lines.

DECIDE — THE VERDICT

Is it worth driving to the cheaper gas station?

Back to the question that opened this section — now with the math to settle it.

Solution. Break it into pieces (Fermi style):

- Savings at B: 13 gal × $0.20/gal = **$2.60**.
- Round trip to B: 24 miles ÷ 28 mpg ≈ 0.86 gal × $3.09/gal ≈ **$2.65** in fuel — before counting ~25 minutes of your time.

The decision: you spend $2.65 to save $2.60 — a net loss, and you haven't been paid for the drive. The $0.20/gallon headline is real; the *savings* are an illusion once the trip itself is part of the equation. Fermi estimation's real job is killing attractive-looking bad deals before they cost you.

AI AI Learning Lab 1.4

AI Level 1 — Understand.

Explain the Fermi estimation method in 4 steps as if I'm 15 years old, using a fun example like "how many pencils does our school use each year?" Then give me one Fermi question to try. I'll show my breakdown, and you critique each piece.

AI Level 2 — Apply.

Give me 3 Fermi problems of increasing difficulty. For each one, I'll show my breakdown and my final order of magnitude. Check whether each of my pieces is reasonable and tell me exactly where my reasoning breaks down.

AI Level 3 — Matter.

The gas-station decision killed an attractive bad deal in five lines of Fermi-style estimation: $2.60 saved, $2.65 spent. This story matters because life is full of "$0.20 cheaper!" headlines designed to bypass your judgment. Discuss with me: WHY does Fermi thinking matter to a citizen, not just a student — what are the implications for spotting misleading ads, viral claims, or political promises? Then the harder question: Fermi estimates can also be wrong — what are the implications of acting confidently on a bad estimate? Ask me for one claim I've seen this week that deserves a Fermi check.

Level 4 — Decide. *If this were your money and your car, what would you choose?*

> **Example decision problem.** A warehouse club costs $60/year. It's 15 miles away (30-mile round trip), your car gets 28 mpg, gas is $3.20/gal. You'd save about $8 per trip shopping twice a month. Should you join?

AI Level 4 — Decide.

I'm deciding whether a $60/year warehouse club membership is worth it. It's 15 miles away, my car gets 28 mpg, gas is $3.20/gal, and I'd save about $8 per trip going twice a month. Be my Fermi sparring partner: make ME break the decision into pieces and estimate each one — then attack my weakest piece like a debater. Give a verdict with the yearly bottom line. Finally, ask me: what single number would flip your answer?

Check your verdict: each trip costs $30 \div 28 \times \$3.20 \approx \$3.43$ in gas; 24 trips a year cost $\approx \$82.29$. Savings: $24 \times \$8 = \$192$. Net: $\$192 - \$82.29 - \$60 \approx +\$49.71$/year — Fermi says join. (Sometimes the trip IS worth it — the method decides, not the headline.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 1.4.

Turn the tables: ask your AI to Fermi-estimate something about *your* life — "how many text messages will I send by age 80?" or "how many hours will I sleep in my lifetime?" Then critique *its* breakdown: which piece is weakest? Where did it pull a number from thin air? Write down the one assumption you would demand evidence for. Estimating is easy; knowing what you don't know is the skill.

TRY IT

Fermi-estimate the number of drops of water in your school's swimming

pool (or the nearest lake, if there is no pool). Four lines maximum. Then ask your AI to do the same and compare breakdowns — whose pieces are better?

Practice Problems 1.4

1. Estimate how many text messages the students at your school send in one day. Show your breakdown.
2. About how many basketballs would fit inside your school gym? Show your reasoning.
3. Estimate the number of hairs on a typical human head. (Hint: estimate the scalp area and the number of hairs per square centimeter.)

1.5 Averages and Weighted Averages

THE DECISION

Where should your next A go?

Your grades: Math C (2.0, 4 credits), English B (3.0, 3 credits), Science B (3.0, 3 credits), Art B (3.0, 2 credits), PE A (4.0, 1 credit).

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

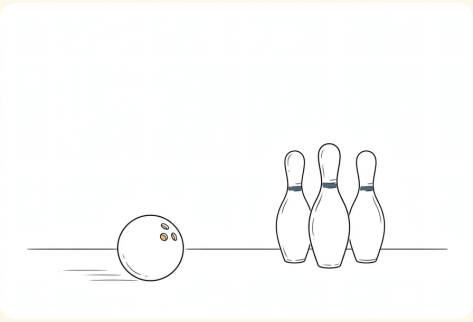
- **Hint 1:** Compute your current GPA first — but weighted by credits, not a simple average.

- **Hint 2:** Now recompute it twice: once with Math raised to a B, once with Art raised to an A.
- **Hint 3:** Compare the two gains — the course with more credits moves the GPA more per grade step.

The Simple Average

STORY

The Bowling Night.



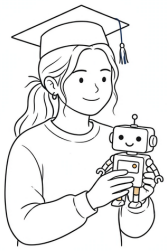
Theo bowled three games: 120, 150, and 180. His average is (120 + 150 + 180) ÷ 3 = **150** — every game counts the same, so the math is just add and divide. *If his fourth game is a 90, where does the average land — and why does one bad game drag it down so far?*

The **average** (or **mean**) of a set of numbers is their sum divided by how many there are. Your three test scores of 80, 90, and 100 average to (80 + 90 + 100) ÷ 3 = **90**. Simple — but only fair when every value counts equally. The moment some values matter more than others, the simple average starts lying.

The Weighted Average

STORY

Elena's Scholarship.



Elena needs a 3.2 GPA to keep her robotics scholarship. Her grades: Math C (2.0, 4 credits), English A (4.0, 2 credits), Art A (4.0, 2 credits). She averaged the three grade points — (2.0 + 4.0 + 4.0) ÷ 3 = 3.33 — and celebrated. Then the counselor weighted by credits: (2.0×4 + 4.0×2 + 4.0×2) ÷ 8 = 24 ÷ 8 = **3.0**. Below the cutoff. The simple average had lied to her, because her heaviest course counted the least in her own math. *Which of your grades is secretly carrying the most weight?*

Often, some values count more than others. A **weighted average** multiplies each value by its **weight** (importance), then divides by the total weight:

Weighted average = Σ(value × weight) ÷ Σ(weights)

Weights are usually percents that add to 100%, but they can be anything — headcounts, dollars, hours. The weights encode a *judgment* about what matters. Two people with the same scores can have different weighted averages if they disagree about the weights — and that disagreement is about values, not arithmetic.

KEY IDEA

Never average averages without weighting.

If Class A (10

students) averages 80 and Class B (30 students) averages 70, the school average is NOT 75. The big class counts three times as much: (80×10 + 70×30) ÷ 40 = 72.5. Averaging the two means pretends both classes are the same size. This mistake appears in news reports, sports statistics, and business dashboards constantly.

Worked Example 1 — Weighted test scores. Your scores are Math 80 (weighted 70%), English 90 (weighted 20%), and PE 100 (weighted 10%). Find the weighted average and compare it with the simple average.

Solution. Simple average = (80 + 90 + 100) ÷ 3 = **90**. Weighted average = 80(0.70) + 90(0.20) + 100(0.10) = 56 + 18 + 10 = **84**. The weighted average is lower because your lowest score (Math) carries the most weight. Which number is "fair" depends on what the weights are *for* — and that is a decision, not just a calculation.

Worked Example 2 — When groups have different sizes. Three classes took the same test: Class 1: average 80, with 10 students. Class 2: average 70, with 30 students. Class 3: average 90, with 20 students. What is the true overall average?

Solution. Averaging the three means gives (80 + 70 + 90) ÷ 3 = 80 — but that's wrong, because the classes aren't the same size. Weight by headcount: (80×10 + 70×30 + 90×20) ÷ (10 + 30 + 20) = 4,700 ÷ 60 ≈ **78.33**. The big, low-scoring class pulls the true average down. Never average averages without weighting.

Worked Example 3 — The coffee mixture. How many pounds of blend A ($9.50/lb) should be mixed with 2 pounds of blend B ($11.75/lb) to make a blend worth $10.00/lb?

Solution. Let x = pounds of blend A. The total value must match: 9.50x + 11.75(2) = 10.00(x + 2) 9.5x + 23.5 = 10x + 20 3.5 = 0.5x → **x = 7 pounds**. Check: (7 × $9.50 + 2 × $11.75) ÷ 9 = $90 ÷ 9 = $10.00/lb. ✓

DECIDE — THE VERDICT

Where should your next A go?

Back to the question that opened this section — now with the math to settle it.

Solution. Simple average of grade points: (2 + 3 + 3 + 3 + 4) ÷ 5 = **3.00**. Weighted by credits: (2×4 + 3×3 + 3×3 + 3×2 + 4×1) ÷ 13 = 36 ÷ 13 ≈ **2.77**. Same transcript, two GPAs — and many schools and scholarships use the weighted one.

Now the decision: you have one strong study push left this semester. Where does it buy the most GPA?

- Raise Math C → B (4 credits): (12 + 9 + 9 + 6 + 4) ÷ 13 = 40 ÷ 13 ≈ 3.08 — a gain of **+0.31**.
- Raise Art B → A (2 credits): (8 + 9 + 9 + 8 + 4) ÷ 13 = 38 ÷ 13 ≈ 2.92 — a gain of **+0.15**.

The decision: the same effort is worth *twice* as much GPA in Math as in Art — because the weights say Math matters more. Weighted averages don't just report the past; they tell you where future points are cheapest. Study accordingly.

AI AI Learning Lab 1.5

AI Level 1 — Understand.

Explain the difference between a simple average and a weighted average using a course-grade example, as if I'm 15. Then quiz me with 3 questions, one at a time, checking each answer before moving on.

AI Level 2 — Apply.

Create 3 weighted-average problems: one about course grades, one about mixing two products, and one about combining groups of different sizes. Hide the answers. I'll solve each and show my steps; check my work and explain any mistake.

AI Level 3 — Matter.

The GPA decision showed the same transcript producing two different GPAs — and the weighted one deciding your future. This story matters because weights are everywhere: college rankings, hiring scores, sports MVPs. Discuss with me: WHY does "weights encode values" matter — who gets to choose the weights, and what are the implications for the people being measured? What happens when the weights are hidden? Ask me: if you could set the weights for one real decision in your life — a college admission, a team tryout — what would they be, and what does that say about what YOU value?

Level 4 — Decide. *If these were your 5 hours, where would you put them?*

> **Example decision problem.** Your History grade: tests average 82 (weight 70%), quizzes 90 (20%), homework 90 (10%) — current grade 84.4. You have 5 study hours this weekend. Each hour of test prep raises your test average 2 points; each hour of homework raises homework 2 points; each hour of quiz prep raises quizzes 3 points. Where should the hours go?

AI Level 4 — Decide.

I have 5 study hours and must allocate them across tests (avg 82, weight 70%), quizzes (90, 20%), and homework (90, 10%). Each test-prep hour gains 2 test points, each homework hour gains 2 homework points, each quiz hour gains 3 quiz points. Make me compute the weighted payoff PER HOUR of each option myself — don't just tell me. Then make me commit to an allocation and defend it. Finally, flip it: what if homework were weighted 40% instead of 10%? Would my decision change?

Check your verdict: per hour, tests gain 2 × 0.70 = 1.4 points, quizzes 3 × 0.20 = 0.6, homework 2 × 0.10 = 0.2. All 5 hours on tests: 84.4 + 7.0 = 91.4 — the heaviest weight wins again.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 1.5.

Give your AI three grading schemes: your real one, a friend's (or an imaginary one), and one you design to be deliberately unfair. Ask: "Which scheme is fairest, and why?" Then argue back — because here is the secret the math can't tell you: *fairness is a decision, not a calculation.* Weights encode values. Write two sentences about what your own course weights say about what your school values most.

TRY IT

Look up your school's actual grading weights (syllabus, handbook, or

ask a teacher). Compute your current weighted average in one class. Now ask: if you could reassign the weights, what would you change — and what does that say about what *you* value?

Practice Problems 1.5

1. Test scores of 88, 92, and 76 carry weights of 30%, 30%, and 40%. Find the weighted average.
2. Two classes take the same quiz: 25 students average 82, and 15 students average 90. What is the combined average?
3. How many pounds of $8/lb trail mix must be blended with 5 pounds of $12/lb trail mix to make a $9/lb blend?
4. Your course grade is homework 95 (20%), quizzes 80 (30%), tests 88 (50%). What is your final grade? Would the simple average of the three scores be higher or lower?

1.6 Check Digits: Catching Errors in ID Numbers

THE DECISION

Should a small business add check digits to its order numbers?

A bakery's loyalty cards use 5-digit member numbers, typed by hand at the register. About 2% of entries are mistyped, and each mistype costs roughly $4 in staff time (void the sale, re-enter, call a manager). The shop handles 500 member transactions a day. A software update adding a Luhn-style check digit would catch essentially every single-digit typo at the moment of entry.

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Start with the mistype count per day — 2% of 500 transactions.
- **Hint 2:** Turn that into dollars per year — a rates problem: dollars per day becomes dollars per year.

- **Hint 3:** Compare the yearly cost of doing nothing against a one-time software update.

Why Check Digits Exist

STORY

The Wrong Fine.

Sofia opened her library account and found a $12 overdue fine — for a book she had never borrowed. The librarian, Mr. Alvarez, had typed member number 204813 as 208413: a single wrong digit, tapped at a busy register. It took three days of emails to untangle. One check digit at the end of every member number would have made the computer flash "invalid" the instant the wrong digit was typed. *How many of the library's 40 monthly typos would that erase?*

Every barcode, ISBN, and credit card number ends with a **check digit** — an extra digit computed from all the others. When a cashier types a number by hand, the check digit lets the computer instantly detect typos. It's simple arithmetic doing quality control, and it catches billions of errors a year without anyone noticing.

The idea is always the same: combine the digits with some weights, and choose the last digit so the total satisfies a simple rule (usually "must be a multiple of 10"). If even one digit is mistyped, the rule breaks — and the computer rejects the number.

UPC Barcodes

STORY

The Self-Checkout Beep.

Nora's cereal box wouldn't scan, so she typed the 12-digit barcode by hand — and the machine buzzed "invalid code." She had mistyped a single digit, and the register knew instantly. *How can a machine detect your typo without knowing what you meant to type?*

A 12-digit UPC code (the barcode on groceries) is validated like this:

1. Add the digits in the **odd** positions (1st, 3rd, 5th, ...).

2. Multiply that sum by 3.
3. Add the digits in the **even** positions.
4. The check digit is whatever makes the total a multiple of 10.

Why multiply by 3? Because weights of 1 and 3 catch the most common human error — typing a single wrong digit. (Whether they catch *swapped* digits is a subtler question. See the Deep-Dive Challenge.)

Worked Example 1 — Verifying a UPC. Check the code 0-36000-29145-2 (the last digit, 2, is the check digit).

Solution. Odd positions: 0, 6, 0, 2, 1, 5 → sum = 14. Even positions: 3, 0, 0, 9, 4 → sum = 16. Total = 3 × 14 + 16 = 58. The next multiple of 10 is 60, so the check digit is 60 − 58 = **2**. ✓ It matches — the code is valid. If any single digit were mistyped, the total would no longer land on a multiple of 10, and the error would be caught.

ISBN-13 (Book Numbers)

STORY

The Book Order.

Mr. Alvarez ordered 30 copies of a novel for his class and emailed the 13-digit ISBN to the bookstore. His finger slipped on one digit — but the bookstore's computer rejected the number before anyone shipped the wrong book. *What hidden arithmetic inside those 13 digits caught the mistake?*

Every book's 13-digit ISBN works similarly, with alternating weights of 1 and 3: multiply the 1st digit by 1, the 2nd by 3, the 3rd by 1, and so on, add them up, and the check digit brings the total to a multiple of 10.

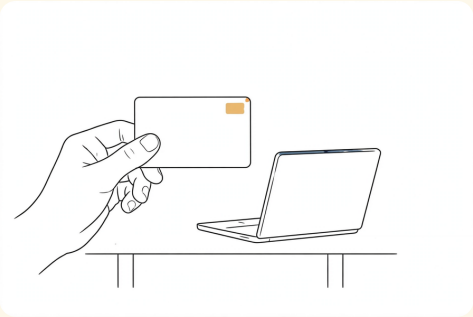
Worked Example 2 — Completing an ISBN. Find the check digit for 978-1-4028-9462-?.

Solution. Weighted sum: 9(1) + 7(3) + 8(1) + 1(3) + 4(1) + 0(3) + 2(1) + 8(3) + 9(1) + 4(3) + 6(1) + 2(3) = 9 + 21 + 8 + 3 + 4 + 0 + 2 + 24 + 9 + 12 + 6 + 6 = 104. 104 mod 10 = 4, so the check digit is 10 − 4 = **6**. The complete ISBN is 978-1-4028-9462-6. (This very book will get one when it's published!)

The Luhn Algorithm (Credit Cards)

STORY

The Online Typo.



Lena was buying concert tickets when her finger slipped and she typed a 6 as a 9 in her card number. The site flashed "invalid card number" before she even clicked Pay — no bank contacted, no charge attempted. *How did the website know the number was wrong using only the digits themselves?*

Credit cards use the **Luhn algorithm**: starting from the rightmost digit (the check digit) and moving left, double every *second* digit; if a doubled value exceeds 9, add its digits (equivalently, subtract 9). Add everything up — a valid number sums to a multiple of 10.

Why does doubling help? Because doubling spreads single-digit errors across the total: changing one digit changes the sum by a nonzero amount that is never a multiple of 10, so the check fails and the typo is caught.

Worked Example 3 — Verifying a card number. Is 4539 1488 0343 6467 a valid number under the Luhn check?

Solution. From the right, doubling every second digit: 7, (6→3), 4, (6→3), 3, (4→8), 3, (0→0), 8, (8→7), 4, (1→2), 9, (3→6), 5, (4→8). Sum: 7+3+4+3+3+8+3+0+8+7+4+2+9+6+5+8 = 80. 80 is a multiple of 10, so the number is **valid**. (Don't worry — this is a made-up example number, not a real card!)

DECIDE — THE VERDICT

Should a small business add check digits to its order numbers?

Back to the question that opened this section — now with the math to settle it.

Solution. Errors per day: 500 × 0.02 = 10. Cost per day: 10 × $4 = $40. Cost per year: $40 × 365 ≈ **$14,600**.

The decision: $14,600 a year evaporates into typo-fixing — against a one-time software update. The check digit doesn't sell a single extra croissant; it *stops a leak*. That is the quiet power of error-detecting math: it protects decisions (and dollars) you already made.

AI AI Learning Lab 1.6

AI Level 1 — Understand.

Explain what a check digit is and why barcodes need one, as if I'm 15. Then walk me through verifying one UPC code step by step — let ME do each step and confirm before moving on.

AI Level 2 — Apply.

Give me 3 check-digit problems: one UPC to verify, one ISBN-13 check digit to compute, and one Luhn check on a made-up card number. Hide the answers. I'll show every step of my work; check each step and tell me exactly where I go wrong.

AI Level 3 — Matter.

The bakery decision showed $14,600 a year evaporating into typos — stopped by one silent check digit. This story matters because check digits are invisible math protecting you hundreds of times a day: every barcode, every card swipe, every ISBN. Discuss with me: WHY does invisible math matter — what are the implications of a world WITHOUT error detection (wrong medicine doses? lost money? misdirected shipments?)? Who does the quiet work of preventing errors in the systems you rely on? Ask me: what's one system in my life where a tiny error would be catastrophic — and does it have a check digit?

Level 4 — Decide. *If you were the library director, what would you recommend?*

> **Example decision problem.** Your library mistypes 40 member numbers a month (like Sofia's $12 fine in the Wrong Fine story). Each takes 20 minutes of staff time at $18/hour to untangle. A check-digit system costs $800 one-time and would catch nearly all typos. Should the library buy it?

AI Level 4 — Decide.

I'm advising my library: 40 mistyped member numbers a month, 20 minutes each at $18/hour to fix, and a one-time $800 check-digit system that catches nearly all typos. Walk me through the yearly cost of doing nothing — make me do each multiplication myself. Then give a recommendation with the payback time. Finally, stress-test it: at what system price would your recommendation flip?

Check your verdict: 480 typos/year × 20 min = 160 hours × $18 = $2,880/year down the drain. Payback: $800 ÷ $2,880 ≈ 0.28 years ≈ 3.3 months — buy it.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 1.6.

Ask your AI to invent a 12-digit UPC, compute its check digit, and show its work. Verify the AI's arithmetic yourself (remember the golden rule!). Then introduce a deliberate typo into the code and see if YOU can catch it with the check-digit method before the AI tells you the answer. Finally, ask the AI: "What kinds of errors does this method NOT catch?" — and make it prove its answer with an example.

TRY IT

Pick up any packaged item near you. Find the UPC barcode. Verify its

check digit by hand. Then ask your AI to verify it too — and compute the percentage error between its total and yours (it should be 0%).

Practice Problems 1.6

1. Verify the check digit of the UPC 0-36000-29145-2 using the method in Worked Example 1. Then change one digit and show that the check fails.
2. Compute the check digit for the ISBN-13 prefix 978-0-13-110362-?.
3. Use the Luhn algorithm to check whether 4539 1488 0343 6467 is valid. (You should get a sum of 80.)
4. A cashier mistypes one digit of a 12-digit UPC. Explain in your own words why the check digit will catch the mistake.

Vocabulary

- **Ratio** — a comparison of two quantities, written a : b or a/b.
- **Rate** — a ratio comparing quantities with different units (e.g., miles per hour).
- **Unit rate** — a rate with denominator 1 (e.g., dollars per *one* pound).
- **Unit price** — a unit rate for money per quantity (e.g., dollars per ounce).
- **Proportion** — an equation stating that two ratios are equal.
- **Percent** — "per hundred"; a number expressed as parts per 100 (e.g. 37.5% = 37.5 per 100).
- **Percentage point** — an absolute difference between percents (50% vs. 40% is 10 percentage points).
- **Percent change** — (new − old) ÷ old × 100%.

- **Absolute error** — |estimated value − actual value|.
- **Percentage error** — absolute error ÷ actual value × 100% (also called absolute percentage error).
- **Exponent (index)** — the number telling how many times a base is multiplied by itself.
- **Base** — the number being multiplied in a power.
- **Scientific notation** — writing a number as $a \times 10^n$ with $1 \leq a < 10$.
- **Order of magnitude** — the power of ten nearest to a quantity; the scale of an estimate.
- **Estimate** — an approximate value, often to the nearest power of ten.
- **Fermi problem** — an estimation problem solved by breaking it into smaller estimable pieces.
- **Mean (average)** — sum of values divided by the number of values.
- **Weighted average** — Σ(value × weight) ÷ Σ(weights); values count in proportion to their weights.
- **Weight** — the relative importance assigned to a value in a weighted average.
- **Check digit** — an extra digit computed from the other digits of an ID number, used to detect errors.
- **Luhn algorithm** — the check-digit method used for credit card numbers.
- **Prompt** — the instruction or question you give an AI chatbot; the main tool for directing AI learning in this book.

Summative Task: The 3-Paragraph Decision Essay

The prompt. Your friend just got their first credit card and says: "The bank set the minimum payment at $50, so paying $50 a month must be a sensible plan." You know the numbers from this chapter: a $2,400 balance at 24% APR (2% per month). Write a 3-paragraph essay: should your friend pay the $50 minimum, or stretch to $200 a month? Defend your recommendation with this chapter's math — and be honest about when your advice would change.

Planning your essay (do this before you write). 1. **Compute first.** Run both plans yourself: $2,400 at 2% per month. The $50 minimum → ≈163 months, ≈$8,130 total, ≈$5,730 interest. $200 a month → ≈14 months, ≈$2,772 total, ≈$372 interest. Don't take these on faith — verify them with your own calculation. 2. **Thesis template.** "I recommend ___ because ***, even though*** ." Fill it in for YOUR side — either side is defensible if the numbers back it. 3. **Evidence checklist.** Every claim needs a number behind it: months, totals, and interest for each plan, plus at least one idea from this chapter (percent, unit rate, or average) that strengthens the case. 4. **Flip test.** Name one specific condition that would change your

recommendation — a 0% balance-transfer offer? a lost job? If you can't name one, your argument isn't finished.

Paragraph 1 — Your decision and claim. State your recommendation in one clear sentence, then one or two sentences on why it matters. (Example thesis — either side is fine: "Pay $200 a month, because the $50 minimum costs an extra $5,730 in interest." Or the other side: "Pay the $50 minimum, because the extra $150 a month belongs in emergency savings.")

Paragraph 2 — Mathematical evidence. Show the numbers: months to payoff, total paid, and interest under each plan (≈163 months / $8,130 total / $5,730 interest vs. ≈14 months / $2,772 total / $372 interest). Use unit rates, percent, or averages from this chapter wherever they strengthen the case. Every claim needs a number behind it.

Paragraph 3 — Implications, limitations, and what would change your mind. Who profits when borrowers don't do this math? What are the limits of your analysis (for instance: what if your friend loses income and *can't* pay $200)? Name one specific condition that would flip your recommendation — for example, a 0% balance-transfer offer — and explain why.

Length: 200–300 words. Three paragraphs, no more, no fewer.

Using AI. Write your first draft entirely on your own — no AI. Then you may use AI as a revision coach: ask it to check your logic, flag weak evidence, and suggest clearer sentences. Never let it write paragraphs for you. And remember this book's golden rule: AI makes mistakes — verify everything, especially any numbers it gives you.

Rubric and exemplar essays are in the Teacher Guide.

Answer Key — Practice Problems

1.1 — 1. 8 cups of white paint. 2. 30 miles per gallon; 420 miles on 14 gallons. 3. 3 : 4; yes — both reduce to 2/3. 4. 35 boxes/hour; 280 boxes in 8 hours. 5. 6 apples: $0.40/apple; 10 apples: $0.38/apple — the 10-apple bag is the better buy.

1.2 — 1. 37.5%. 2. $45.00. 3. $|6 - 5.2| \div 5.2 \times 100\% \approx 15.4\%$. 4. 100 → 110 → 99: a net 1% decrease. 5. ($200 + $350) ÷ $800 = 68.75%.

1.3 — 1. $5^5 = 3{,}125$. 2. $4^{-2} = 1/16$; a negative exponent means "take the reciprocal" — divide 1 by the positive power. 3. $7^0 = 7^3 \div 7^3 = 343 \div 343 = 1$; each step down in the exponent divides by 7, so the pattern 343, 49, 7, … continues to 1. 4. 4.5×10^{-4}. 5. 6×10^7. 6. 2.34×10^6; percentage error = 340,000 ÷ 2,340,000 × 100% ≈ 14.5%.

1.4 — Sample approaches (answers will vary; any reasonable breakdown earns credit): 1. Students × texts per student: e.g., 1,500 students × 50 texts/day ≈ 75,000 texts/day. 2. Gym volume ÷ basketball volume: e.g., (30 × 20 × 8) m^3 ÷ (0.24 m)^3 ≈ 4,800 ÷ 0.0138 ≈ 350,000 basketballs — on the order of a few hundred thousand. 3. Scalp ≈ 600 cm^2 × ~150–200 hairs/cm^2 ≈ 10^5 hairs (about a hundred thousand).

1.5 — 1. 88(0.30) + 92(0.30) + 76(0.40) = 26.4 + 27.6 + 30.4 = 84.4. 2. (25 × 82 + 15 × 90) ÷ 40 = 3,400 ÷ 40 = 85. 3. 8x + 12(5) = 9(x + 5) → x = 15 pounds. 4. 95(0.20) + 80(0.30) + 88(0.50) = 19 + 24 + 44 = 87; the simple average (95 + 80 + 88) ÷ 3 = 87.67 is slightly higher.

1.6 — 1. Odd-position sum 14, even-position sum 16, total 58 → check digit 2 ✓. Changing any single digit changes the total so it no longer reaches a multiple of 10 (try it!). 2. Weighted sum 73 → check digit 7: 978-0-13-110362-7. 3. Luhn sum = 80 → valid. 4. Sample: the check digit is chosen so the weighted total is a multiple of 10; changing one digit changes the total by a nonzero amount that isn't a multiple of 10, so the check fails and the typo is caught.

Level 4 example checks — **1.1:** big bag usable = 15 oz → $0.50/oz; small bag $0.42/oz — buy the small bag. **1.2:** ≈32 months, ≈$786 total vs $600 cash — the "discount" costs ≈$186 extra; don't open the card at minimum payments. **1.3:** passphrase ≈4.4 hours to crack; 10-char complex ≈1,700 years — (B) wins the math ~3.4 million times over, but pick the one you'll actually keep unique. **1.4:** net ≈ +$49.71/year — join the club. **1.5:** test-prep hour = +1.4 weighted points (best); all 5 hours on tests → 91.4. **1.6:** $2,880/year in typo-fixing; $800 system pays back in ≈3.3 months — buy it.

CHAPTER 2

Geometric Ratios

A Note on AI Learning Labs in This Chapter

Chapter 1 introduced the AI Learning Labs and the golden rule — **AI makes mistakes, so verify everything**. That rule matters *more* in this chapter, because trigonometry runs on decimals your calculator produces and an AI merely predicts. Your standing assignment continues: whenever an AI gives you a numerical answer, compute it yourself and find the percentage error between the two (§1.2). A chatbot that confidently says sin(35°) = 0.71 is close — but "close" is exactly what percentage error was invented to judge.

About This Chapter

Some distances you can measure with a tape — and some you can't. How tall is that tree without climbing it? How wide is the river without crossing it? How long a beam do you need to brace a solar panel at exactly the right tilt? Geometry answers these questions with ratios, and the decisions that follow — where to build, what to buy, how to design — are what this chapter is about.

We begin with the Pythagorean theorem and the distance formula, then meet the three trigonometric ratios (sine, cosine, tangent) and use them to solve right triangles in real situations: heights, shadows, ramps, and angles of elevation and depression. From there we go further: the area of a triangle from two sides and an angle, the Sine Rule and Cosine Rule for triangles with no right angle, and finally problems in three dimensions — the longest rod that fits in a box, the angle of a space diagonal, and guy-wires holding up a tower.

By the end of this chapter, you will be able to:

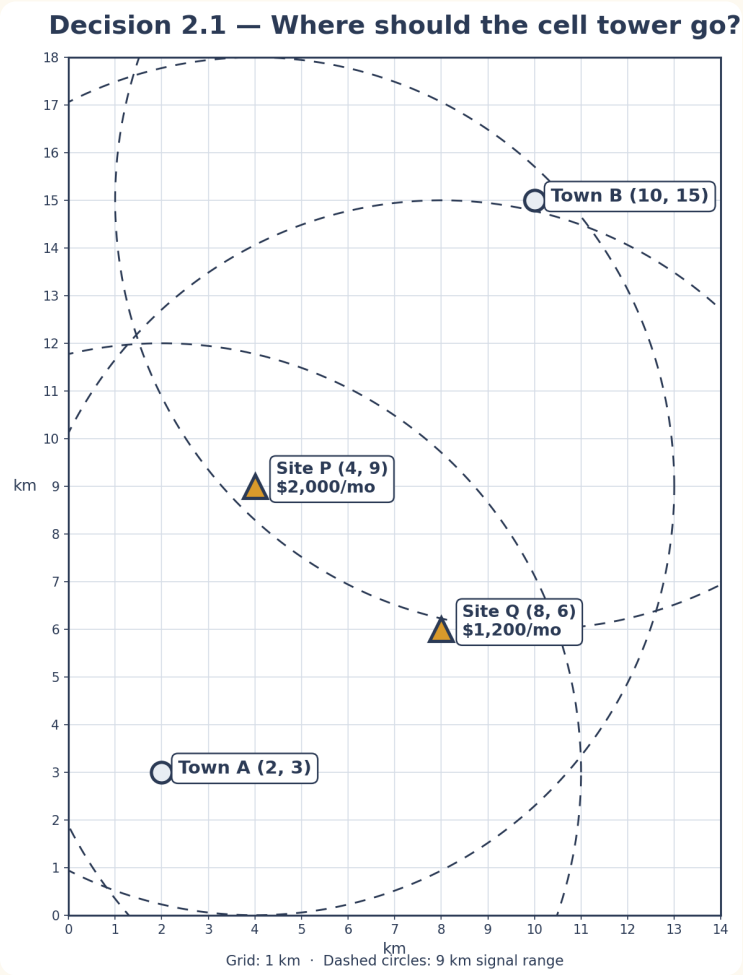
- Use the Pythagorean theorem and the distance formula to find unknown lengths.
- Compute sine, cosine, and tangent for an angle, and use inverse trigonometric functions to recover angles.
- Solve right-triangle problems involving angles of elevation and depression.
- Find the area of a triangle using the formula ½ab·sin C.
- Apply the Sine Rule and the Cosine Rule to solve non-right triangles, including inaccessible-distance problems.

- Solve three-dimensional problems involving space diagonals and angles.
- Use AI prompts at four levels (understand, apply, matter, decide) to learn trigonometry — and verify AI-generated trig values with your calculator and percentage error.

2.1 The Pythagorean Theorem and the Distance Formula

THE DECISION

Where should the cell tower go?



Two towns need coverage: Town A at (2, 3) and Town B at (10, 15) on the county map (units: km). A tower's signal reaches 9 km. Two sites are available: Site P at (4, 9) leasing for $2,000/month, and Site Q at (8, 6) leasing for $1,200/month. (Building a brand-new tower instead would cost $50,000 — keep that number in mind.)

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

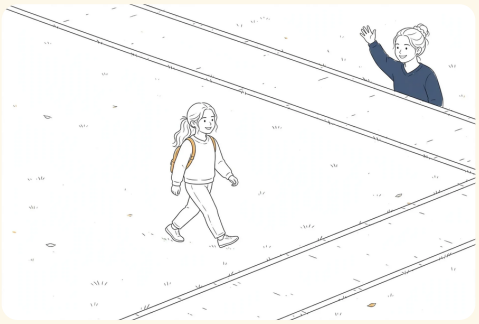
- **Hint 1:** For each site, find the distance to each town — that's four distances.

- **Hint 2:** Compare every distance with the 9 km signal range — coverage is a yes-or-no test before price matters.
- **Hint 3:** A cheaper lease means nothing if a town falls outside the circle — check both towns against 9 km for each site, then decide.

Right Triangles

STORY

The Park Shortcut.



Every day, Amara walks 300 m east along Main Street, then 400 m north to the library — 700 m total. Her friend Kai cuts diagonally across the park instead: $\sqrt{(300^2 + 400^2)} = \sqrt{250{,}000} =$ **500 m**. Kai saves 200 m each way. At a walking speed of 80 m per minute, that is 2.5 minutes saved each way — 5 minutes a day, 25 minutes a school week. *Where is the hidden hypotenuse in your daily walk?*

A **right triangle** has one angle of exactly 90°. The side opposite the right angle is the **hypotenuse** — always the longest side. The other two sides are the **legs**.

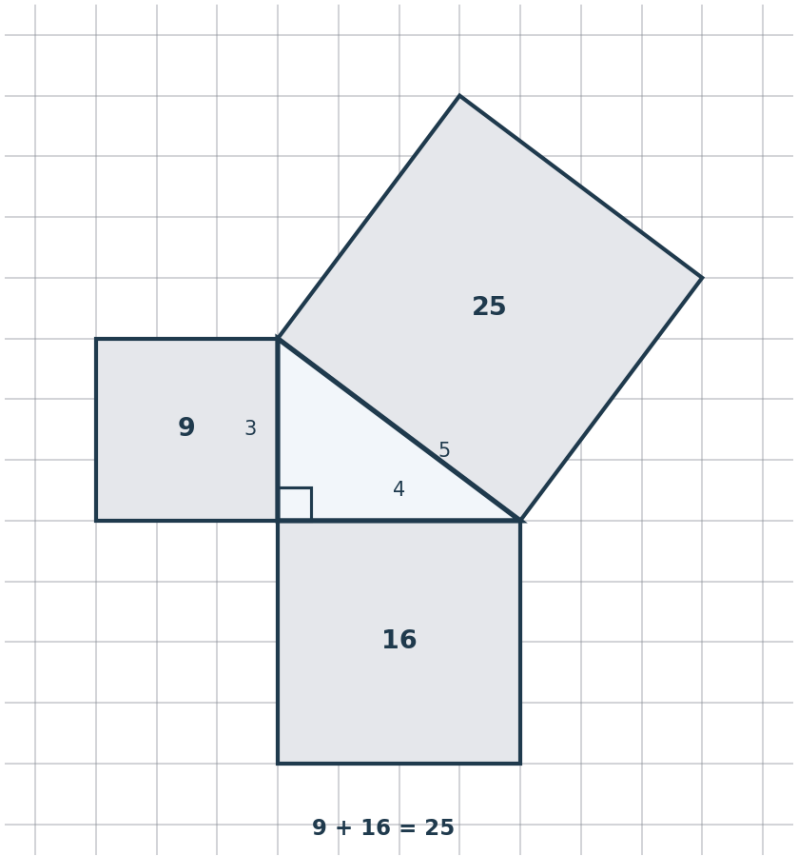
The **Pythagorean theorem** relates the three sides: if the legs have lengths a and b and the hypotenuse has length c, then

$$\mathbf{a^2 + b^2 = c^2}$$

This is the single most useful fact in this chapter. Everything else — including the distance formula and trigonometry — grows out of it. When this chapter feels complicated, come back to this sentence: it is all Pythagoras wearing different costumes.

Fig. 2.1

Fig. 2.1 — The theorem as areas



$a^2 + b^2 = c^2$ is a statement about areas, not just lengths.

Why does the formula work? Picture a square built on each side of the right triangle. The theorem says the two smaller squares *exactly fill* the largest one: 9 + 16 = 25. It is a statement about areas, not just lengths — and that picture is worth more than a memorized formula, because it tells you *why* squaring appears at all.

The theorem also works backwards, and carpenters depend on it. If three lengths satisfy $a^2 + b^2 = c^2$, the triangle *must* contain a right angle. A 3-4-5 triangle (or any multiple: 6-8-10, 9-12-15) is a portable square corner: stretch a rope into 3-4-5 segments, and the angle between the 3 and 4 sides is guaranteed 90°. That is why the 3-4-5 triangle appears everywhere in construction and surveying.

Worked Example 1 — The 3-4-5 triangle. A right triangle has legs of 3 cm and 4 cm. Find the hypotenuse.

Solution. $c^2 = 3^2 + 4^2 = 9 + 16 = 25$, so c = **5 cm**.

Worked Example 2 — Finding a leg. A 10-foot ladder leans against a wall with its base 6 feet from the wall. How high up the wall does it reach?

Solution. The ladder is the hypotenuse (c = 10) and the ground distance is one leg (a = 6). Solve for the other leg: $b^2 = c^2 - a^2 = 100 - 36 = 64$, so b = **8 feet**.

CAUTION

Which side is c?

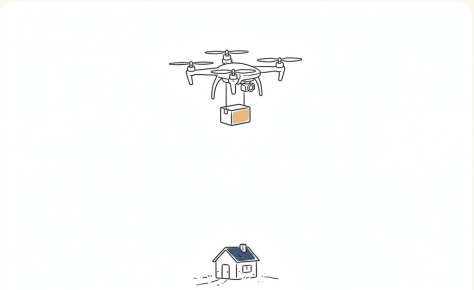
The most common Pythagoras mistake is plugging

numbers into $a^2 + b^2 = c^2$ without asking which side is the hypotenuse. The ladder problem above is the classic trap: 10 is the *hypotenuse*, not a leg. Always label the triangle first — hypotenuse gets c, always.

The Distance Formula

STORY

The Drone Delivery.



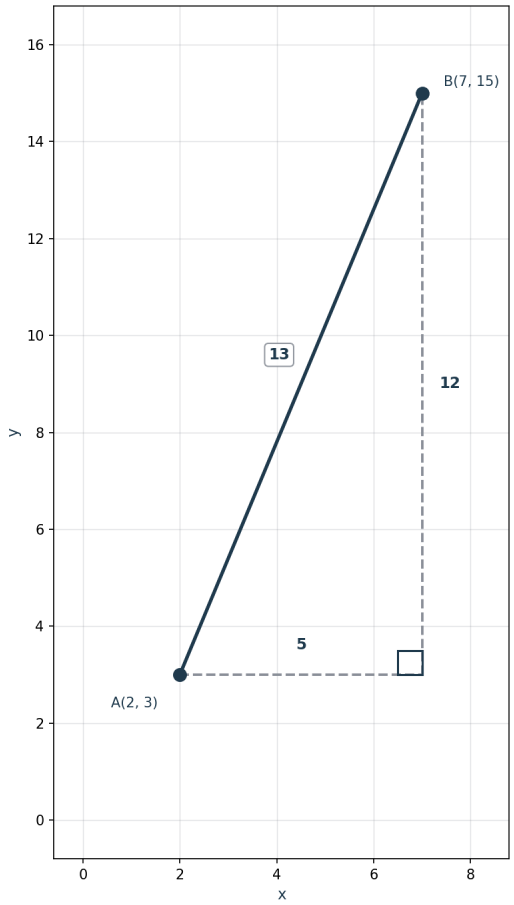
A delivery drone lifts off from the warehouse at map point (0, 0) and flies straight to a house at (9, 12) — blocks east, blocks north. The pilot needs no streets, only the distance formula: $\sqrt{(9^2 + 12^2)} = \sqrt{225} =$ **15 blocks**. *If each block is 100 m and the drone flies at 12 m/s, how many seconds is the trip?*

The Pythagorean theorem also measures distances on a coordinate grid. The horizontal and vertical gaps between two points are the legs of a right triangle, and the distance between the points is the hypotenuse:

distance $= \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

Fig. 2.2

Fig. 2.2 — Distance as a hidden right triangle



The gaps are the legs; the distance is the hypotenuse.

Two things worth noticing. First, only the *gaps* matter — where the points sit on the grid is irrelevant. Second, the order of subtraction cannot hurt you: $(2 - 7)^2 = (-5)^2 = 25$, exactly the same as $(7 - 2)^2$. Squaring erases the sign, so "x_2 minus x_1" and "x_1 minus x_2" give the same distance. That is not a coincidence; it is the geometry telling you distance has no direction.

Worked Example 3 — Distance between two points. Find the distance between A(2, 3) and B(7, 15).

Solution. The horizontal gap is $7 - 2 = 5$ and the vertical gap is $15 - 3 = 12$. Distance $= \sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} =$ **13 units**. Notice the 3-4-5 pattern hiding inside: 5-12-13 is another Pythagorean triple.

TRY IT

Pick any two points on graph paper — your choice. Compute the

distance with the formula, then measure it with a ruler (agree on a scale first). How close are you? If they disagree, which one do you trust, and why?

DECIDE — THE VERDICT

Where should the cell tower go?

Back to the question that opened this section — now with the math to settle it.

Solution. Use the distance formula from each site to each town:

- Site P → A: $\sqrt{((4-2)^2 + (9-3)^2)} = \sqrt{40} \approx$ **6.3 km** ✓ in range.
- Site P → B: $\sqrt{((4-10)^2 + (9-15)^2)} = \sqrt{72} \approx$ **8.5 km** ✓ in range.
- Site Q → A: $\sqrt{((8-2)^2 + (6-3)^2)} = \sqrt{45} \approx$ **6.7 km** ✓ in range.
- Site Q → B: $\sqrt{((8-10)^2 + (6-15)^2)} = \sqrt{85} \approx$ **9.2 km** ✗ out of range.

The decision: the cheaper site fails the only test that matters — it leaves Town B uncovered. Covering B from Site Q would need a second tower ($50,000 to build), dwarfing the $800/month lease savings. The distance formula just saved the county from a $50,000 mistake: pick Site P.

AI AI Learning Lab 2.1

AI Level 1 — Understand.

Explain the Pythagorean theorem to me as if I'm 15, using the "squares on each side" picture — not just the formula. Then ask me 3 questions, one at a time, to check whether I can tell which side is the hypotenuse in different diagrams.

AI Level 2 — Apply.

Give me 4 problems: two "find the hypotenuse," one "find a leg," and one distance-formula problem with coordinates. Hide the answers. I'll solve each and show my work; then check my work step by step and tell me exactly where I went wrong if I made a mistake.

AI Level 3 — Matter.

The cell-tower decision showed the cheaper site leaving Town B with no signal — a $50,000 mistake the distance formula prevented. This story matters because public decisions like this are made with YOUR tax money, and "cheapest bid wins" is how governments usually buy. Discuss with me: WHY does this matter to the people of Town B — what's at stake beyond money (emergency calls? jobs? property values?)? What are the implications when decision-makers can't do the math themselves and just trust the lowest price? Ask me: what's one public decision in our county I'd now want to check the math on?

Level 4 — Decide. *If you were advising the council, what would you recommend?*

> **Example decision problem.** Two neighborhoods need fire-station coverage: N1 at (1, 2), N2 at (9, 8) on the town map (km). Trucks average 60 km/h and must reach both within 6 minutes (a 6 km range). Site X at (5, 5) costs $300,000 to build; Site Y at (3, 7) costs $210,000. Which site?

AI Level 4 — Decide.

I'm advising the town council on a fire-station site. Neighborhoods at (1, 2) and (9, 8); trucks average 60 km/h and must reach both within 6 minutes — that's a 6 km range. Site X at (5, 5) costs $300,000; Site Y at (3, 7) costs $210,000. Make me compute all four distances with the distance formula myself, then test each against the 6 km range. Then role-play the debate with me: you play the budget hawk AND the fire chief, and I cast the deciding vote with written reasons.

Check your verdict: X covers both (5.0 km, 5.0 km). Y covers N1 ($\sqrt{29} \approx 5.39$ km) but misses N2 ($\sqrt{37} \approx 6.08$ km) by about 80 m — roughly 5 seconds of driving. The decision: is 5 seconds worth $90,000? There is no formula for that — only judgment.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 2.1.

Ask your AI: "Invent a real-world Pythagorean theorem word problem with whole-number answers, and solve it." Then do the detective work: check every number in its solution with your own calculation. Does $a^2 + b^2$ really equal c^2? Compute the percentage error (§1.2) between the AI's answer and yours. AI chatbots are fluent storytellers and sloppy arithmeticians — this chapter is where you learn to catch them.

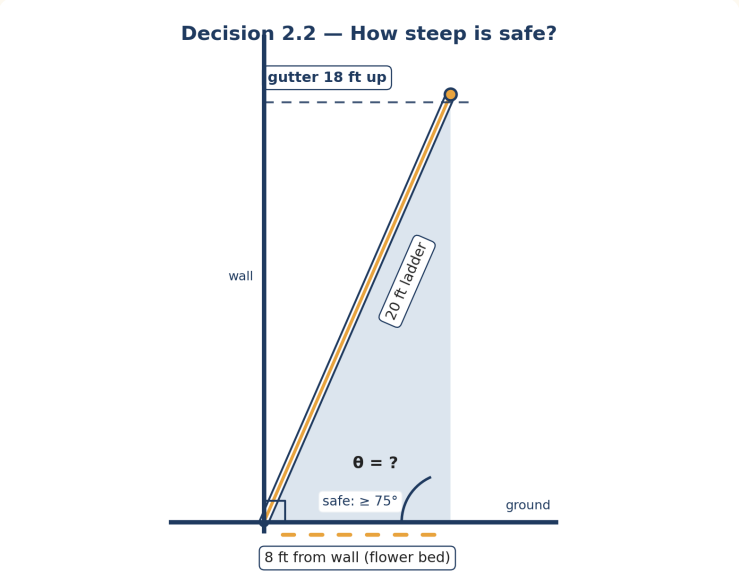
Practice Problems 2.1

1. A 10-foot ladder stands with its base 6 feet from a wall. How high does it reach? (This is Worked Example 2 — write the equation yourself.)
2. A right triangle has legs of 5 cm and 12 cm. Find the hypotenuse.
3. Find the distance between R(1, 4) and S(7, 12) using the distance formula. Show the horizontal and vertical gaps first.
4. A rectangular garden is 9 m by 12 m. How long is its diagonal?

2.2 Trigonometric Ratios: Sine, Cosine, Tangent

THE DECISION

How steep is safe? — the ladder decision.



Safety guidelines say a ladder should lean at about 75° (the "4-to-1 rule": 1 foot out for every 4 feet up). You have a 20-foot ladder and need to reach a gutter 18 feet up. A flower bed forces the ladder's base 8 feet from the wall. Is this setup safe?

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

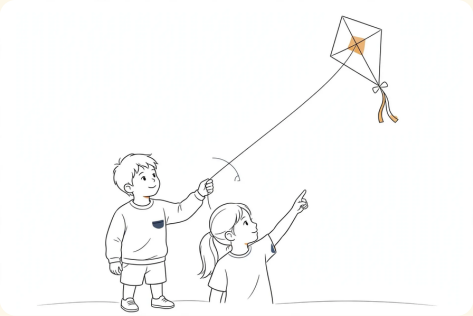
Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Sketch the triangle: the 20-foot ladder is the hypotenuse and the 18-foot height is one leg — what is unknown?
- **Hint 2:** This is a trig-ratio question — which ratio connects the base angle to the sides you know?
- **Hint 3:** Solve for the actual base angle, then ask: is it at or above the recommended 75°?

SOH-CAH-TOA

STORY

The Kite Argument.



Diego flew his kite on 100 m of string, and the string made a 40° angle with the ground. His sister Luna insisted the kite was "about 100 meters up — that's how much string is out!" Diego disagreed: height = 100 × sin 40° ≈ 100 × 0.643 ≈ **64 m**. The string is the hypotenuse; the height is only the opposite side. Luna owed him a soda. *When someone confuses the hypotenuse with the height, which ratio settles the argument?*

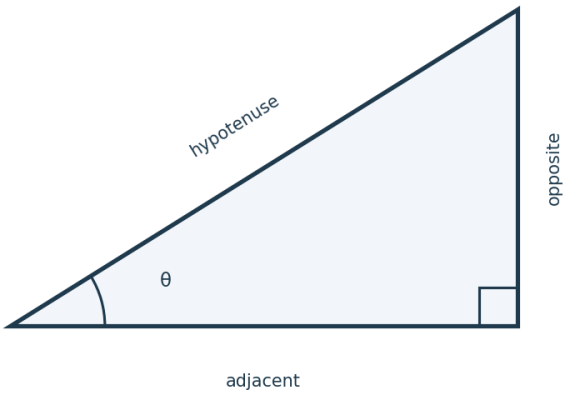
Pick one of the acute angles in a right triangle and call it θ (theta). The side across from θ is the **opposite** side, the side next to θ (that isn't the hypotenuse) is the **adjacent** side, and the hypotenuse stays the hypotenuse. The three trigonometric ratios are:

- **Sine:** sin θ = opposite / hypotenuse — **S**ine = **O**pposite / **H**ypotenuse
- **Cosine:** cos θ = adjacent / hypotenuse — **C**osine = **A**djacent / **H**ypotenuse
- **Tangent:** tan θ = opposite / adjacent — **T**angent = **O**pposite / **A**djacent

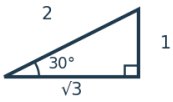
The memory aid is **SOH-CAH-TOA**.

Fig. 2.3

Fig. 2.3 — Naming the sides



30-60-90 reference



sin(30°) = 1/2

cos(30°) = √3/2

tan(45°) = 1

The names depend on your chosen angle θ — pick a different angle and 'opposite' and 'adjacent' swap.

Here is the deep idea students usually miss: these ratios depend **only on the angle**, not on the size of the triangle. Double every side and opposite/hypotenuse does not change — the 2's cancel. Every right triangle with a 30° angle is *similar* to every other (same angles, scaled sides), so every one of them has sin(30°) = 0.5. That is why your calculator can "look up" the sine of an angle: the answer was fixed by geometry long before you picked up the triangle.

Two special triangles are worth knowing exactly, no calculator needed. In a 45-45-90 triangle the legs are equal (say 1 and 1), so the hypotenuse is $\sqrt{2}$ and $\sin(45°) = \cos(45°) = 1/\sqrt{2} \approx 0.707$, $\tan(45°) = 1$. In a 30-60-90 triangle the sides are 1, $\sqrt{3}$, 2, so $\sin(30°) = 1/2$ exactly. These are the triangles your calculator is secretly thinking about.

CAUTION

Degree mode.

Before you touch a trig button, check that your

calculator is in *degree* mode, not radian mode. sin(30°) = 0.5 — but sin(30) in radian mode ≈ −0.988, a completely different (and wrong-for-this- chapter) answer. This one setting has sunk more homework assignments than any actual misunderstanding.

Worked Example 1 — Reading the ratios. In a right triangle, the side opposite θ is 5, the side adjacent to θ is 12, and the hypotenuse is 13. Find sin θ, cos θ, and tan θ.

Solution. sin θ = 5/13 ≈ **0.385**, cos θ = 12/13 ≈ **0.923**, tan θ = 5/12 ≈ **0.417**.

Using a Calculator — and Going Backwards

STORY

The Slide.

The park's new slide is 12 ft long, and its top platform stands 8 ft off the ground. The designer needs the angle where the slide meets the ground — going backwards from ratio to angle: $\sin^{-1}(8 \div 12) \approx$ **41.8°**. *Her calculator has an inverse button and a degree/radian switch — which setting does she need, and what goes wrong on the other one?*

Your calculator gives the ratio for any angle: tan(20°) ≈ 0.364, sin(30°) = 0.5 exactly, cos(40°) ≈ 0.766.

Inverse trigonometric functions go the other way: if sin x = 0.5, then x = $\sin^{-1}(0.5)$ = **30°**. The $^{-1}$ button (often labeled arcsin, arccos, arctan) recovers the angle from the ratio. Think of it as the "undo" button: sin turns an angle into a ratio, $\sin^{-1}$ turns a ratio back into an angle. (In this chapter the answers are always acute angles; Chapter 1's "undo" idea from inverse operations applies here too.)

Worked Example 2 — Finding an angle. If tan x = 1, what is the acute angle x?

Solution. x = $\tan^{-1}(1)$ = **45°**. (Check: in a 45-45-90 triangle the legs are equal, so opposite/adjacent = 1. ✓)

Worked Example 3 — Finding a missing side. A right triangle has a 25° angle, and the side adjacent to it is 20 cm long. Find the opposite side and the hypotenuse.

Solution. Use tangent for the opposite side: tan(25°) = opposite / 20, so opposite = 20 × tan(25°) ≈ 20 × 0.4663 ≈ **9.33 cm**. Use cosine for the hypotenuse: cos(25°) = 20 / hypotenuse, so hypotenuse = 20 / cos(25°) ≈ 20 / 0.9063 ≈ **22.07 cm**.

TRY IT

Cover the solutions above. For Worked Example 3, decide *before*

calculating: which ratio uses the side you know (adjacent) and the side you want (opposite)? Choosing the right ratio is the whole skill — the calculator does the arithmetic.

DECIDE — THE VERDICT

How steep is safe? — the ladder decision.

Back to the question that opened this section — now with the math to settle it.

Solution. At the recommended 75°: height reached = 20 × sin(75°) ≈ 20 × 0.9659 ≈ **19.3 ft** ✓ (clears the 18-ft gutter), and the base sits at 20 × cos(75°) ≈ **5.2 ft** from the wall. But your base is 8 feet out: the actual angle is $\cos^{-1}(8/20) = \cos^{-1}(0.4) \approx$ **66.4°** — noticeably shallower than 75°. Height reached: 20 × sin(66.4°) ≈ **18.3 ft** — still reaches the gutter, but with less margin and a greater slip risk.

The decision: the trig says it *reaches* but *isn't safe*. Move the flower bed (or the ladder) to get the base to ~5 feet, or rent a longer ladder. "It reaches" and "it's safe" are two different calculations — and only one of them keeps you off the ground.

AI AI Learning Lab 2.2

AI Level 1 — Understand.

I keep mixing up opposite and adjacent. Explain how to label the sides of a right triangle relative to a chosen angle θ, using a simple labeled diagram described in words. Then describe 3 triangles in words, one at a time, and ask me to identify which side is opposite, which is adjacent, and which ratio (sin, cos, or tan) connects two given sides.

AI Level 2 — Apply.

Give me 5 right-triangle problems mixing "find the side" and "find the angle" — include the degree-mode trap once (warn me AFTER I answer). Hide all answers. I'll show my setup (which ratio I chose and why) before calculating; check my setup first, then my arithmetic.

AI Level 3 — Matter.

The ladder decision drew a line between "it reaches" and "it's safe" — 66.4° vs 75°, same ladder, different outcome. This story matters because the most dangerous mistakes aren't calculation errors; they're answering the wrong question. Discuss with me: WHY does this distinction matter beyond ladders — where else do people confuse "it works" with "it's safe" (medicine doses? car repairs? electrical work?)? What are the implications of a DIY culture that optimizes for "reaches" and never computes "safe"? Ask me: what's one "it works" shortcut in my life I should re-examine?

Level 4 — Decide. *If you were on the committee, what would you recommend?*

Example decision problem. Your church's front steps rise 30 inches. The walkway fits a 25-foot ramp. ADA rules allow at most a 1:12 slope. Does the ramp comply? If not, what should the building committee do?

AI Level 4 — Decide.

I'm on the church building committee: the steps rise 30 inches, we have room for a 25-foot ramp, and ADA allows a maximum 1:12 slope. Walk me through the compliance check — make me compute the actual slope and the ramp angle myself, and compare both against the ADA limit. Then brainstorm options with me (longer ramp? switchback? lift?), weighing cost against access, and help me write a one-paragraph recommendation.

Check your verdict: slope = 2.5 ÷ 25 = 0.10 = 1:10, steeper than the 1:12 limit — it fails (angle ≈ 5.7° vs the allowed ≈ 4.8°). A compliant ramp needs 30 inches × 12 = 30 ft of run: extend it, switchback, or install a lift.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 2.2.

Ask your AI for the values of sin(35°), cos(35°), and tan(35°) to 4 decimal places. Then compute each one on your own calculator and find the percentage error (§1.2) for all three. Write two sentences: is this AI trustworthy at trigonometry? What does your answer imply about using AI for *any* numerical work?

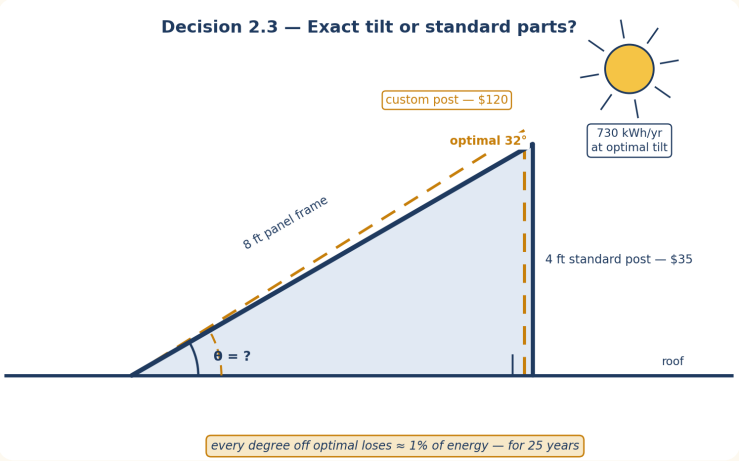
Practice Problems 2.2

1. In a right triangle the side opposite θ is 5, adjacent is 12, hypotenuse is 13. Write sin θ, cos θ, and tan θ as fractions and as decimals.
2. Find the acute angle x in each case: (a) sin x = 0.5, (b) tan x = 1, (c) cos x ≈ 0.866.
3. A right triangle has a 25° angle with adjacent side 20 cm. Find the opposite side and the hypotenuse (see Worked Example 3).
4. A 15-foot ladder makes a 70° angle with the ground. How high up the wall does it reach, and how far is its base from the wall?

2.3 Solving Right Triangles in the Real World

THE DECISION

Exact tilt or standard parts? — the solar panel decision.



The school is mounting 10 solar panels on the roof. Each panel frame is 8 feet long, and engineers say the optimal tilt is 32°. But there is a catch: the exact tilt needs custom-length support posts at $120 each, while standard 4-foot posts cost only $35. The standard posts force a slightly different tilt — and every degree off optimal costs about 1% of the panel's energy output, year after year, for the panels' 25-year life. Each panel produces about 730 kWh/year at the optimal tilt, and electricity costs $0.15/kWh.

Which posts do you buy — and how would you *prove* your choice is right? *Hold that question: the math you need comes next.*

HINTS

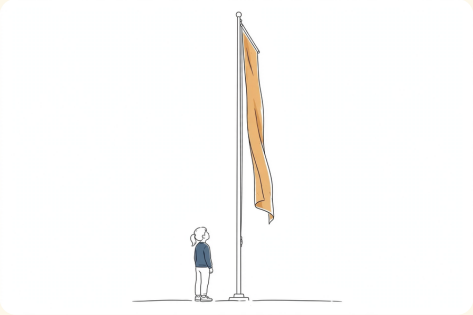
Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Two separate costs to compute: what the standard posts cost you in lost energy, and what the custom posts cost you in extra money.
- **Hint 2:** The 4-foot posts force a specific tilt — use trig on the 8-foot frame to find it, then the energy lost per degree off optimal.
- **Hint 3:** Turn both options into dollars, then divide the extra build cost by the yearly savings — and compare that payback time with the 25-year panel life.

Angles of Elevation and Depression

STORY

The Flagpole.

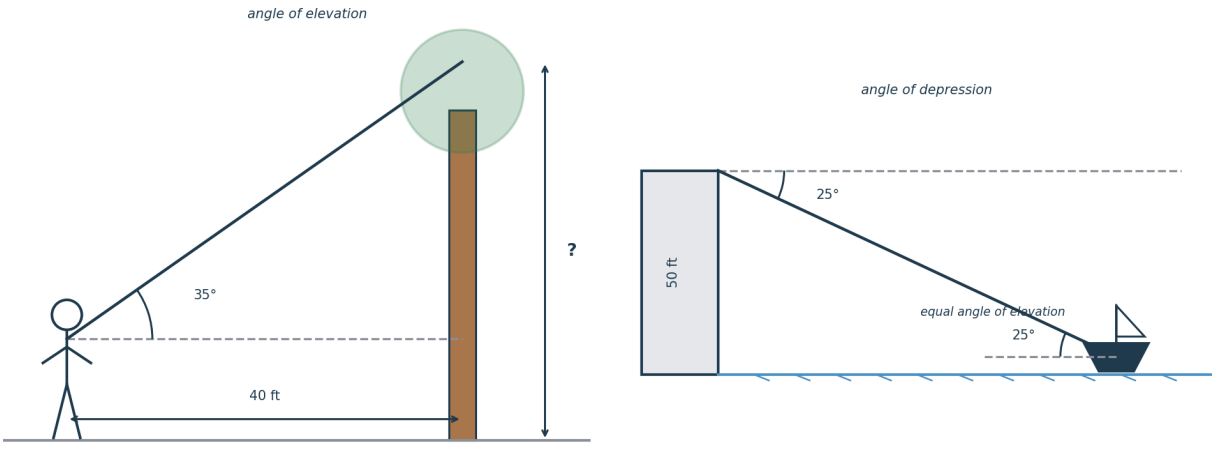


The science club needed the height of the school flagpole to order a new banner, but nobody was allowed to climb it. Maya stood 20 m from the base and measured the angle of elevation to the top: 35°. Height = 20 × tan 35° ≈ 20 × 0.700 ≈ **14.0 m** (plus Maya's eye height, about 1.6 m — so the pole is really about 15.6 m). They ordered a 16-meter banner with confidence — no ladder required. *What tall thing near you could you measure the same way?*

When you look *up* at something, the angle between your horizontal line of sight and the object is the **angle of elevation**. When you look *down*, it's the **angle of depression**. The two are equal for the same pair of points — they are alternate interior angles cut by the line of sight across two parallel horizontal lines (yours and theirs). That single geometric fact is what makes depression problems solvable: the angle you are *given* (depression, up at the cliff) equals the angle inside the triangle you can *use* (elevation, up from the boat).

Fig. 2.4

Fig. 2.4 — Elevation and depression



Depression from above equals elevation from below — parallel horizontal lines.

KEY IDEA

Draw first, calculate second.

Nearly every word problem in

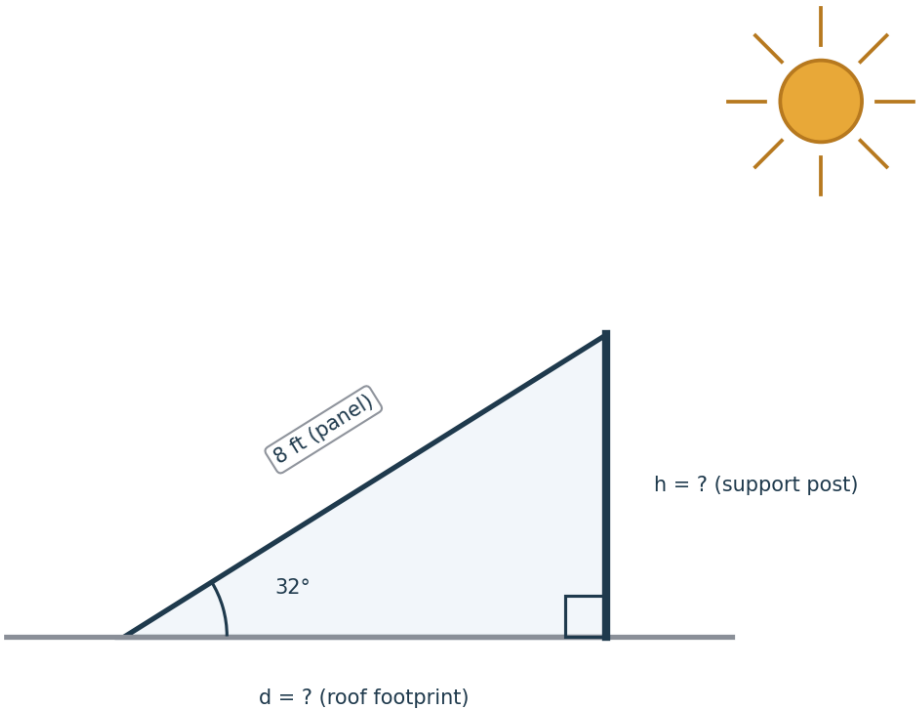
this section is solved the moment the triangle is correctly labeled. The three questions to answer before touching the calculator: (1) Where is the right angle? (2) Where is θ? (3) Which sides do I know, and which do I want? Students who skip the sketch almost always plug numbers into the wrong ratio. The sketch *is* the mathematics.

Worked Example 1 — The tree. You're standing 40 feet from a tree and measure the angle of elevation to its top as 35°. How tall is the tree?

Solution. The ground distance (40 ft) is adjacent to the 35° angle and the height is opposite: tan(35°) = height / 40, so height = 40 × tan(35°) ≈ 40 × 0.7002 ≈ **28.0 feet**.

Fig. 2.5

Fig. 2.5 — The solar panel triangle



One triangle, two unknowns — sine gives h, cosine gives d.

Worked Example 2 — The solar array. The school is mounting a solar panel on the roof. The panel frame is 8 feet long (the hypotenuse), tilted at the optimal angle of 32°. Find the height h of the vertical support post and the horizontal roof footprint d.

Solution. Relative to the 32° angle, h is opposite and d is adjacent: h = 8 × sin(32°) ≈ 8 × 0.5299 ≈ **4.24 feet**; d = 8 × cos(32°) ≈ 8 × 0.8480 ≈ **6.78 feet**.

> **Decision Point: One fixed angle, four seasons.** Panels capture the most winter sun at a 45° tilt but need only 15° in summer. These panels are permanently fixed. Peak school energy use happens during the school year (fall through spring), not summer break. What single angle would you choose — and what information would change your mind?
>
> **Decision Point: Whose angle wins?** The solar engineer wants a steep 40° tilt for maximum winter energy. The safety director wants under 20° to cut wind resistance in storms. The facilities lead needs at least 4 rows of panels, which means keeping row spacing — and shadow length — small. Sketch the triangle each person is imagining. How would you negotiate a compromise?

Worked Example 3 — Angle of depression. From the top of a 50-foot cliff you spot a boat. The angle of depression to the boat is 25°. How far is the boat from the base of the cliff?

Solution. The angle of depression (25°) equals the angle of elevation from the boat up to you. The cliff height (50 ft) is opposite that angle and the ground distance is adjacent: tan(25°) = 50 / distance, so distance = 50 / tan(25°) ≈ 50 / 0.4663 ≈ **107.2 feet**.

DECIDE — THE VERDICT

Exact tilt or standard parts? — the solar panel decision.

Back to the question that opened this section — now with the math to settle it.

Solution. Energy value per panel at 32°: 730 × $0.15 = **$109.50/year**. Lost to the 30° tilt: 2% × $109.50 ≈ **$2.19/panel/year** → $21.90/year for all ten. Extra cost of custom posts: 10 × ($120 − $35) = **$850**. Payback time for the custom posts: $850 ÷ $21.90/year ≈ **39 years** — longer than the panels' 25-year lifespan.

The decision: buy the standard posts and pocket the $850. The 32° optimum is mathematically real and economically irrelevant — the triangle gave you both angles, but only the *money* calculation tells you which one to build. Precision has a price; always ask whether it pays.

AI AI Learning Lab 2.3

AI Level 1 — Understand.

Explain the difference between angle of elevation and angle of depression using a concrete scenario (a lifeguard tower and a swimmer). Then explain WHY they are equal using parallel lines — don't just state it. Quiz me with 2 scenarios, one at a time, asking me to identify which angle is which.

AI Level 2 — Apply.

Give me 4 word problems: a tree height, a shadow/building problem, an angle-of-depression problem, and a ramp problem. For each one, FIRST ask me to describe the triangle in words (right angle location, θ location, known side, wanted side) before I calculate anything. Only let me compute after my labeling is correct.

AI Level 3 — Matter.

The solar decision proved the 32° optimum was real — and economically irrelevant: a 39-year payback on 25-year panels. This story matters because "optimal" is the most seductive word in engineering. Discuss with me: WHY does "mathematically optimal ≠ worth building" matter — what are the implications for green-energy decisions, where the planet benefits but the math doesn't pay? When SHOULD we build the thing that doesn't pay back? Ask me: what's one "optimal" choice in my life that might not be worth its price?

Level 4 — Decide. *If this were your house and your money, what would you choose?*

> **Example decision problem.** Two home upgrades, both lasting 25 years. (A) A $200 smart thermostat saving 10% of your $1,800/year energy bill. (B) $8,000 double-pane windows saving 15%. Which pays for itself — and which would you choose?

AI Level 4 — Decide.

I'm deciding on home energy upgrades: (A) a $200 smart thermostat saving 10% of my $1,800/year bill, (B) $8,000 windows saving 15%. Both last 25 years. Make me compute both payback periods myself. Then debate me: the windows also cut noise and add resale value — things the payback formula can't see. Help me decide, and make me write down which numbers decided it and which judgment calls did.

Check your verdict: (A) saves $180/year → payback ≈ 1.1 years — yes. (B) saves $270/year → payback ≈ 29.6 years > 25-year life — no on pure math. The method is the same one that settled the solar panels.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 2.3.

Describe this triangle to your AI *in words only* (no diagram): "A 10-foot ramp rises 3 feet off the ground." Ask it to find the angle the ramp makes with the ground. Then critique its answer like a teacher grading a student: did it pick the right ratio? Did it confuse opposite with hypotenuse? Write a 3-sentence "grading report." Then solve it yourself — $\sin\theta = 3/10$ gives $\theta \approx 17.5°$ — and compute the percentage error if the AI disagrees.

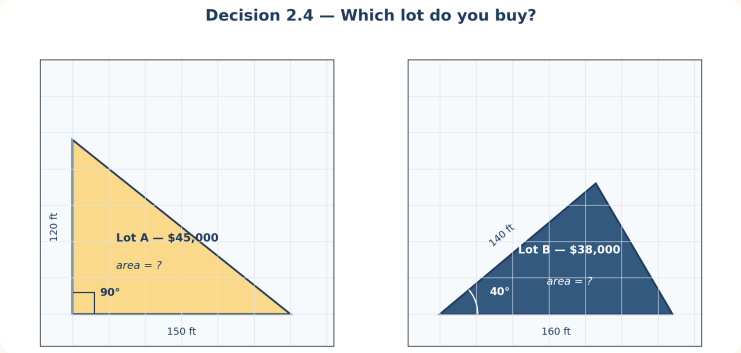
Practice Problems 2.3

1. A kite string is 100 m long and makes a 55° angle of elevation with the ground. How high is the kite?
2. A building casts a 30 m shadow when the sun's angle of elevation is 40°. How tall is the building?
3. From an 80-foot fire tower, the angle of depression to a car is 30°. How far is the car from the base of the tower?
4. A 12-foot wheelchair ramp rises 3 feet. What angle does it make with the ground? (Hint: you know opposite and hypotenuse — which ratio is that?)

2.4 Area of a Triangle Using Sine

THE DECISION

Which lot do you buy?



Two triangular lots are for sale:

- **Lot A:** sides 120 ft and 150 ft with a 90° included angle. Price: $45,000.
- **Lot B:** sides 140 ft and 160 ft with a 40° included angle. Price: $38,000. Lot B is cheaper and its sides are longer — but which is the better buy per square foot of usable land?

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Price per square foot needs two numbers — the price is given, but the area is not.
- **Hint 2:** This is a ½ab·sin C problem — and the given angle is the included angle, exactly what the formula needs.
- **Hint 3:** Compute both areas, divide each price by its area — the lot with longer sides may surprise you.

STORY

The Sail.

Noah's sailing club needed a new triangular sail: two edges 12 ft and 15 ft long, with a 50° angle between them. Sailcloth costs $8 per square foot, and the club had budgeted $500. Area = ½ × 12 × 15 × sin 50° ≈ 90 × 0.766 ≈ **68.9 ft²** — about $551 of cloth. Noah went back to the club treasurer for another $51. *Have you ever budgeted for a project before computing its true size?*

You know "half base times height" — but what if you don't know the height? If you know two sides and the angle *between* them (the **included angle**), the height can be written with sine, and the height problem disappears.

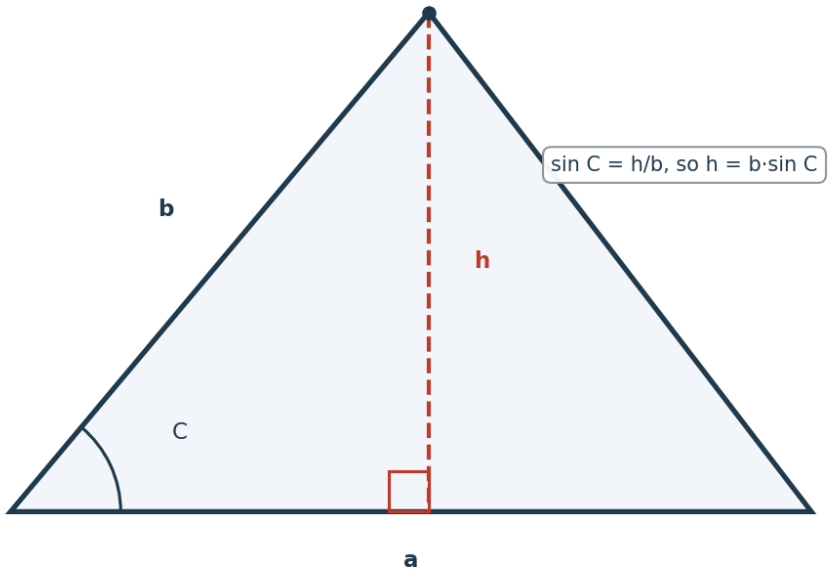
Here is the whole derivation in two lines. Drop the altitude h onto side a from the opposite vertex. In the small right triangle this creates, sin C = h / b (opposite over hypotenuse), so h = b·sin C. Plug into the familiar formula:

Area = ½ × a × b × sin C

where C is the angle between sides a and b. Any side can play the role of "base" — just use the angle between your two chosen sides.

Fig. 2.6

Fig. 2.6 — Where the sine comes from



The altitude is just b wearing a sine costume.

KEY IDEA

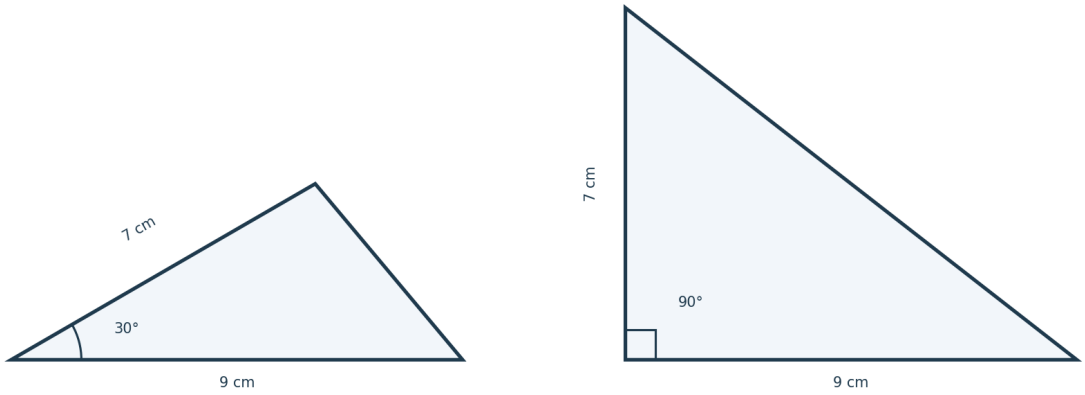
The included angle is load-bearing.

The formula demands the

angle *between* your two sides — not just any angle in the triangle. Use the wrong angle and the formula silently gives you the wrong area. And notice something powerful: for fixed side lengths, the area is largest when the included angle is 90°, because sin C maxes out at 1. Same materials, more area — a genuinely useful fact when laying out a garden or a solar array.

Fig. 2.7

Fig. 2.7 — Same sides, different area



Identical sides, double the area — the angle decides.

Worked Example 1 — Basic use. Two sides of a triangle are 8 cm and 10 cm, with an included angle of 40°. Find the area.

Solution. Area = ½ × 8 × 10 × sin(40°) = 40 × 0.6428 ≈ **25.7 cm²**.

Worked Example 2 — The included angle matters. Sides of 9 m and 14 m include an angle of 65°. Find the area.

Solution. Area = ½ × 9 × 14 × sin(65°) = 63 × 0.9063 ≈ **57.1 m²**.

TRY IT

Take sides 10 and 10. Compute the area with included angles of 30°,

60°, 90°, and 120°. What pattern do you see? Why does the area *fall* again after 90°? (Hint: what is sin(120°)?)

DECIDE — THE VERDICT

Which lot do you buy?

Back to the question that opened this section — now with the math to settle it.

Solution. Area = ½ab·sin C:

- Lot A: ½ × 120 × 150 × sin(90°) = 9,000 × 1 = **9,000 ft²** → $45,000 ÷ 9,000 = **$5.00/ft²**.
- Lot B: ½ × 140 × 160 × sin(40°) = 11,200 × 0.6428 ≈ **7,199 ft²** → $38,000 ÷ 7,199 ≈ **$5.28/ft²**.

The decision: Lot A — bigger *and* cheaper per square foot, despite the smaller sticker price and shorter sides on Lot B. The long sides of Lot B are mostly angle wasted: a 40° included angle strangles the area. When someone sells you "more land" with long sides, ask about the angle between them — it decides how much land you're actually buying.

AI AI Learning Lab 2.4

AI Level 1 — Understand.

I know "half base times height" but I don't see where sine comes in. Using a triangle with sides a and b and the angle C between them, walk me through the derivation: drop the altitude, find h with sine, and substitute it into the area formula. Ask me to explain each step back to you before continuing.

AI Level 2 — Apply.

Give me 4 area problems using ½ab·sin C with different units and contexts (a garden plot, a sail, a solar panel, a triangular pennant flag). Hide the answers. For one of them, deliberately give me the WRONG angle (not the included one) in the problem statement — see if I catch it before calculating.

AI Level 3 — Matter.

The lot decision exposed the trick: longer sides, lower price — and LESS land per dollar, because a 40° angle strangles the area. This story matters because it's how things are sold: the impressive number on the sign isn't the number that matters. Discuss with me: WHY does this matter to buyers — what are the implications for real-estate transparency, and who benefits when buyers can't do this math? Where else are you sold "more" that's actually less — phone storage? food packages? Ask me: what's the last thing I bought where the headline number lied?

Level 4 — Decide. *If you were ordering for the team, what would you pick?*

Example decision problem. Pizza for the team party: (A) a 16-inch round pizza for $18; (B) a 12 × 18-inch rectangular pizza for $20. Which gives more pizza per dollar?

AI Level 4 — Decide.

I'm ordering pizza for the team: a 16-inch round for $18 vs a 12 × 18-inch rectangle for $20. Make me compute each pizza's area and the price per square inch myself. Then decide with me — and argue the other side: what would have to be true (taste? cleaner slicing? better leftovers?) for the loser to win anyway?

Check your verdict: round = $\pi \times 8^2 \approx 201.1$ in² → ≈ \$0.0895/in²; rectangle = 216 in² → ≈ \$0.0926/in². The round wins per dollar — same "price per unit of area" thinking that settled the lots.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 2.4.

Ask your AI: "Two sides are 7 cm and 9 cm. Find the area if the included angle is 30°, and again if it's 90°." Verify both answers yourself (15.75 cm² and 31.5 cm²). Then ask the follow-up that matters: "Explain *why* the 90° triangle has exactly double the area, using the sine values." If the AI's explanation is hand-wavy, rewrite it in your own precise words — precision is the assignment.

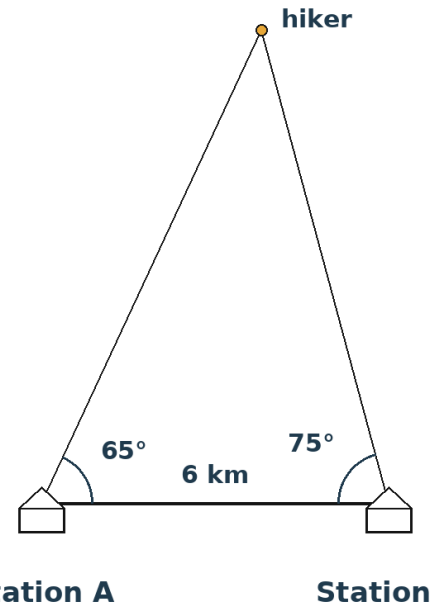
Practice Problems 2.4

1. Two sides are 6 cm and 9 cm with an included angle of 60°. Find the area.
2. Two sides are 10 m and 10 m with an included angle of 90°. Find the area. (Check your answer against ½ × base × height.)
3. Find the area of an equilateral triangle with side 4 cm.
4. Two triangles both have sides 8 cm and 11 cm. One has an included angle of 25°, the other 90°. Which has the greater area, and by how much?

2.5 The Sine Rule

THE DECISION

Which station dispatches the rescue team?



A hiker is lost. Two ranger stations sit 6 km apart on a straight trail. Station A measures the angle to the hiker's emergency signal as 65°; Station B measures 75°. The team should launch from the nearer station — but which one is nearer? Nobody can see the hiker.

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Sketch the triangle: the 6 km trail is one side, the hiker is the third vertex, and you know two angles.
- **Hint 2:** This is a Sine Rule setup — side a / sin A = side b / sin B = side c / sin C.
- **Hint 3:** Find the third angle first (angles sum to 180°), then solve for both unknown distances and compare them.

STORY

The Boat.

Two lifeguards, Ana and Ben, stood 300 m apart on a

straight beach and spotted a drifting boat. Ana measured 55° between the shoreline and the boat; Ben measured 70°. The third angle was 180° − 55° − 70° = 55°, so by the Sine Rule Ana's distance to the boat was 300 × sin 70° ÷ sin 55° ≈ 300 × 0.940 ÷ 0.819 ≈ **344 m**. Ana grabbed the rescue board — she knew exactly how far she would swim. *Could you have found Ben's distance just as fast?*

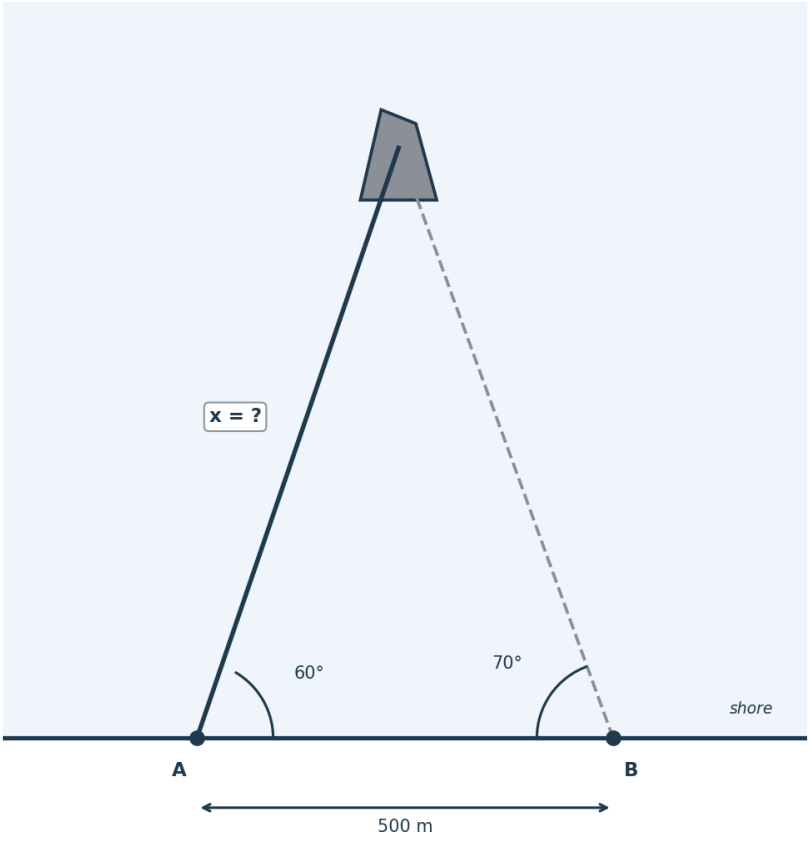
The Sine Rule relates every side of a triangle to the sine of its opposite angle:

a / sin A = b / sin B = c / sin C

Use it when you know **two angles and any side** (ASA or AAS) — find the third angle first (angles sum to 180°), then solve for the unknown sides. It also handles one of the oldest practical problems in surveying: measuring something you cannot reach. Walk a baseline you *can* measure, take two angles, and let the rule find the inaccessible distance. No boat, no climbing, no trespassing required.

Fig. 2.8

Fig. 2.8 — The inaccessible distance



Two angles and a baseline unlock a distance you can never walk.

Worked Example 1 — Two angles and a side. In triangle ABC, ∠A = 40°, ∠B = 50°, and side a = 12 cm (opposite ∠A). Find side b.

Solution. b / sin(50°) = 12 / sin(40°), so b = 12 × sin(50°) / sin(40°) ≈ 12 × 0.7660 / 0.6428 ≈ **14.3 cm**. (Since ∠C = 90°, you could check this with right-triangle trig: b = 12 × tan(50°) ≈ 14.3. ✓ Two different roads, same destination — a good habit is checking one method against the other.)

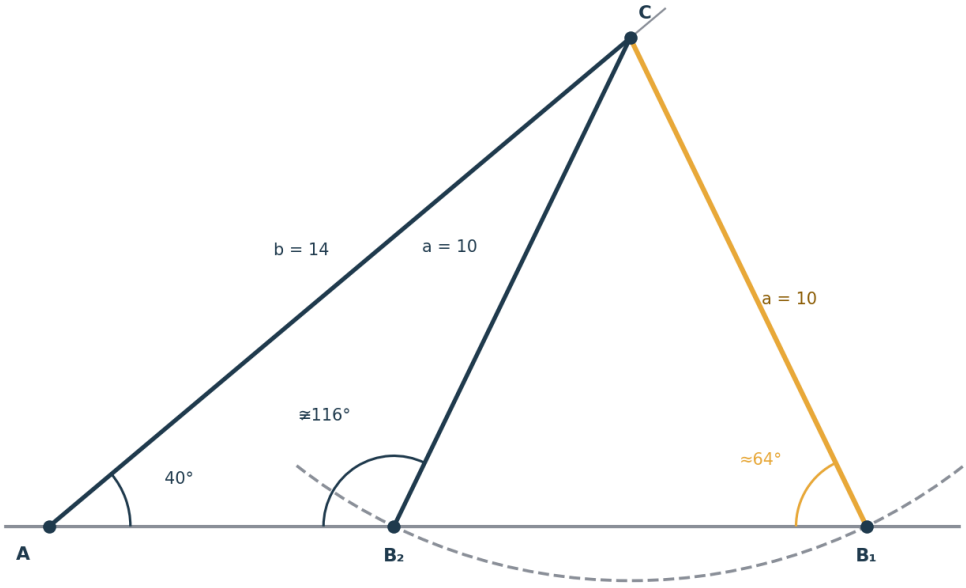
Worked Example 2 — The inaccessible distance. A surveyor wants the distance from point A on shore to a lighthouse rock she cannot reach. She walks 500 m along the straight shoreline to point B. From A the angle to the rock is 60°; from B it is 70°. How far is the rock from A?

Solution. The triangle's third angle (at the rock) is 180° − 60° − 70° = 50°. The 500 m baseline is opposite that 50° angle. Let x be the distance from A (opposite the 70° angle at B): x / sin(70°) = 500 / sin(50°), so x = 500 × 0.9397 / 0.7660 ≈ **613 m**.

Watch out: the ambiguous case. If you know two sides and an angle *opposite one of them* (SSA), there can be **two** different triangles fitting the data — the given side can swing to two positions, like a door that latches in two places. Always check: if sin B computes to a value with two possible angles (B and 180° − B), test whether both keep the angle sum under 180°. Practice Problem 3 below is exactly this trap.

Fig. 2.9

Fig. 2.9 — The ambiguous case (SSA)



Same SSA data, two valid triangles — the given side swings to two positions.

KEY IDEA

Which rule when?

This is the decision skill of the chapter.

Ask in order: (1) Is there a right angle? → SOH-CAH-TOA (§2.2–2.3). (2) Do I know two angles and a side? → Sine Rule (§2.5). (3) Do I know two sides and the *included* angle, or all three sides? → Cosine Rule (§2.6). If you can answer these three questions, you can solve any triangle in this book.

DECIDE — THE VERDICT

Which station dispatches the rescue team?

Back to the question that opened this section — now with the math to settle it.

Solution. The triangle's angles: 65° at A, 75° at B, so 180° − 65° − 75° = **40°** at the hiker. The 6 km baseline is opposite the 40° angle. Sine Rule:

- Distance A → hiker (opposite the 75° angle): 6 × sin(75°) ÷ sin(40°) = 6 × 0.9659 ÷ 0.6428 ≈ **9.0 km**.
- Distance B → hiker (opposite the 65° angle): 6 × sin(65°) ÷ sin(40°) = 6 × 0.9063 ÷ 0.6428 ≈ **8.5 km**.

The decision: dispatch from Station B — half a kilometer closer, which at rescue hiking speed is about 6–8 minutes. Nobody walked anything; two angle readings and the Sine Rule just bought the hiker those minutes. This is triangulation — locating from *angles*. Your phone's GPS uses its cousin, trilateration, which locates from *distances* to satellites instead.

AI AI Learning Lab 2.5

AI Level 1 — Understand.

Explain the Sine Rule as a proportion: why does a/sin A always equal b/sin B in the same triangle? Give me an intuitive reason, not just algebra. Then give me 2 ASA problems, one at a time, and make me find the third angle FIRST before solving — stop me if I skip that step.

AI Level 2 — Apply.

Give me 3 problems: one ASA, one AAS, and one inaccessible-distance surveyor problem with a baseline and two angles. Hide the answers. I'll set up each proportion and solve; check my setup (did I match each side to its OPPOSITE angle?) before checking my arithmetic.

AI Level 3 — Matter.

The rescue decision bought a hiker 6–8 minutes with two angle readings — dispatch from Station B, 8.5 km instead of 9.0. This story matters because it's the same math that finds your phone from cell towers — and the same math that fails when the measurements are wrong. Discuss with me: WHY does triangulation matter — what are the implications of a 2° error in an angle reading (where does the hiker "move" to)? Who else depends on this math working — pilots? 911 dispatchers? Ask me: what's one technology I use daily that secretly runs on the Sine Rule?

Level 4 — Decide. *If you were the dispatcher, who would you send?*

> **Example decision problem.** A crash is reported. An ambulance 7 km away drives at 60 km/h. A helicopter 12 km away flies at 180 km/h but needs 3 minutes to spin up. Which do you dispatch?

AI Level 4 — Decide.

I'm the 911 dispatcher: ambulance 7 km away at 60 km/h, helicopter 12 km away at 180 km/h but needing 3 minutes to spin up. Make me compute both ETAs myself, including the spin-up. Then the twist: it's a tie — so walk me through what breaks the tie (injury severity? weather? cost?), role-playing the radio call with me. I make the final call and defend it in two sentences.

Check your verdict: ambulance = 7 min; helicopter = 4 + 3 = 7 min — a dead heat. The math can't break this tie; triage judgment has to.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 2.5.

Ask your AI to solve the ambiguous case above (∠A = 40°, a = 10, b = 14) and to state how many triangles fit. Many AIs confidently report only ONE angle (64.1°) and miss the second (115.9°). Did yours? Write a verdict: does your AI understand the ambiguous case? Either way, solve it fully yourself — sin B = 14·sin(40°)/10 ≈ 0.8999, so ∠B ≈ 64.1° or 115.9° — and explain in one paragraph why the "obvious" single answer is a trap.

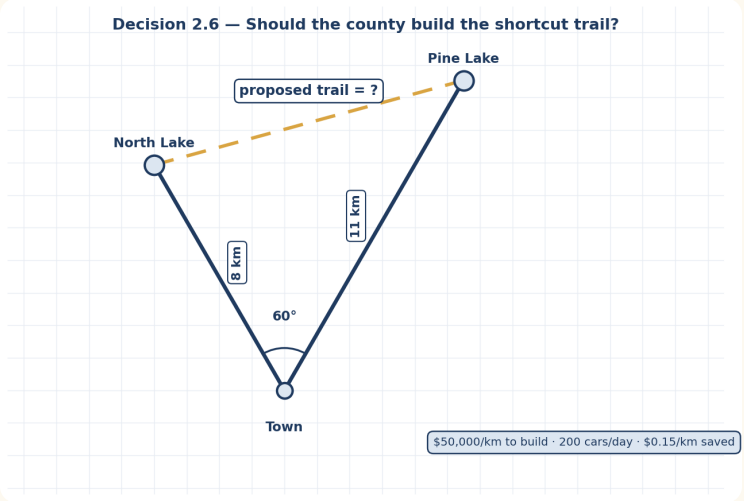
Practice Problems 2.5

1. In a triangle, ∠A = 30°, ∠B = 70°, and side a = 8 cm. Find ∠C and sides b and c.
2. A surveyor walks a 500 m baseline along a river. The angles to a tree on the far bank are 60° (from A) and 70° (from B). How far is the tree from point A?
3. In a triangle, ∠A = 40°, side a = 10 cm, side b = 14 cm. Find the possible value(s) of ∠B. (How many triangles fit this data?)
4. ∠A = 55°, ∠B = 65°, side b = 20 m. Find side c.

2.6 The Cosine Rule

THE DECISION

Should the county build the shortcut trail?



Two roads leave town at a 60° angle: one runs 8 km to North Lake, the other 11 km to Pine Lake. Drivers heading between the two lakes currently go back through town — 19 km each way. A proposed straight trail through the forest would connect the two trailheads directly. Building it costs $50,000 per km. About 200 cars a day would use it, each saving fuel and wear worth about $0.15 per km. Does the trail pay for itself?

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** The trail is the third side of a triangle with two known sides and the angle between them.
- **Hint 2:** This is a Cosine Rule problem: $c^2 = a^2 + b^2 - 2ab \cdot \cos C$.
- **Hint 3:** Once you have the trail length, find the kilometers saved per trip, turn that into dollars per day, then divide the build cost by the daily savings.

STORY

The Ridge Trail.

The hiking club debated a shortcut: two existing trails leave base camp at a 110° angle — one 6 km, one 8 km — and a proposed ridge trail would connect their far ends directly. Before arguing about cost, someone had to know the length: $c^2 = 6^2 + 8^2 - 2 \times 6 \times 8 \times \cos 110°$ $= 100 - 96 \times (-0.342) \approx 132.8$, so $c \approx$ **11.5 km**. Eleven and a half kilometers of wilderness trail — the debate got serious fast. *What would you want to know before voting yes or no?*

When you know **two sides and the included angle** (SAS), or **all three sides** (SSS), the Sine Rule can't get started — there is no complete side-angle pair to anchor the proportion. The Cosine Rule can:

$$\mathbf{c^2 = a^2 + b^2 - 2ab \cdot \cos C}$$

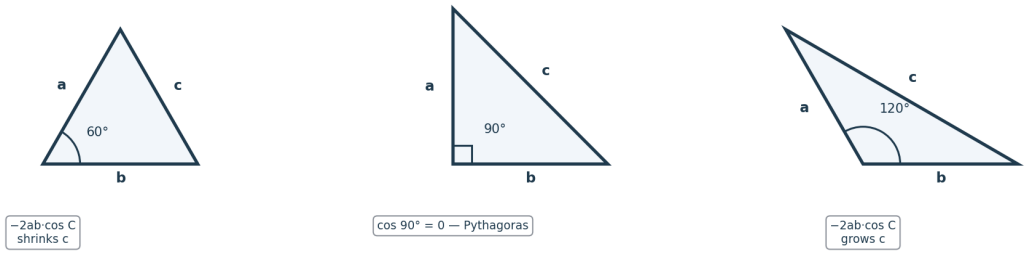
where C is the angle between sides a and b, opposite side c. To find an angle from three sides, rearrange:

$$\mathbf{\cos C = (a^2 + b^2 - c^2) / (2ab)}$$

Now look at the formula with fresh eyes: it is the Pythagorean theorem with a *correction term*. When C = 90°, cos(90°) = 0, the correction vanishes, and you get $c^2 = a^2 + b^2$. When C is acute, cos C is positive and the correction *shrinks* c^2 (the side opposite a sharp angle is shorter than Pythagoras would predict). When C is obtuse, cos C is negative and the correction *grows* c^2. One formula, three behaviors — and Pythagoras is just the special case where the angle happens to be square.

Fig. 2.10

Fig. 2.10 — Cosine Rule as corrected Pythagoras



One formula: the correction term bends Pythagoras to fit any angle.

Worked Example 1 — Two sides and the included angle. Triangle FGH has h = 12.9 cm, g = 3.7 cm, and the included angle ∠F = 47°. (Recall: side h is opposite ∠H, so the sides around ∠F are g and h.) Find side f.

Solution. $f^2 = 12.9^2 + 3.7^2 - 2(12.9)(3.7)\cos(47°) = 166.41 + 13.69 - 95.46 \times 0.6820 \approx 180.10 - 65.10 = 115.00$, so f ≈ **10.7 cm**.

CAUTION

"Included" is doing real work.

The Cosine Rule only works in

the form $c^2 = a^2 + b^2 - 2ab\cdot\cos C$ when C is the angle *between* sides a and b. Students' #1 error here is grabbing a non-included angle and plugging it in anyway. Label the triangle, circle the included angle, and only then reach for the formula.

Worked Example 2 — Clean numbers. Two sides are 7 m and 10 m with an included angle of 60°. Find the third side.

Solution. $c^2 = 7^2 + 10^2 - 2(7)(10)\cos(60°) = 49 + 100 - 140 \times 0.5 = 79$, so c ≈ **8.89 m**.

Worked Example 3 — Three sides, find an angle. A triangle has sides 5, 7, and 8. Find the angle opposite the side of length 8.

Solution. $\cos C = (5^2 + 7^2 - 8^2) / (2 \times 5 \times 7) = (25 + 49 - 64) / 70 = 10/70 \approx 0.1429$, so $C = \cos^{-1}(0.1429) \approx$ **81.8°**.

TRY IT

Before calculating Worked Example 3, predict: is the angle

opposite the 8-side acute or obtuse? (Hint: compare 8^2 with $5^2 + 7^2$.) The Cosine Rule's correction term makes this a one-second judgment call — train the instinct.

DECIDE — THE VERDICT

Should the county build the shortcut trail?

Back to the question that opened this section — now with the math to settle it.

Solution. Trail length (Cosine Rule): $c^2 = 8^2 + 11^2 - 2(8)(11)\cos(60°) = 64 + 121 - 176 \times 0.5 = 97$, so $c = \sqrt{97} \approx$ **9.85 km**. Savings per trip between the lakes: $19 - 9.85 =$ **9.15 km**. Daily savings: $200 \times 9.15 \times \$0.15 \approx$ **\$274.50/day**. Trail cost: $9.85 \times \$50{,}000 \approx$ **\$492,500**. Payback: \$492,500 ÷ \$274.50/day ≈ 1,794 days ≈ **4.9 years**.

The decision: if the county plans on a 5+ year horizon, build it; if the budget cycle is 3 years, don't. The Cosine Rule measured the shortcut, but the *payback period* makes the decision. Notice what the math can't tell you: the forest you'd cut through has a value too — and somebody has to decide what that's worth.

AI AI Learning Lab 2.6

AI Level 1 — Understand.

Show me that the Cosine Rule becomes the Pythagorean theorem when the angle is 90°. Then explain in plain words what the −2ab·cos C term does when the angle is acute vs. obtuse. Quiz me with 3 quick "which is bigger than Pythagoras predicts?" questions, one at a time.

AI Level 2 — Apply.

Give me 4 problems: two SAS (find the third side) and two SSS (find an angle). Hide the answers. For each, FIRST ask me which form of the rule I'm using and which angle is included — don't let me calculate until the setup is right.

AI Level 3 — Matter.

The trail decision hinged on a 4.9-year payback — build it for a 5-year horizon, skip it for 3. And the math stayed silent about the forest. This story matters because payback thinking now runs public spending everywhere. Discuss with me: WHY does this matter — what are the implications of letting a payback formula decide what gets built? What does the math leave out, and who speaks for the things with no price — the forest? the quiet? Ask me: what's one thing in my town I'd protect even if the payback said no — and what does that say about my values?

Level 4 — Decide. *If you sat on the county board, how would you vote?*

> **Example decision problem.** A second trail proposal: upgrade 5 km + 7 km of existing fire roads ($20,000/km) instead of building the new 9.85 km direct trail ($50,000/km). The two-leg route totals 12 km; drivers currently do 19 km each way between the lakes, 200 cars/day, saving $0.15/km. Which proposal should the county choose?

AI Level 4 — Decide.

The county now has two trail proposals: (A) the new 9.85 km direct trail at $50,000/km; (B) upgrading 12 km of existing fire roads at $20,000/km. Drivers currently do 19 km each way between the lakes; 200 cars/day save $0.15/km. Make me compute BOTH build costs and BOTH daily savings myself. Then help the council choose: cheaper to build vs better for drivers — who should decide, and what would you recommend? Make me write the deciding argument in three sentences.

Check your verdict: (A) costs $492,500, saves 9.15 km/trip → $274.50/day. (B) costs $240,000, saves 7 km/trip → $210/day. (B) is cheaper to build; (A) is better for every driver, every day. The numbers frame it — the board decides it.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 2.6.

Give your AI this SSS triangle: sides 9, 12, 15. Ask for all three angles. Then check its work yourself — you should find 90°, 36.9°, and 53.1° (note: $9^2 + 12^2 = 15^2$, so this is a right triangle in disguise!). The interesting question: did the AI *notice* it's a right triangle, or did it grind through the Cosine Rule blindly? Write 3 sentences on which approach is smarter and why recognizing structure beats running formulas.

Practice Problems 2.6

1. Two sides are 5 cm and 8 cm with an included angle of 60°. Find the third side.
2. A triangle has sides 9, 12, and 15. Find the angle opposite the side of length 15, then the angle opposite the side of length 9.
3. Two sides of 10 m and 14 m include an angle of 120°. Find the third side.
4. A hiker walks 5 km from camp, turns, and walks 9 km more, with a 60° angle between the two legs of the trip. How far is she from camp?

2.7 Problems in Three Dimensions

THE DECISION

Will the kayak fit?

You're choosing between two kayaks: 12.5 ft ($900) and 14 ft ($1,150 — tracks better in open water). Your garage is 12 ft deep and 8 ft wide, and the kayak must live there diagonally on the floor. Does the 14-footer fit?

What would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** The longest straight line on the garage floor is the diagonal of the 12-by-8 rectangle.
- **Hint 2:** Find that diagonal with the Pythagorean theorem — the sides are the legs, the diagonal is the hypotenuse.
- **Hint 3:** Compare the diagonal with 14 feet — then ask yourself how much clearance you'd want before spending the extra $250.

STORY

The Curtain Rod.

Mia bought a 10-foot curtain rod and carried it home on the bus — but she needed to drive it to her sister's apartment, and it would not fit in her car. Could it fit diagonally in her sister's storage closet, which is 8 ft by 6 ft by 5 ft? The space diagonal is $\sqrt{(8^2 + 6^2 + 5^2)} = \sqrt{125} \approx$ **11.2 ft**. The rod fit with more than a foot to spare, and Mia never had to saw it in half. *What is the longest object that could fit in your bedroom?*

Real objects have depth, and trigonometry works there too. The master strategy for every 3D problem in this section is one sentence: **find the right triangle hiding inside the 3D shape.** The shape is three-dimensional; the triangle you actually solve is always flat.

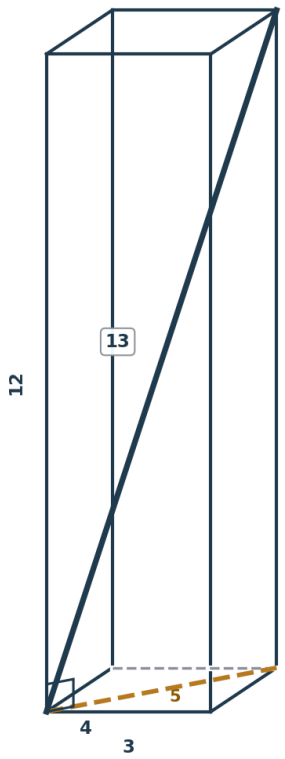
For a box with length l, width w, and height h, apply Pythagoras twice — first across the base to get the base diagonal, then up through space:

space diagonal = $\sqrt{(l^2 + w^2 + h^2)}$

The derivation is worth seeing once: base diagonal d satisfies $d^2 = l^2 + w^2$. The space diagonal s, the base diagonal d, and the height h form a right triangle (the height is perpendicular to the base), so $s^2 = d^2 + h^2 = l^2 + w^2 + h^2$. Two applications of Pythagoras, zero new ideas — which is exactly the point. New dimensions, same theorem.

Fig. 2.11

Fig. 2.11 — The space diagonal



Two flat right triangles hiding in one box: 3-4-5 on the base, 5-12-13 through space.

Worked Example 1 — The longest rod. What is the longest thin rod that fits inside a box 3 m by 4 m by 12 m?

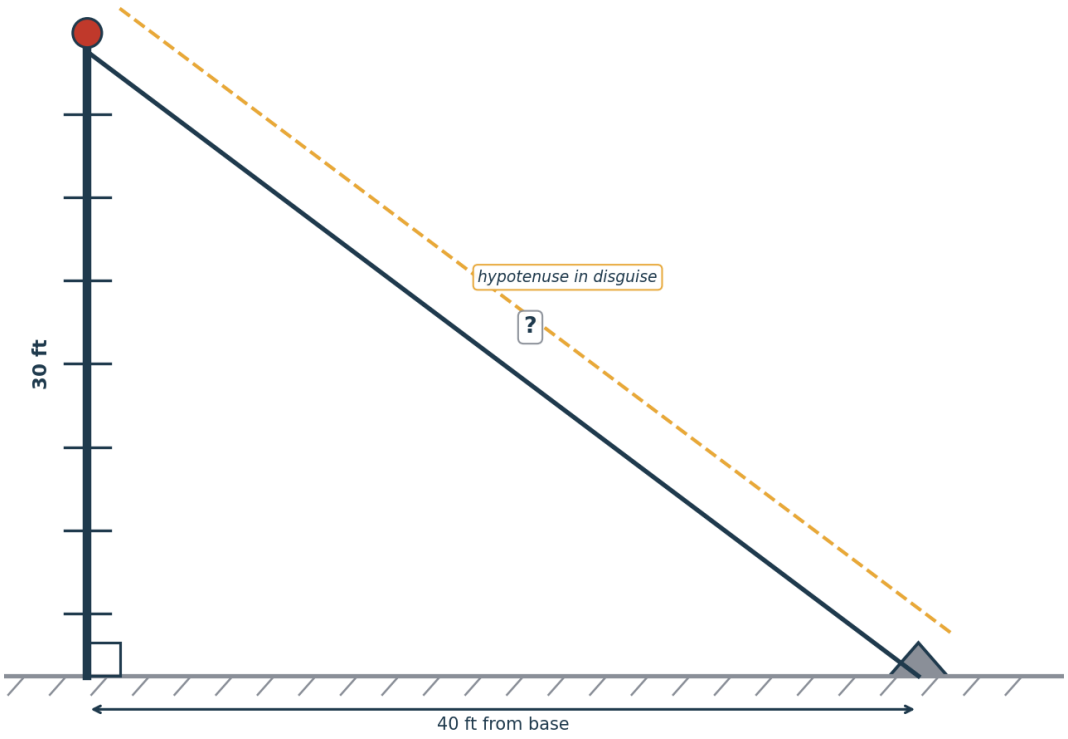
Solution. The longest rod lies along the space diagonal: $d = \sqrt{(3^2 + 4^2 + 12^2)} = \sqrt{(9 + 16 + 144)} = \sqrt{169} =$ **13 m**. (The 3-4-5 pattern strikes again: the base diagonal is 5, and 5-12-13 is another triple. This box was practically designed to show off.)

Worked Example 2 — The angle of the diagonal. At what angle does that 13 m space diagonal meet the base of the box?

Solution. The diagonal, the base diagonal (5 m), and the height (12 m) form a right triangle. The angle θ with the base satisfies $\tan\theta = 12/5 = 2.4$, so $\theta = \tan^{-1}(2.4) \approx$ **67.4°**. Steep — most of the rod's length is going *up*, not across. (Sanity check with §2.2 instinct: the opposite side (12) dwarfs the adjacent (5), so θ should be well past 45°. It is.)

Fig. 2.12

Fig. 2.12 — Guy-wire as a hidden hypotenuse



Towers, tents, and cranes are all right triangles wearing work clothes.

Worked Example 3 — The guy-wire. A 30-foot radio tower is held by a guy-wire anchored 40 feet from the tower's base. How long is the wire?

Solution. The tower, the ground, and the wire form a right triangle: wire = $\sqrt{(30^2 + 40^2)} = \sqrt{(900 + 1600)} = \sqrt{2500} =$ **50 feet**. Guy-wires, tent ropes, and crane cables are all hypotenuses in disguise.

DECIDE — THE VERDICT

Will the kayak fit?

Back to the question that opened this section — now with the math to settle it.

Solution. The floor diagonal: $\sqrt{(12^2 + 8^2)} = \sqrt{208} \approx$ **14.42 ft**. The 14-ft kayak fits — with 14.42 − 14 = 0.42 ft ≈ **5 inches** to spare.

The decision: buy the 14-footer *if* you trust your tape measure more than the brochure — 5 inches of clearance is real but unforgiving, and it assumes perfectly square corners and no storage boxes in the way. The space diagonal just turned a $250 upgrade decision into a measurement question: go re-measure the garage before you order. (And measure the door, too — fitting *inside* is only half the problem.)

AI AI Learning Lab 2.7

AI Level 1 — Understand.

Explain the space diagonal of a box by building it in two steps: first the base diagonal, then the space diagonal — each as its own right triangle. Use a 3 m by 4 m by 12 m box and show me both Pythagorean triples hiding inside. Then ask me to find the space diagonal of a 6-8-10 box on my own.

AI Level 2 — Apply.

Give me 4 3D problems: two "longest rod in a box," one "angle the diagonal makes with the base," and one "largest box that fits through a doorway." Hide the answers. For each, ask me FIRST to name the hidden right triangle (what are its three sides?) before I calculate.

AI Level 3 — Matter.

The kayak decision came down to 5 inches — trust your tape measure more than the brochure, and measure the door too. This story matters because "will it fit" is one of life's most repeated expensive questions: furniture, appliances, moving trucks. Discuss with me: WHY does this matter — what are the implications of buying before measuring (return fees? a sawed-in-half bookshelf? a kayak that lives in the rain?)? The math also warned that fitting *inside* is only half the problem. Ask me: what's the biggest thing I've ever bought that didn't fit — and what would I measure first next time?

Level 4 — Decide. *If this were your move, would you carry it or disassemble it?*

Example decision problem. You're moving a rigid 9-foot ladder around a hallway corner: hallway A is 4 ft wide, hallway B is 3 ft wide. The longest ladder that fits around such a corner is (w1^(2/3) + w2^(2/3))^(3/2) feet. Does your 9-ft ladder make the turn — or do you disassemble it?

AI Level 4 — Decide.

I'm moving and must get my rigid 9-foot ladder around a corner from a 4-ft-wide hallway into a 3-ft-wide hallway. The longest ladder that fits is (w1^(2/3) + w2^(2/3))^(3/2) feet. First, have the AI explain WHY this formula works — what's the geometry? — then make ME verify its arithmetic step by step with my own calculator (remember the golden rule). If it fits, I carry it around the corner; if not, I disassemble. Decide with me, and finish by computing the percentage error between the AI's answer and mine.

Check your verdict: (4^(2/3) + 3^(2/3))^(3/2) ≈ (2.52 + 2.08)^1.5 ≈ 9.87 ft > 9 ft — it fits, with about 10 inches to spare. Carry it (slowly).

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 2.7.

Ask your AI: "What's the longest pole that fits in a 2 m by 3 m by 6 m crate?" It should find 7 m (√(4 + 9 + 36) = 7). Then push further: "Now suppose the pole must stay 10 cm away from every wall — what's the longest pole NOW?" Watch how it handles the follow-up: does it shrink the box dimensions first (correct instinct), or flail? Solve the follow-up yourself — effective box 1.8 × 2.8 × 5.8, diagonal ≈ 6.69 m — and grade the AI's reasoning, not just its number.

Practice Problems 2.7

1. Find the space diagonal of a box 6 cm by 8 cm by 10 cm.
2. What is the longest pole that fits in a crate 2 m by 3 m by 6 m?
3. For a box 6 m by 8 m by 10 m, at what angle does the space diagonal meet the base? (First find the base diagonal.)
4. A 25-foot flagpole has a guy-wire anchored 15 feet from its base. How long is the wire, and what angle does it make with the ground?

Vocabulary

- **Right triangle** — a triangle with one 90° angle.
- **Hypotenuse** — the side opposite the right angle; the longest side.
- **Legs** — the two sides forming the right angle.
- **Pythagorean theorem** — $a^2 + b^2 = c^2$ for a right triangle.
- **Pythagorean triple** — three whole numbers satisfying $a^2 + b^2 = c^2$ (e.g., 3-4-5, 5-12-13).
- **Converse of the Pythagorean theorem** — if $a^2 + b^2 = c^2$, the triangle contains a right angle.
- **Distance formula** — $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, Pythagoras on a grid.
- **Opposite / Adjacent** — sides relative to a chosen acute angle θ.
- **Sine (sin θ)** — opposite / hypotenuse (SOH).
- **Cosine (cos θ)** — adjacent / hypotenuse (CAH).
- **Tangent (tan θ)** — opposite / adjacent (TOA).
- **SOH-CAH-TOA** — memory aid for the three trig ratios.
- **Similar triangles** — same angles, proportional sides; why trig ratios depend only on the angle.
- **Inverse trig functions** — $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$; recover the angle from the ratio.
- **Angle of elevation** — angle above the horizontal when looking up.
- **Angle of depression** — angle below the horizontal when looking down.
- **Included angle** — the angle between two given sides.
- **Included side** — the side between two given angles.
- **Sine Rule** — $a/\sin A = b/\sin B = c/\sin C$.
- **Cosine Rule** — $c^2 = a^2 + b^2 - 2ab\cdot\cos C$.
- **Ambiguous case (SSA)** — two sides and a non-included angle can give zero, one, or two valid triangles.
- **Space diagonal** — the longest segment inside a box, joining opposite corners: $\sqrt{l^2 + w^2 + h^2}$.

Summative Task: The 3-Paragraph Decision Essay

The prompt. You are a county board member, and the vote is tonight. The shortcut trail through the forest would pay for itself in about 4.9 years — but it cuts through woods the math never priced. Write a 3-paragraph essay: vote YES or NO on building the trail. Use this chapter's geometry, then argue the part the math cannot decide.

Planning your essay (do this before you write). 1. **Compute first.** Run the numbers yourself: $c^2 = 8^2 + 11^2 - 2(8)(11)\cos 60° = 97$, so $c \approx 9.85$ km; savings per trip: $19 - 9.85 = 9.15$ km; daily savings: $200 \times 9.15 \times \$0.15 \approx \$274.50$/day; build cost: $9.85 \times \$50{,}000 \approx \$492{,}500$; payback: ≈1,794 days ≈ 4.9 years. Don't take these on faith — verify them with your own calculation. 2. **Thesis template.** "I vote ___ because ***, even though*** ." Fill it in for YOUR side — either side is defensible if the numbers back it. 3. **Evidence checklist.** Every claim needs a

number behind it: trail length, daily savings, construction cost, and payback period, plus at least one idea from this chapter (the Cosine Rule, the Sine Rule, or right-triangle geometry) that strengthens the case. 4. **Flip test.** Name one specific condition that would flip your vote — 400 cars a day instead of 200? an endangered species found in the forest? If you can't name one, your argument isn't finished.

Paragraph 1 — Your decision and claim. State your vote in one clear sentence, then one or two sentences on why it matters. (Example thesis — either side is fine: "Vote yes, because $274.50 a day in savings pays for the trail in under 5 years." Or the other side: "Vote no, because a 4.9-year payback that destroys a forest is the wrong kind of profit.")

Paragraph 2 — Mathematical evidence. Show the numbers: the trail length from the Cosine Rule ($c^2 = 8^2 + 11^2 - 2(8)(11)\cos 60° = 97$, so $c \approx 9.85$ km), the 9.15 km saved per trip between the lakes, ≈$274.50 saved per day, the ≈$492,500 construction cost, and the ≈4.9-year payback. Every claim needs a number behind it.

Paragraph 3 — Implications, limitations, and what would change your mind. What value does the forest have that never entered the equations? Who gains the savings and who loses the woods — are they the same people? Name one specific condition that would flip your vote — for example, 400 cars a day instead of 200, or an endangered species found in the forest — and explain why.

Length: 200–300 words. Three paragraphs, no more, no fewer.

Using AI. Write your first draft entirely on your own — no AI. Then you may use AI as a revision coach: ask it to check your logic, flag weak evidence, and suggest clearer sentences. Never let it write paragraphs for you. And remember this book's golden rule: AI makes mistakes — verify everything, especially any numbers it gives you.

Rubric and exemplar essays are in the Teacher Guide.

Answer Key — Practice Problems

2.1 — 1. 8 feet ($b^2 = 100 - 36 = 64$). 2. 13 cm (5-12-13). 3. Horizontal gap 6, vertical gap 8; $\sqrt{(36 + 64)} = 10$ units. 4. $\sqrt{(81 + 144)} = 15$ m.

2.2 — 1. $\sin\theta = 5/13 \approx 0.385$; $\cos\theta = 12/13 \approx 0.923$; $\tan\theta = 5/12 \approx 0.417$. 2. (a) 30°, (b) 45°, (c) ≈ 30°. 3. Opposite ≈ 9.33 cm; hypotenuse ≈ 22.07 cm. 4. Height = $15 \cdot \sin(70°) \approx 14.10$ ft; base = $15 \cdot \cos(70°) \approx 5.13$ ft.

2.3 — 1. $100 \cdot \sin(55°) \approx 81.9$ m. 2. $30 \cdot \tan(40°) \approx 25.2$ m. 3. $80/\tan(30°) \approx 138.6$ ft. 4. $\sin\theta = 3/12 = 0.25 \rightarrow \theta \approx 14.5°$.

2.4 — 1. ½ × 6 × 9 × sin(60°) ≈ 23.4 cm². 2. 50 m². 3. ½ × 4 × 4 × sin(60°) ≈ 6.93 cm². 4. The 90° triangle: 44.0 cm² vs 18.6 cm² — 25.4 cm² greater.

2.5 — 1. ∠C = 80°; b = 8·sin(70°)/sin(30°) ≈ 15.04 cm; c = 8·sin(80°)/sin(30°) ≈ 15.76 cm. 2. ≈ 613 m from A. 3. sin B = 14·sin(40°)/10 ≈ 0.8999 → ∠B ≈ 64.1° **or** 115.9°: two triangles fit. 4. ∠C = 60°; c = 20·sin(60°)/sin(65°) ≈ 19.11 m.

2.6 — 1. $c^2 = 25 + 64 - 80(0.5) = 49 \rightarrow c = 7$ cm. 2. Angle opposite 15: cos C = (81 + 144 − 225)/(2·9·12) = 0 → 90° (a 9-12-15 right triangle!); angle opposite 9: $\cos^{-1}(0.8) \approx 36.9°$. 3. $c^2 = 100 + 196 - 280\cdot\cos(120°) = 436 \rightarrow c \approx 20.88$ m. 4. $d^2 = 25 + 81 - 90\cdot\cos(60°) = 61 \rightarrow d \approx 7.81$ km.

2.7 — 1. √(36 + 64 + 100) = √200 ≈ 14.14 cm. 2. √(4 + 9 + 36) = 7 m. 3. Base diagonal = 10 m; tan θ = 10/10 = 1 → θ = 45°. 4. Wire = √(625 + 225) = √850 ≈ 29.15 ft; angle = $\tan^{-1}(25/15) \approx 59.0°$.

Level 4 example checks — **2.1:** X covers both (5.0 km each); Y misses N2 by ≈80 m (≈5 s) — is 5 seconds worth $90,000? Your call. **2.2:** slope 1:10 > 1:12 limit — fails; a compliant ramp needs 30 ft of run. **2.3:** thermostat payback ≈1.1 yr (yes); windows ≈29.6 yr > 25-yr life (no). **2.4:** round ≈$0.0895/in² beats rectangle ≈$0.0926/in² — buy the round. **2.5:** 7 min vs 7 min — tie; triage judgment breaks it. **2.6:** (A) $492,500 build / $274.50/day saved; (B) $240,000 build / $210/day saved — cheaper to build vs better for drivers. **2.7:** max ≈9.87 ft > 9 ft — carry it around the corner.

CHAPTER 3

Function Models & Decision Making

A Note on AI Learning Labs in This Chapter

Chapter 1 taught you the golden rule — **AI makes mistakes, so verify everything** — and Chapter 2 made you practice it on decimals. This chapter raises the stakes: functions are *models*, and a model is a claim about the real world. When an AI hands you a "cost function" or a "profit function," your job is not to admire it — it is to interrogate it. Does each input give exactly one output? Does the slope's sign make sense? Does the vertex land where the story says it should? Every AI Learning Lab in this chapter keeps the four levels (understand, apply, matter, decide), and the standing assignment continues: check every AI number yourself, and score it with percentage error (§1.2).

About This Chapter

A function is a machine: feed it an input, and it returns exactly one output. That simple idea runs the modern world — phone plans, shipping rates, tax brackets, and profit forecasts are all functions. This chapter turns functions into decision tools. You will learn what makes a relation a function, how to read and write function notation, and how linear, quadratic, and exponential functions model real situations — then you will use those models to decide: which company to hire, whether to launch a business, what price to charge, whether the truck fits under the bridge, which bank gets your savings, and whether to buy the car now or wait.

By the end of this chapter, you will be able to:

- Decide whether a relation is a function using mappings and the vertical line test; identify domain and range.
- Read, write, and evaluate function notation f(x), and solve for inputs from outputs.
- Interpret slope as a rate of change and the y-intercept as a starting value; graph linear functions and find equations from data.
- Build cost, revenue, and profit functions; find break-even points and use them to make go/no-go decisions.
- Find the vertex and axis of symmetry of a quadratic; convert between standard, vertex, and factored forms.

- Use quadratic models to maximize revenue or profit and to analyze projectile motion — and defend the resulting decision.
- Distinguish exponential growth from decay; write, evaluate, and graph y = a·b^x.
- Use exponential models — compound interest, depreciation, doubling time and half-life — to make buy/wait/invest decisions.
- Use AI prompts at four levels (understand, apply, matter, decide) to learn functions — and verify AI-generated models with your own calculations.

3.1 What Is a Function?

THE DECISION

Which shipping company can you trust with your business?

You sell handmade soap online and ship about 20 packages a month. Two local shippers quote you for a 2-pound package:

- **ShipFast:** "$8 base fee plus $3 per pound — always."
- **ShipCheap:** "For 2 pounds it's usually around $10 to $16 — depends on the day." (On Monday they charged you $12 for 2 pounds; on Wednesday, $15 for the same 2 pounds.)

Which shipper would you sign a contract with — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Make a small table for each company: for a 2-pound package, how many *different* prices appear?
- **Hint 2:** A function gives each input exactly one output. Which company's price list breaks that rule?
- **Hint 3:** If you ship 20 packages a month, which pricing lets you predict your total bill in advance — and why does that matter for a business?

What Is a Function?

A **function** is a rule that assigns to each input **exactly one** output. "Exactly one" is the whole game: one input, one answer, no surprises, no ties. ShipFast's pricing is a function — a 2-pound package costs $14, today, tomorrow, and next Tuesday. ShipCheap's pricing is *not* a function — the same 2-pound input produced $12 one day and $15 another.

The set of allowed inputs is the **domain**; the set of possible outputs is the **range**. For ShipFast's pricing $C(w) = 8 + 3w$, the domain is weights $w \geq 0$ (you can't ship a negative-weight package), and the range is costs $C \geq 8$ (the base fee is the cheapest possible bill).

STORY

The Vending Machine.

At school, the vending machine's B4 button always drops the same chips — press B4, get chips; press B4 again, get the same chips. Imagine if B4 gave you chips on Monday, candy on Tuesday, and nothing on Wednesday. You would stop trusting the machine, because B4 no longer *means* anything. A function is a vending machine that never lies: each button (input) delivers exactly one snack (output).

Domain and Range

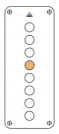
The **domain** answers "what inputs are allowed?" and the **range** answers "what outputs are possible?" Both are part of the function's identity — the same formula with a different domain is a different deal.

Consider $A(t) = 50t$, the dollars you earn in t hours at $50/hour. As pure algebra, t could be anything. But in the real story, $t \geq 0$ (no negative hours) and maybe $t \leq 40$ (a work week). The domain is $0 \leq t \leq 40$, and the range is $0 \leq A \leq 2{,}000$. Change the story — a 60-hour overtime week — and the domain and range change with it, even though the formula didn't.

STORY

The Elevator.

The hotel elevator has buttons for floors 1 through 10. Press 7, and you arrive at floor 7 — every time. But press 11, and nothing happens: 11 is not in the domain. The domain is {1, 2, ..., 10}; the range is the same set. *What would it mean if pressing 7 sometimes took you to floor 7 and sometimes to floor 3?*



Mapping Diagrams

A **mapping diagram** draws the function as arrows: inputs in one oval, outputs in another, one arrow per input. It makes "exactly one output" visible at a glance.

Function: {1, 2, 3} → arrows: 1→A, 2→A, 3→B. (Two inputs may share an output — that's fine. The rule bans one input with *two* outputs, not two inputs with one.)

Not a function: arrows 1→A *and* 1→B. Input 1 has two outputs — the diagram branches, and the rule is broken.

STORY

The Locker Assignment.

On the first day of school, each student is assigned exactly one locker: Maya→114, Jordan→115, Rio→114 (Maya and Rio share — allowed). But imagine the list showed Maya→114 *and* Maya→207. On Monday morning, where does Maya go? A mapping where one input points two ways isn't an assignment — it's a mess. *Why is "two students, one locker" fine, but "one student, two lockers" a problem?*

The Vertical Line Test

On a graph, the "exactly one output" rule becomes a picture: a relation is a function exactly when **no vertical line hits its graph more than once**. Why? A vertical line is the set of all outputs for a *single* input x. Two hits means one input with two outputs — not a function.

- y = 2x + 1: every vertical line hits exactly once → function.

• $x^2 + y^2 = 25$ (a circle): the vertical line $x = 0$ hits $(0, 5)$ and $(0, -5)$ → not a function.

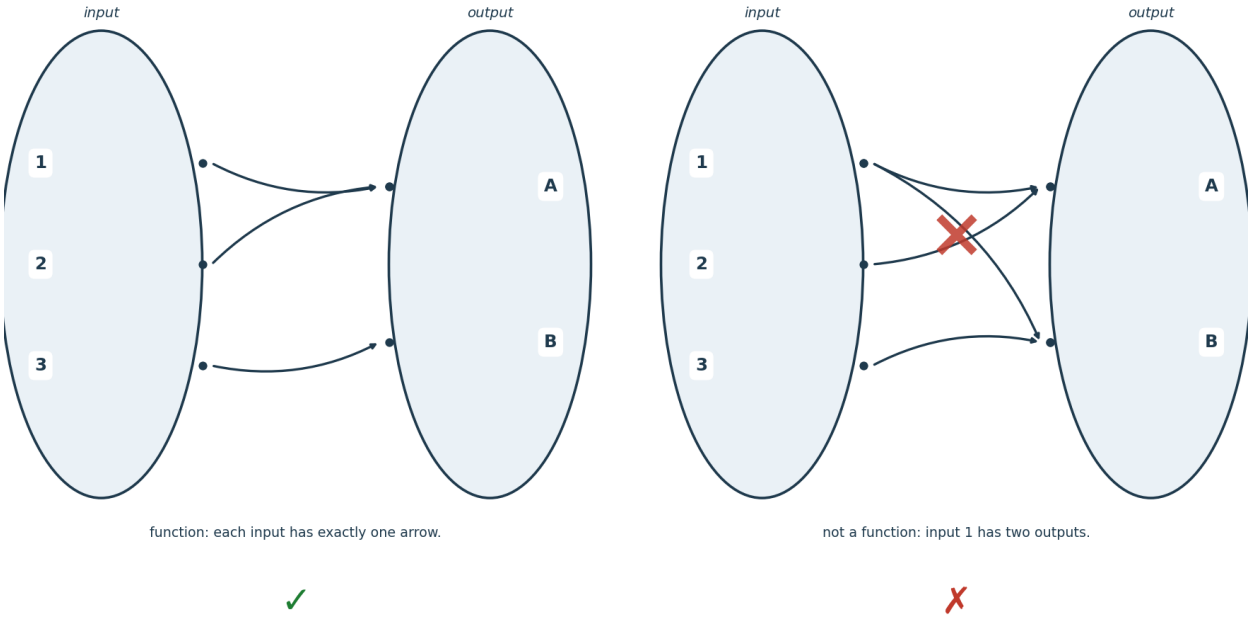
STORY

The Speed Camera.

A speed camera photographs your car at one instant and records one speed: at 3:04 p.m., you were going 42 mph. It cannot record 42 *and* 55 at the same instant — time is the input, speed is the output, and physics allows only one. *If the camera's graph showed two speeds at 3:04 p.m., would you trust the ticket?*

Fig. 3.1

Fig. 3.1 — Mapping diagrams: function vs. not.



function: each input has exactly one arrow. ✓

not a function: input 1 has two outputs. ✗

A function never branches.

Worked Example 1 — The price table (Hint 1). Build the table Hint 1 asked for: for a 2-pound package, which prices has each company actually charged — and how many *different* prices is that?

Solution. ShipFast: C(w) = 8 + 3w, so C(2) = 8 + 3(2) = **$14** — one price, every time. ShipCheap: **$12** on Monday, **$15** on Wednesday — two different prices for the same 2-pound input.

- ShipFast: one input → one output. Its price list **is a function** of weight — Hint 2's test, passed.
- ShipCheap: one input → two outputs. Its price list **is not a function** — Hint 2's test, failed.

Hint 3's payoff: with ShipFast, 20 packages at 2 pounds cost 20 × $14 = **$280**, known in advance. With ShipCheap, "20 × something between $10 and $16" is a guess, not a budget. A function is a price list you can plan on.

Worked Example 2 — Domain and range in context. ShipFast charges C(w) = 8 + 3w dollars for a package weighing w pounds. What are the domain and range in this story?

Solution. Weight can't be negative, so the domain is w ≥ 0. The cheapest bill is the base fee at w = 0: C(0) = $8, and cost grows from there, so the range is C ≥ 8. (Pure algebra would allow all real numbers — the *story* restricts the domain.)

DECIDE — THE VERDICT

Which shipping company can you trust with your business?

Back to the question that opened this section — now with the math to settle it.

Solution. ShipFast: C(w) = 8 + 3w. Every weight maps to exactly one price: C(2) = 8 + 6 = **$14**, C(5) = **$23**, C(10) = **$38**. It is a function — predictable. ShipCheap: the input "2 pounds" mapped to $12 *and* $15. It fails the definition of a function — unpredictable.

The decision: sign with ShipFast. For 20 packages a month at ~2 pounds, your bill is 20 × $14 = **$280**, known in advance. ShipCheap might average less — or might not; you can't budget a maybe. In business, a function is worth more than a gamble: predictability *is* the product.

AI AI Learning Lab 3.1

AI Level 1 — Understand.

Explain what a function is as if I'm 15, using the vending-machine idea: each button gives exactly one snack. Then give me 3 relations (as sets of ordered pairs), one at a time, and ask me whether each is a function and why. Check each answer before moving on.

AI Level 2 — Apply.

Give me 5 "function or not?" problems mixing ordered pairs, tables, and graphs (describe the graphs in words). Hide the answers. I'll solve each, state the domain and range where it's a function, and show my reasoning; then check my work step by step.

AI Level 3 — Matter.

The shipping decision showed that a non-function price list — $12 one day, $15 the next for the same package — is untrustworthy, even if it sometimes looks cheaper. But this runs deeper than shipping. Discuss with me: WHY does "one input, exactly one output" matter for trust in everyday life — think about grading (one student, one grade), medicine (one symptom set, one diagnosis process), or the law (one crime, one sentence range)? What are the implications when a system that *should* be a function starts behaving like ShipCheap — who gets hurt, and who benefits from the unpredictability? End by asking me for one system in MY life I now want to test for function-ness.

Level 4 — Decide. *If you were choosing, which membership would you pick?*

> **Example decision problem.** Gym A charges A(v) = 25 + 8v dollars per month, where v is your number of visits (a $25 base plus $8 per visit). Gym B charges a flat $60/month. You expect to go about 6 times a month. Which gym should you join?

AI Level 4 — Decide.

I need to decide between two gyms. Gym A charges A(v) = 25 + 8v per month (v = visits); Gym B is a flat $60/month. I expect ~6 visits a month. Act as my decision coach: make me evaluate both functions at v = 6 myself, then make me find the crossover point where they cost the same (solve 25 + 8v = 60). Then ask me: if my habit slips to 3 visits, does my decision change? Make me state my final choice with reasons — don't decide for me.

Check your verdict: A(6) = 25 + 48 = $73 vs. B's $60 — Gym B wins at 6 visits, saving $13/month. Crossover at v = 4.375, so 5+ visits favors B and 4 or fewer favors A. The decision depends on honest self-knowledge about your habits.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.1.

Ask your AI: "Invent a 6-question 'function or not?' quiz using mappings, tables, and graphs, with an answer key." Then do the detective work: verify *every* answer yourself with the mapping rule or the vertical line test. If the AI mislabeled even one, write a one-paragraph explanation of exactly which input broke the rule and how. AI chatbots are fluent at inventing examples and sloppy at checking them — in this chapter, *you* are the checker.

Practice Problems 3.1

1. Is {(2, 5), (3, 5), (4, 7)} a function? If so, state its domain and range.
2. Is {(1, 1), (1, 2), (2, 3)} a function? Explain which input breaks the rule.
3. A table shows x: 0, 1, 2, 3 and y: 4, 4, 7, 4. Is y a function of x? Explain.
4. The graph of $x^2 + y^2 = 25$ is a circle. Use the vertical line test to decide whether y is a function of x.
5. Is $y = x^2$ a function of x? Is $x = y^2$ a function of x? Explain the difference.

3.2 Function Notation

THE DECISION

Which summer job should you take?

Two landscaping companies offer you summer work, and both write the offer as a function of h, your hours per week:

- **GreenGrow:** P(h) = 18h + 200 ($18/hour plus a $200 weekly equipment bonus).
- **QuickCut:** Q(h) = 22h ($22/hour, no bonus).

You expect to work about 30 hours a week — but some weeks could reach 55 hours.

Which job would you choose — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** P(30) means "replace every h in 18h + 200 with 30." Do the same for Q(30).
- **Hint 2:** Compare the two paychecks at 30 hours. Now try 55 hours — does the winner change?
- **Hint 3:** Set the functions equal (18h + 200 = 22h) and solve for h. That crossover hour tells you exactly when the better job flips.

Reading f(x): "f of x"

The notation **f(x)** is read "**f of x**" — it means "the output of the function named f when the input is x." The letter f is the function's *name* (like a person's name); the x in parentheses is the *input*. So f(3) means "the output of f when the input is 3."

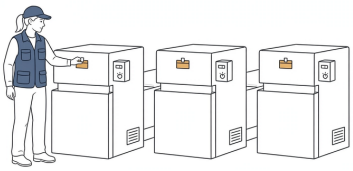
This notation is compact and powerful. Instead of writing "the weekly pay, in dollars, when you work h hours," you write **P(h)** — and P(30) is a complete question: *what is the pay for 30 hours?*

STORY

The Name Tag.

At the warehouse, every machine has a name tag:

"Press A," "Press B." The tag *f* on a machine means "this machine's output." f(3) is just asking machine f: "what do you produce from input 3?" *If g(5) = 15, what is machine g telling you about input 5?*



Evaluating Functions

To **evaluate** f(a), substitute a for every x and compute. From this section's warm-up:

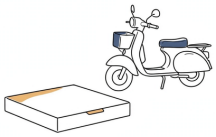
- f(x) = −3x + 12, find f(3): f(3) = −3(3) + 12 = −9 + 12 = **3**.
- g(x) = 2x + 5, find g(5): g(5) = 2(5) + 5 = **15**.

The parentheses are doing real work: they mark the input slot. f(x + 1) means "put x + 1 into the machine," which is *not* the same as f(x) + 1. Pay attention to what goes inside the parentheses — that's the input.

STORY

The Delivery Fee.

Tony's pizza app charges D(m) = 3 + 1.5m dollars to deliver m miles. D(4) = 3 + 6 = **$9** — four miles costs nine dollars. His friend claims "D(4) means D times 4." *Is D(4) multiplication? What does the 4 actually do inside the parentheses?*



Solving for the Input

Functions also run backwards: given an output, find the input. If f(x) = 2x − 7 and f(x) = 15, then 2x − 7 = 15, so 2x = 22 and **x** = **11**. Check: f(11) = 22 − 7 = 15 ✓. This "reverse evaluation" is how you answer "how many hours until I earn $1,000?" — set the function equal to the target and solve.

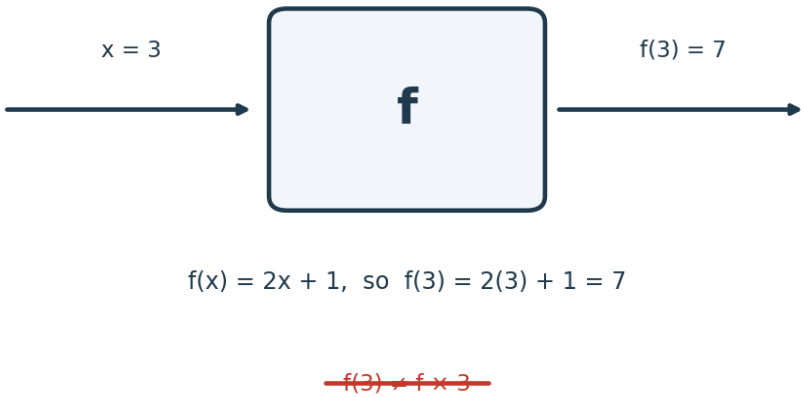
STORY

The $1,000 Goal.

Priya's tutoring pays T(h) = 25h dollars for h hours. She needs $1,000 for a laptop: 25h = 1,000 → **h** = **40**. Forty hours of tutoring buys the laptop. *Her friend says "just evaluate T(1000)" — why is that the wrong move, and what should she do instead?*

Fig. 3.2

Fig. 3.2 — The function machine



f(x) names the machine (f) and the input (x) — it is not multiplication.

Worked Example 1 — Evaluating the two offers (Hint 1). Evaluate P(30) and Q(30): replace every h with 30 and compute.

Solution. P(30) = 18(30) + 200 = 540 + 200 = **$740**. Q(30) = 22(30) = **$660**. The parentheses are input slots — 30 goes in, a paycheck comes out. At 30 hours, GreenGrow pays $80 more per week.

Worked Example 2 — The crossover (Hints 2 and 3). Set the two functions equal and solve for h — the exact hour where the better job flips. Then check a 55-hour week.

Solution. 18h + 200 = 22h → 200 = 4h → **h = 50**. At exactly 50 hours both pay the same: P(50) = 18(50) + 200 = **$1,100** = Q(50). Check a 55-hour week (Hint 2): P(55) = 990 + 200 = **$1,190**; Q(55) = 22(55) = **$1,210** — QuickCut takes the lead by $20. The winner *changes with the hours*: below 50, GreenGrow; above 50, QuickCut.

Worked Example 3 — Notation in a story. A catering company charges C(p) = 200 + 45p dollars for a party with p guests. What does C(30) mean, and what is it? How many guests can you invite with a $2,000 budget?

Solution. C(30) is the cost for 30 guests: C(30) = 200 + 45(30) = 200 + 1,350 = **$1,550**. For the budget: 200 + 45p = 2,000 → 45p = 1,800 → **p = 40 guests**.

DECIDE — THE VERDICT

Which summer job should you take?

Back to the question that opened this section — now with the math to settle it.

Solution. Evaluate both pay functions:

- At 30 hours: P(30) = 18(30) + 200 = 540 + 200 = **$740**; Q(30) = 22(30) = **$660**. GreenGrow wins by $80.
- At 55 hours: P(55) = 990 + 200 = **$1,190**; Q(55) = 22(55) = **$1,210**. QuickCut wins by $20.
- Crossover: 18h + 200 = 22h → 4h = 200 → **h = 50**. At exactly 50 hours both pay $1,100.

The decision: at your expected 30 hours, take GreenGrow — $80 more per week. But function notation revealed the flip: past 50 hours, QuickCut wins. If overtime-heavy weeks are likely, renegotiate or reconsider. The notation didn't just answer "which job?" — it answered "*when* does the answer change?"

AI AI Learning Lab 3.2

AI Level 1 — Understand.

Explain function notation f(x) as if I'm 15: what the f names, what the x in parentheses means, and why f(3) is NOT f times 3. Then quiz me with 3 evaluations, one at a time, and check each before moving on.

AI Level 2 — Apply.

Give me 5 function-notation problems: 3 evaluations (including one with a negative input) and 2 "find the input" problems. Hide the answers. I'll show my substitution step by step; then check my work and pinpoint any substitution error.

AI Level 3 — Matter.

The summer-job decision turned on reading P(30) vs. Q(30) — two little expressions that held a whole summer's wages. But notation is power, and power cuts both ways. Discuss with me: WHY does it matter that workers can read formulas like P(h) = 18h + 200 — who benefits when employees *can't* decode their own pay stub, and what are the implications for wage transparency? Flip it: when is a formula *less* honest than plain words (fine print? hidden fees dressed as variables?)? Ask me: what's one formula in MY life — a bill, a grade scheme, a game rule — I should learn to read?

Level 4 — Decide. *If it were your $60, how many hours would you ride?*

> **Example decision problem.** A bike rental shop charges R(h) = 12 + 7h dollars for h hours ($12 flat plus $7/hour). You have $60. How many full hours can you ride, and how much money is left over?

AI Level 4 — Decide.

I'm deciding how long to rent a bike: R(h) = 12 + 7h, and I have $60. Act as my decision coach. Make me test h = 6 and h = 7 myself and compare each total to $60. Then ask me: is it smarter to ride 6 hours and keep the change, or is there a better use of the leftover? Make me state my decision with the arithmetic shown — don't decide for me.

Check your verdict: R(6) = 12 + 42 = $54 ≤ $60; R(7) = $61 > $60. Six full hours, $6 left over — enough for a snack, not for a seventh hour.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.2.

Ask your AI: "Write a realistic weekly-pay function from a job ad (base + per-hour or per-item), then ask me to evaluate it at three different input values." Solve them yourself first — then make the AI show its work and compare line by line. Compute the percentage error (§1.2) on any disagreement — treat your hand-computed answer as the true value. If the AI's substitution slipped (a dropped negative sign is the classic), write a one-paragraph "substitution safety checklist" for your notebook.

Practice Problems 3.2

1. If f(x) = 4x − 9, find f(6).
2. If g(x) = −2x + 5, find g(−3). Watch the signs.
3. If h(x) = x^2 + 1, find h(4) and h(−4). What do you notice?
4. If f(x) = 3x + 8, find x so that f(x) = 50.
5. A party costs C(p) = 200 + 45p dollars for p guests. Find C(30). How many guests can you invite with a $2,000 budget?

3.3 Linear Functions

THE DECISION

Which moving company should you hire?

You're moving 400 miles for college. Two companies quote you as functions of m, the miles:

- **MoveEasy:** M(m) = 300 + 2.50m ($300 base plus $2.50 per mile).
- **QuickMove:** Q(m) = 500 + 1.75m ($500 base plus $1.75 per mile).

Which company would you hire for your 400-mile move — and how would you prove it? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Write each quote as a function and evaluate both at m = 400.
- **Hint 2:** The number in front of m is the per-mile rate (the slope); the constant is the starting cost (the intercept). Which company climbs faster?
- **Hint 3:** Set the two functions equal and solve for m — the crossover mile tells you exactly when the cheaper company flips.

Slope: Rate of Change

A **linear function** has the form **f(x) = ax + b**, and its graph is a straight line. The number **a** is the **slope**: how much the output changes for each 1-unit increase in the input — the *rate of change*.

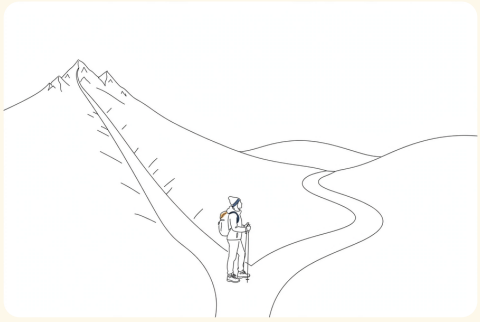
Slope = (change in output) ÷ (change in input) = $(y_2 - y_1) \div (x_2 - x_1)$.

For MoveEasy's M(m) = 300 + 2.50m, the slope 2.50 means "each extra mile adds $2.50." For QuickMove, each extra mile adds only $1.75 — a gentler climb. Steeper slope = faster-growing cost.

STORY

The Hiking Trail.

Trail A climbs 800 feet over 2 miles (slope = 400 ft/mile); Trail B climbs 800 feet over 4 miles (slope = 200 ft/mile). Same mountain, different slopes — Trail A is the thigh-burner. *If a trail sign said "elevation gain: 0 feet per mile," what would the hike feel like?*



The y-Intercept: Where You Start

The number **b** is the **y-intercept**: the output when the input is zero — where the line starts. MoveEasy's intercept is \$300 (you owe \$300 before the truck moves an inch); QuickMove's is \$500. A lower intercept with a steeper slope versus a higher intercept with a gentler slope — that tension *is* the decision in this section.

The **x-intercept** is where the output hits zero (solve ax + b = 0). For cost functions it's rarely reached — but for profit functions, the x-intercept is the break-even point (§3.4).

STORY

The Taxi Meter.

The meter reads \$3.50 the moment you sit down — before a single wheel turns. That \$3.50 is the y-intercept: the starting value. Then \$2.10 per mile ticks on — that's the slope. *Two taxis: one starts at \$3.50 with \$2.10/mile, another at \$5.00 with \$1.60/mile. For a 2-mile ride, which is cheaper — and at what distance do they tie?*

Graphing Lines

To graph f(x) = ax + b: plot the y-intercept (0, b), then use the slope as "rise over run" — from any point, move right 1 and up a. Two points determine the line; a third checks your work.

From this section's warm-up: find f(x) = ax + b with f(2) = 1 and f(−3) = 11. Slope a = (11 − 1) ÷ (−3 − 2) = 10 ÷ (−5) = **−2**. Then 1 = −2(2) + b gives **b = 5**, so f(x) = −2x + 5. Check: f(−3) = 6 + 5 = 11 ✓. Two points, one line, zero mystery.

STORY

The Lemonade Stand.

Ana's stand profits P(c) = 2c − 30 dollars

for c cups sold ($2 per cup, $30 stand cost). The graph starts at (0, −30) — day one, she's $30 in the hole — and climbs with slope 2. *Where does her line cross zero, and what does that crossing mean for Ana?*

Fig. 3.3

Two linear cost models cross once.

Worked Example 1 — Evaluate both quotes (Hint 1). M(m) = 300 + 2.50m and Q(m) = 500 + 1.75m. Evaluate each at m = 400.

Solution. M(400) = 300 + 2.50(400) = 300 + 1,000 = **$1,300**. Q(400) = 500 + 1.75(400) = 500 + 700 = **$1,200**. For your 400-mile move, QuickMove is $100 cheaper — but don't stop at the totals. The hints ask *why* the winner wins.

Worked Example 2 — Who climbs faster? (Hint 2). Read the slope and the intercept off each quote, then build a small table of costs at m = 0, 100, 200, 300, 400 and watch the gap change.

Solution. MoveEasy: slope **2.50** ($/mile), intercept **$300**. QuickMove: slope **1.75** ($/mile), intercept **$500**. MoveEasy climbs faster (2.50 > 1.75) but starts lower (300 < 500).

m = 0: M = $300, Q = $500 (Q costs $200 more) m = 100: M = $550, Q = $675 (Q costs $125 more) m = 200: M = $800, Q = $850 (Q costs $50 more) m = 300: M = $1,050, Q = $1,025 (M costs $25 more) m = 400: M = $1,300, Q = $1,200 (M costs $100 more)

The steeper line starts cheaper and loses ground every mile — the table shows the lead flipping somewhere between 200 and 300 miles.

Worked Example 3 — The crossover (Hint 3). Set M(m) = Q(m) and solve: the exact mile where the cheaper company flips.

Solution. 300 + 2.50m = 500 + 1.75m → 0.75m = 200 → m = 200 ÷ 0.75 ≈ **267 miles**. Below 267 miles, MoveEasy's lower intercept wins; above 267, QuickMove's gentler slope wins. Your move is 400 miles — past the flip.

DECIDE — THE VERDICT

Which moving company should you hire?

Back to the question that opened this section — now with the math to settle it.

Solution. Evaluate at 400 miles:

- M(400) = 300 + 2.50(400) = 300 + 1,000 = **$1,300**.
- Q(400) = 500 + 1.75(400) = 500 + 700 = **$1,200**.

Crossover: 300 + 2.50m = 500 + 1.75m → 0.75m = 200 → **m ≈ 267 miles**. Below 267 miles MoveEasy wins (lower intercept); above it, QuickMove's gentler slope takes over.

The decision: hire **QuickMove** — $100 cheaper for your 400-mile move. The slope told you the *rate* of the race; the intercept told you who got the head start. Read both, and the graph picks the winner before you spend a dime.

AI AI Learning Lab 3.3

AI Level 1 — Understand.

Explain slope as "rise over run" and as a rate of change, as if I'm 15. Use a taxi meter ($3.50 start + $2.10/mile): which number is the slope, which is the intercept, and what does each mean in plain English? Then quiz me on 3 graphs you describe in words, one at a time.

AI Level 2 — Apply.

Give me 5 problems: 2 slope-from-two-points, 1 find-the-equation (like f(2) = 1, f(−3) = 11), and 2 intercept problems. Hide the answers. I'll show every step; then check my work and flag any sign error precisely.

AI Level 3 — Matter.

The moving-company decision was really about reading slopes: 2.50 vs. 1.75 dollars per mile decided a $100 difference. But slopes run the world quietly — wages per hour, rent per square foot, emissions per mile. Discuss with me: WHY does it matter that ordinary people can read a slope — who benefits when consumers *can't* translate "per mile" or "per month" into total cost? What are the implications for fairness when companies advertise the intercept ("only $300 to start!") and hide the slope? Ask me: what's one advertised deal in MY life whose slope I should compute?

Level 4 — Decide. *If you were booking the rental, which company wins?*

> **Example decision problem.** Company A rents cars at A(d, m) = 55d + 0.30m ($55/day plus $0.30/mile); Company B charges B(d) = 85d flat per day. For a 4-day trip driving 500 miles, which is cheaper, and by how much?

AI Level 4 — Decide.

I'm deciding between two rental-car companies for a 4-day, 500-mile trip: A(d, m) = 55d + 0.30m vs. B(d) = 85d flat. Act as my decision coach: make me compute both totals myself (careful: A has TWO inputs). Then ask me what would flip the decision — fewer miles? more days? Make me state my choice with the numbers shown.

Check your verdict: A = 55(4) + 0.30(500) = 220 + 150 = $370; B = 85(4) = $340. Company B wins by $30. Note the structure: B's flat daily rate is a pure intercept play — it wins exactly when mileage is high.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.3.

Ask your AI: "Invent two competing phone plans as linear functions and tell me which is cheaper for 8 GB/month." Then verify everything yourself: find both functions' slopes and intercepts, compute the 8-GB totals, and find the crossover point. If the AI's crossover is off by even a mile (or a megabyte), compute the percentage error (§1.2) and write a one-paragraph ruling on which plan *really* wins — with the crossover as your evidence.

Practice Problems 3.3

1. Find the slope of the line through (2, 5) and (6, 13).
2. For f(x) = −3x + 7, state the slope and the y-intercept.
3. Write the equation of the line through (1, 4) with slope 2.
4. Find the x-intercept of y = 3x + 9.
5. Sketch the graph of y = −(1/2)x + 3 by finding both intercepts. (Describe the points; graph paper recommended.)

3.4 Linear Models for Decisions

THE DECISION

Should you launch the T-shirt business?

You want to sell custom T-shirts at school. The numbers:

- Fixed costs: **$2,400** (screen-printing setup, website, initial stock).
- Cost per shirt: **$6** (blank shirt + printing).
- Selling price: **$15** per shirt.
- Your honest estimate: you can sell **300 shirts** this semester.

Would you launch — risking $2,400 up front — and how would you prove it's a good call? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Write the total cost as a function C(x) and the total revenue as a function R(x), where x is shirts sold.
- **Hint 2:** Profit is revenue minus cost. Break-even is where profit equals zero — set the functions equal and solve.
- **Hint 3:** Compare your 300-shirt estimate with the break-even number. How much cushion do you have? What happens at 250 shirts?

Cost, Revenue, and Profit Functions

Three functions run every business:

- **Cost:** C(x) = (fixed costs) + (cost per item)·x.
- **Revenue:** R(x) = (price per item)·x.
- **Profit:** P(x) = R(x) − C(x).

For the T-shirts: C(x) = 2,400 + 6x, R(x) = 15x, so P(x) = 15x − (2,400 + 6x) = **9x − 2,400**. Every shirt sold contributes $9 toward covering the $2,400 fixed cost — that $9 is called the **contribution margin**, and it is the slope of the profit line.

STORY

The Bake Sale.

The chess club's bake sale has $60 in fixed costs (table fee, ingredients bought in bulk) and each cookie costs $0.40 to make, selling for $1.50. P(x) = 1.10x − 60: every cookie contributes $1.10 toward the $60 hole. *If they sell 50 cookies, are they in profit yet — and how do you know without computing the whole total?*

The Break-Even Point

The **break-even point** is where profit is zero — cost and revenue cross. Set R(x) = C(x) and solve. Below it you lose money; above it you earn.

For the T-shirts: 15x = 2,400 + 6x → 9x = 2,400 → **x ≈ 267 shirts**. Sell 267 and you exactly cover costs (check: R(267) = $4,005; C(267) = 2,400 + 1,602 = $4,002 — the $3 gap is rounding; the true crossing is at 266.67). Shirt 267 is the first one that truly earns.

STORY

The Car Wash.

The band's car wash charges $10 per car; soap and supplies cost $2 per car, plus a $96 pressure-washer rental. Break-even: 10x = 96 + 2x → 8x = 96 → **12 cars**. The 12th car pays off the washer; the 13th car is pure profit. *The forecast says rain at 3 p.m. — if they can wash only 10 cars, how much do they lose?*

Comparing Two Models

Linear models also settle "which option?" contests. From this section's class activity: a cereal company finds $40,000 in ads sells 100,000 boxes and $60,000 sells 200,000 boxes. The ad-sales model: slope = (200,000 − 100,000) ÷ (60,000 − 40,000) = **5 boxes per dollar**; using (40,000, 100,000): 100,000 = 5(40,000) + b → **b = −100,000**. So f(x) = 5x − 100,000. To sell 300,000 boxes: 5x − 100,000 = 300,000 → 5x = 400,000 → **x = $80,000** in ads. The negative intercept even tells a story: with zero ads, the model predicts −100,000 boxes — i.e., the model only makes sense past $20,000 in spending (where f(x) = 0).

STORY

The Ad Budget.

The cereal company's data said $40k → 100k boxes, $60k → 200k boxes. The marketing manager wants 300k boxes and guesses "$100,000 should do it." *The linear model says $80,000 — who's right, and what assumption is the model secretly making?*

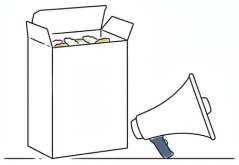
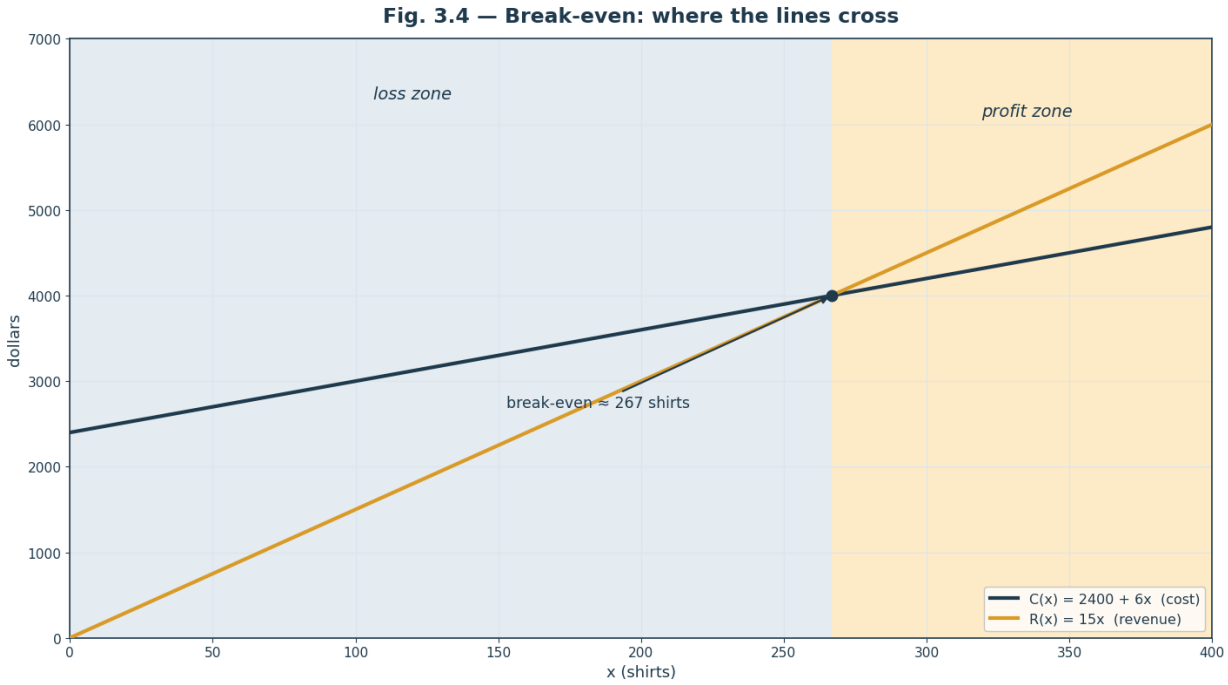


Fig. 3.4



Left of the crossing you bleed; right of it you earn.

Worked Example 1 — Write the functions, find break-even (Hints 1 and 2). Write C(x) and R(x) for the T-shirt business, form P(x), then set revenue equal to cost.

Solution. C(x) = 2,400 + 6x; R(x) = 15x (Hint 1). Profit: P(x) = R(x) − C(x) = 15x − (2,400 + 6x) = **9x − 2,400**. Break-even (Hint 2): 15x = 2,400 + 6x → 9x = 2,400 → x = 2,400 ÷ 9 ≈ 266.67. You can't sell two-thirds of a shirt, so round up: **267 shirts**. Check: R(267) = 15(267) = $4,005; C(267) = 2,400 + 6(267) = $4,002 — the $3 gap is rounding; the true crossing sits at 266.67.

Worked Example 2 — Your estimate vs. break-even (Hint 3). For the T-shirt business (C(x) = 2,400 + 6x, R(x) = 15x), what is the profit at 300 shirts? At 250? How much cushion do you have?

Solution. P(x) = 9x − 2,400. P(300) = 2,700 − 2,400 = **+$300**. P(250) = 2,250 − 2,400 = **−$150** (a $150 loss). Your cushion is 300 − 267 = **33 shirts**: miss your estimate by more than 33 and the profit turns into a loss. Fifty shirts is the difference between +$300 and −$150.

DECIDE — THE VERDICT

Should you launch the T-shirt business?

Back to the question that opened this section — now with the math to settle it.

Solution. C(x) = 2,400 + 6x; R(x) = 15x; P(x) = 9x − 2,400.

- Break-even: 9x = 2,400 → x = 266.67 → **267 shirts**.
- At your 300-shirt estimate: P(300) = +**$300 profit**.
- At 250 shirts: P(250) = **−$150 loss**.

The decision: launch — but eyes open. Your cushion is only 33 shirts (300 − 267): an 11% shortfall in sales flips profit to loss. The model says *go* if you trust your 300-shirt estimate, and *wait* if you don't. That's what break-even analysis is for: it doesn't remove the risk, it measures it.

AI AI Learning Lab 3.4

AI Level 1 — Understand.

Explain cost, revenue, and profit functions as if I'm 15, using a lemonade stand. Then explain the break-even point three ways: as an equation, as a graph crossing, and as a sentence a business owner would say. Quiz me with 2 quick checks, one at a time.

AI Level 2 — Apply.

Give me 4 break-even problems with different numbers (a bake sale, a car wash, a lawn-mowing service, a 3D-printing shop). Hide the answers. I'll write C(x), R(x), P(x) and solve for each break-even; then check my algebra step by step.

AI Level 3 — Matter.

The T-shirt decision came down to a 33-shirt cushion — an 11% sales shortfall turns $300 profit into a loss. That's the mathematics of risk, and real businesses live or die on cushions like it. Discuss with me: WHY does break-even analysis matter beyond passing a math test — who gets hurt when a founder *can't* compute it (think: food-truck owners, Etsy sellers, your neighbor's lawn-care startup)? What are the implications of the "contribution margin" idea — that every sale first pays off the past (fixed costs) before it builds the future? Ask me: what's one thing I might someday sell, and what would its fixed costs be?

Level 4 — Decide. *If it were your $80, would you set up the stand?*

Example decision problem. A lemonade stand costs $80 fixed (stand, cooler, cups in bulk); each cup costs $0.50 to make and sells for $2. You expect to sell 100 cups on Saturday. Should you do it?

AI Level 4 — Decide.

I'm deciding whether to run a lemonade stand Saturday: $80 fixed, $0.50/cup cost, $2/cup price, expecting 100 cups. Act as my decision coach: make me write C(x), R(x), P(x) myself, find the break-even, and compute profit at 100 cups AND at 60 cups (bad weather). Then ask me: how many cups would I need to make this *clearly* worth my Saturday? Make me state the decision with numbers.

Check your verdict: P(x) = 1.50x − 80; break-even at x = 53.33 → 54 cups. At 100 cups: P = $70 profit. At 60 cups: P = $10 — technically profit, but $10 for a Saturday? The math says yes; judgment sets the bar.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.4.

Ask your AI: "I'm thinking of selling custom phone cases at school. Fixed costs $500, cost $4/case, price $12/case. Build my profit function and tell me the break-even." Then audit it like an accountant: recompute C(x), R(x), P(x) yourself, verify the break-even, and test two scenarios the AI didn't mention (what if the price drops to $10? what if fixed costs were really $800?). Write a one-paragraph "investment memo" with your own verdict — fund it or kill it — citing your numbers, not the AI's.

Practice Problems 3.4

1. A business has C(x) = 1,200 + 8x and R(x) = 20x. Find the break-even point.
2. For the business in problem 1, find the profit at x = 250.
3. A hot-dog cart has $150 fixed costs, costs $1.20 per hot dog, and sells at $3.50. Write P(x) and find the break-even (round up to whole hot dogs).
4. Two suppliers: A(x) = 500 + 30x and B(x) = 900 + 20x (x = units). At what quantity do they cost the same? Which is cheaper for 100 units?
5. If the T-shirt price in this section dropped from $15 to $12 (costs unchanged), what is the new break-even? What does the change tell you about pricing power?

3.5 Quadratic Functions

THE DECISION

Can the truck pass under the arch?

A historic stone bridge has a parabolic arch: 40 feet wide at the base and 10 feet tall at the center. Its shape follows $y = -x^2/40 + 10$ (x = feet from the center, y = feet above the road). A delivery truck is 8 feet wide and 8 feet tall and must pass under the center of the arch.

Would you drive the truck through — and how would you prove it's safe? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** The arch's highest point is the vertex of the parabola — find it — the vertex is at x = 0; evaluate y there.
- **Hint 2:** The truck's roof corners sit at x = −4 and x = +4 feet from center. Evaluate the arch's height at those x-values.
- **Hint 3:** Compare the arch's height at x = 4 with the truck's 8-foot height. How much clearance is there — and is it enough for you?

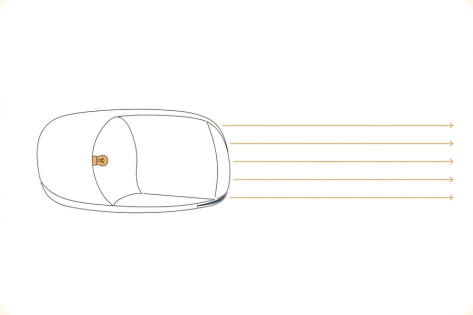
The Parabola's Shape

A **quadratic function** has the form $\mathbf{f(x) = ax^2 + bx + c}$ ($a \neq 0$), and its graph is a **parabola** — a U-shape. If $a > 0$ the parabola opens upward (a valley, with a minimum); if $a < 0$ it opens downward (a hill, with a maximum). The bridge arch opens downward ($a = -1/40 < 0$): it is a hill, and its peak is what the truck must clear.

STORY

The Headlight.

A car's headlight reflector is a parabola. Every ray from the bulb bounces off the curved mirror and exits *parallel* — that's why the beam is a beam instead of a glow. The curve's exact shape decides where the light goes. *What would happen to the beam if the reflector were a V-shape (two lines) instead of a parabola?*



Vertex and Axis of Symmetry

The **vertex** is the parabola's turning point — its peak or valley. The **axis of symmetry** is the vertical line through the vertex; the parabola mirrors itself across it. For $f(x) = ax^2 + bx + c$:

> **Vertex x-coordinate** $= -b/(2a)$; the axis of symmetry is $x = -b/(2a)$.

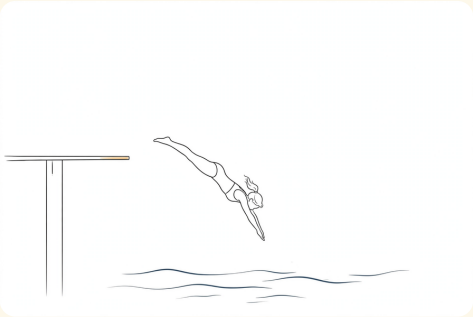
For the arch $y = -x^2/40 + 10$: $a = -1/40$, $b = 0$, so the vertex is at $x = -0 \div (2(-1/40)) = \mathbf{0}$, and $y = 10$. Vertex: **(0, 10)** — 10 feet at the center, exactly as advertised. Axis: $x = 0$.

From this section's warm-up: $f(x) = 2x^2 - 8x + 3$. Vertex at $x = -(-8)/(2\cdot 2) = \mathbf{2}$; $f(2) = 8 - 16 + 3 = \mathbf{-5}$. Vertex: $(2, -5)$, a valley ($a = 2 > 0$). A table of values shows the symmetry: $x = 0,1,2,3,4$ gives $y = 3,-3,-5,-3,3$ — mirror images around $x = 2$.

STORY

The Diver.

A diver's height off the board follows a parabola: up, peak, down — and the rise mirrors the fall around the peak instant. The vertex is the moment of weightlessness at the top. *If the diver's peak is 17 feet at 0.75 seconds, how high is she at 0.5 seconds — and what does symmetry tell you about her height at 1.0 second?*



Three Forms of a Quadratic

Quadratics wear three useful disguises:

- **Standard form:** $y = ax^2 + bx + c$ — read a, b, c straight off; the y-intercept is c.

- **Vertex form:** $y = a(x - h)^2 + k$ — the vertex (h, k) stares at you. (The arch is already here: $y = -(1/40)(x - 0)^2 + 10$.)
- **Factored form:** $y = a(x - r_1)(x - r_2)$ — the x-intercepts (roots) r_1, r_2 stare at you.

Example: $y = (x - 1)(x - 5) = x^2 - 6x + 5$. Roots at x = 1 and 5; the vertex sits halfway between at x = 3, y = (2)(−2) = **−4**. Three forms, one parabola — pick the form that answers your question.

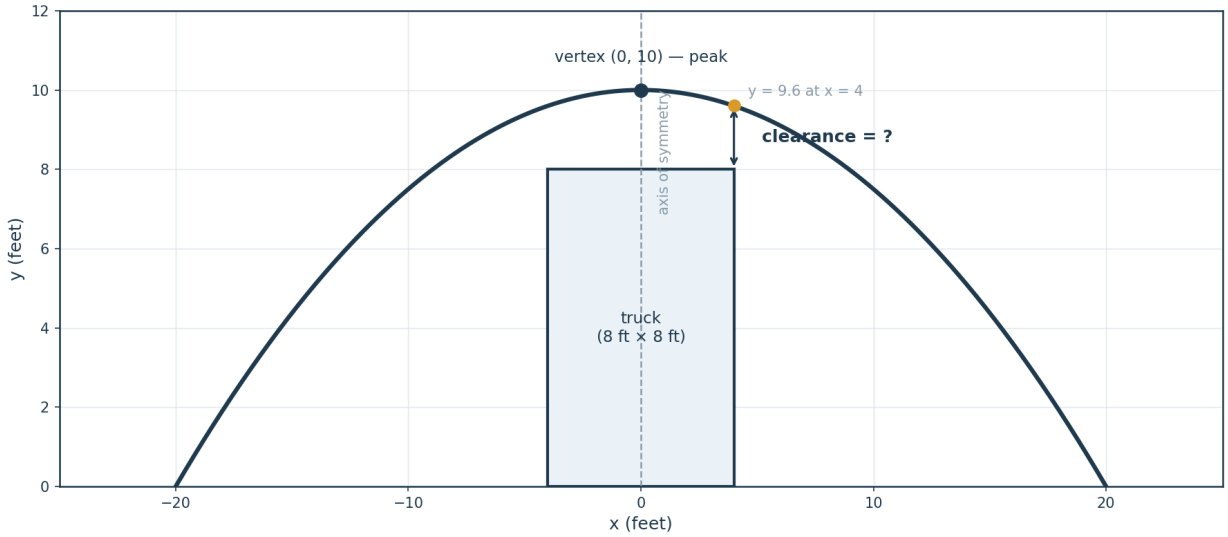
STORY

The Three Disguises.

A detective gets three descriptions of one suspect: "wears a red hat" (standard form — easy to spot the y-intercept), "standing at the corner of 3rd and Main" (vertex form — location, location), "was seen entering at 1 p.m. and leaving at 5 p.m." (factored form — the crossings). *Same suspect, three descriptions — which one would you want if your question was "where does it hit the ground?"*

Fig. 3.5

Fig. 3.5 — The arch and the truck



The vertex gives the peak; evaluating at x = 4 gives the clearance.

KEY IDEA

The vertex is the answer to "biggest or smallest?"

Whenever

a story asks for a maximum (revenue, height, area) or a minimum (cost, time, material), the mathematics is: write the quadratic, find $-b/(2a)$. The vertex turns optimization into arithmetic.

Worked Example 1 — The arch's vertex (Hint 1). The arch is $y = -x^2/40 + 10$. Read off a and b, compute $-b/(2a)$, then evaluate y there to get the peak.

Solution. $a = -1/40$, $b = 0$. Vertex $x = -0 \div (2(-1/40)) = \mathbf{0}$. $y(0) = -0 + 10 = \mathbf{10}$. Vertex: **(0, 10)** — the peak is 10 feet, dead center, exactly as the story promised. ($a < 0$, so it opens downward: a hill, a maximum.)

Worked Example 2 — Height at the truck's roof corners (Hint 2). The truck is 8 feet wide, so its roof corners sit at $x = -4$ and $x = +4$ feet from center. Evaluate the arch's height at those x-values.

Solution. $y(4) = -(4)^2/40 + 10 = -16/40 + 10 = -0.4 + 10 = \mathbf{9.6\ feet}$. By symmetry, $y(-4) = 9.6$ feet too. Above each roof corner, the arch is 9.6 feet high.

Worked Example 3 — The clearance (Hint 3). Compare the arch's height at $x = 4$ with the truck's 8-foot height. How much clearance is there?

Solution. Clearance = arch height − truck height = $9.6 - 8 = \mathbf{1.6\ feet}$. The truck fits — with less than two feet to spare. But whether 1.6 feet is *enough* is a judgment call, not arithmetic: that's the DECIDE box's job.

DECIDE — THE VERDICT

Can the truck pass under the arch?

Back to the question that opened this section — now with the math to settle it.

Solution. The arch: $y = -x^2/40 + 10$, $a = -1/40 < 0$ (opens downward — a peak). Vertex at $x = -0/(2(-1/40)) = 0$, $y = 10$: **(0, 10)**. The truck's roof corners are at $x = \pm 4$: $y(4) = -16/40 + 10 = -0.4 + 10 = \mathbf{9.6\ feet}$. The truck is 8 feet tall: clearance $= 9.6 - 8 = \mathbf{1.6\ feet}$.

The decision: drive through — but slowly. The math says 1.6 feet of clearance, and the vertex form told you exactly where the peak is. A good driver adds judgment to the arithmetic: 1.6 feet is enough, but there's no room for a taller load or a bumpy road. The parabola gave you the number; you supply the caution.

AI AI Learning Lab 3.5

AI Level 1 — Understand.

Explain the vertex and axis of symmetry as if I'm 15, using a hill (maximum) and a valley (minimum). Show me how $-b/(2a)$ finds the vertex's x-coordinate for $f(x) = 2x^2 - 8x + 3$, step by step. Then quiz me on 2 more, one at a time.

AI Level 2 — Apply.

Give me 5 vertex problems: 2 from standard form, 1 from vertex form (identify the vertex by reading), 1 from factored form (roots first, then vertex), and 1 "is it a max or min?" Hide the answers. I'll show all steps; then check my work, especially my sign handling in $-b/(2a)$.

AI Level 3 — Matter.

The truck decision turned on 1.6 feet of clearance under a parabola — a number nobody could guess by looking. Parabolas hide in bridges, headlights, satellite dishes, and basketball shots. Discuss with me: WHY does it matter that someone *computed* the arch instead of eyeballing it — what are the implications when engineers guess at curves (bridge strikes? collapsed roofs under snow?)? Who bears the cost when the math is skipped? Ask me: what's one curved structure in MY town I'd now want to see the equation of?

Level 4 — Decide. *If you were the diver's coach, would you clear the trick?*

> **Example decision problem.** A diver leaves an 8-foot board; her height follows $h(t) = -16t^2 + 24t + 8$ feet, t seconds after takeoff. Her trick needs 15 feet of clearance to be safe. Should the coach clear it?

AI Level 4 — Decide.

I'm the coach: the diver's height is $h(t) = -16t^2 + 24t + 8$, and the trick needs 15 ft of clearance. Act as my decision coach: make me find the vertex (time and height) myself with $-b/(2a)$, then compare with 15 ft. Then ask me: how much margin is enough for a human life — would I clear it at 15.5 ft peak? Make me state the decision with reasons.

Check your verdict: vertex at $t = -24 \div (2(-16)) = 0.75$ s; $h(0.75) = -16(0.5625) + 18 + 8 = -9 + 26 = 17$ ft. Peak 17 ft > 15 ft needed — clear it, with a 2-foot margin. (She hits the water at $t \approx 1.78$ s.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.5.

Ask your AI: "Graph $f(x) = -x^2 + 6x - 5$, state the vertex, axis of symmetry, and whether it's a max or min." Then verify it all yourself: compute $-b/(2a)$, evaluate f at the vertex, and check two symmetric x-values (e.g., $x = 1$ and $x = 5$). If the AI's vertex is wrong, compute the percentage error (§1.2) between its y-value and yours, and write a one-paragraph "vertex verification routine" for your notebook: formula, substitution, symmetry check.

Practice Problems 3.5

1. For $y = x^2 - 4x + 3$, find the vertex and the axis of symmetry.
2. For $y = -2x^2 + 8x$, find the vertex. Is it a maximum or a minimum?
3. For $y = (x + 2)(x - 6)$, find the roots and the vertex.
4. Write $y = x^2 + 2x + 5$ in vertex form. State the vertex.
5. A ball's height is $h(t) = -16t^2 + 48t$ feet. Find its maximum height and when it occurs.

3.6 Quadratic Models for Decisions

THE DECISION

What price should you charge for the phone cases?

You sell custom phone cases at school. Right now: price **$20**, sales **100 cases/week**. You've tracked a pattern: each **$1 price increase** loses you **8 sales** per week (and each $1 decrease gains 8).

Should you keep the price at $20 — or change it? If you change it, to what? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Write sales as a function of price: start at 100 cases at $20, then subtract 8 for every dollar above $20.

- **Hint 2:** Revenue = price × quantity. Multiply out to get a quadratic in p — which way does it open?
- **Hint 3:** The best price is the vertex: p = −b/(2a). Compare the max revenue with today's $2,000/week.

Revenue as a Quadratic

Revenue is price times quantity — and when quantity *depends* on price, their product is quadratic. Let p = price. Quantity: q(p) = 100 − 8(p − 20) = 100 − 8p + 160 = **260 − 8p**. Revenue:

$$R(p) = p \cdot q(p) = p(260 - 8p) = \mathbf{-8p^2 + 260p}.$$

Since a = −8 < 0, the parabola opens downward: revenue has a *maximum*. p = −b/(2a) = −260 ÷ (2(−8)) = −260 ÷ (−16) = **$16.25**. Maximum revenue: R(16.25) = −8(16.25)² + 260(16.25) = −2,112.50 + 4,225 = **$2,112.50**/week. At $16.25 you sell q = 260 − 8(16.25) = **130 cases** — check: 130 × 16.25 = $2,112.50 ✓.

CAUTION

Revenue ≠ profit.

This model maximizes *revenue* — it ignores

costs. If each case costs you $5 to make, profit is P(p) = R(p) − 5·q(p), a *different* quadratic with a *different* vertex. Always ask: am I maximizing the right thing?

STORY

The Lemonade Price.

Ben sells lemonade at $2/cup: 50 cups/day.

He notices each $0.50 increase loses 10 cups. R(p) = p(50 − 20(p − 2)) = −20p² + 90p; vertex at p = −90 ÷ (−40) = **$2.25**, R = **$101.25** vs. today's $100. *$1.25 more per day — is the extra math worth reprinting the sign?*

Projectile Motion

Gravity writes quadratics. An object launched upward follows:

$$h(t) = -16t^2 + v_0t + h_0 \text{ (feet, t in seconds)}$$

where v_0 is the initial upward speed and h_0 the starting height. The -16 is half of gravity's 32 ft/s^2 pull. The vertex gives the peak; the positive root gives the landing time.

Fireworks show: a shell launches with $h(t) = -16t^2 + 160t$. Vertex at $t = -160 \div (2(-16)) =$ **5 s**; $h(5) = -16(25) + 800 =$ **400 feet**. It lands when $-16t^2 + 160t = 0 \rightarrow t(-16t + 160) = 0 \rightarrow$ **t = 10 s** ($t = 0$ is launch). The show's drone films at 350 feet — directly in the shell's path. *Decision: move the drone above 400 feet, or move it aside.*

STORY

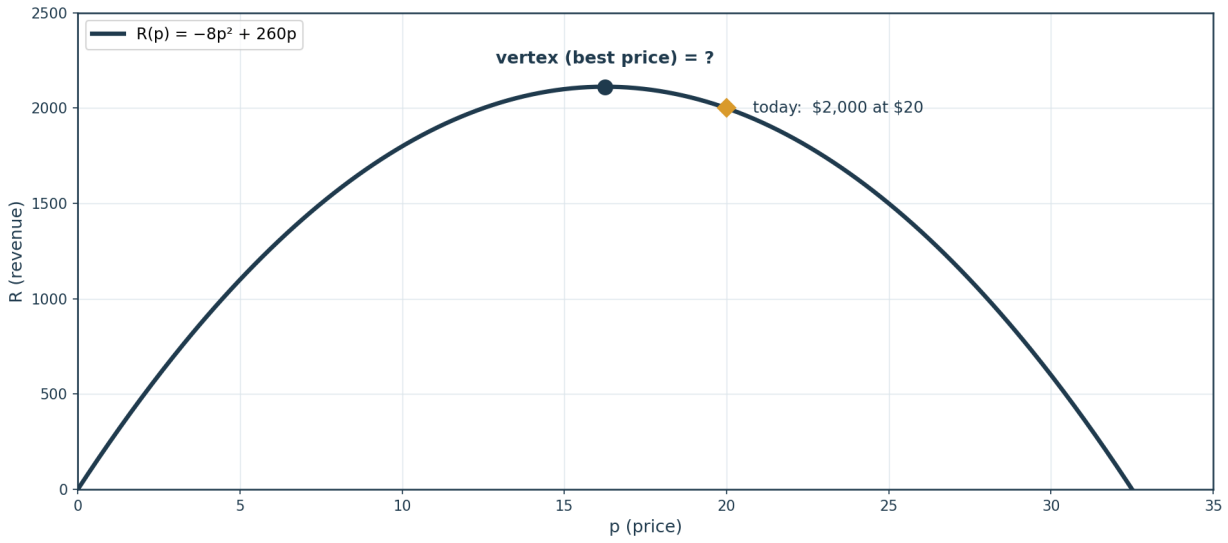
The Fireworks.

The crew chief's rule: "No drone within 50 feet of the shell's peak." The shell peaks at 400 feet; the drone hovers at

1. *The pilot says "it's below the peak, so we're fine" — what is wrong with that reasoning, and where should the drone actually go?*

Fig. 3.6

Fig. 3.6 — The revenue hill



Revenue is a hill — the vertex is the price you'd never guess.

Worked Example 1 — Max revenue, from the demand pattern (Hints 1–3). Start where Hint 1 starts: write sales as a function of price. Then build R(p), take the vertex, and compare with today's $2,000/week.

Solution. Hint 1: at $20 you sell 100 cases; each $1 above $20 loses 8 sales. $q(p) = 100 - 8(p - 20) = 100 - 8p + 160 =$ **$260 - 8p$**. Hint 2: $R(p) = p \cdot q(p) = p(260 - 8p) =$ **$-8p^2 + 260p$**; $a = -8 < 0$, so the parabola opens downward and a maximum exists. Hint 3: $p = -b/(2a) = -260 \div (2(-8)) = -260 \div (-16) =$ **16.25**. $R(16.25) = -8(264.0625) + 4{,}225 = -2{,}112.50 + 4{,}225 =$ **$2,112.50$**. At $16.25 you sell $q = 260 - 8(16.25) =$ **130 cases** — check: $130 \times 16.25 = \$2{,}112.50$ ✓. Today's revenue: $100 \times \$20 = \$2{,}000$ — the optimum beats it by **$112.50/week**.

Worked Example 2 — Projectile peak and landing. $h(t) = -16t^2 + 32t + 5$. Find the maximum height, when it occurs, and when the object hits the ground.

Solution. Vertex at $t = -32 \div (2(-16)) =$ **1 s**; $h(1) = -16 + 32 + 5 =$ **21 feet** (maximum). Landing: $-16t^2 + 32t + 5 = 0 \rightarrow t = [-32 \pm \sqrt{(32^2 - 4(-16)(5))}] \div (2(-16)) = [-32 \pm \sqrt{1344}] \div (-32)$. $\sqrt{1344} \approx 36.66$; $t = (-32 - 36.66) \div (-32) \approx$ **2.15 s** (the negative root, ≈ -0.15 s, is before launch — discard it).

Worked Example 3 — Profit, not revenue. Phone cases cost $5 each to make. Write the profit function and find the profit-maximizing price. (Compare with the revenue-maximizing $16.25!)

Solution. q(p) = 260 − 8p. P(p) = R(p) − 5q(p) = (−8p² + 260p) − 5(260 − 8p) = −8p² + 260p − 1,300 + 40p = **−8p² + 300p − 1,300**. Vertex: p = −300 ÷ (2(−8)) = −300 ÷ (−16) = **$18.75**. Max profit: P(18.75) = −8(351.5625) + 300(18.75) − 1,300 = −2,812.50 + 5,625 − 1,300 = **$1,512.50**/week. Note: the profit-maximizing price ($18.75) is *higher* than the revenue-maximizing price ($16.25) — costs punish volume.

DECIDE — THE VERDICT

What price should you charge for the phone cases?

Back to the question that opened this section — now with the math to settle it.

Solution. R(p) = −8p² + 260p opens downward (a = −8 < 0). Vertex: p = −260 ÷ (−16) = **$16.25**; R(16.25) = **$2,112.50**/week at 130 cases. Today: $20 × 100 = $2,000/week.

The decision: lower the price to **$16.25**. It's counterintuitive — charge *less*, earn *more* — because the 30 extra cases outweigh the $3.75 per-case cut. But read the [CAUTION] above: this maximizes revenue. With $5/case costs, the *profit* optimum is $18.75 (Worked Example 3). The quadratic doesn't just give you an answer — it forces you to ask the right question first: revenue, or profit?

AI AI Learning Lab 3.6

AI Level 1 — Understand.

Explain why revenue = price × quantity becomes a quadratic when quantity depends on price, as if I'm 15. Use the phone-case numbers. Then explain projectile motion $h(t) = -16t^2 + v_0t + h_0$ the same way: what does each term mean physically? Quiz me with 2 checks, one at a time.

AI Level 2 — Apply.

Give me 4 problems: 2 max-revenue (write R(p), find the vertex) and 2 projectile (find the peak; find the landing time). Hide the answers. I'll show the −b/(2a) step every time; then check my work and verify my quadratic-formula arithmetic digit by digit.

AI Level 3 — Matter.

The phone-case decision said "charge less, earn more" — and then the profit calculation moved the optimum again, from $16.25 to $18.75. The model answers whatever question you ask it, even the wrong one. Discuss with me: WHY does this matter for real decisions — what are the implications when a business optimizes revenue while ignoring costs (think: startups chasing growth, restaurants filling seats at a loss)? Who gets fooled when the *model* is right but the *question* is wrong? Ask me: what's one decision in MY life where I might be maximizing the wrong thing?

Level 4 — Decide. *If you were promoting the concert, what would you charge?*

> **Example decision problem.** You promote a concert: at $50/ticket you sell 400 tickets, and each $5 price increase loses 30 sales. What ticket price maximizes revenue, how many tickets is that, and how much more do you earn vs. today?

AI Level 4 — Decide.

I'm promoting a concert: $50 → 400 tickets; each $5 increase loses 30 sales. Act as my decision coach: make me write q(p) and R(p) myself (careful: the $5 steps mean q drops 6 per $1), find the vertex, and compute the revenue gain over today's $20,000. Then challenge me: what if the venue holds only 300 — does the optimum change? Make me state the pricing decision with reasons.

Check your verdict: $q(p) = 400 - 6(p - 50) = 700 - 6p$; $R(p) = -6p^2 + 700p$; vertex at $p = -700 \div (-12) \approx \$58.33$; $q \approx 350$ tickets; $R \approx \$20{,}416.67$ — about $416.67 more than today's $20,000. (If the venue caps at 300, the constraint binds: price at the $q = 300$ point, $p = \$66.67$, $R = \$20{,}000$ — the math bends to reality.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.6.

Ask your AI: "A farmer has 60 feet of fencing for a rectangular garden against a barn (one side free). What dimensions maximize the area?" Then verify it yourself: write $A(w) = w(60 - 2w) = -2w^2 + 60w$, find the vertex ($w = 15$, length 30, area 450 ft²), and check the AI's numbers. Then push further than the AI did: what if the farmer has a $500 budget and fencing costs $4/foot — write the *area* function and find the dimensions that maximize the area. (Hint: $500 buys 125 feet of fence, so $A(w) = w(125 - 2w) = -2w^2 + 125w$; the vertex is at $w = 31.25$, length 62.5, area $\approx$ 1,953 ft².) Write a one-paragraph verdict: did the AI maximize the right thing?

Practice Problems 3.6

1. Revenue is $R(p) = -5p^2 + 200p$. Find the price that maximizes revenue and the maximum revenue.
2. A garden with 60 ft of fencing against a barn has area $A(x) = -2x^2 + 60x$ (x = width). Find the dimensions and maximum area.
3. A ball's height is $h(t) = -16t^2 + 32t + 5$ feet. Find its maximum height, when it occurs, and when it hits the ground (nearest hundredth).
4. Profit is $P(x) = -3x^2 + 120x - 500$. Find the production level that maximizes profit and the maximum profit.
5. A ball is thrown with $h(t) = -16t^2 + 64t$. Find its maximum height and when it hits the ground.

3.7 Exponential Functions

THE DECISION

Bank A or Bank B?

You have **$10,000** to save for 10 years. Bank A offers **5% simple interest** — the same dollars added every year. Bank B offers **4.8% compound interest** — the balance multiplies every year. The bigger rate *looks* better.

Which bank leaves you richer after 10 years — and by how much? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Simple interest *adds* the same amount yearly; compound interest *multiplies* the whole balance yearly. Write both 10-year formulas before touching a calculator.
- **Hint 2:** Simple: 10000(1 + 0.05 × 10). Compound: 10000(1.048)^10. Which operation wins over a decade — repeated adding or repeated multiplying?
- **Hint 3:** Compute both totals and find the difference. Then ask the deeper question: in *which year* does the compound account pull ahead? (Work through the years one by one — which value passes which first?)

Exponential Growth: Multiplying, Not Adding

Linear functions add the same amount each step. Exponential functions *multiply* by the same factor each step — and multiplication leaves addition in the dust. Start with $100 growing 10% per year: year 1 → $110, year 2 → $121, year 3 → $133.10. The *increases* themselves increase ($10, then $11, then $12.10), because each year's growth earns its own growth.

> **Exponential growth** means y = a·b^x with **b > 1**: every step multiplies the output by b.

STORY

The Rumor.

On Monday Zoe tells a secret to 3 friends and swears them to secrecy — badly. Each day, everyone who knows tells 3 new people. Day 1: 3 know. Day 2: 9. Day 3: 27. By day 5, 3^5 = **243** students know "the secret." *The school has 1,200 students — on which day does everyone know? (3^6 = 729, 3^7 = 2,187…)*

Exponential Decay: The Reverse Gear

Flip the factor below 1 and the same machinery runs in reverse. A 200 mg dose of medicine halves in your body every 4 hours: after 4 h → 100 mg, after 8 h → 50 mg, after 12 h → **25 mg**. The drops shrink (100, then 50, then 25) but the *factor* never changes.

> **Exponential decay** means y = a·b^x with **0 < b < 1**: every step multiplies the output by b — shrinking it by the same proportion.

STORY

The Medicine.

Leo takes 200 mg of allergy medicine at 8 AM; his body clears half of it every 4 hours. By 8 PM (12 hours later) only 200 × (1/2)^3 = **25 mg** remain — consistent with a "take every 12 hours" label. *How much is left at midnight, 16 hours after the dose?*

The General Form y = a·b^x

Every exponential function has the same skeleton:

> **y = a·b^x**, where a is the starting value (when x = 0, y = a) and b is the growth factor.

- If b > 1 → growth. Write b = 1 + r: a 5% growth rate means b = 1.05.
- If 0 < b < 1 → decay. Write b = 1 − r: a 12% decay rate means b = 0.88.

The graph always passes through **(0, a)** and hugs the x-axis as a horizontal asymptote — growth explodes upward to the right, decay melts toward zero. (If you ever meet the number **e ≈ 2.71828**, it's just nature's favorite growth factor — continuous compounding's home base. You won't need it for the decisions in this chapter.)

CAUTION

The #1 exponential error.

A "5% growth rate" does NOT mean

b = 0.05 — it means **b = 1.05**. The factor is 1 *plus* the rate (growth) or 1 *minus* the rate (decay). Using 0.05 as b would shrink your money to 5% of itself every year — a very different decision.

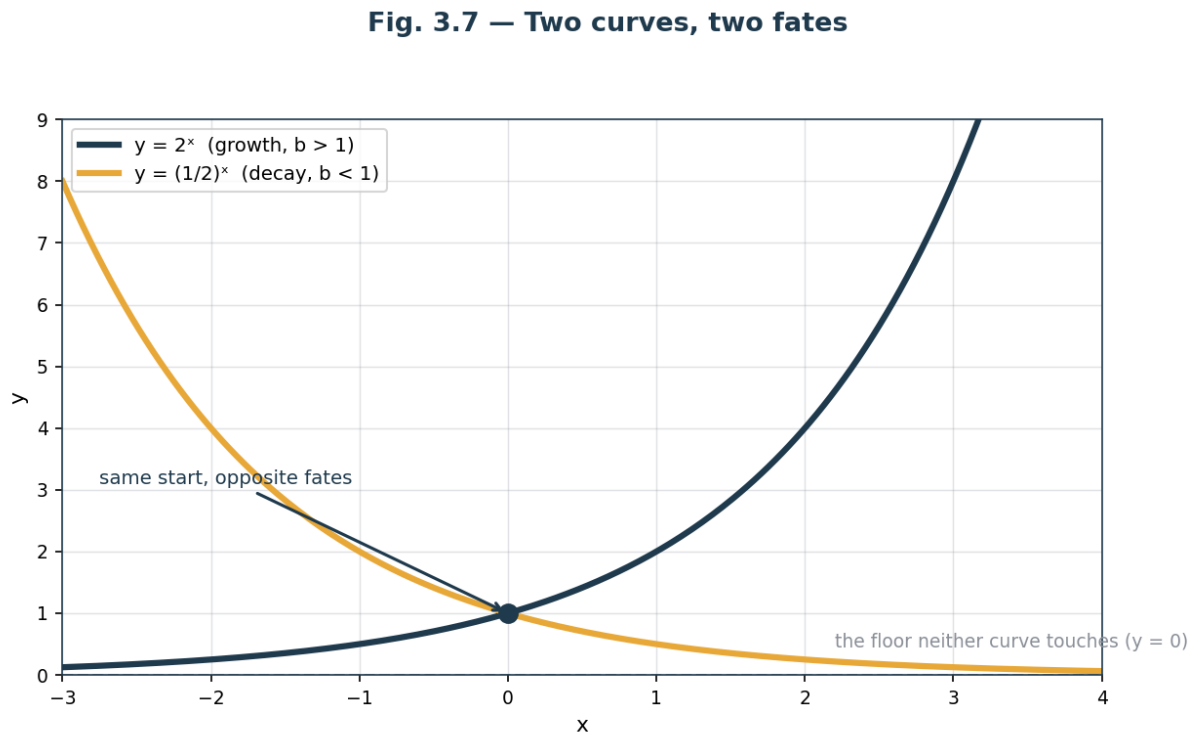
STORY

The Follower Count.

Jay's video channel has 500 followers and

grows 20% per week: y = 500(1.2)^x. After 6 weeks: 500 × 1.2^6 ≈ 500 × 2.986 ≈ **1,493 followers**. *At this rate, in which week does he pass 5,000? (Solve 500 × 1.2^x = 5,000 — it takes about 12.6 weeks, so during the 13th week.)*

Fig. 3.7



Same form, y = a·b^x — everything depends on whether b sits above or below 1.

Worked Example 1 — Write both formulas first (Hint 1). Before touching a calculator, write Bank A's and Bank B's 10-year balance formulas. What operation does each one repeat?

Solution. Bank A (simple — *adds* the same dollars every year): A(t) = 10,000(1 + 0.05t). Bank B (compound — *multiplies* the whole balance every year): A(t) = 10,000(1.048)^t. Same $10,000 start, same 10 years — but one formula adds and the other multiplies. That's the whole contest, written before a single number is computed.

Worked Example 2 — Compute both totals (Hints 2 and 3). Evaluate both formulas at t = 10 and find the difference.

Solution. Bank A: 10,000(1 + 0.05 × 10) = 10,000 × 1.5 = **$15,000**. Bank B: 10,000 × 1.048^10 ≈ 10,000 × 1.598133 = **$15,981.33**. Difference: 15,981.33 − 15,000 = **$981.33**. The "smaller" 4.8% wins — repeated multiplying beat repeated adding over a decade.

Worked Example 3 — When does compound pull ahead? (Hint 3). Work through the years one by one — in which year does the compound account first pass the simple one?

Solution. Compute both balances year by year:

Year Simple (5%) Compound (4.8%) 1 $10,500 $10,480.00 2 $11,000 $10,983.04 3 $11,500 $11,510.23 4 $12,000 $12,062.72 5 $12,500 $12,641.73

For two years the simple account leads. In **year 3** the compound account pulls ahead ($11,510.23 > $11,500) — and the lead only grows after that. The crossover was hiding inside the formulas from Worked Example 1.

DECIDE — THE VERDICT

Bank A or Bank B?

Back to the question that opened this section — now with the math to settle it.

Solution. Bank A (simple): 10000(1 + 0.05 × 10) = 10000 × 1.5 = **$15,000**. Bank B (compound): 10000(1.048)^10 ≈ 10000 × 1.598133 = **$15,981.33**.

The decision: choose **Bank B**. The "smaller" 4.8% beats the "bigger" 5% by **$981.33** — because multiplying beats adding. Notice the crossover: the compound account pulls ahead around year 3 (1.1510 vs. 1.1500) and the lead only grows. The judgment lesson: *never compare two rates without asking "simple or compound?"* — the words matter more than the numbers.

AI AI Learning Lab 3.7

AI Level 1 — Understand.

Explain exponential growth vs. decay as if I'm 15, using the rumor (3^x) and the medicine ((1/2)^x) examples. Then explain why the graph of y = a·b^x always passes through (0, a) and what the horizontal asymptote means in plain words. Quiz me with 2 checks, one at a time.

AI Level 2 — Apply.

Give me 4 problems: 2 "classify as growth or decay and give the rate" and 2 compound-interest computations (write A = P(1+r)^t first, then compute). Hide the answers. I'll show the b = 1 ± r step every time; then check my arithmetic digit by digit.

AI Level 3 — Matter.

Bank B won with the SMALLER rate — 4.8% compound beat 5% simple by $981.33. The advertisement shouts the rate; the fine print whispers "simple" or "compound." Discuss with me: WHY does this matter for real life — what are the implications when borrowers can't tell simple from compound (think: credit cards, payday loans, "0% financing" deals)? Who profits from that confusion, and who pays for it? Ask me: what's one rate *I've* seen advertised where the compounding words mattered?

Level 4 — Decide. *If you were saving for college, which account would you pick?*

> **Example decision problem.** You have $8,000 for college in 8 years. Account X pays 7% compounded annually; Account Y pays 5.5% simple interest. Which leaves you richer, and by how much?

AI Level 4 — Decide.

I have $8,000 for college in 8 years: 7% compounded annually vs. 5.5% simple. Act as my decision coach: make me write both formulas myself, compute both 8-year totals, and state the decision with the dollar difference. Then challenge me: at what simple rate would Account Y exactly tie Account X? Make me solve for it.

Check your verdict: X: 8000(1.07)^8 ≈ 8000 × 1.718186 ≈ $13,745.49; Y: 8000(1 + 0.055 × 8) = $11,520. Compound wins by ≈ $2,225.49. Tie rate: 1 + 8r = 1.718186 → r ≈ 8.98% simple — Bank Y would need to advertise nearly 9% to match 7% compound.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.7.

Ask your AI: "If I invest $1,000 at 8% compounded annually, how many years until I have $4,000?" Then verify it yourself: solve 1.08^t = 4 (t = ln 4 ÷ ln 1.08 ≈ 18.01 years — the Rule of 72 says 72 ÷ 8 = 9 years to *double*, so quadrupling takes about twice that: 18 ✓). Then push further than the AI did: what if the 8% compounds *monthly* instead of annually — what is the effective annual rate? ((1 + 0.08/12)^12 − 1 ≈ 8.30%.) Write a one-paragraph verdict: did the AI distinguish nominal from effective rates?

Practice Problems 3.7

1. Evaluate f(x) = 5(2)^x at x = 3.

2. For g(t) = 120(0.85)^t: growth or decay? Give a, b, and the percent rate per time unit.
3. $3,000 is invested at 5% compounded annually for 8 years. Find the balance.
4. A 150 g sample has a half-life of 6 years (it halves every 6 years). How much remains after 18 years?
5. Which grows faster, y = 100(1.1)^x or y = 100(1.2)^x? Explain in one sentence.

3.8 Exponential Models for Decisions

THE DECISION

Buy the car now, or wait a year?

A 3-year-old sedan you want costs **$18,000** today. Cars like it lose about **12% of their value per year** — so waiting a year drops the price to about $15,840, saving you **$2,160**. But you need wheels *now*: the rental you'd need for that year costs **$400/month**.

Should you buy now or wait? The $2,160 saving is real — but so is the rental bill. Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Depreciation is exponential decay. Write the car's value after one year: 18000 × (1 − 0.12).
- **Hint 2:** Waiting has two dollar amounts, not one: the depreciation saving *and* the cost of going carless for a year. List both.
- **Hint 3:** Compare the $2,160 saving against the 12-month rental total — which number is bigger, and what does that tell you about "wait for the price drop"?

Compound Interest: Money That Multiplies

Compound interest is exponential growth wearing a suit: A = P(1 + r)^t. Each year the *whole balance* — including last year's interest — earns the rate. Over decades the curve bends upward so hard it looks like a hockey stick, which is why starting early beats contributing more later.

STORY

The Birthday Money.

When Lily was born, her grandmother put

$1,000 in a 5% compound account and never touched it. At Lily's 18th birthday: 1000 × 1.05^18 ≈ **$2,406.62**. Left alone until she's 30: 1000 × 1.05^30 ≈ **$4,321.94** — the *second* 12 years earned more than the first 18. *How many years until the original $1,000 triples?*

Depreciation: Value That Melts Away

Assets decay too. A car's value often follows V(t) = P(1 − r)^t — the same exponential form, now with b < 1. The first year's dollar loss is always the biggest, because the percentage bites the largest balance.

STORY

The New Car Smell.

Chloe's $28,000 car loses 15% of its value

each year: V(t) = 28000(0.85)^t. After 3 years it's worth 28000 × 0.85^3 ≈ **$17,195.50** — over a third gone. *In which year does its value drop below half the purchase price? (Solve 0.85^t < 0.5 — about 4.27 years, so during the 5th year.)*

Doubling Time and Half-Life

Two shortcut numbers run exponential decisions:

> **Doubling time** (growth): how long until it doubles. **Rule of 72**: years ≈ 72 ÷ (percent rate). At 7%, ≈ 10.3 years (exact: 10.24). **Half-life** (decay): how long until half remains. After n half-lives, (1/2)^n is left.

STORY

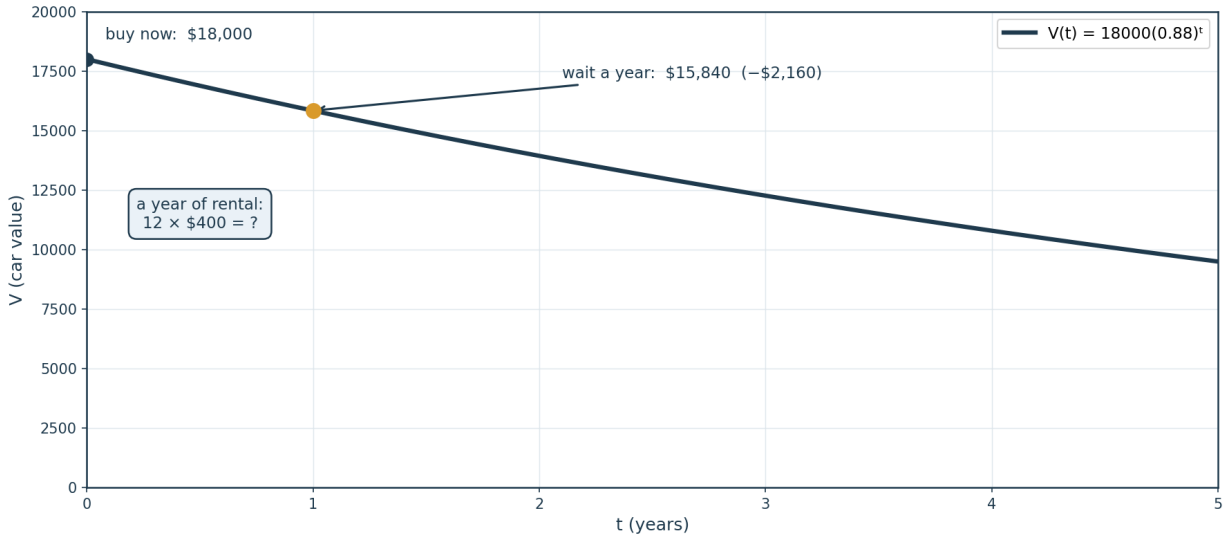
The Bacteria Plate.

A lab culture starts with 100 bacteria

and doubles every 20 minutes: after 2 hours (6 doublings), 100 × 2^6 = **6,400**. *The researcher needs 1,000,000 for the experiment — how many more doublings is that, and how long will she wait? (2^n > 156.25 → about 8 more doublings, roughly 2 hours 40 minutes.)*

Fig. 3.8

Fig. 3.8 — Wait or buy?



Depreciation is only half the decision — the cost of waiting is the other half.

Worked Example 1 — The car's value after one year (Hint 1). Depreciation is exponential decay. Write the car's value after one year as 18000 × (1 − 0.12) and compute the saving from waiting.

Solution. V(1) = 18,000 × 0.88 = **$15,840**. The one-year depreciation saving is 18,000 − 15,840 = **$2,160**. That number is real — but it is only half the decision (Hint 2).

Worked Example 2 — Both dollar amounts side by side (Hints 2 and 3). Waiting has two dollar amounts, not one: the depreciation saving *and* the cost of going carless for a year. List both, then compare.

Solution. Saving from waiting: **$2,160** (Worked Example 1). Cost of going carless: 12 months × $400/month = **$4,800**. Net effect of waiting: $2,160 − $4,800 = **−$2,640** — waiting costs $2,640 more than it saves. Hint 3's answer: the rental bill is bigger. The arithmetic is settled; whether *you* need a car every day is the judgment the DECIDE box makes.

Worked Example 3 — Doubling time, exact and estimated. How long does money take to double at 7% compounded annually?

Solution. Rule of 72: 72 ÷ 7 ≈ **10.3 years**. Exact: solve 1.07^t = 2 → t = ln 2 ÷ ln 1.07 ≈ 0.6931 ÷ 0.0677 ≈ **10.24 years**. The rule of thumb is within a tenth of a year — good enough for a decision, then verify with the logarithm.

DECIDE — THE VERDICT

Buy the car now, or wait a year?

Back to the question that opened this section — now with the math to settle it.

Solution. One year of depreciation: 18000 × 0.88 = **$15,840** — a real saving of **$2,160**. One year of rental: 12 × $400 = **$4,800**. Net effect of waiting: $2,160 − $4,800 = **−$2,640** — waiting costs $2,640 more than it saves.

The decision: buy **now**. Waiting "saves" $2,160 on the sticker but costs $4,800 in rental — you pay $2,640 for the privilege of waiting. But note what the model assumed: you need a car *every day*. Work from home and bike on weekends? At $150/month for occasional rentals ($1,800/year < $2,160 saving), the verdict flips to *wait*. The exponential model doesn't decide — it prices each scenario so *you* can.

AI AI Learning Lab 3.8

AI Level 1 — Understand.

Explain compound interest A = P(1+r)^t and depreciation V = P(1−r)^t as if I'm 15, using the birthday money and the new-car stories. Then explain doubling time, half-life, and the Rule of 72 in plain words. Quiz me with 2 checks, one at a time.

AI Level 2 — Apply.

Give me 4 problems: 2 depreciation computations (write V = P(1−r)^t first) and 2 doubling-time/half-life problems (one with the Rule of 72, one exact with logarithms). Hide the answers. I'll show every substitution step; then check my arithmetic carefully.

AI Level 3 — Matter.

The car decision came down to a hidden assumption: "you need a car every day." Change that assumption and the verdict flipped. The model answered the question it was asked — including the parts nobody said out loud. Discuss with me: WHY does this matter beyond car shopping — what are the implications when models bake in unstated assumptions (think: loan approvals, insurance pricing, hiring algorithms)? Who decides what goes into the model, and who lives with what was left out? Ask me: what's one assumption hiding inside a decision *I'm* facing?

Level 4 — Decide. *If you were the principal, would you buy the Chromebooks now?*

> **Example decision problem.** Your school can buy 30 Chromebooks now at $300 each ($9,000 total), or wait a year when the price drops 10% — but students then go a year without them. The principal values a year of student access at $1,500. Buy now or wait?

AI Level 4 — Decide.

My school can buy 30 Chromebooks at $300 each now ($9,000), or wait a year for a 10% price drop — but students lose a year of access, valued at $1,500. Act as my decision coach: make me compute the wait-a-year price and the saving myself, then weigh it against the $1,500 access value, and state the buy/wait decision with reasons. Then challenge me: at what access value would the decision flip?

Check your verdict: wait-a-year price 9000 × 0.9 = $8,100 — a $900 saving against $1,500 of lost access value. Buy now (waiting nets −$600). The flip point: waiting wins only if a year of access is worth less than $900.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 3.8.

Ask your AI: "Should I buy a $15,000 car now or wait 2 years if it depreciates 12% per year and I'd spend $150/month on buses and rideshares while waiting?" Then verify it yourself: waiting saves $15000 \times (1 - 0.88^2) = \$3{,}384$ but the transit costs $24 \times \$150 = \$3{,}600$ — buy now, by $216. Then push further than the AI did: at what monthly transit cost would waiting *break even*? ($3,384 ÷ 24 = $141/month — below that, wait; above it, buy.) Write a one-paragraph verdict: did the AI separate the depreciation saving from the waiting cost, or blur them together?

Practice Problems 3.8

1. A $25,000 car depreciates 10% per year. Find its value after 3 years.
2. A town of 80,000 people grows 2.5% per year. Find the population after 12 years (nearest whole person).
3. How long does money take to double at 6%? Give the Rule-of-72 estimate and the exact value.
4. Investment A: $5,000 at 4% compounded annually for 10 years. Investment B: $5,000 at 5% simple interest for 10 years. Which wins, and by how much? (The answer may surprise you.)
5. Your $900 phone loses 25% of its value each year. After how many whole years is it first worth less than $200?

Vocabulary

- **Function** — a rule assigning each input exactly one output.
- **Domain** — the set of allowed inputs of a function.
- **Range** — the set of possible outputs of a function.
- **Mapping diagram** — arrows from inputs to outputs showing a function's pairings.
- **Vertical line test** — a graph is a function iff no vertical line hits it more than once.
- **Function notation** — f(x), read "f of x": the output of function f at input x.
- **Evaluate** — substitute an input value and compute the output.
- **Linear function** — f(x) = ax + b; its graph is a straight line.
- **Slope** — a in f(x) = ax + b; the rate of change, (change in y) ÷ (change in x).
- **y-intercept** — the output when x = 0; the point (0, b).
- **x-intercept** — the input where the output is zero; found by solving ax + b = 0.
- **Rate of change** — how much the output changes per unit of input; the slope's real-world meaning.
- **Cost function** — C(x) = fixed costs + (per-item cost)·x.
- **Revenue function** — R(x) = (price)·x.
- **Profit function** — P(x) = R(x) − C(x).
- **Contribution margin** — revenue per item minus cost per item; the slope of the profit line.
- **Break-even point** — where R(x) = C(x) and profit is zero.
- **Quadratic function** — $f(x) = ax^2 + bx + c$ ($a \neq 0$); its graph is a parabola.
- **Parabola** — the U-shaped graph of a quadratic; opens up if a > 0, down if a < 0.
- **Vertex** — the parabola's peak (maximum) or valley (minimum), at $x = -b/(2a)$.
- **Axis of symmetry** — the vertical line $x = -b/(2a)$ through the vertex.
- **Standard form** — $y = ax^2 + bx + c$.

- **Vertex form** — $y = a(x - h)^2 + k$, with vertex (h, k).
- **Factored form** — $y = a(x - r_1)(x - r_2)$, with x-intercepts r_1, r_2.
- **Projectile motion** — $h(t) = -16t^2 + v_0t + h_0$, height in feet under gravity.
- **Exponential function** — $f(x) = a \cdot b^x$ ($b > 0$, $b \neq 1$); constant percent change per step.
- **Exponential growth** — $b > 1$; each step multiplies the output by b.
- **Exponential decay** — $0 < b < 1$; each step shrinks the output by the factor b.
- **Growth/decay rate** — r where $b = 1 + r$ (growth) or $b = 1 - r$ (decay).
- **Compound interest** — $A = P(1 + r)^t$; interest earns interest.
- **Simple interest** — $A = P(1 + rt)$; interest on the principal only.
- **Depreciation** — exponential decay of an asset's value, $V = P(1 - r)^t$.
- **Doubling time** — time for a growing quantity to double; $\approx 72 \div$ (percent rate) by the Rule of 72.
- **Half-life** — time for a decaying quantity to shrink to half.

Summative Task: The 3-Paragraph Decision Essay

The prompt. Your cousin offers to invest $1,000 in your T-shirt business. The model says break-even is 267 shirts — and your cushion is only 33 shirts above that. Write a 3-paragraph essay advising your cousin: invest, or walk away? Use this chapter's function models — and name exactly what would change your recommendation.

Paragraph 1 — Your decision and claim. State your advice in one clear sentence, then one or two sentences on why it matters. (Example thesis — either side is fine: "Invest, because a 267-shirt break-even leaves a real cushion if the 300-shirt estimate holds." Or the other side: "Walk away for now, because an 11% sales shortfall turns the whole venture into a loss.")

Planning your essay (do this before you write). 1. **Compute first.** Run the numbers yourself: $C(x) = 2{,}400 + 6x$, $R(x) = 15x$, $P(x) = 9x - 2{,}400$; break-even at 267 shirts; $P(300) = +\$300$ but $P(250) = -\$150$; the 33-shirt cushion is $33 \div 300 = 11\%$ of your 300-shirt estimate. Don't take these on faith — verify them with your own calculation. 2. **Thesis template.** "I recommend ___ because ***, even though*** ." Fill it in for YOUR side — either side is defensible if the numbers back it. 3. **Evidence checklist.** Every claim needs a number behind it: break-even, profit at 300 shirts, loss at 250 shirts, and the 11% cushion, plus at least one idea from this chapter (function notation, break-even, or percent change) that strengthens the case. 4. **Flip test.** Name one specific condition that would change your recommendation — 150 pre-orders in hand? fixed costs cut to $1,200? If you can't name one, your argument isn't finished.

Paragraph 2 — Mathematical evidence. Show the numbers: C(x) = 2,400 + 6x, R(x) = 15x, P(x) = 9x − 2,400; break-even at 267 shirts; P(300) = +$300 but P(250) = −$150. Explain what the 33-shirt cushion means as a percent of your 300-shirt estimate. Every claim needs a number behind it.

Paragraph 3 — Implications, limitations, and what would change your mind. What is your 300-shirt estimate based on — and how much do you trust it? What does the model leave out (your time? unsold inventory? a competitor)? Name one specific condition that would flip your advice — for example, 150 pre-orders in hand, or fixed costs cut to $1,200 — and explain why.

Length: 200–300 words. Three paragraphs, no more, no fewer.

Using AI. Write your first draft entirely on your own — no AI. Then you may use AI as a revision coach: ask it to check your logic, flag weak evidence, and suggest clearer sentences. Never let it write paragraphs for you. And remember this book's golden rule: AI makes mistakes — verify everything, especially any numbers it gives you.

Rubric and exemplar essays are in the Teacher Guide.

Answer Key — Practice Problems (odd-numbered)

3.1 — 1. Yes, a function. Domain: {2, 3, 4}; range: {5, 7}. 3. Yes — each x has exactly one y (repeated outputs are allowed). 5. $y = x^2$ is a function of x (each x gives one y); $x = y^2$ is not (e.g., x = 4 gives y = 2 and y = −2).

3.2 — 1. 15. 3. h(4) = 17 and h(−4) = 17 — different inputs can share an output and it is still a function. 5. C(30) = $1,550; 40 guests on a $2,000 budget.

3.3 — 1. 2. 3. y = 2x + 2. 5. Intercepts (0, 3) and (6, 0); the line falls gently left to right.

3.4 — 1. x = 100 units. 3. P(x) = 2.3x − 150; break-even at 66 hot dogs (65 still loses $0.50). 5. New break-even: 400 shirts — cutting the price from $15 to $12 raises the break-even by half (267 → 400), because each shirt now contributes only $6 toward fixed costs instead of $9.

3.5 — 1. Vertex (2, −1); axis x = 2. 3. Roots −2 and 6; vertex (2, −16). 5. Maximum 36 ft at t = 1.5 s.

3.6 — 1. p = $20; max revenue $2,000. 3. Max 21 ft at t = 1 s; hits the ground at t ≈ 2.15 s. 5. Max 64 ft at t = 2 s; hits the ground at t = 4 s.

3.7 — 1. 40. 3. $4,432.37. 5. y = 100(1.2)^x — the larger base multiplies faster every step.

3.8 — 1. $18,225. 3. Rule of 72: 12 years; exact: t = ln 2 ÷ ln 1.06 ≈ 11.90 years. 5. 900(0.75)^t < 200 → t ≈ 5.23 years — during the 6th year.

Level 4 example checks — **3.1:** A(6) = $73 vs. B's $60 — join Gym B at 6 visits/month; crossover at v = 4.375, so 4 or fewer visits favors A. **3.2:** R(6) = $54 (fits), R(7) = $61 (over) — ride 6 hours, keep $6. **3.3:** A = $370 vs. B = $340 — Company B wins by $30. **3.4:** P(x) = 1.5x − 80; break-even 54 cups; P(100) = $70, P(60) = $10 — profitable but thin. **3.5:** vertex at t = 0.75 s, 17 ft > 15 ft needed — clear the trick (2-ft margin). **3.6:** q(p) = 700 − 6p, R(p) = $-6p^2 + 700p$; max at p ≈ $58.33 with ≈350 tickets and ≈$20,416.67 — about $417 more than today's $20,000. **3.7:** X: 8000(1.07)^8 ≈ $13,745.49 vs. Y: $11,520 — compound wins by ≈$2,225.49; tie needs ≈8.98% simple. **3.8:** waiting saves $900 but sacrifices $1,500 of access value — buy now (waiting nets −$600); flips if access is worth less than $900.

CHAPTER 4

Probability

A Note on AI Learning Labs in This Chapter

Chapter 3 taught you to interrogate models. This chapter teaches you to interrogate *certainty itself*. Probability is the one topic where AI chatbots sound the most confident and are wrong the most often: they will tell you a test is "95% accurate so you're 95% likely sick" or that a lottery is "a good investment because someone has to win." Both are wrong, and both collapse the moment you demand the sample space. So this chapter's standing rule: **never accept a probability without asking "probability OF WHAT, out of WHAT?"** Every AI Learning Lab keeps the four levels (understand, apply, matter, decide), and every Level 4 ends the way the chapter begins — with a decision that is yours to make.

About This Chapter

Probability is the mathematics of not knowing — and of deciding anyway. Every day you bet on uncertain outcomes: you carry an umbrella, buy insurance, trust a medical test, or enter a raffle. This chapter turns those bets into calculations. You will learn to list sample spaces, to separate theoretical probability from experimental results, to read probability from data, to combine events with tree and Venn diagrams, to tell independent events from dependent ones, and to compute expected value — then you will use all of it to decide: which raffle to enter, whether the carnival game is rigged, whether to choose surgery, how many alarms to set, what a positive test really means, whether to accept a shipment, and whether the lottery ticket beats the savings account.

By the end of this chapter, you will be able to:

- List a sample space and compute basic probabilities as (favorable outcomes) ÷ (total outcomes); convert between probability and odds.
- Distinguish theoretical probability (reasoning) from experimental probability (data), and explain the Law of Large Numbers.
- Read probabilities from real data and use the complement rule, P(not E) = 1 − P(E).
- Build tree diagrams for AND probabilities and use the addition rule for OR probabilities, including Venn diagrams for overlapping events.

- Tell independent events from dependent events, and compute conditional probabilities like P(sick | positive test).
- Compute probabilities with and without replacement, and explain why sampling works as a decision tool.
- Compute expected value and use it to judge fairness — and to know when expected value is not the whole story.
- Use AI prompts at four levels (understand, apply, matter, decide) to learn probability — and force every AI claim to show its sample space.

4.1 Basic Probability & Sample Space

THE DECISION

Which raffle tickets do you want to hold?

The school fundraiser sells two raffles for the same $100 gift card, and a friend offers you a straight trade:

- **Raffle A:** 200 tickets sold total. You currently hold 5.
- **Raffle B:** 500 tickets sold total. Your friend holds 20 and will trade all 20 for your 5.

Do you make the trade? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** In Raffle A, what fraction of all tickets is yours? Write it as (your tickets) ÷ (total tickets).
- **Hint 2:** Do the same for Raffle B. Convert both fractions to percentages.
- **Hint 3:** The bigger percentage wins. But ask yourself: does the *number of tickets you hold* matter more, or the *fraction*?

What Is Probability?

A **probability** is a number between 0 and 1 (or 0% and 100%) that measures how likely an event is. The basic formula:

P(event) = (favorable outcomes) ÷ (total possible outcomes)

P = 0 means impossible; P = 1 means certain. Everything else is a fraction in between. Raffle A gives you 5 ÷ 200 = 0.025 = 2.5%. That is all a probability ever is: your slice divided by the whole pie.

STORY

The Marble Jar.

A jar holds 7 red marbles and 3 blue marbles,

well mixed. You close your eyes and grab one. P(red) = 7 ÷ 10 = 0.7 = 70%. Nobody needs to run an experiment — the jar's *composition* tells you the chance. *If someone added 10 more blue marbles without telling you, would your 70% still be true?*

The Sample Space

The **sample space** is the list of *all* possible outcomes of an experiment. You cannot compute a probability until you know what "total possible outcomes" means — and writing the sample space is how you find out.

Flip two coins. The sample space is {HH, HT, TH, TT} — four outcomes. So P(at least one head) = 3 ÷ 4 = 75%. Most probability mistakes are really sample-space mistakes: people count the outcomes wrong, then divide by the wrong total.

STORY

The Lunch Menu.

The cafeteria offers 3 mains (pizza, burger,

salad) and 2 sides (fries, fruit). How many lunches are possible? List them: pizza-fries, pizza-fruit, burger-fries, burger-fruit, salad-fries, salad-fruit — 6 total. P(pizza with fries) = 1 ÷ 6. *The menu never said "6 lunches" — you built that number by listing. Where else do the totals hide instead of being given?*

Odds vs. Probability

Odds and probability describe the same situation differently. Probability compares favorable outcomes to the *total*; odds compare favorable outcomes to *unfavorable* ones.

P(rain) = 30% means 30 rainy outcomes out of 100 total. The **odds** of rain are 30 to 70, or "3 to 7." Gamblers and bookmakers quote odds; mathematicians quote probability. Convert: odds of a to b → P = a ÷ (a+b).

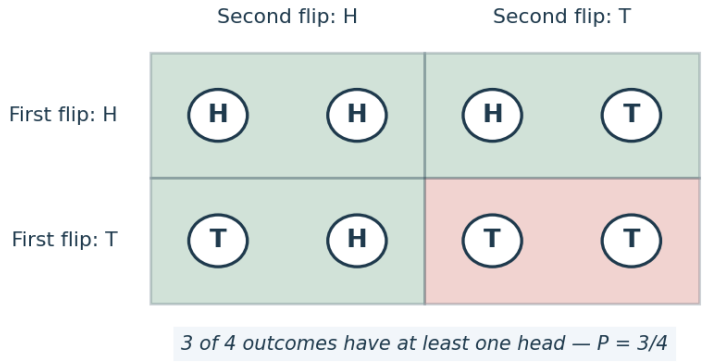
STORY

The Weather App.

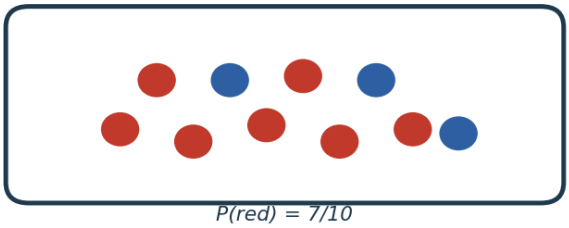
Your app says "30% chance of rain." Your friend says "the odds are 7 to 3 against rain." They are saying the same thing — 30 rainy futures out of 100. *Why do you think betting sites say "3 to 7" instead of "30%"? Which phrasing makes the rain feel less likely?*

Fig. 4.1

Fig. 4.1 — Sample space of two coin flips



3 of 4 outcomes have at least one head — P = 3/4



P(red) = 7/10

Sample space of two coin flips.

Worked Example 1 — The Jumble. Joel's MP3 player holds 2 classical songs, 13 rock songs, and 5 rap songs, and plays one at random. Find P(rock) and P(rap).

Solution. Total songs = 2 + 13 + 5 = 20. P(rock) = 13 ÷ 20 = 0.65 = **65%.** P(rap) = 5 ÷ 20 = 0.25 = **25%.** (P(classical) = 2 ÷ 20 = 10% — check: 65 + 25 + 10 = 100%. ✓)

Worked Example 2 — Die and coin. You roll a die and flip a coin. List the sample space and find P(even and heads).

Solution. Die: {1..6}, coin: {H, T} → 6 × 2 = 12 outcomes: {(1,H), (2,H), …, (6,T)}. Even-and-heads outcomes: (2,H), (4,H), (6,H) — 3 of them. P = 3 ÷ 12 = **1/4 = 25%.**

DECIDE — THE VERDICT

Which raffle tickets do you want to hold?

Back to the question that opened this section — now with the math to settle it.

Solution. Raffle A: 5 ÷ 200 = 0.025 = **2.5%**. Raffle B: 20 ÷ 500 = 0.04 = **4%**. The trade raises your chance from 2.5% to 4% — 1.6 times as likely.

The decision: make the trade. Twenty tickets *sounds* like a lot more than five — but 500 total tickets dilutes them. Probability doesn't care how many tickets you hold; it cares about your *fraction* of the whole. Whenever someone brags about a big number of chances, ask for the total.

AI AI Learning Lab 4.1

AI Level 1 — Understand.

Explain probability as "my slice divided by the whole pie" as if I'm 15. Then explain sample space, and the difference between probability and odds, each with one tiny example. After each of the three ideas, give me one quick check question and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 basic probability problems: two that need a sample space listed first, two "favorable ÷ total" computations, and one probability-to-odds conversion. Hide the answers. I'll solve all five and show my sample spaces; then check my work step by step.

AI Level 3 — Matter.

The raffle decision showed that 20 tickets can be worse than 5 tickets once you see the total — the fraction matters, not the count. Discuss with me: WHY does quantifying chance as a fraction matter for fairness in real life — think about lotteries, military drafts, or who gets audited? What are the implications when probabilities stay hidden and only raw counts get advertised — who benefits, and who gets fooled? End by asking me for one advertisement or claim in MY life I should re-check as a fraction.

Level 4 — Decide. *If it were your $3, which game would you play?*

Example decision problem. Two carnival games cost $3 per play with a $10 prize. Game A: flip a coin twice — win if both are heads. Game B: draw one marble from a

bag with 8 red and 12 blue marbles — win if it's red. Which game gives you the better chance?

AI Level 4 — Decide.

I need to choose between two $3 carnival games with a $10 prize. Game A: flip a coin twice, win on two heads. Game B: draw one marble from 8 red + 12 blue, win on red. Act as my decision coach: make me list the sample space for Game A myself and compute both probabilities as fractions, then ask me which I'd pick and why — don't decide for me.

Check your verdict: Game A: {HH, HT, TH, TT}, P(HH) = 1/4 = 25%. Game B: 8 ÷ 20 = 2/5 = 40%. Game B wins — 1.6 times the chance for the same $3. (Expected-value thinking comes in §4.7; for now, chance alone decides.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 4.1.

Ask your AI: "Invent a 6-question sample-space quiz — coins, dice, marbles — with an answer key." Then do the detective work: for *every* question, list the full sample space yourself and verify the AI's denominator. If the AI counted the outcomes wrong even once, write a one-paragraph explanation of which outcomes it missed or double-counted. AI chatbots are fluent at dividing and sloppy at listing — in this chapter, *you* are the one who lists.

Practice Problems 4.1

1. You roll a fair six-sided die. What is P(rolling an even number)?
2. A standard deck has 52 cards. What is P(drawing a heart)?
3. List the sample space for flipping two coins. What is P(at least one head)?
4. The odds in favor of your team are 3 to 1. What is the probability your team wins?
5. Your class has 28 students; 12 play soccer. If one student is picked at random, what is P(the student plays soccer)? Give a percent.

4.2 Theoretical vs Experimental Probability

THE DECISION

Do you pool your $100 on the "hot" dice?

At the carnival, a dice game costs $2 per play: roll two dice, win $10 if you roll doubles. The theoretical probability of doubles is 6 out of 36. But your friend just played 60 times and saw doubles **12 times** — that's 20%, well above the theoretical 16.7%. "These dice are hot!" she says. "Pool $100 with me and we'll play all night."

Do you join her? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** List the doubles in the two-dice sample space: (1,1), (2,2), … — how many are there out of 36?
- **Hint 2:** In 60 rolls, how many doubles would you *expect* at the theoretical rate? Compare 10 expected vs. 12 observed — is that a big gap or a small one?
- **Hint 3:** At the theoretical rate, what's your expected winnings per $2 play? (Hint: (1/6) × $10, minus the $2 cost.)

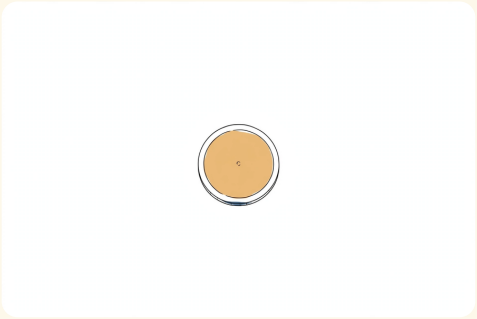
Theoretical Probability

Theoretical probability is probability by reasoning — no experiment needed. If every outcome is equally likely, P(event) = (favorable) ÷ (total), computed from the structure of the situation. Doubles on two dice: 6 favorable out of 36 total → 6/36 = **1/6 ≈ 16.7%**. You know this before a single die is rolled.

STORY

The Unflipped Coin.

A fair coin has P(heads) = 1/2 — and you know it without flipping once. The coin's *symmetry* is the whole argument: two sides, no reason to favor either. *What would have to be true about the coin for you to stop trusting the 1/2?*



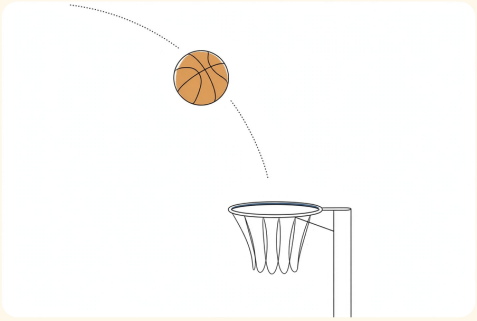
Experimental Probability

Experimental probability is probability by data: (times it happened) ÷ (times you tried). Flip a coin 50 times, get 27 heads → experimental P(heads) = 27/50 = **54%**. It measures the *world*, not the *reasoning* — and the world includes biased coins, sticky dice, and small samples.

STORY

The Free-Throw Line.

Your teammate made 37 of her last 50 free throws. Her experimental probability is 37/50 = **74%** — that's the number you'd bet on for her next shot, not the league average. *Why is her 74% more useful to you than the "theoretical" 50/50 of make-or-miss?*



The Law of Large Numbers

Here is the bridge: as the number of trials grows, experimental probability *settles toward* theoretical probability. Twelve doubles in 60 rolls (20%) looks hot; but 60 rolls is a puddle, not an ocean. With 6,000 rolls, the experimental rate would hug 1/6 — the noise washes out.

This is why casinos always win: they don't need rigged games, just *many* trials. A 51%-to-49% edge means nothing over 10 bets and everything over 10 million.

STORY

The Casino Floor.

A roulette wheel pays out slightly less than

fair odds on every spin — the house edge is about 5%. One lucky tourist can absolutely win tonight. But across a million spins a year, the casino's 5% is as reliable as rent. *The casino doesn't predict YOUR spin. What does it predict instead — and why is that enough?*

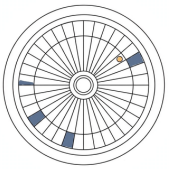
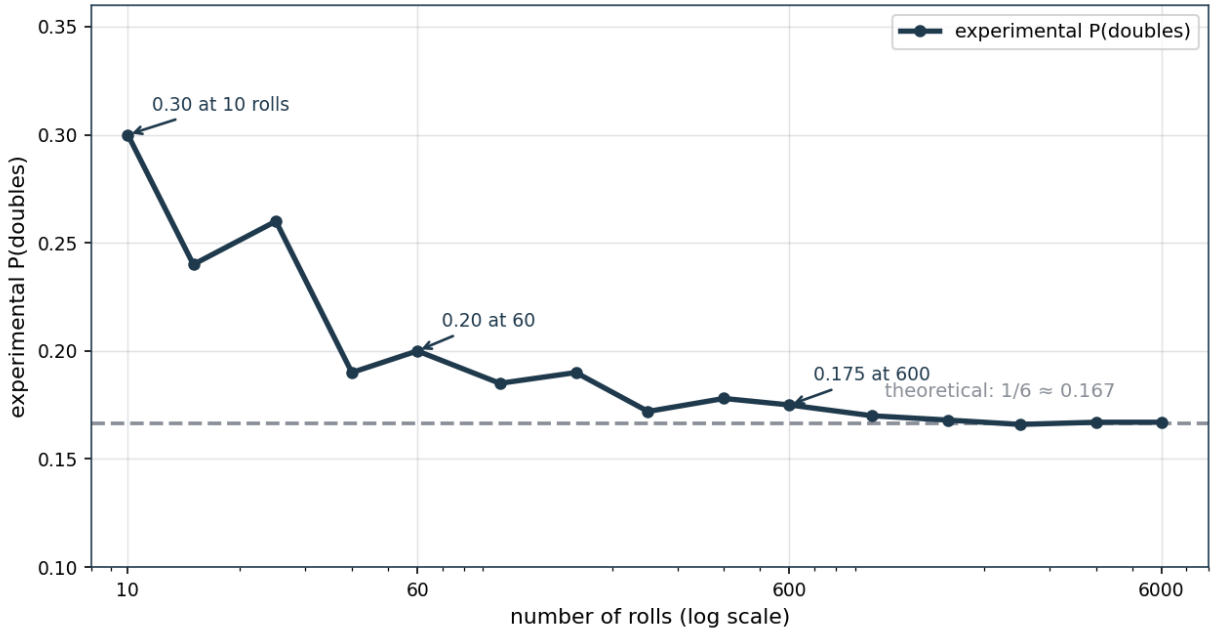


Fig. 4.2

Fig. 4.2 — The Law of Large Numbers in action



More trials, less noise — the experimental rate converges to the theoretical 1/6.

Worked Example 1 — List the doubles. Roll two dice. List the doubles in the sample space and find P(doubles).

Solution. The sample space has 36 outcomes. The doubles are (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) — 6 of them. P(doubles) = 6 ÷ 36 = **1/6 ≈ 16.7%**. This is the theoretical probability from the DECISION box — the number your friend's "hot dice" would have to beat.

Worked Example 2 — Reading an experiment. A coin is tossed 50 times: 27 heads, 23 tails. What is the experimental P(heads)? How does it compare to theoretical?

Solution. Experimental P(heads) = 27 ÷ 50 = **54%.** Theoretical is 50%. The 4-point gap is ordinary noise in 50 trials — not evidence of a biased coin.

Worked Example 3 — The game's expected value. The carnival game pays $10 for doubles at $2 per play. What is your expected net per play at the theoretical rate?

Solution. Expected winnings = (1/6) × $10 ≈ $1.67. Minus the $2 cost: $1.67 − $2.00 = **−$0.33 per play**. Every play costs you 33 cents *on average* — the game is a slow leak, not an opportunity. (Expected value gets its full treatment in §4.7.)

DECIDE — THE VERDICT

Do you pool your $100 on the "hot" dice?

Back to the question that opened this section — now with the math to settle it.

Solution. Expected doubles in 60 rolls = 60 × (1/6) = **10**. Your friend saw 12 — two extra doubles in 60 rolls is noise, not heat. And the game's expected value is −$0.33 per play *at the theoretical rate*; a "hot" streak of 12-in-60 doesn't change the dice, it changes your confidence.

The decision: don't pool the $100. The hot hand is a story your brain tells about randomness — the Law of Large Numbers says 60 rolls is far too few to declare the dice biased, and the negative expected value says the game charges you for the privilege of playing. If you love the game, bring $10 for fun, not $100 for profit.

AI AI Learning Lab 4.2

AI Level 1 — Understand.

Explain theoretical vs. experimental probability as if I'm 15, using the doubles-on-dice example (6/36 by reasoning vs. 12/60 by watching). Then explain the Law of Large Numbers with the casino idea. After each of the three ideas, give me one check question and wait for my answer.

AI Level 2 — Apply.

Give me 5 problems: two asking for theoretical probabilities (dice, cards), two asking for experimental probabilities from given trial data, and one asking whether an observed result (e.g., 70 sixes in 200 rolls) looks biased. Hide the answers; I'll solve and show reasoning; then check my work step by step.

AI Level 3 — Matter.

The dice decision showed that 12 doubles in 60 rolls looks "hot" but is just noise — and casinos get rich off people who can't tell the difference. Discuss with me: WHY does the hot-hand fallacy persist, and what are its implications — think about gambling addiction, "due for a win" lottery thinking, or investors chasing last year's hot stock? Who profits when people mistake noise for pattern, and what does the Law of Large Numbers imply about *their* business model? End by asking me for one "hot streak" belief in MY life I should re-examine.

Level 4 — Decide. *If it were your dollar, would you keep spinning?*

> **Example decision problem.** A spinner has 4 equal sectors, one red. Each spin costs $1; landing on red pays $5. The theoretical P(red) is 1/4. You watch a friend spin 20 times and win only 3 times (15%). Do you play?

AI Level 4 — Decide.

I'm deciding whether to play a $1 spinner game (4 equal sectors, one red, pays $5 on red). My friend won only 3 of 20 spins (15%). Act as my decision coach: make me compute the theoretical probability and the expected value per spin myself, then make me compare 3 observed wins to the expected 5 in 20 spins and judge whether that's suspicious — don't decide for me.

Check your verdict: theoretical P = 1/4 = 25%; expected value = 0.25 × $5 − $1 = +$0.25 per spin — a rare player-favorable game. In 20 spins you'd expect 5 wins; 3 observed is ordinary noise. Play — and don't let a small sample scare you off a positive expectation.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 4.2.

Ask your AI: "Simulate (or describe the results of) rolling two dice 600 times and report how often doubles came up." Then do the detective work: compute the experimental probability from its numbers, compare it to 1/6, and compute the gap. If the AI claims the result "proves" anything from a tiny trial count, write a one-paragraph explanation of why the Law of Large Numbers forbids that conclusion. AI chatbots love declaring patterns in noise — *you* are the one who counts the trials.

Practice Problems 4.2

1. A coin is tossed 80 times: 44 heads, 36 tails. What is the experimental probability of heads?

2. What is the theoretical probability of rolling an odd number on a fair die?
3. A die is rolled 200 times and shows a six 70 times. What is the experimental P(six)? Does the die look biased? Explain.
4. In your own words, explain the Law of Large Numbers — and why a casino with a 5% edge doesn't fear a lucky tourist.
5. Amy catches the 7:45 bus; in 79 days she arrived on time 53 times. What is the experimental probability she arrives on time?

4.3 Probability from Data

THE DECISION

Surgery — or wait and hope?

Your doctor recommends surgery for your knee. From the clinic's records: **920 of the last 1,000** similar patients recovered fully after this surgery. Without surgery, patient records show **60%** recover on their own within a year — the other 40% stay limited.

Do you choose surgery? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Convert the surgery record to a percent: 920 out of 1,000 is what percent?
- **Hint 2:** Now compute the complement: what percent did NOT recover fully? (Subtract from 100%.)
- **Hint 3:** Compare 92% vs. 60%. Then ask the question the numbers can't answer: what does "not recovered" actually *look like* for each option?

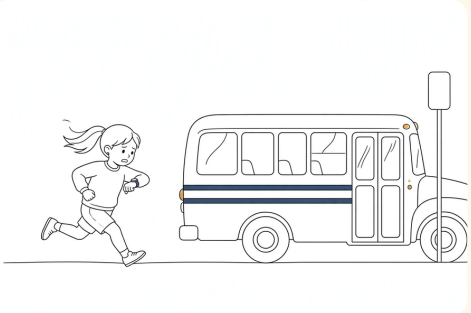
Reading Probability from Data

Sometimes nobody hands you a clean sample space — you get *records*. **Probability from data** is experimental probability grown up: (times it happened in the records) ÷ (total records). Amy's bus: on time 53 of 79 days → P(on time) ≈ 53 ÷ 79 ≈ **67.1%**. No dice, no coins — just history, used as a forecast.

STORY

Amy's Bus.

Amy catches the 7:45 a.m. bus. Over 79 school days, she arrived on time 53 times. Tomorrow's forecast isn't a weather report — it's 53/79 ≈ 67%. *If Amy has a can't-miss exam, is 67% good enough, or does she need a backup plan?*



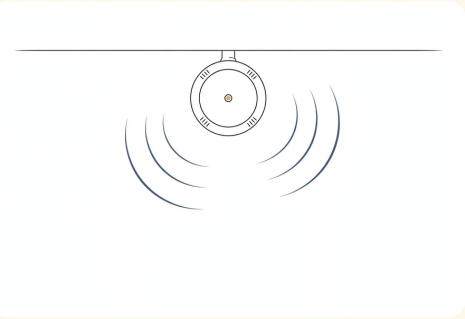
The Complement Rule

The **complement** of an event is "everything else" — and its probability is blissfully easy: **P(not E) = 1 − P(E)**. If P(surgery succeeds) = 92%, then P(it doesn't) = 8%. Complements turn hard "at least one" problems into easy subtractions, as §4.4 will show.

STORY

The Fire Alarm.

Your smoke alarm's test record: it sounded in 997 of 1,000 tests → P(works) = 99.7%. The complement, P(fails) = 0.3%, is the number that should keep you up at night — small, but not zero. *Why do engineers obsess over the 0.3% instead of celebrating the 99.7%?*



From Data to Decision

Data gives you probabilities; it doesn't make the decision. The 92% surgery success rate beats 60% natural recovery — but "8% didn't recover" is not one thing: it could mean "needed a second surgery" or "worse than before." The numbers tell you *which option wins*; only asking questions tells you *what winning costs*.

STORY

The Surgeon's Choice.

Two surgeons quote you success rates:

Dr. Lee, 92% over 1,000 patients; Dr. Park, 39 of 40 (≈98%). The bigger sample is more trustworthy — 40 patients is a puddle (remember §4.2). *If Dr. Park's 98% came from 4,000 patients instead, would your choice change? What else would you ask before deciding?*

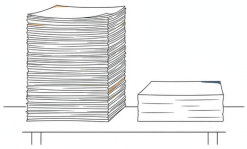
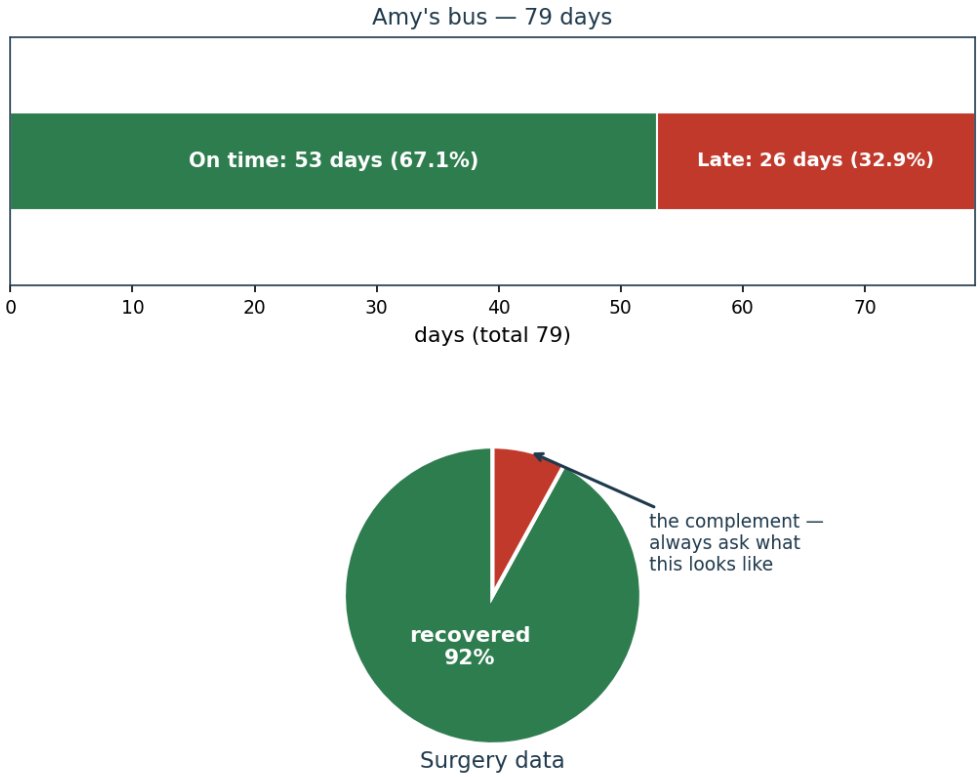


Fig. 4.3

Fig. 4.3 — Amy's bus: 79 days of data



Data becomes probability by division; the complement becomes a question.

Worked Example 1 — Defective bulbs. A factory tests 1,200 light bulbs; 15 are defective. Find P(defective) and P(not defective).

Solution. P(defective) = 15 ÷ 1,200 = 0.0125 = **1.25%**. P(not defective) = 1 − 0.0125 = 0.9875 = **98.75%**.

Worked Example 2 — Rain tomorrow. Historical data: on 42 of the last 120 days with today's weather pattern, it rained. Find P(rain) and P(no rain).

Solution. P(rain) = 42 ÷ 120 = 0.35 = **35%**. P(no rain) = 1 − 0.35 = **65%**.

DECIDE — THE VERDICT

Surgery — or wait and hope?

Back to the question that opened this section — now with the math to settle it.

Solution. Surgery: 920/1,000 = **92%** full recovery; complement **8%** not fully recovered. Waiting: 60% recover, 40% stay limited. On the numbers, surgery wins by 32 points — and the 8% failure still beats the 40% limitation rate of doing nothing.

The decision: choose surgery — *after* one question. The 8% is not a number, it's a set of human outcomes: ask the surgeon what "not fully recovered" meant for those 80 patients. If it means "needed physical therapy," sign the consent form. If it means "worse than before," get a second opinion. Probability from data tells you which door to open; the complement tells you to peek through it first.

AI AI Learning Lab 4.3

AI Level 1 — Understand.

Explain "probability from data" as if I'm 15, using Amy's bus (53/79) as the example. Then explain the complement rule with the fire alarm (99.7% → 0.3%). After each idea, give me one check question and wait for my answer.

AI Level 2 — Apply.

Give me 5 problems using data: three "compute the probability from the records" and two "use the complement rule." Include one with a small sample I should distrust (like Dr. Park's 40 patients). Hide the answers; I'll solve and show my division; then check my work step by step.

AI Level 3 — Matter.

The surgery decision showed that 92% success still leaves an 8% you must interrogate — and that 98% from 40 patients is weaker than 92% from 1,000. Discuss with me: WHY do published "success rates" mislead — think about what counts as "success," who was excluded from the records, or survivorship bias? What are the implications for medical decisions, product reviews ("4.9 stars from 12 reviews"), or election polls — and who benefits when we see the percent but not the denominator? End by asking me for one statistic in MY life I should re-check with its denominator.

Level 4 — Decide. *If it were your shoes, would you carry the umbrella?*

> **Example decision problem.** Weather records: on 42 of the last 120 days with today's pattern, it rained (35%). If it rains and you have no umbrella, you'll buy a $12 emergency poncho. Carrying your umbrella costs you nothing but the hassle. Do you bring it?

AI Level 4 — Decide.

I'm deciding whether to bring my umbrella. Records say 35% rain on days like today; a poncho costs $12 if I'm caught; carrying the umbrella is free. Act as my decision coach: make me compute the expected cost of going without (0.35 × $12) myself, compare it to the hassle of carrying, and state my decision with reasons — don't decide for me.

Check your verdict: expected poncho cost = 0.35 × $12 = $4.20 vs. $0 for the umbrella you already own. Bring the umbrella — the math says the hassle is cheaper than the risk. (This is expected-value thinking warming up for §4.7.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 4.3.

Ask your AI: "Find a real-world claim of the form 'X% effective' or 'X% success rate' — a drug, a product, a method." Then do the detective work: what was the denominator (how many trials)? What exactly counted as "success"? What does the complement — the (100 − X)% — actually look like in human terms? Write a one-paragraph verdict on whether the claim earns your trust. AI chatbots quote percentages the way parrots quote poetry — *you* are the one who asks for the denominator.

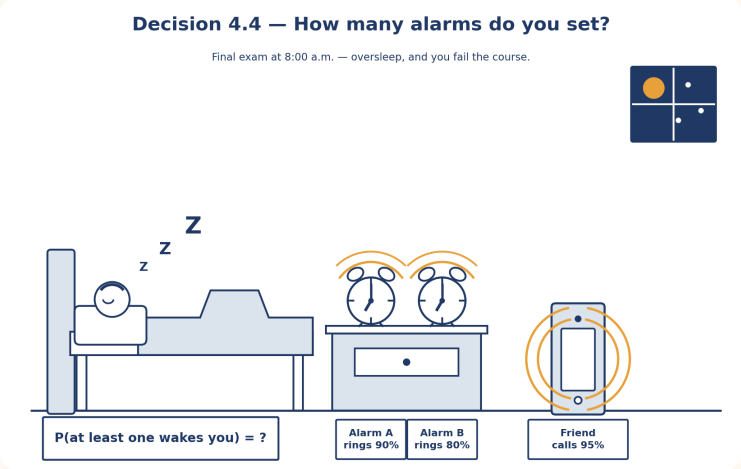
Practice Problems 4.3

1. A factory tests 2,500 bulbs; 40 are defective. What is P(a random bulb is defective)? What is P(it works)?
2. On 51 of the last 150 days with today's weather pattern, it rained. What is P(rain) tomorrow?
3. If P(rain) = 35%, what is P(no rain)? Which rule did you use?
4. A survey finds 340 of 500 voters support the proposal. What percent support it?
5. A player made 74 of 100 free throws. What is P(she makes the next one)? What is P(she misses)?

4.4 Compound Probability: AND & OR

THE DECISION

How many alarms do you set for the final exam?



Final exam at 8:00 a.m. — miss it and you fail the course. Your track records: **Alarm A rings 90%** of mornings; **Alarm B rings 80%** of mornings. If you set both, what's the chance at least one wakes you? And is that enough — or do you also ask your friend (who reliably calls 95% of the time) for a wake-up call?

How safe is safe enough? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** "At least one rings" is hard to count directly. Flip it: what's the chance BOTH alarms fail?
- **Hint 2:** P(A fails) = 10%, P(B fails) = 20%. Multiply: 0.1 × 0.2.
- **Hint 3:** Subtract from 1 to get "at least one rings." Then add the friend (5% fail rate) and recompute — how much safety does one text message buy?

AND: Both Happen

A **compound event** combines simple events. For **AND** — both happen — multiply along the branches of a **tree diagram**: P(A and B) = P(A) × P(B), *when the events are independent* (more on that word in §4.5).

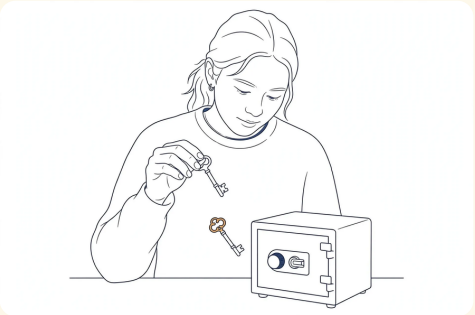
Two coins: P(both heads) = (1/2) × (1/2) = **1/4**. Each flip is a branch; each path through the tree is a compound outcome.

STORY

The Two Keys.

The safe needs Key A AND Key B — you carry both,

but each key has a 10% chance of being left at home. P(both keys with you) = 0.9 × 0.9 = 81%. *The safe doesn't care that you "usually" remember. What single change would push 81% above 95%?*



OR: At Least One

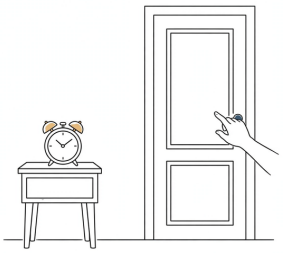
For **OR**, the complement is your best friend: **P(at least one) = 1 − P(neither)**. Counting "A or B or both" directly is a mess; counting "neither" is one multiplication.

Two alarms: P(both fail) = 0.1 × 0.2 = 0.02 → P(at least one rings) = 1 − 0.02 = **98%**. When events *can't* both happen (mutually exclusive), it's even simpler: just add. Jumble: P(rock or rap) = 13/20 + 5/20 = **90%** — a song can't be both.

STORY

The Backup Alarm.

Your phone alarm failed twice this month — each time, your roommate's knocking saved you. You didn't plan redundancy; you lucked into it. P(at least one) thinking is *planned* luck: 1 − P(everything fails). *What's one system in your life running on luck instead of planned redundancy?*



Mutually Exclusive vs. Inclusive

Two events are **mutually exclusive** if they can't both happen (heads or tails on one flip) — then P(A or B) = P(A) + P(B). They're **inclusive** if they can overlap — then you must subtract the double-counted middle: **P(A or B) = P(A) + P(B) − P(A and B)**. A **Venn diagram** shows the overlap as the lens in the middle.

In an Algebra 2 class of 30: 15 play basketball, 16 play baseball, 7 play both. P(basketball or baseball) = 15/30 + 16/30 − 7/30 = 24/30 = **80%**. Skip the subtraction and you'd claim 31/30 — more than certain, which is how you know you double-counted.

STORY

The Sports Survey.

Seven students play both sports — they were counted in the 15 *and* in the 16. The Venn diagram's overlapping lens is where double-counting lives. *A club reports "40 members like pizza, 35 like burgers, 60 members total." What must be true about the overlap?*

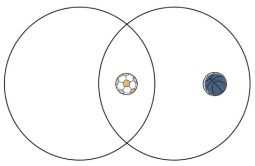
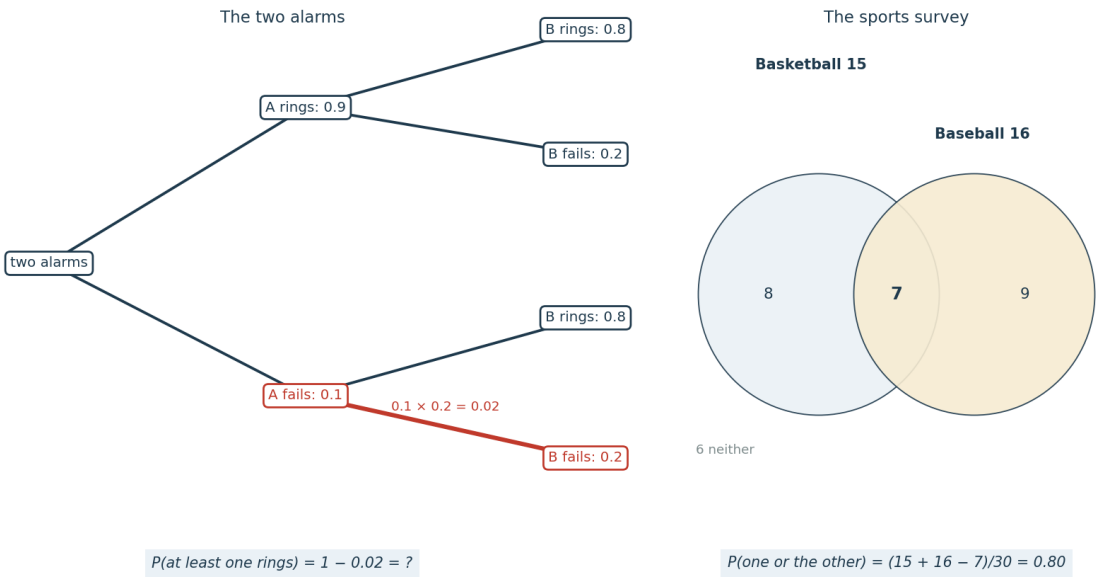


Fig. 4.4

Fig. 4.4 — The two alarms & the sports survey



Tree diagram: the two alarms + Venn diagram: the sports survey.

Worked Example 1 — Rock or rap (mutually exclusive). Joel's Jumble: 2 classical, 13 rock, 5 rap. Find P(rock or rap).

Solution. A song can't be both → mutually exclusive. P = 13/20 + 5/20 = 18/20 = **0.9 = 90%**.

Worked Example 2 — Heart or face card (inclusive). From a standard deck, find P(heart or face card).

Solution. 13 hearts, 12 face cards, 3 cards are both (J, Q, K of hearts). P = 13/52 + 12/52 − 3/52 = 22/52 = **11/26 ≈ 42.3%**.

DECIDE — THE VERDICT

How many alarms do you set for the final exam?

Back to the question that opened this section — now with the math to settle it.

Solution. Both alarms: P(both fail) = 0.1 × 0.2 = 0.02 → P(at least one rings) = **98%**. Add the friend (fails 5%): P(all three fail) = 0.1 × 0.2 × 0.05 = 0.001 → P(at least one) = **99.9%**.

The decision: send the text. Ninety-eight percent sounds safe until you realize 2% is 1 morning in 50 — over a school year, that's 3 or 4 mornings. The friend's call costs one text message and buys the jump from 98% to 99.9%. Redundancy is cheap; a failed final is not. (Engineers call this "defense in depth" — your exam deserves it too.)

AI AI Learning Lab 4.4

AI Level 1 — Understand.

Explain AND vs. OR probability as if I'm 15: AND multiplies along tree branches (two keys), OR is easiest via the complement (backup alarm). Then explain mutually exclusive vs. inclusive with the sports survey Venn diagram. One check question after each idea; wait for my answers.

AI Level 2 — Apply.

Give me 5 compound probability problems: two AND (tree diagram), two OR (one via complement, one via addition), and one Venn diagram overlap problem like the sports survey. Hide the answers; I'll solve and show my trees/diagrams; then check my work step by step.

AI Level 3 — Matter.

The alarm decision showed that "at least one" thinking — 1 minus the chance everything fails — is how you engineer safety. Discuss with me: WHY does redundancy thinking matter beyond alarms — think about airplane backup systems, hospital staffing, or election recount rules? What are the implications of the opposite mindset, "everything must work" thinking — who pays when there's no backup, and who decided the backup wasn't worth it? End by asking me for one no-backup system in MY life I should redesign.

Level 4 — Decide. *If it were your group project, would you trust them?*

> **Example decision problem.** Tom comes to class on time 90% of days; John 70%. For tomorrow's group meeting: (a) what's the probability at least one is on time? (b) what's the probability BOTH are on time? Should you schedule the critical decision for the first five minutes?

AI Level 4 — Decide.

I'm deciding whether to trust Tom (90% on time) and John (70% on time) for our group meeting. Act as my decision coach: make me compute (a) P(at least one on time) via the complement and (b) P(both on time) via multiplication myself, then ask me whether the critical vote should go in the first five minutes — don't decide for me.

Check your verdict: (a) $1 - 0.1 \times 0.3 = 97\%$ — someone will be there. (b) $0.9 \times 0.7 = 63\%$ — don't bet the critical vote on it. Trust the pair for attendance; protect the agenda from their overlap.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 4.4.

Ask your AI: "Draw (describe) the tree diagram for flipping three coins and find P(at least 2 heads) and P(exactly one tail)." Then do the detective work: list all 8 outcomes yourself — HHH, HHT, HTH, THH, HTT, THT, TTH, TTT — and verify both answers (at least 2 heads: 4/8 = 1/2; exactly one tail: 3/8). If the AI's tree is missing a branch or its fractions don't sum to 1, write a one-paragraph diagnosis. AI chatbots prune trees carelessly — *you* count every leaf.

Practice Problems 4.4

1. You roll a fair die. What is P(rolling a 2 or a 5)?
2. From a standard deck, what is P(king or queen)?
3. From a standard deck, what is P(spade or ace)? Show the subtraction.
4. You flip two coins. What is P(both heads)? Draw the tree.
5. In a class of 30, 15 play basketball and 16 play baseball; 4 play neither. How many play both? What is P(a random student plays both)?

4.5 Independent vs Dependent Events

THE DECISION

You tested positive. Do you panic?

A rapid test for a virus: 95% of sick people test positive (true-positive rate), and 5% of healthy people *also* test positive (false-positive rate). The virus is present in **1%** of the population. You just tested positive.

What is the probability you're actually sick — and do you panic, or just schedule a retest? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Imagine 10,000 people. How many are sick? How many are healthy?

- **Hint 2:** Of the sick group, how many test positive? Of the healthy group, how many test positive? Add them: total positives.
- **Hint 3:** What fraction of ALL the positives are actually sick? (Sick positives ÷ total positives.)

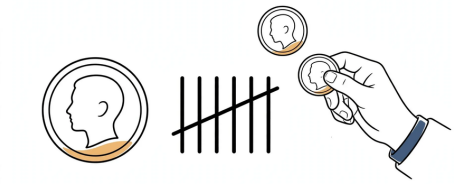
Independent Events

Two events are **independent** if one doesn't change the other's probability. Flip a coin, get heads — the next flip is still 50/50. Roll a die twice — the first roll doesn't touch the second. For independent events, P(A and B) = P(A) × P(B), no adjustments needed.

STORY

The Coin Has No Memory.

The coin landed heads five times in a row. "Tails is due!" — no. The coin doesn't remember, doesn't owe you, doesn't care. P(heads) on flip six is still 1/2. *Casinos post the last 20 roulette results on a glowing board. Why would they advertise history for a memoryless game?*



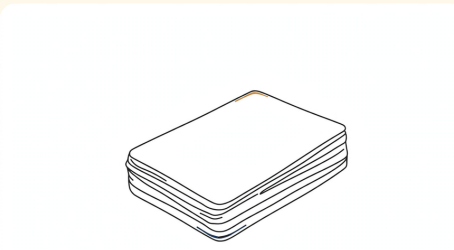
Dependent Events

Events are **dependent** when the first outcome changes the second's probabilities. Draw a card from a deck and *don't* put it back: P(ace) was 4/52; now it's 3/51. The tree's branches change weight as you move down — §4.6 is the full tour, but the idea starts here.

STORY

The Shrinking Deck.

You're drawing cards for a poker hand with no replacement. After three hearts hit the table, P(the next is a heart) drops from 13/52 to 10/49. The deck *remembers* — every card dealt rewrites the odds for the next. *Card counters in blackjack track exactly this. What are they really tracking?*



Conditional Probability: "Given That..."

Conditional probability is probability *after* news arrives: P(A | B) reads "P of A *given* B happened." The positive test is conditional probability in disguise — you're not asking P(sick), you're asking P(sick | tested positive). And the answer humbles everyone: most positives are healthy people, because healthy people vastly outnumber sick ones.

The tree: 10,000 people → 100 sick, 9,900 healthy → positives: 95 sick + 495 healthy = 590 → P(sick | positive) = 95 ÷ 590 ≈ **16.1%**.

STORY

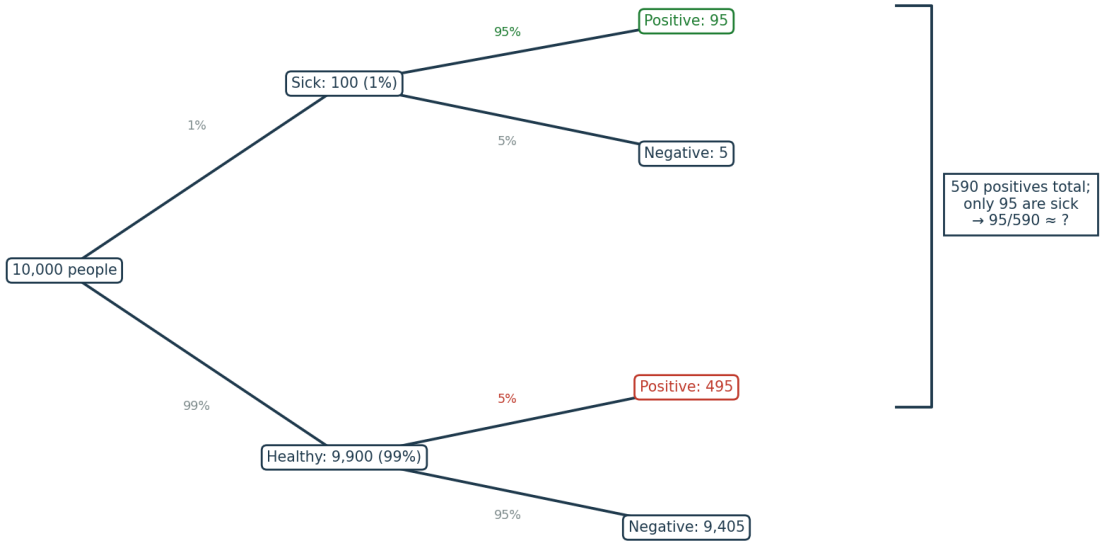
The Positive Test.

A "95% accurate" test sounds like a 95%

verdict. But accuracy is measured *per group*, and the groups are wildly unequal: 9,900 healthy vs. 100 sick. The 5% false-positive rate fires on the huge group — 495 false alarms drown out 95 true ones. *This is the base-rate fallacy, and it fooled doctors before it fooled you. What other "95% accurate" claims deserve the 10,000-person treatment?*

Fig. 4.5

Fig. 4.5 — The positive test, as a tree of 10,000 people



A 95% accurate test can still be wrong 5 times out of 6 — when the base rate is 1%.

Worked Example 1 — Two aces, no replacement. From a standard deck, draw two cards without replacement. Find P(both aces).

Solution. First draw: 4/52. Second (dependent — one ace is gone): 3/51. P = (4/52) × (3/51) = 12/2652 = **1/221 ≈ 0.45%**.

Worked Example 2 — The 99% test. Same virus (1% base rate), but a better test: 99% true-positive, 1% false-positive. Find P(sick | positive).

Solution. 10,000 people → 100 sick, 9,900 healthy. Positives: 99 sick + 99 healthy = 198. P = 99 ÷ 198 = **50%**. Even a "99% accurate" test is a coin flip here — the base rate dominates.

DECIDE — THE VERDICT

You tested positive. Do you panic?

Back to the question that opened this section — now with the math to settle it.

Solution. P(sick | positive) = 95 ÷ 590 ≈ **16.1%**. Five out of six positives are false alarms. And if the retest *also* comes back positive, keep walking down the tree: of the 590 first positives, 95 are truly sick and 495 are false alarms. The second test catches about 95% of the 95 (≈90) and re-flags about 5% of the 495 (≈25) → P(sick | two positives) ≈ 90 ÷ 115 ≈ **78.3%**. Now it's serious.

The decision: don't panic — retest. A single positive is *information*, not a diagnosis; it moves you from 1% to 16%, not from 1% to 95%. Two positives move you to 78% — that's when you call the doctor and clear your week. Conditional probability is the math of updating your mind: the disciplined version of "don't jump to conclusions."

AI AI Learning Lab 4.5

AI Level 1 — Understand.

Explain independent vs. dependent events as if I'm 15 (coin with no memory vs. shrinking deck). Then explain conditional probability with the positive-test tree of 10,000 people — walk me through the 95 and 495. One check question after each idea; wait for my answers.

AI Level 2 — Apply.

Give me 5 problems: two deciding independent vs. dependent, one two-aces-style without-replacement computation, and two conditional probability problems (one medical-test style with a tree, one simpler like "drawn card is a face card — P(it's a king?)"). Hide the answers; I'll solve and show my trees; then check my work step by step.

AI Level 3 — Matter.

The positive-test decision showed that "95% accurate" can mean 16% certainty — the base rate swallows the accuracy. Discuss with me: WHY does the base-rate fallacy fool even experts — think about mammograms, job-interview "tests," or predictive policing? What are the implications when a test's accuracy is advertised but its base rate is hidden — who gets mislabeled, who profits from the confusion, and what would honest labeling look like? End by asking me for one "accuracy" claim in MY life I should run through the 10,000-person tree.

Level 4 — Decide. *If it were your turn, which version would you play?*

> **Example decision problem.** A bag holds 5 red and 5 blue marbles. A game: draw two marbles, win $20 if both are red. Entry $5. Version X: with replacement (put the first marble back). Version Y: without replacement. The house lets you choose the version. Which do you pick?

AI Level 4 — Decide.

I'm choosing between two versions of a marble game (5 red, 5 blue, win $20 on two reds, $5 entry): with replacement or without. Act as my decision coach: make me compute both probabilities myself — $(5/10)^2$ for with-replacement, $(5/10)\times(4/9)$ without — then ask me which the house would prefer to offer and why, and make me state my pick with reasons — don't decide for me.

Check your verdict: with replacement: (5/10) × (5/10) = 25%; without: (5/10) × (4/9) = 20/90 ≈ 22.2%. Pick with-replacement — and notice the house offered you the choice because most players can't tell the versions apart. (On expected value: 0.25 × $20 − $5 = $0 — the with-replacement game is exactly fair; the other is a tax on innumeracy.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 4.5.

Ask your AI: "Explain the Monty Hall problem — three doors, one prize — and tell me whether to switch." Then do the detective work: draw the full tree yourself. If you pick door 1: with probability 1/3 the prize is there (staying wins); with probability 2/3 it's behind door 2 or 3, the host opens the other one, and switching wins. Verify that switching wins 2/3 of the time. If the AI says "it's 50/50 after one door opens," write a one-paragraph refutation using your tree. AI chatbots *feel* the 50/50 — *you* prove the 2/3.

Practice Problems 4.5

1. A coin has landed heads five times in a row. What is P(heads) on the next flip? Are the flips independent or dependent?
2. From a standard deck, draw two cards without replacement. What is P(both face cards)?
3. A bag holds 6 red and 4 blue marbles. Draw two without replacement. What is P(red then blue)?
4. Same bag, but *with* replacement. What is P(red then blue)?

5. A test has 90% true-positive and 5% false-positive rates for a condition in 2% of people. You test positive. What is P(you have the condition)?

4.6 Sampling With & Without Replacement

THE DECISION

Do you accept the shipment?

Your club ordered 50 phone chargers for a fundraiser. The supplier claims "at most 2% defective" — that's at most **1 bad charger in 50**. You randomly pull 2 chargers to test (without replacement) and **1 is defective**. If the supplier's claim were true, how surprising is that?

Do you accept the shipment — or test more? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** If exactly 1 of 50 is defective, what's P(the first charger you pull is good)?
- **Hint 2:** After pulling one good charger, 49 remain with the 1 defective still hiding. What's P(the second is also good)? Multiply.
- **Hint 3:** That's P(no defectives found). Subtract from 1 for P(at least one found). Is 4% the kind of thing that "just happens"?

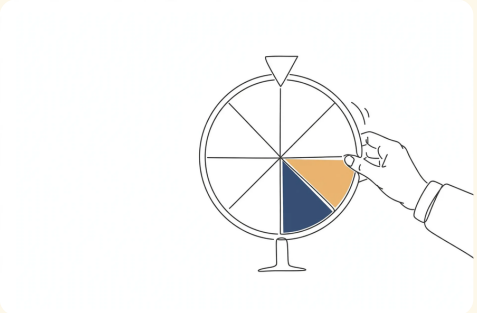
With Replacement

With replacement, you put each drawn item back — the probabilities *reset* every draw. Draw a heart (13/52), replace it, draw again: still 13/52. Every draw is independent, every branch of the tree identical. Replacement is the mathematician's laboratory: clean, but rarely how the real world works.

STORY

The Reset Button.

A spinner game lets you spin, then the wheel resets — every spin is a fresh 1/4. *Real life rarely has a reset button: name a situation where someone pretends draws are "with replacement" when they aren't.*



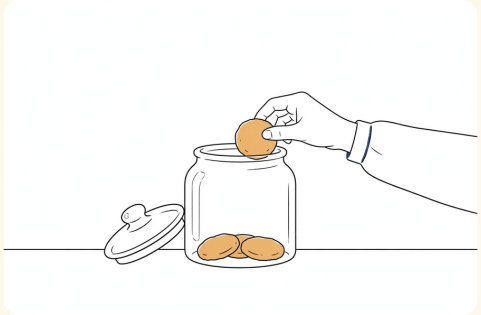
Without Replacement

Without replacement, each draw changes the next: numerators and denominators both shrink. The classic: 5 red, 3 blue, 2 green marbles — draw 3 without replacement. P(all 3 red) = (5/10) × (4/9) × (3/8) = 60/720 = **1/12 ≈ 8.3%**. The tree's branches get thinner as you descend.

STORY

The Cookie Jar.

The jar holds 10 cookies; you take one without looking, then your sister takes one. P(you get the chocolate-chip one) was 3/10; for her it's now 2/9 or 3/9 depending on your luck. *She insists on picking first next time. Does going first actually help?*



Sampling as a Decision Tool

Here's the power move: a *small* sample, drawn properly, can *judge* a big claim. Finding 1 defective in 2 draws when the claim allows only 1 in 50 isn't proof — but at 4% probability, it's evidence. This is hypothesis testing wearing a disguise: assume the claim is true, compute how weird your sample is, and decide.

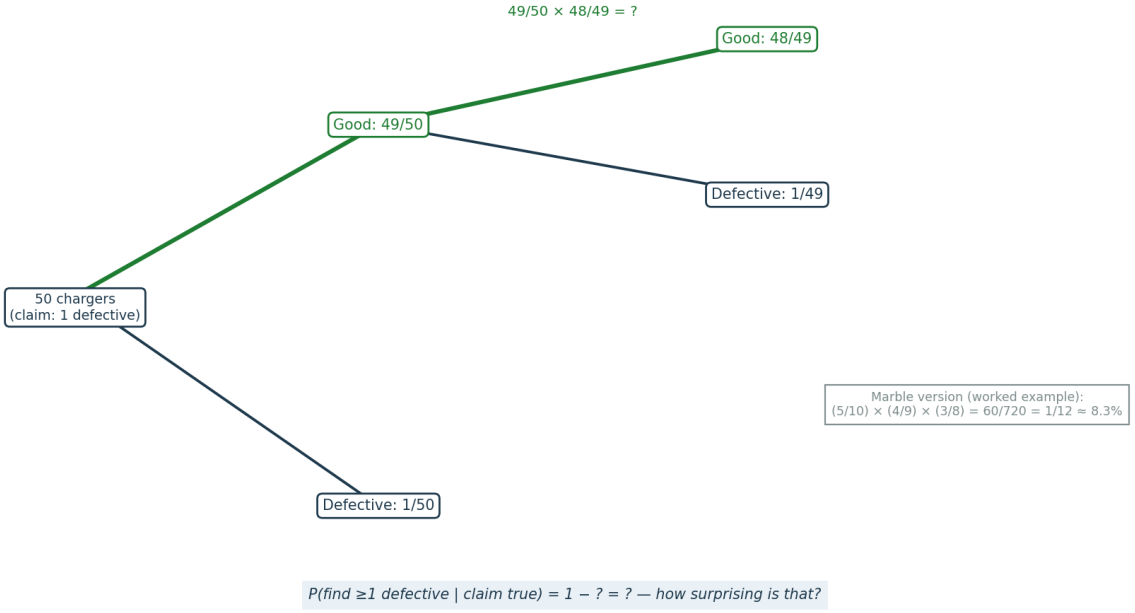
STORY

The Inspector.

A food inspector can't taste every can on the shelf — she samples 5. If the factory's "1% spoiled" claim were true, finding 2 spoiled in 5 would be a ~0.1% miracle. She doesn't need the miracle explained; she needs the shipment held. *What makes her sample fair — and what would make it a witch hunt?*

Fig. 4.6

Fig. 4.6 — Sampling without replacement: the chargers



Sampling without replacement: the chargers.

Worked Example 1 — Three red marbles. A bag holds 5 red, 3 blue, 2 green marbles. Three are drawn without replacement. Find P(all three are red).

Solution. P = (5/10) × (4/9) × (3/8) = 60/720 = **1/12 ≈ 8.33%**. Each draw shrinks both the reds remaining and the total.

Worked Example 2 — At least one defective. A shipment of 50 chargers actually contains 3 defectives. You test 2 without replacement. Find P(at least one defective).

Solution. Use the complement: P(neither defective) = (47/50) × (46/49) = 2162/2450 ≈ 0.8824. P(at least one) = 1 − 0.8824 = **≈ 11.76%**.

DECIDE — THE VERDICT

Do you accept the shipment?

Back to the question that opened this section — now with the math to settle it.

Solution. Under the supplier's claim (1 defective in 50): P(both draws good) = (49/50) × (48/49) = 48/50 = 0.96, so P(find ≥1 defective) = 1 − 0.96 = **4%**. You found a defective in just 2 draws — an event with only a 4% chance if the claim were true.

The decision: don't accept — test 10 more. Four percent isn't proof the supplier lied (it happens 1 time in 25), but it's far too suspicious to bet the fundraiser on. This is the whole logic of quality control in one shipment: you can't check everything, so you check a little, and let the sample interrogate the claim. If 10 more draws show zero defectives, reconsider; if they show another, reject the shipment.

AI AI Learning Lab 4.6

AI Level 1 — Understand.

Explain with vs. without replacement as if I'm 15, using the cookie jar (probabilities shrink) vs. the spinner reset (probabilities reset). Then explain how a small sample can judge a big claim, using the charger shipment. One check question after each idea; wait for my answers.

AI Level 2 — Apply.

Give me 5 sampling problems: two with replacement, two without (including one "at least one" via the complement), and one where I must judge a claim from a sample like the charger problem. Hide the answers; I'll solve and show my tree multiplications; then check my work step by step.

AI Level 3 — Matter.

The shipment decision showed that 2 draws can put a 50-charger claim on trial — sampling is judgment, not just arithmetic. Discuss with me: WHY does sampling matter for power — think about food inspectors, election polls, or drug testing? What are the implications when sampling is done badly or dishonestly — a rigged poll, a cherry-picked audit? Who gets protected by fair sampling, and who gets exposed? End by asking me for one claim in MY life I could test with a small honest sample.

Level 4 — Decide. *If it were your dollar, would you buy the ticket?*

> **Example decision problem.** A lottery: pick 2 numbers from 1–10 (without replacement, order doesn't matter). A $1 ticket wins $40 if your pair exactly matches the drawn pair. Do you play?

AI Level 4 — Decide.

I'm deciding whether to buy a $1 lottery ticket: pick 2 numbers from 1–10, win $40 on an exact match. Act as my decision coach: make me count the possible pairs myself (C(10,2)), compute the win probability and the expected value (prize × P − cost), and state my decision with reasons — don't decide for me.

Check your verdict: possible pairs = C(10,2) = 45; P(win) = 1/45 ≈ 2.2%; expected value = $40 × (1/45) − $1 ≈ −$0.11 per ticket. Skip it — the lottery keeps 11 cents of every dollar. (Full expected-value machinery is §4.7 — you're already doing it.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 4.6.

Ask your AI: "Is it true that in a group of 23 people, there's over a 50% chance two share a birthday? Prove it." Then do the detective work: compute it yourself via the complement — P(no shared birthday) = (365/365) × (364/365) × … × (343/365), and 1 minus that. Verify the answer is ≈ 50.7%. If the AI hand-waves ("it's just a paradox"), write a one-paragraph explanation of *why* the complement multiplication is the right tool. AI chatbots quote the birthday paradox like trivia — *you* derive it.

Practice Problems 4.6

1. From a standard deck, draw two cards without replacement. What is P(both hearts)?
2. Same draw, but *with* replacement. What is P(both hearts)?
3. A committee of 5 is chosen from 20 students (8 girls, 12 boys). What is P(all 5 are girls)?
4. A bag holds 4 red and 6 blue marbles. Draw two without replacement. What is P(one red and one blue, in either order)?
5. Same bag, but with replacement. What is P(one red and one blue, in either order)?

4.7 Expected Value & Fair Decisions

THE DECISION

The $5 scratch-off — or the savings account?

A scratch-off ticket costs $5. The lottery commission publishes the odds: P(win $100) = 1/500, P(win $10) = 1/50, otherwise $0. Your alternative: put the $5 in your savings account, where it just sits.

Which grows your money? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** In 500 tickets, how many $100 winners are there? How many $10 winners? What's the total prize money?
- **Hint 2:** 500 tickets cost 500 × $5. Subtract the total prizes — how much do ticket buyers lose altogether?
- **Hint 3:** Divide that total loss by 500. That's your expected loss per ticket. Now compare with the savings account.

What Is Expected Value?

Expected value (EV) is the long-run average outcome per trial: **EV = Σ (value × probability)** — each outcome's value times its chance, added up. It doesn't predict *your* ticket; it predicts the average over *thousands* of tickets. The fair coin game (win $2 on heads, lose $2 on tails): EV = 0.5($2) + 0.5(−$2) = **$0** — a *fair game*.

STORY

The Fair Coin Game.

You and your friend bet $2 on each coin flip, 100 flips. Nobody "expects" to finish even — but the *average* finish across many such matches is $0. *If the bet were win $3 / lose $2, who would want to play 100 flips — and why would the other refuse?*

EV as a Decision Tool

For repeatable small-stakes decisions, EV is the whole game: positive EV → play; negative EV → walk away. The die game: win $12 on a six, lose $2 otherwise. EV = (1/6)($12) + (5/6)(−$2) = $2 − $1.67 = +**$0.33 per roll**. Play it all night — the Law of Large Numbers (§4.2) is your employee.

STORY

The Insurance Puzzle.

Your $20,000 car has a 1% annual chance

of being totaled. Expected loss: 0.01 × $20,000 = **$200**. The insurer charges $300. EV says: skip insurance, save $100 a year on average! Yet you buy it. *Why would a rational person pay $300 to avoid a $200 expectation? Hold that — the next section answers.*

When EV Isn't Enough

EV assumes you can survive the worst case and play again. The million-dollar flip: heads wins $2M, tails loses $1M. EV = 0.5($2M) + 0.5(−$1M) = +**$500,000** — enormously positive. Would you play? If losing $1M ruins you, no. **Ruin risk** breaks EV: insurance exists because one $20,000 loss can wreck a life that $100/year of savings can't rebuild. Use EV for small repeatable bets; buy certainty when ruin is on the table.

STORY

The Million-Dollar Flip.

A billionaire offers you the flip

above. EV says play — and for *her*, it's correct: she can lose $1M thirty times and laugh. For you, one tails ends the game. *Same math, opposite decisions. What differs — the numbers, or the player?*

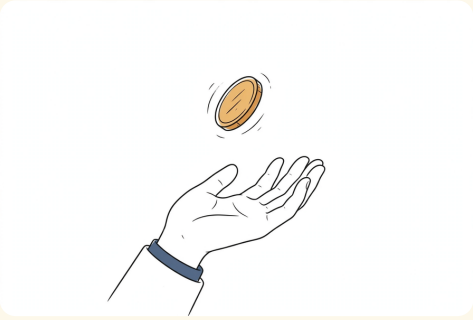
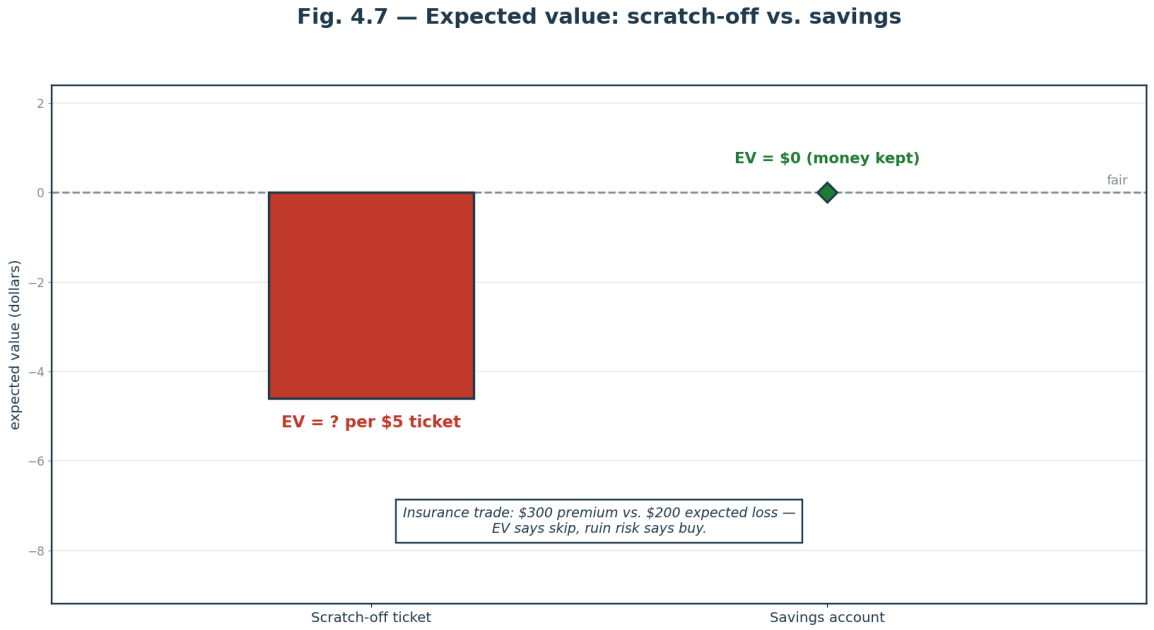


Fig. 4.7



Expected value prices the average; ruin risk prices the worst case.

Worked Example 1 — The die game. Win $12 on a six, lose $2 otherwise. Find the EV per roll.

Solution. EV = (1/6)($12) + (5/6)(−$2) = $2 − $10/6 = $2 − $1.67 = **+$1/3 ≈ +$0.33**. Positive — a player-favorable game.

Worked Example 2 — The raffle. 1,000 tickets at $2 each; one $1,000 prize. Find the EV of one ticket.

Solution. EV = (1/1000)($1,000) − $2 = $1 − $2 = **−$1**. Every ticket donates $1 to the fundraiser — which is the point, if you knew that going in.

DECIDE — THE VERDICT

The $5 scratch-off — or the savings account?

Back to the question that opened this section — now with the math to settle it.

Solution. Per 500 tickets: prizes = 1 × $100 + 10 × $10 = $200. Cost = 500 × $5 = $2,500. Buyers lose $2,300 total → EV = −$2,300 ÷ 500 = **−$4.60 per ticket**. The savings account's EV is $0 (your $5 stays $5).

The decision: savings — it's not close. The ticket returns 40 cents per $5 *on average*; the lottery keeps $4.60 as its business model. But remember the insurance puzzle: EV rules repeatable small bets. If someone offered you "double or nothing on your life savings," EV might say play — and ruin risk would say run. Probability tells you the average; wisdom tells you whether you can afford it.

AI AI Learning Lab 4.7

AI Level 1 — Understand.

Explain expected value as if I'm 15: "each outcome's value times its chance, added up" — use the fair coin game (EV $0). Then explain why positive EV means "play" and negative means "walk," with the die game (+$0.33). One check question after each idea; wait for my answers.

AI Level 2 — Apply.

Give me 5 expected-value problems: two simple coin/die games, one raffle, one insurance-style (premium vs. expected loss), and one where I must decide play-or-skip from the EV. Hide the answers; I'll compute each EV and state the decision; then check my work step by step.

AI Level 3 — Matter.

The scratch-off decision showed EV = −$4.60 — yet millions buy tickets weekly, while the insurance puzzle showed people *overpay* vs. EV to avoid ruin. Discuss with me: WHY do people buy negative-EV lottery tickets — hope, entertainment, innumeracy? And what are the implications — who plays the lottery most, where does the $4.60 per ticket actually go, and is a state lottery a tax on hope? Contrast with insurance: when is paying *above* EV rational? End by asking me for one negative-EV "ticket" in MY life I should price honestly.

Level 4 — Decide. *If it were your night, how many rolls would you play?*

Example decision problem. The die game from this section: win $12 on a six, lose $2 otherwise (EV +$0.33 per roll). A friend offers you 100 rolls at these stakes. Do you accept — and what changes if the stakes were $1,200/$200 instead of $12/$2?

AI Level 4 — Decide.

I'm deciding whether to accept 100 rolls of the die game (win $12 on six, lose $2 otherwise). Act as my decision coach: make me recompute the EV myself, estimate the 100-roll total, then ask me what changes if stakes were 100× bigger — don't decide for me.

Check your verdict: EV = +$1/3 per roll → 100 rolls ≈ +$33 expected. Accept at $12/$2 stakes — small, repeatable, positive. At $1,200/$200 stakes, decline: one bad run costs thousands, and ruin risk overrides EV. Same game, different player — §4.7's whole lesson in one question.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 4.7.

Ask your AI: "Design a carnival game with dice or a spinner that is exactly fair (EV = $0), and prove it." Then do the detective work: recompute the EV yourself from its probabilities and payouts. If the EV isn't actually zero — if the AI rounded a probability, forgot the entry cost, or mispriced a payout — write a one-paragraph audit showing the true EV and who it favors. AI chatbots design "fair" games the way they design everything: confidently, approximately. *You* are the auditor.

Practice Problems 4.7

1. Coin flip: win $3 on heads, lose $3 on tails. What is the EV? Is the game fair?
2. Die roll: win $6 on a six, lose $1 otherwise. What is the EV? Would you play?
3. A raffle sells 800 tickets at $3 each with a single $1,200 prize. What is the EV of one ticket? Buy or skip?
4. A game: 60% chance to win $10, 40% chance to lose $15. What is the EV? Play or skip?
5. Design your own fair die game (EV = $0): state the payouts and prove the EV is zero.

Vocabulary

- **Probability** — a number from 0 to 1 measuring how likely an event is; P(event) = (favorable outcomes) ÷ (total outcomes).
- **Sample space** — the list of all possible outcomes of an experiment.
- **Favorable outcomes** — the outcomes in the sample space that make the event happen.
- **Odds** — favorable outcomes compared to unfavorable ones (a to b); P = a ÷ (a + b).
- **Theoretical probability** — probability by reasoning, assuming equally likely outcomes.
- **Experimental probability** — (times it happened) ÷ (times tried), from real trials or records.
- **Law of Large Numbers** — as trials grow, experimental probability settles toward theoretical probability.
- **Probability from data** — experimental probability computed from historical records.
- **Complement** — "everything else"; P(not E) = 1 − P(E).
- **Compound event** — an event combining two or more simple events.
- **Tree diagram** — branches showing sequential outcomes and their probabilities; multiply along a path for AND.
- **Venn diagram** — overlapping circles showing shared outcomes between events.
- **Mutually exclusive events** — cannot both happen; P(A or B) = P(A) + P(B).
- **Inclusive events** — can overlap; P(A or B) = P(A) + P(B) − P(A and B).
- **Independent events** — one doesn't change the other's probability; the coin has no memory.
- **Dependent events** — the first outcome changes later probabilities (drawing without replacement).
- **Conditional probability** — P(A | B): probability of A given that B happened.
- **Base rate** — how common a condition is in the population before any test; it dominates test results.
- **With replacement** — drawn items are put back; probabilities reset.
- **Without replacement** — drawn items stay out; probabilities shift.
- **Expected value (EV)** — Σ (value × probability); the long-run average per trial.
- **Fair game** — EV = $0; neither side has an edge.
- **Ruin risk** — the chance a single loss ends the game; EV ignores it.

Summative Task: The 3-Paragraph Decision Essay

The prompt. Your rapid test came back positive. The clinic says: stay home until the lab retest in 3 days. A friend in the same situation asks your advice: treat the positive as a diagnosis — or as information? Write a 3-paragraph essay advising your friend, using this chapter's conditional probability. Your numbers: the test catches 95% of sick people, falsely

flags 5% of healthy people, and the virus is in 1% of the population — so P(sick | positive) ≈ 16.1%.

Planning your essay (do this before you write). 1. **Compute first.** Run the numbers yourself: 10,000 people → 100 sick → 95 positives; 9,900 healthy → 495 positives; 95 ÷ 590 ≈ 16.1%. Then the retest tree: about 90 of the 95 retest positive, about 25 of the 495 retest positive → 90 ÷ 115 ≈ 78.3%. Don't take these on faith — verify them with your own calculation. 2. **Thesis template.** "I recommend ___ because ***, even though*** ." Fill it in for YOUR side — either side is defensible if the numbers back it. 3. **Evidence checklist.** Every claim needs a number behind it: the 16.1% first-positive rate, the 78.3% two-positive rate, the 495-vs-95 base-rate picture, plus at least one idea from this chapter (conditional probability, base rate, or the complement) that strengthens the case. 4. **Flip test.** Name one specific condition that would change your recommendation — a second positive? Symptoms appearing? A higher base rate in your town? If your advice wouldn't flip under any condition, it's a slogan, not a decision.

Paragraph 1 — Your decision and claim. State your advice in one clear sentence, then one or two sentences on why it matters. (Use the thesis template: "I recommend ___ because ***, even though*** ." Both sides are arguable — the "treat it as a diagnosis" side leans on the cost of spreading the virus; the "treat it as information" side leans on the 16.1% number.)

Paragraph 2 — Mathematical evidence. Show the 10,000-person tree: 100 sick → 95 positives; 9,900 healthy → 495 positives; 95 ÷ 590 ≈ 16.1%. Explain why the 495 false positives drown out the 95 true ones (the base rate). Every claim needs a number behind it.

Paragraph 3 — Implications, limitations, and what would change your mind. What does "stay home 3 days" cost your friend — and what does a missed infection cost others? What are this calculation's limits (is the 1% base rate right for *your* town?)? Name one specific condition that would flip your advice — for example, a second positive (≈78.3%), or symptoms appearing — and explain why.

Length: 200–300 words. Three paragraphs, no more, no fewer.

Using AI. Write your first draft entirely on your own — no AI. Then you may use AI as a revision coach: ask it to check your logic, flag weak evidence, and suggest clearer sentences. Never let it write paragraphs for you. And remember this book's golden rule: AI makes mistakes — verify everything, especially any numbers it gives you. (Ask it for P(sick | positive) and watch whether it remembers the base rate.)

Rubric and exemplar essays are in the Teacher Guide.

Answer Key — Practice Problems (odd-numbered)

4.1 — 1. 3/6 = 1/2 = 50%. 3. {HH, HT, TH, TT}; P(at least one head) = 3/4 = 75%. 5. 12/28 = 3/7 ≈ 42.9%.

4.2 — 1. 44/80 = 55%. 3. 70/200 = 35% vs. theoretical 1/6 ≈ 16.7% — yes, it looks biased (more than double the expected rate). 5. 53/79 ≈ 67.1%.

4.3 — 1. P(defective) = 40/2500 = 1.6%; P(works) = 98.4%. 3. 65% — the complement rule. 5. P(make) = 74%; P(miss) = 26%.

4.4 — 1. 2/6 = 1/3 ≈ 33.3%. 3. 13/52 + 4/52 − 1/52 = 16/52 = 4/13 ≈ 30.8%. 5. Both = 15 + 16 + 4 − 30 = 5 students; P = 5/30 = 1/6 ≈ 16.7%.

4.5 — 1. 1/2 — independent; the coin has no memory. 3. (6/10) × (4/9) = 24/90 = 4/15 ≈ 26.7%. 5. Out of 10,000 people: 200 sick, 9,800 healthy. Sick positives = 0.90 × 200 = 180; healthy positives = 0.05 × 9,800 = 490. Total positives = 670. P(condition | positive) = 180 ÷ 670 ≈ **26.9%** — the base rate still dominates, even with a different test.

4.6 — 1. (13/52) × (12/51) = 156/2652 = 1/17 ≈ 5.9%. 3. C(8,5) ÷ C(20,5) = 56 ÷ 15504 ≈ 0.36%. 5. 2 × (4/10) × (6/10) = 48%.

4.7 — 1. EV = 0.5($3) + 0.5(−$3) = $0 — fair. 3. EV = (1/800)($1200) − $3 = −$1.50 — skip (unless you're donating). 5. Answers vary — verify that Σ(value × probability) = 0.

Level 4 example checks — **4.1:** A: 1/4 = 25%; B: 8/20 = 40% — play Game B. **4.2:** EV = 0.25 × $5 − $1 = +$0.25/spin — play; 3 wins vs. 5 expected in 20 spins is noise. **4.3:** 0.35 × $12 = $4.20 > $0 — bring the umbrella. **4.4:** (a) 1 − 0.1 × 0.3 = 97%; (b) 0.9 × 0.7 = 63% — don't schedule the critical vote in the first five minutes. **4.5:** with-replacement $(5/10)^2$ = 25% (EV $0) vs. without (5/10)(4/9) ≈ 22.2% (EV ≈ −$0.56) — pick with-replacement. **4.6:** C(10,2) = 45 pairs; P = 1/45 ≈ 2.2%; EV = $40/45 − $1 ≈ −$0.11 — skip. **4.7:** EV = +$1/3 per roll → ≈ +$33 over 100 rolls — accept at $12/$2 stakes; decline at 100× stakes (ruin risk overrides EV).

CHAPTER 5

Statistics

A Note on AI Learning Labs in This Chapter

Chapter 4 taught you to interrogate certainty. This chapter teaches you to interrogate *averages* — the single most quoted, and most abused, number in public life. AI chatbots love averages: they will tell you the "average salary" of a career, the "average rent" in a city, or the "average score" of a school, and every one of those numbers can be technically true and practically a lie. So this chapter's standing rule: **never accept an average without asking "average of WHAT — and what is hiding behind it?"** Every AI Learning Lab keeps the four levels (understand, apply, matter, decide), and every Level 4 ends the way the chapter begins — with a decision that is yours to make.

About This Chapter

Statistics is the mathematics of describing crowds with a few numbers — and of catching the lies those numbers can tell. Every day someone quotes you an average: a job ad, a rent listing, a medical claim, a poll. This chapter gives you the tools to see through the quote. You will learn to classify data as categorical or numerical (discrete vs. continuous) and choose the summary each type deserves; to compute and compare mean, median, and mode; to present data with frequency tables and dot plots; to read box-and-whisker plots and hunt outliers; to group data into histograms; to measure spread with range, IQR, and standard deviation; to spot misleading graphs; to judge correlation without confusing it with causation; and to fit lines of best fit, read the correlation coefficient, and respect the limits of prediction — then you will use all of it to decide: which survey questions give real answers, which job to take, which player makes the roster, whether a medicine is safe, when to leave for the bus, which investment fits your life, whether the toothpaste is worth double, whether the tutoring is worth $500, and whether the $12,500 used car is a fair deal.

Running through the whole chapter is **The Personal Data Project** (§5.10): you will write two survey questions about something you actually care about, collect at least 20 responses from classmates, and apply every tool in this chapter — dot plots, box plots, histograms, standard deviation, correlation — to your own data, one milestone at a time. By the final poster, the numbers won't be the book's. They'll be yours.

By the end of this chapter, you will be able to:

- Classify data as categorical or numerical (discrete vs. continuous), and choose the right summary for each type — never averaging labels.
- Compute mean, median, and mode, and explain which "average" fits which situation — especially when an outlier is pulling the mean.
- Build frequency tables and dot plots, and describe a distribution's center, spread, and shape.
- Construct box-and-whisker plots, find quartiles and the IQR, and flag outliers with the 1.5×IQR rule.
- Group discrete and continuous data into class intervals, draw histograms, and estimate the mean from grouped data.
- Measure dispersion with range, IQR, and standard deviation, and explain why two data sets with the same mean can mean very different things.
- Detect misleading statistical graphs: truncated axes, cherry-picked ranges, and dishonest pictures.
- Read scatter plots, judge correlation strength, draw a line of best fit — and refuse to confuse correlation with causation.
- Fit and use a line of best fit to predict, compute residuals to judge individual cases, interpret the correlation coefficient r — and tell safe interpolation from dangerous extrapolation.
- Use AI prompts at four levels (understand, apply, matter, decide) to learn statistics — and force every AI "average" to show the data behind it.

5.1 What Kind of Data Is It?

THE DECISION

Which survey questions do you approve?

You're on the student council, and the principal will redesign the cafeteria based on your survey of 200 students. Three questions are proposed:

- **Q1:** "Which lunch line do you use most?" — Pizza / Hot meal / Salad bar / Sandwich.
- **Q2:** "How many minutes do you wait in line?" — any number.

- **Q3:** "How satisfied are you?" — Very / Somewhat / Not at all.

Results come back. Q1: Pizza 90, Hot meal 50, Salad bar 40, Sandwich 20. Q2: average wait 8.5 minutes. Q3: the analyst codes Very = 3, Somewhat = 2, Not at all = 1, computes an "average satisfaction" of 2.25 (up from 2.10 last year), and announces **"satisfaction is up 7%!"**

The principal asks you: which questions give real answers — and which "average" is a mirage? *Hold that question: the math you need comes next.*

> HINTS
>
> **Stuck? Work through these in order — do the math yourself.**

- **Hint 1:** For Q1, can you add "Pizza + Salad bar"? What single number honestly summarizes 90, 50, 40, 20?
- **Hint 2:** For Q2, does "8.5 minutes" describe a real wait? What would a couple of 25-minute waits do to that number?
- **Hint 3:** The "average satisfaction" is 2.25. Say it in words: a student is "2.25 satisfied." Does that sentence mean anything? What changed, in plain people-count, between last year and this year?

Categorical Data

Categorical data sorts things into buckets: lunch lines, blood types, eye colors, zip codes. You can count how many fall in each bucket, but arithmetic on the buckets is nonsense — "Pizza + Salad bar" is not a thing, and the "average" of two zip codes is a joke.

For categorical data, the honest summaries are **counts, percents, and the mode** (the biggest bucket). Q1's answer: Pizza, 90 of 200 students (**45%**) — the mode. That single fact tells the principal more than any average could: expand the pizza line.

STORY

The Blood Drive.

A clinic records 200 donors' blood types: O 92, A 70, B 28, AB 10. A new intern codes O = 1, A = 2, B = 3, AB = 4 and reports "the average blood type is 1.79, down from 1.84 last year." The head nurse stares at him. *What would you stock more of — and what does "1.79" even mean in blood?*

Numerical Data: Discrete

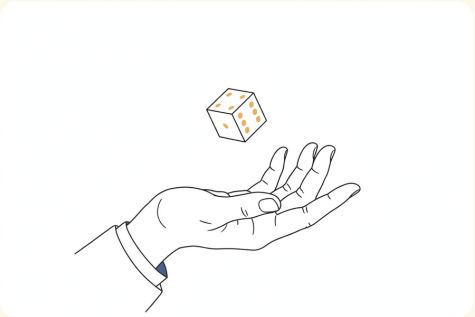
Numerical data is measured in numbers — and it splits in two. **Discrete** data is counted in whole chunks: number of siblings, goals scored, buses taken to school. Between 2 and 3 siblings there is no 2.5 siblings; the gaps are real.

Discrete data takes every summary: mean, median, and mode all make sense, because the numbers are genuinely numbers. "Average 2.2 children per family" is fine — nobody thinks a 0.2 child exists; it's a rate, not a person.

STORY

The Dice Bet.

Your friend rolls a die 60 times and bets you the average will be exactly 3.5. If you bet against him, you'd almost certainly win a dollar. The mean of many rolls does approach 3.5, but no single roll ever shows 3.5: discrete data can average to values it never takes. *When is an average allowed to be a value that never occurs?*



Numerical Data: Continuous

Continuous data is measured, not counted: height, weight, wait time, temperature. Between 8 and 9 minutes there are 8.1, 8.15, 8.153 minutes — you can always slice finer, limited only by your instrument.

Because continuous data flows smoothly, the mean and median are its natural summaries — but watch for skew. Q2's mean wait is 8.5 minutes, yet the median is 7: a few 25-minute horror waits

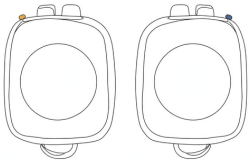
drag the mean up while most students wait 5–9 minutes. The median describes the typical student; the mean describes the total pain. Both are true; they answer different questions.

STORY

The Track Timer.

Two stopwatches time the same sprint: 11.42

and 11.47 seconds. Nobody argues about which is "right" — continuous data is always approximate, and more decimals just mean a better instrument. *If the coach recorded times as "fast / medium / slow" instead, what information would die?*



Why the Type Decides the Math

Here is the chapter's first commandment: **the data type chooses the math, not the other way around.** Feed categorical data into a mean and you get a number that is technically computable and completely empty — like the "average satisfaction" of 2.25, or a mean jersey number of 23.4.

The test is one question: *can you do arithmetic on the values and have the answer mean something?* Waiting 8.5 minutes: yes. Being "2.25 satisfied": no — so report what changed in real terms instead: "Very satisfied" rose from 30% to 40% of students, a 10-percentage-point gain of actual humans.

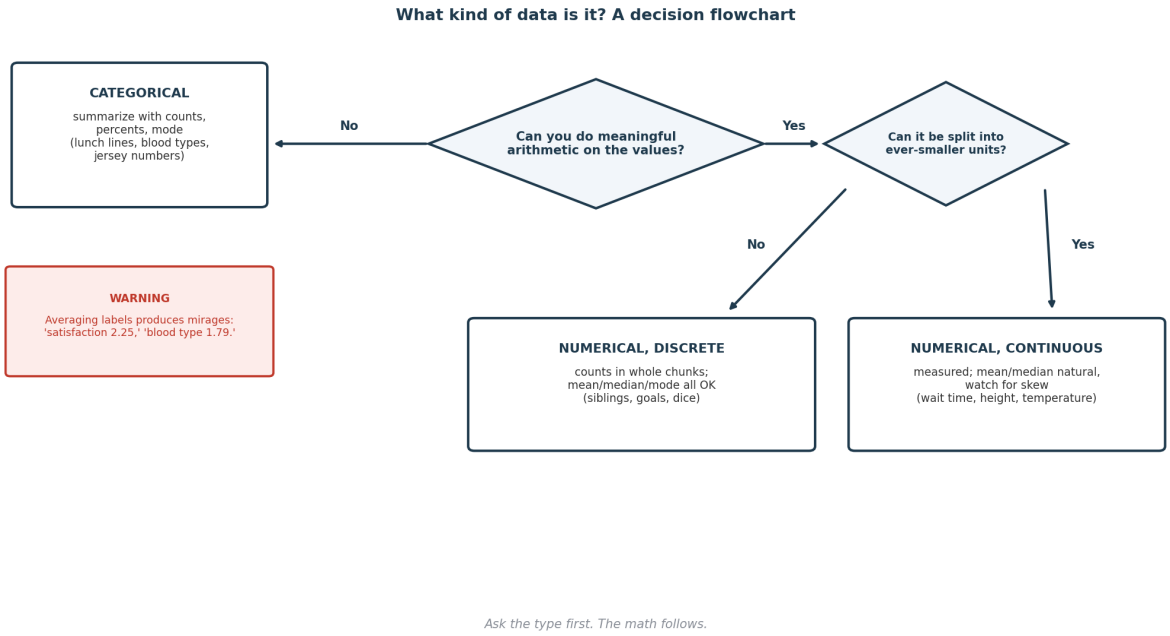
STORY

The Jersey Numbers.

A coach lists her players' numbers — 4,

11, 23, 23, 32 — and the team app proudly displays "mean jersey number: 18.6." The players laugh: nobody wears 18.6, and 23 isn't "more" than 4 the way 32 points is more than 4 points. Jersey numbers are labels wearing number costumes — categorical data in disguise. *What other numbers in your life are secretly labels?*

Fig. 5.1



Ask the type first. The math follows.

Ask the type first. The math follows.

Worked Example 1 — Classify and choose. Classify each variable, then name the best single summary: (a) eye color of 30 students, (b) number of text messages sent yesterday, (c) time to run 100 meters.

Solution. (a) Categorical — labels, no arithmetic. Best: mode / percents ("40% brown eyes"). (b) Numerical, discrete — counted in whole messages. Best: mean or median ("average 22 texts"). (c) Numerical, continuous — measured ever-finer. Best: mean or median ("median 14.2 seconds" if a few slow runners skew it).

Worked Example 2 — Spot the mirage. A report says "average satisfaction rose 7%, from 2.10 to 2.25." The raw data: last year 60 Very / 100 Somewhat / 40 Not at all; this year 80 Very / 90 Somewhat / 30 Not at all (200 students each year). Is the 7% claim honest?

Solution. The arithmetic checks: (80×3 + 90×2 + 30×1) ÷ 200 = 450 ÷ 200 = 2.25; last year (180 + 200 + 40) ÷ 200 = 2.10; (2.25 − 2.10) ÷ 2.10 ≈ 7.1%. But satisfaction words were never numbers — "2.25" is a mirage. The honest report: "Very satisfied" went from 60 to 80 students, 30% → 40%. Same direction, but now it counts humans, not hallucinations.

DECIDE — THE VERDICT

Which survey questions do you approve?

Back to the question that opened this section — now with the math to settle it.

Solution. **Approve Q1** (categorical): report the mode — Pizza, 90 of 200 (**45%**). The principal's move: expand the pizza line, not the salad bar. **Approve Q2** (numerical, continuous): report the median, 7 minutes, not the mean of 8.5 — the mean is inflated by a few 25-minute waits, and those outliers are a separate problem to investigate (see §5.4). **Reject the "satisfaction up 7%" headline** (categorical words force-fed into a mean): report instead that "Very satisfied" rose from 30% to 40% — ten percentage points of real students.

The decision: hand the principal two actions (expand pizza, hunt the long waits) and one corrected headline. The survey was fine; the *analysis* needed a data-type check. From here on, every average in this chapter gets interrogated the same way: average of WHAT?

AI AI Learning Lab 5.1

AI Level 1 — Understand.

Explain categorical vs. numerical data (discrete vs. continuous) as if I'm 15, using the lunch-line survey: Pizza/Hot/Salad/Sandwich vs. minutes waited. Then quiz me one at a time on 8 variables (blood type, siblings, height, zip code, goals scored, temperature, satisfaction rating, bus number) — I classify each, you check before the next.

AI Level 2 — Apply.

Give me 10 variables to classify as categorical, discrete numerical, or continuous numerical — include two traps (numbers that are secretly labels). Then for each, make me name the best summary (mode? median? mean?) and why. Hide the answers; check my work step by step.

AI Level 3 — Matter.

The survey decision showed that coding words as numbers (Very = 3 ...) produces a "2.25 satisfaction" that means nothing — yet reports do this constantly. Discuss with me: WHY is a fake-precise number like "up 7%" so persuasive, and who benefits when satisfaction, happiness, or "average ratings" get averaged — think about product reviews, school ratings, or polls? What are the implications when nobody asks "average of WHAT?" End by asking me for one averaged-words claim in MY life I should re-check.

Level 4 — Decide. *Your friend ran a class survey — which of her "averages" do you trust?*

> **Example decision problem.** 30 students answered two questions. Favorite subject: Math 12, Science 10, English 8. Books read this month: four students read 0, six read 1, eight read 2, seven read 3, three read 4, two read 5. Your friend codes Math = 1, Science = 2, English = 3 and reports "average favorite subject = 1.87" and "average books read = 2.17." Which average is legitimate, and what should she report instead for the other?

AI Level 4 — Decide.

My friend surveyed 30 students: favorite subject (Math 12, Science 10, English 8) and books read (0:4, 1:6, 2:8, 3:7, 4:3, 5:2). She reports "average favorite subject = 1.87" and "average books = 2.17." Act as my decision coach: make me classify each variable's data type myself, make me recompute both numbers, then ask me which one survives the "average of WHAT?" test and what she should report instead — don't decide for me.

Check your verdict: Books are numerical discrete — total 65 ÷ 30 ≈ 2.17, legitimate (median 2 confirms it). Favorite subject is categorical — (12×1 + 10×2 + 8×3) ÷ 30 ≈ 1.87 is a mirage: "1.87" isn't a subject. Report the mode instead: Math, 40%. Trust the books average; delete the other one.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.1.

Ask your AI: "Write a short hospital report on 200 blood donors (O 92, A 70, B 28, AB 10) that codes the types as 1–4 and reports a 'mean blood type.'" Then do the detective work: explain in one paragraph exactly why the mean is meaningless (what breaks — the arithmetic or the meaning?), compute the correct summary yourself, and rewrite the report's headline honestly. AI chatbots will average anything you hand them — in this chapter, *you* decide what deserves averaging.

⚑ PROJECT MILESTONE

M1: Write your two questions

The project starts now (full brief: §5.10). Draft TWO *numerical* survey questions about something YOU care about — sports, music, sleep, games, food — that might be *related* to each other (you'll hunt for correlation in M6: gaming hours vs. sleep hours, practice time vs. free-throw percent). ⚠️ Get your teacher's approval before collecting a single response. A bad question poisons every number after it.

Practice Problems 5.1

1. Classify each as categorical, discrete numerical, or continuous numerical: (a) eye color, (b) number of siblings, (c) 100m sprint time, (d) zip code, (e) noon temperature, (f) goals scored this season.
2. A coach lists jersey numbers 4, 11, 23, 23, 32 and reports "mean jersey 18.6." What is wrong, and what should she report instead?
3. Fifty customers rate a product: 30 give 5 stars, 10 give 4, 5 give 3, 3 give 2, 2 give 1. Compute the "mean rating." Then explain one way this number misleads and one way it is still useful.
4. Give an example of a data set where the mean is misleading but the median is honest. Classify your data's type and explain why the mean fails.
5. You are surveying "How do students get to school?" Write one categorical, one discrete numerical, and one continuous numerical question on this topic.

5.2 Descriptive Statistics: Mean, Median, Mode

THE DECISION

Which café job do you take?

Two cafés near school are hiring for 20 hours a week:

- **Café A:** $11/hour, flat. That's $220 every week, guaranteed.
- **Café B:** $9/hour plus tips. Their flyer brags: *"Our crew averages $240 a week!"* A friend who works there shares her last 10 weekly paychecks (base + tips): $200, $200, $205, $210, $210, $220, $225, $230, $240, $460.

Which job do you take? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Add up the 10 paychecks and divide by 10. Is the flyer's "$240 average" telling the truth?
- **Hint 2:** Sort the paychecks smallest to largest. What is the middle value — the paycheck a *typical* week looks like?
- **Hint 3:** Which paycheck amount shows up most often? Now ask: which of your three answers best describes "a normal week" — and is $220 guaranteed better or worse than that?

The Mean

The **mean** is the familiar "average": add everything up, divide by how many there are.

Mean = (sum of values) ÷ (number of values)

The 10 paychecks sum to $2,400; the mean is 2400 ÷ 10 = **$240**. The flyer is telling the truth — and still misleading you, as you'll see.

STORY

The Test Average.

Your class average on the quiz is 82%. But

one student scored 100 and another scored 41 — everyone else sits near

1. The mean 82% describes *nobody in particular*: it is the class's total points redistributed equally, a fiction of perfect fairness. *When a teacher says "the average is 82," who is that number actually about?*

The Median

The **median** is the middle value when data is sorted. Half the values sit above it, half below. For an even number of values, average the two middle ones.

Sorted paychecks: 200, 200, 205, 210, 210, 220, 225, 230, 240, 460. The two middle values are 210 and 220, so the median is (210 + 220) ÷ 2 = **$215**. A typical week at Café B pays $215 — not $240.

STORY

The House Prices.

Nine homes on a street sell for $310k,

$320k, $330k, $340k, $350k, $360k, $370k, $380k — and one mansion for $2.4 million. The mean price is about $573k; the median is $350k. A realtor quoting "average $573k" is describing the mansion's neighborhood, not yours. *If you were buying, which number would you budget around?*

The Mode

The **mode** is the value that appears most often. The paychecks are **bimodal**: **$200 and $210** are both modes — each appears twice, and every other amount appears only once. The modes answer "what's the most common experience?" — useful for sizes, votes, and anything where popularity matters.

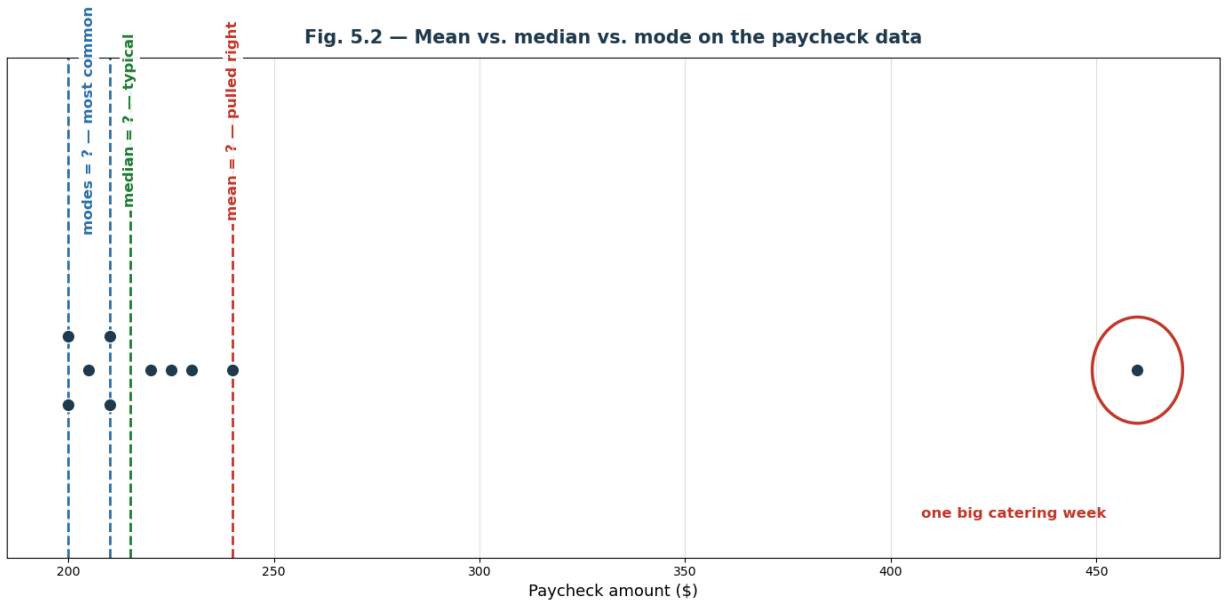
STORY

The Shoe Store.

A shoe store doesn't stock the *mean* shoe

size (8.63 — nobody wears that). It stocks mostly size 9, the mode, because the mode tells you what the most customers actually need. *Where else is "most common" more useful than "average" — pizza toppings? Bus arrival times?*

Fig. 5.2



Mean vs. median vs. mode on the paycheck data.

Worked Example 1 — Mean, median, mode with an outlier. Find the mean, median, and mode of: 3, 1, 7, 6, 8, 41, 2, 6, 6, 7, 9.

Solution. Sort first: 1, 2, 3, 6, 6, 6, 7, 7, 8, 9, 41. Sum = 96; mean = 96 ÷ 11 ≈ **8.73**. Median (6th of 11) = **6**. Mode (most frequent) = **6**. The single value 41 yanks the mean almost 3 points above where most of the data lives — the median and mode stay put.

Worked Example 2 — Reading a rent ad. Nine apartment listings: $900, $950, $1000, $1050, $1100, $1150, $1200, $1250, $2200. The ad says "average rent $1,200!" Check it.

Solution. Sum = $10,800; mean = 10800 ÷ 9 = **$1,200** — technically true. Median (5th of 9) = **$1,100**. The $2,200 penthouse inflates the mean; a typical renter pays $1,100. Budget around the median, not the ad.

DECIDE — THE VERDICT

Which café job do you take?

Back to the question that opened this section — now with the math to settle it.

Solution. Mean = **$240** (the flyer's number — true). Median = **$215** (the typical week). Mode = **$200** (the most common week). The $460 catering week is a lottery ticket disguised as a paycheck: it lifts the mean $25 above reality.

The decision: take Café A. A guaranteed $220 beats a typical $215 — and it beats it *every single week*, with no catering-week lottery required. Whenever someone advertises an average, ask for the median: the mean tells you what the advertiser wants you to hear.

AI AI Learning Lab 5.2

AI Level 1 — Understand.

Explain mean, median, and mode as if I'm 15, using the paycheck example: ten paychecks where one is $460. Show me how each of the three is computed, then give me one quick check question per measure and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two computing mean/median/mode from raw data, one where I must sort first, one identifying which measure a store owner should use (and why), and one spotting how an outlier moves the mean. Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The café decision showed that a true average ($240) can still mislead when one $460 week pulls it $25 above the typical $215 week. Discuss with me: WHY does quoting only the mean benefit the person quoting it — think about job ads, rent listings, or colleges advertising graduate salaries? What are the implications when medians stay hidden — who benefits, and who signs a contract they wouldn't have signed? End by asking me for one "average" claim in MY life I should re-check with a median.

Level 4 — Decide. *Your cousin is apartment hunting — what do you tell her?*

> **Example decision problem.** Nine listings near campus: $900, $950, $1000, $1050, $1100, $1150, $1200, $1250, $2200. The ad says "average rent $1,200!" Your cousin budgeted $1,200/month. Should she feel safe?

AI Level 4 — Decide.

My cousin budgeted $1,200/month for rent based on an ad saying "average rent $1,200!" The nine listings are 900, 950, 1000, 1050, 1100, 1150, 1200, 1250, 2200. Act as my decision coach: make me compute the mean and the median myself, then ask me which one describes her likely rent and whether her budget is safe — don't decide for me.

Check your verdict: Mean = 10800 ÷ 9 = $1,200 (the ad is truthful). Median = $1,100 — her typical rent. She's safe: the median sits $100 under budget. But warn her: the $2,200 penthouse is doing all the inflating, and "average" will keep lying to her in every city she searches.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.2.

Ask your AI: "Invent a data set of 12 salaries where the mean is $75,000 but the median is under $50,000, with an answer key showing all three measures." Then do the detective work: compute all three measures yourself and verify every number. If the AI's mean doesn't match its own data, write a one-paragraph explanation of how it fooled itself. AI chatbots are fluent at averages and sloppy at arithmetic — in this chapter, *you* are the calculator.

[PROJECT CHECK-IN: collect your data.] No milestone deliverable in this section — this is collection time. With your M1 questions approved, gather at least n = 20 responses (the whole class is better; §5.10 explains why 20 is the floor). Your data fuels every milestone from here on.

Practice Problems 5.2

1. Find the mean, median, and mode of: 4, 8, 15, 16, 23, 42, 8.
2. A data set has values 5, 5, 5, 5, 100. Which measure of center best describes a "typical" value? Explain in one sentence.
3. Fifteen students' quiz scores: twelve scored between 78 and 86, two scored 95, one scored 30. Without computing, will the mean be above or below the median? Why?
4. The mean of 6 numbers is 20. A seventh number, 62, is added. What is the new mean?
5. A basketball team advertises "our players average 6'4"!" The roster: five players 6'1", three players 6'3", one player 7'2". Compute the median height. Is the ad misleading?

5.3 Presenting Data: Frequency Tables & Dot Plots

THE DECISION

Who gets the last roster spot?

You're the student coach, and two players are trying out for the final roster spot. Their points over the last 12 games:

- **Player X:** 8, 10, 12, 9, 11, 10, 12, 8, 11, 10, 9, 10
- **Player Y:** 2, 4, 6, 8, 10, 12, 14, 16, 10, 12, 14, 12

Both average exactly 10 points a game. You can only keep one.

Who do you pick? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Verify the averages: add each player's 12 games and divide by 12. (Both should be 10 — the numbers alone can't decide this.)
- **Hint 2:** For each player, tally how many times each score occurred — a frequency table. Where do X's scores cluster? Where do Y's?
- **Hint 3:** Sketch both as dot plots (dots stacked above each score). Now ask: in a must-win final, do you want the player who *always* scores 10, or the one who might score 16 — or 2?

Frequency Tables

A **frequency table** counts how many times each value occurs. Raw data is noise; a frequency table is the same data, organized.

Player X's 12 games become:

Score	Frequency	Score × Frequency
8	2	16
9	2	18
10	4	40
11	2	22
12	2	24

Score	Frequency	Score × Frequency
Total	**12**	**120**

Player Y's:

Score	Frequency	Score × Frequency
2	1	2
4	1	4
6	1	6
8	1	8
10	2	20
12	3	36
14	2	28
16	1	16
Total	**12**	**120**

Same average — utterly different players. The table makes the difference visible in seconds. The Total row does double duty: it checks your tally (12 games each) and hands you the mean — 120 ÷ 12 = 10 for both players.

STORY

The Texting Survey.

You survey 20 classmates: "On a scale of

1–10, how often do you text?" The raw answers are a jumble — 7, 3, 9, 7, 5… A frequency table turns the jumble into a picture: most answers pile at 7–9, almost nobody at 1–2. *The raw data already contained that pattern — so why couldn't you see it before the table?*

Dot Plots

A **dot plot** stacks one dot per occurrence above each value on a number line. It is a frequency table you can see: clusters, gaps, and lonely dots jump out.

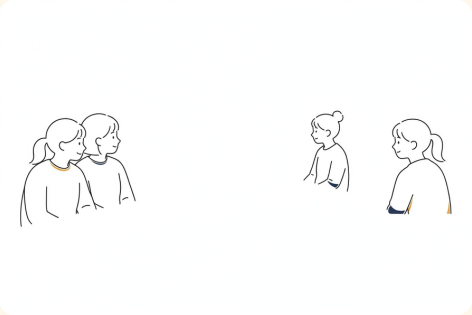
Player X's dot plot is a tight tower around 10. Player Y's is a long, flat spread from 2 to 16. No arithmetic needed — your eyes do the work.

STORY

The Quiz Scores.

Two classes both average 80% on the quiz.

Class A's dot plot is a single clump at 78–82. Class B's is two clumps: half near 95, half near 65. Same average, two different classrooms — one where everyone "gets it," one split in half. *If you were the teacher, would the same lesson plan work for both classes?*



Reading Shape: Center, Spread, and Shape

Every distribution has three features worth naming:

- **Center:** where the data piles up (mean, median — §5.2).
- **Spread:** how far it stretches (range and beyond — §5.6).
- **Shape:** symmetric (mirror halves), **skewed** (a tail dragging one way — like the $460 paycheck), or **bimodal** (two humps, like Class B).

STORY

The Busy Café.

The café's hourly customer counts form a

bimodal dot plot: a hump at 8 a.m., a valley at 2 p.m., a hump at 6 p.m. One "average hour" describes neither rush. The shape *is* the staffing plan: schedule around the humps, not the mean. *What bimodal patterns run your life — sleep? App usage?*

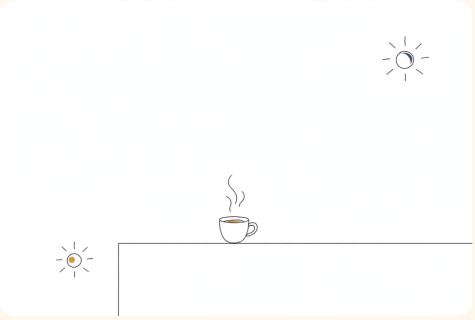
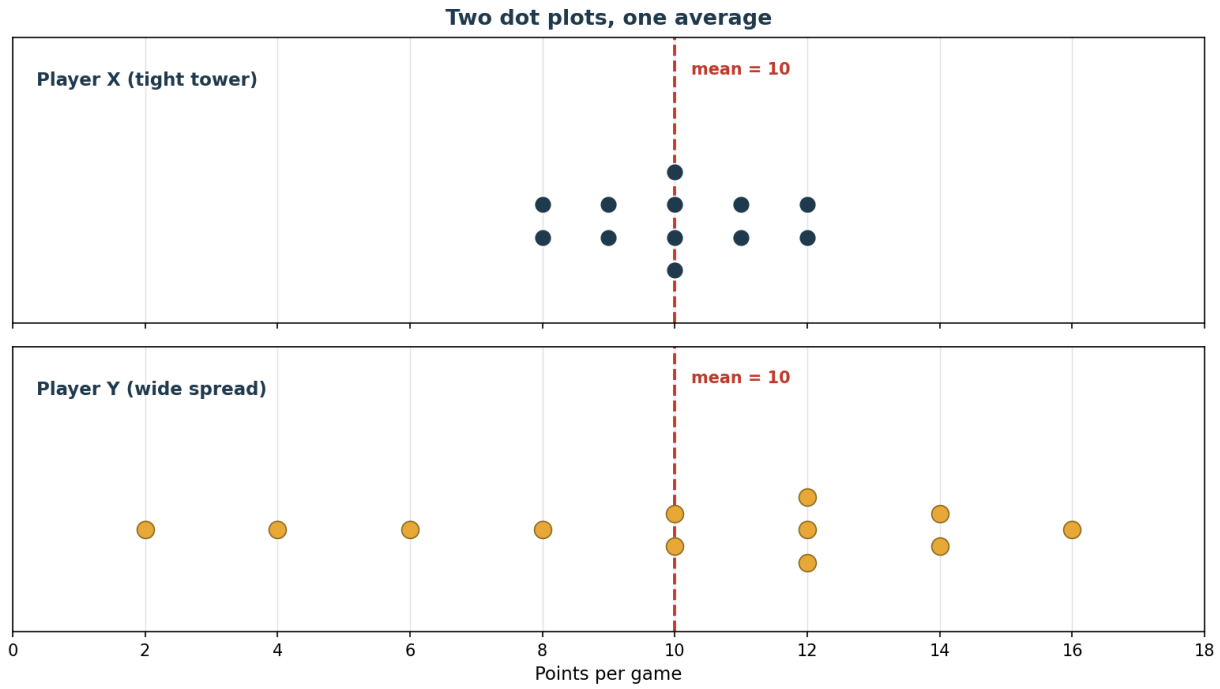


Fig. 5.3



Same average, different players — shape is information the mean throws away.

Worked Example 1 — From raw data to frequency table. Twenty texting-survey answers: 7, 3, 9, 7, 5, 8, 7, 9, 4, 7, 6, 8, 7, 5, 9, 7, 8, 6, 7, 9. Build the frequency table and find the mode.

Solution. Tally each value: 3→1, 4→1, 5→2, 6→2, 7→7, 8→3, 9→4. Check: 1+1+2+2+7+3+4 = 20 ✓. Mode = **7**.

Worked Example 2 — Mean from a frequency table. Using the table above, find the mean without re-listing all 20 values.

Solution. Multiply each value by its frequency and add: (3×1) + (4×1) + (5×2) + (6×2) + (7×7) + (8×3) + (9×4) = 3 + 4 + 10 + 12 + 49 + 24 + 36 = 138. Mean = 138 ÷ 20 = **6.9**.

DECIDE — THE VERDICT

Who gets the last roster spot?

Back to the question that opened this section — now with the math to settle it.

Solution. Both average 10 — the mean is useless here. The dot plots decide: Player X *never* leaves the 8–12 band (consistent, reliable); Player Y ranges from 2 to 16 (streaky — brilliant or invisible).

The decision: keep Player X. In a must-win final you don't need a player who *might* score 16; you need one who *won't* score 2. The average promised "both score 10" — the shape revealed that only one of them actually does.

AI AI Learning Lab 5.3

AI Level 1 — Understand.

Explain frequency tables and dot plots as if I'm 15, using the two basketball players who both average 10. Show me how to build each one, then give me one quick check question per display and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two building frequency tables from raw data, one computing a mean from a frequency table, one sketching a dot plot from a description, and one describing center/spread/shape of a described distribution. Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The roster decision showed that two players with identical 10-point averages are completely different once you see the shape — one reliable, one streaky. Discuss with me: WHY does reducing people to a single average do real harm — think about "average student" labels, or hiring by GPA alone? What are the implications when shape is ignored — who gets misjudged, and what better decision does the full distribution allow? End by asking me for one "average person" claim in MY life I should re-check as a distribution.

Level 4 — Decide. *Two tutors both promise "average improvement of 15 points" — who do you hire?*

Example decision problem. Tutor P's last 10 students improved by: 13, 14, 15, 15, 15, 15, 16, 16, 14, 17 (mean 15). Tutor Q's last 10: 2, 5, 8, 12, 15, 18, 22, 25, 20, 23 (mean 15). Same price. Who do you hire?

AI Level 4 — Decide.

Two tutors both advertise "average improvement of 15 points." Tutor P's last 10 students: 13, 14, 15, 15, 15, 15, 16, 16, 14, 17. Tutor Q's: 2, 5, 8, 12, 15, 18, 22, 25, 20, 23. Same price. Act as my decision coach: make me describe each tutor's distribution shape myself (not just the mean), then ask me which I'd hire and why — don't decide for me.

Check your verdict: Both means = 150 ÷ 10 = 15 ✓. P is a tight tower (13–17) — reliable. Q sprawls from 2 to 25 — a lottery. Hire P unless you specifically need a miracle: Q's "15 average" is built from students who gained almost nothing and students who gained enormously.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.3.

Ask your AI: "Here are 15 quiz scores: [list]. Describe the distribution's center, spread, and shape." Then do the detective work: build the dot plot yourself and check every claim — did it miss a gap? Call a symmetric distribution skewed? If the AI misread the shape even once, write a one-paragraph explanation of what it failed to see. AI chatbots are fluent at describing and sloppy at looking — in this chapter, *you* are the eyes.

⚑ PROJECT MILESTONE

M2: First look at your data

Build a frequency table and a dot plot from your numerical data — by hand first, then check with a spreadsheet. Deliverable: one page with both, plus one sentence — "The shape of my data is ___." (Full brief: §5.10.)

Practice Problems 5.3

1. Build a frequency table for: 2, 5, 5, 3, 2, 5, 4, 3, 2, 5.
2. From your table in #1, find the mean using (value × frequency).
3. Two delivery routes both average 25 minutes. Route A's times cluster at 24–26 minutes; Route B's range from 10 to 45 minutes. Which route do you take to catch a flight? Why?

4. Describe the shape: test scores pile at 90–95 with a tail stretching down to 40. (Symmetric, skewed left, or skewed right?)
5. A dot plot of household sizes shows single dots at 1 and 7, and a tall stack at 3–4. What is the mode, and what does it tell you?

5.4 Interpreting Data: Box-and-Whisker Plots & Outliers

THE DECISION

Do you recommend the new allergy medicine?

A new allergy medicine reports trial results — days until symptoms clear for 12 patients: 3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 8, 30.

The company's brochure says: *"Average recovery: 7.5 days!"* Eleven patients recovered in 8 days or fewer. One patient took 30 days — a severe reaction the brochure calls "an isolated case." Your little brother has the same allergy.

Do you recommend the medicine to your family? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Sort the data and find the median (the middle). Then find the median of the lower half and the median of the upper half.
- **Hint 2:** Subtract: (upper-half median) − (lower-half median). This is the IQR — the range of the middle 50% of patients.
- **Hint 3:** Compute (upper-half median) + 1.5 × IQR. Is 30 above that line? If so, statistics officially calls it an outlier — now ask: does "outlier" mean "safe to ignore" when it's your brother?

Quartiles & the Box-and-Whisker Plot

Quartiles split sorted data into quarters. The median (Q2) splits it in half; **Q1** is the median of the lower half; **Q3** is the median of the upper half.

A **box-and-whisker plot** draws this as a picture: a box from Q1 to Q3 (the middle 50% of the data), a line at the median, and "whiskers" stretching to the smallest and largest *non-outlier* values.

For the medicine data (sorted: 3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 8, 30): median = (5+6)/2 = **5.5**; Q1 = (4+5)/2 = **4.5**; Q3 = (7+7)/2 = **7**. The box runs 4.5 → 7: half the patients recovered in that window.

STORY

The Marathon Pack.

In a marathon, the lead pack finishes

within minutes of each other while stragglers trail by an hour. A box-and-whisker plot of finish times shows a tight box (the pack) and a long whisker (the stragglers). The mean finish time describes neither group — but the box shows you exactly where *most* runners live. *If you were pacing a first-time runner, would you aim at the mean or the box?*

The 1.5×IQR Rule

The **interquartile range (IQR)** = Q3 − Q1 — the spread of the middle 50%. Statisticians flag **outliers** with fences:

- Lower fence = Q1 − 1.5 × IQR
- Upper fence = Q3 + 1.5 × IQR

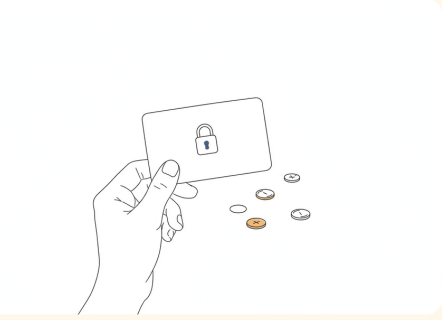
Any value outside the fences is an outlier. For the medicine: IQR = 7 − 4.5 = 2.5; upper fence = 7 + 1.5(2.5) = **10.75**. The 30-day recovery is far outside — officially an outlier.

STORY

The Fraud Alert.

Your bank's fraud system watches your

spending's IQR: most weeks you spend $80–$140 (IQR = $60), so a $900 charge blows past Q3 + 1.5×IQR and the card freezes. The rule doesn't *know* it's fraud — it knows it's not-you. *When has an "outlier" in your life turned out to be the most important data point?*



Outliers: Signal or Noise?

An outlier is a warning, not a verdict. Sometimes it's a measurement error (delete it). Sometimes it's the discovery (study it). The 30-day patient could be a data-entry typo — or the one case that reveals a rare, dangerous reaction. Statistics can *flag* the 30; only investigation can *explain* it.

STORY

The Ozone Hole.

In the 1980s, NASA's software automatically discarded "outlier" ozone readings over Antarctica — they looked like errors. They weren't errors; they were the ozone hole. The most important environmental discovery of the decade was hiding in the data someone threw away. *What "errors" in your world might be discoveries?*

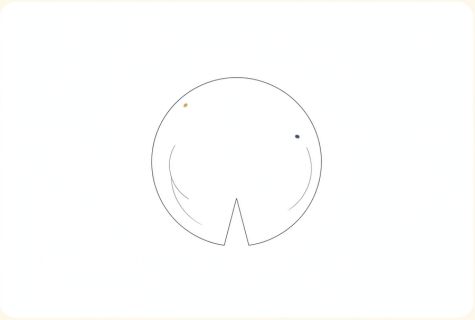
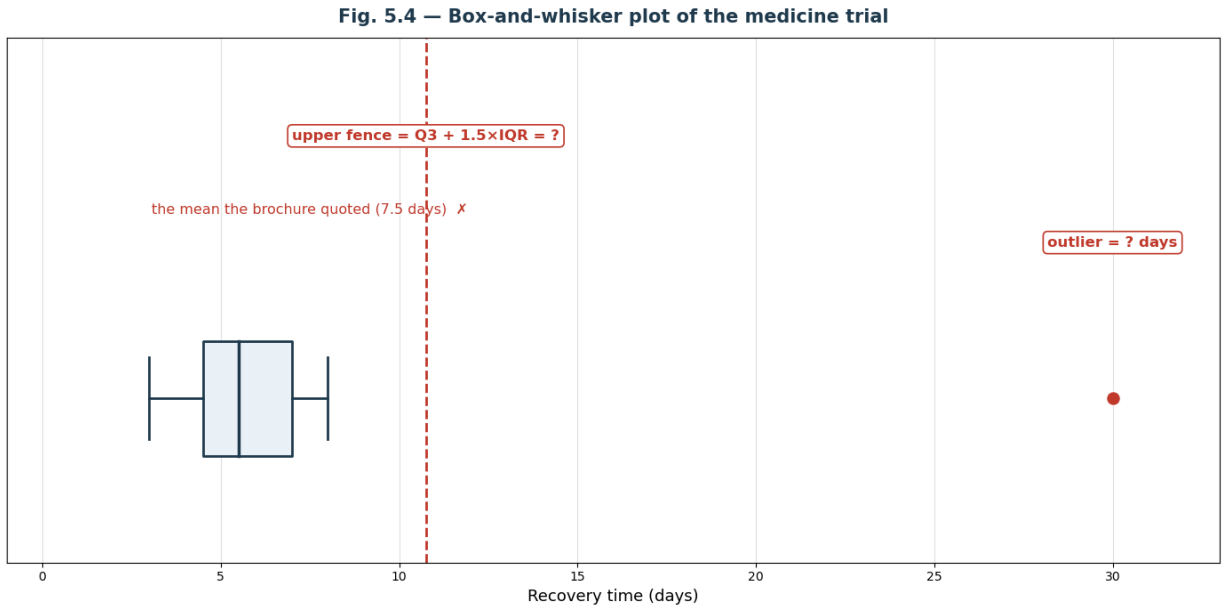


Fig. 5.4



The box shows where most patients are; the dot shows what the average hid.

Worked Example 1 — Five-number summary and outliers. Data: 3, 1, 7, 5, 8, 24, 2, 6, 0, 7, 10. Find the five-number summary (min, Q1, median, Q3, max) and check for outliers.

Solution. Sort: 0, 1, 2, 3, 5, 6, 7, 7, 8, 10, 24. Min = **0**, max = **24**. Median (6th of 11) = **6**. Q1 (median of first five) = **2**. Q3 (median of last five) = **8**. IQR = 8 − 2 = 6. Fences: 2 − 9 = −7; 8 + 9 = 17. **24 > 17 → outlier.** Five-number summary: 0, 2, 6, 8, 24.

Worked Example 2 — The mean vs. the box. For the medicine data, compare the mean (7.5) with the median (5.5).

Solution. Mean = 90 ÷ 12 = **7.5**; median = **5.5**. The single 30-day outlier drags the mean 2 full days above the typical patient — and the brochure quoted the mean. The box (4.5 → 7) tells the honest story: most patients recover in under a week.

DECIDE — THE VERDICT

Do you recommend the new allergy medicine?

Back to the question that opened this section — now with the math to settle it.

Solution. The 1.5×IQR rule confirms it: upper fence 10.75, and 30 is an outlier. The brochure's "7.5 days" is the outlier talking, not the medicine: the median patient recovers in 5.5 days, and 11 of 12 patients recover within 8.

The decision: don't recommend it yet — investigate the 30 first. If that patient had a rare reaction, your brother could be next; if it was a recording error, the medicine is genuinely good (median 5.5 days). An outlier is a question mark, and you never prescribe a question mark to family. "Isolated case" is a conclusion — statistics only earned you a question.

AI AI Learning Lab 5.4

AI Level 1 — Understand.

Explain quartiles, the IQR, and the box-and-whisker plot as if I'm 15, using the medicine data (3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 8, 30). Show me how to find Q1, the median, and Q3, then give me one quick check question per idea and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two finding five-number summaries, two applying the 1.5×IQR rule to flag outliers, and one deciding whether an outlier should be investigated or deleted (with reasoning). Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The medicine decision showed that one 30-day outlier moved the quoted average from 5.5 to 7.5 days — and the brochure quoted the 7.5. Discuss with me: WHY is the choice between "investigate" and "delete" an outlier a moral decision, not just a math one — think about drug trials, airplane safety records, or police stop data? What are the implications when outliers get quietly deleted — who benefits, and what did the ozone hole teach us? End by asking me for one data set in MY life whose outliers deserve investigation.

Level 4 — Decide. *A used car is listed at $6,500 — deal or trap?*

> **Example decision problem.** Twelve used Civics (prices in thousands): 11.5, 12.0, 12.2, 12.5, 12.8, 13.0, 13.2, 13.5, 13.8, 14.0, 14.2, 6.5. The $6,500 car has a salvage title. Is it a bargain?

AI Level 4 — Decide.

I'm eyeing a used Civic listed at $6,500. Eleven similar Civics list at 11.5, 12.0, 12.2, 12.5, 12.8, 13.0, 13.2, 13.5, 13.8, 14.0, 14.2 (thousands). The cheap one has a salvage title. Act as my decision coach: make me find the quartiles and the lower fence myself, then ask me whether $6,500 is a deal or a warning — don't decide for me.

Check your verdict: Sorted with the cheap car: 6.5, 11.5, ..., 14.2. Q1 = 12.1, Q3 = 13.65, IQR = 1.55, lower fence = 12.1 − 2.325 ≈ 9.78. $6,500 < $9,780 → flagged as a low outlier. Median = $12,900 — budget around that. The "deal" is the market telling you something is wrong; outliers have reasons, and a salvage title is this one's.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.4.

Ask your AI: "Here is a data set with a subtle outlier: [use the site's warm-up data 3, 1, 7, 5, 8, 24, 2, 6, 0, 7, 10]. Find the outliers using the 1.5×IQR rule and explain whether each should be investigated or deleted." Then verify its quartiles yourself — many AIs split even-sized halves incorrectly. If it got Q1 or Q3 wrong, write a one-paragraph explanation of exactly where its split went wrong. AI chatbots are fluent at rules and sloppy at sorting — in this chapter, *you* hold the ruler.

⚑ PROJECT MILESTONE

M3: The five-number summary

Draw a box-and-whisker plot of your numerical data; find Q1, median, Q3, IQR; hunt outliers with the 1.5×IQR rule. Deliverable: the plot plus one paragraph — is your outlier a mistake, a superstar, or a story? (Full brief: §5.10.)

Practice Problems 5.4

1. Data: 12, 15, 18, 20, 22, 25, 30, 60. Find Q1, the median, and Q3.
2. For #1, compute the IQR and the fences. Is 60 an outlier?
3. A box plot shows: min 5, Q1 10, median 15, Q3 25, max 40. What percent of the data lies between 10 and 25? Between 5 and 40 (excluding outliers)?
4. A basketball player's game scores have median 18 and mean 14. What does that tell you about the shape — and about one or two terrible games?
5. Your town's household incomes: median $58k, mean $96k. A candidate says "average income is $96k, everyone's doing great!" Write two sentences rebutting with the median.

5.5 Grouped Data: Discrete, Continuous & Histograms

THE DECISION

How early do you leave for the bus?

You recorded the city bus's lateness 40 mornings (minutes past schedule): grouped into intervals —

Lateness	0–5	5–10	10–15	15–20	20–25
Mornings	6	14	11	6	3

Your internship starts at 9:00, and being late twice means you're fired. The bus is *scheduled* to get you there at 8:50.

How early do you leave — do you trust the 8:50 bus? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** How many of the 40 mornings was the bus more than 15 minutes late? Divide by 40 — that's your risk per morning.
- **Hint 2:** The bus scheduled for 8:50 arrives by 9:00 only if it's less than 10 minutes late. How many mornings was it 10+ minutes late?
- **Hint 3:** Estimate the average lateness using interval midpoints (2.5, 7.5, 12.5, ...). Then ask: is the *average* what gets you fired, or the *bad mornings*?

Grouping Data: Discrete vs. Continuous

Discrete data takes separate, countable values (number of siblings, texts per day). **Continuous data** can take any value in a range (height, wait time, rainfall). With lots of data, we **group** values into class intervals — like the bus table's 0–5, 5–10 — and count frequencies per group. Grouping trades detail for clarity.

STORY

The Rainfall.

"Amount of rain" is continuous — 2.37 inches is perfectly possible. "Number of family members" is discrete — 2.37 people is not. A weather station groups rainfall into 0–1, 1–2, 2–3 inches because nobody needs 365 separate measurements to see the pattern. *Your age in years is discrete; your exact age is continuous — which one does the DMV want, and why?*

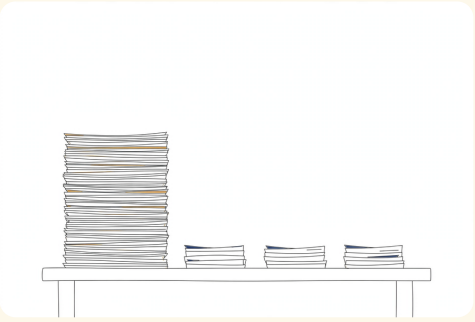
Histograms

A **histogram** is a bar chart of grouped data: bars touch (the data is continuous), and bar *area* shows frequency. The bus histogram rises to the 5–10 minute bar (the **modal class**) and trails right — most mornings are fine, some are disasters.

STORY

The Test Scores.

Your teacher posts a histogram of exam scores: a big bar at 80–90, small bars trailing to 40–50. You find your bar instantly — and so does everyone else. Nobody's exact score is shown, yet the class story is obvious. *What does a histogram reveal that a list of 30 scores hides — and what does it hide that the list reveals?*



Estimating the Mean from Grouped Data

Grouped data lost the exact values — but the **midpoint** of each interval stands in for them:

Estimated mean = Σ(midpoint × frequency) ÷ (total frequency)

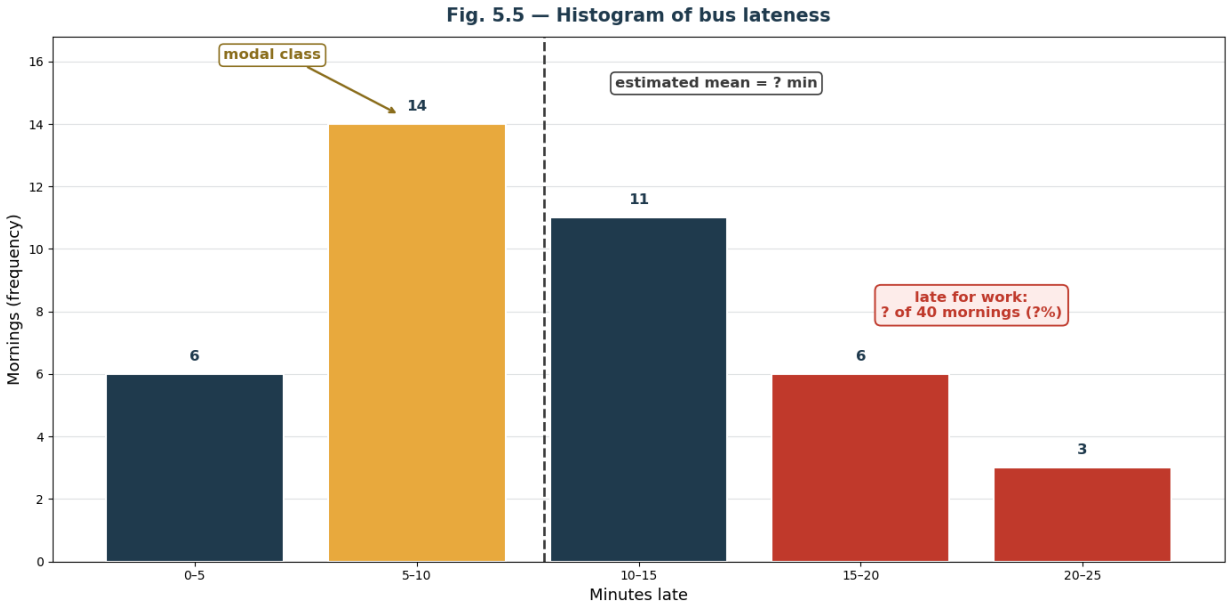
Bus: (2.5×6) + (7.5×14) + (12.5×11) + (17.5×6) + (22.5×3) = 15 + 105 + 137.5 + 105 + 67.5 = 430. Estimated mean = 430 ÷ 40 = **10.75 minutes** late. It's an estimate — the real mean could differ slightly — but it's the best the grouped data allows.

STORY

The Census.

The census reports incomes in $10k brackets, not exact dollars — your privacy survives, and economists still estimate the mean. Every grouped mean is a deal: precision traded for practicality. *When is that trade worth it — and when would you demand the raw data?*

Fig. 5.5



The average is 10.75 minutes — but averages don't get you fired; red mornings do.

Worked Example 1 — Building the table. Twenty students' siblings: 0,1,2,1,3,2,0,2,1,4,2,1,0,3,2,1,2,0,1,2. Group and find the modal class.

Solution. Discrete data — each value is its own group: 0→4, 1→6, 2→7, 3→2, 4→1. Check: 4+6+7+2+1 = 20 ✓. Modal class: **2 siblings**.

Worked Example 2 — Estimated mean. Heights (cm) of 25 students grouped: 150–160: 4, 160–170: 11, 170–180: 8, 180–190: 2. Estimate the mean height.

Solution. Midpoints 155, 165, 175, 185. (155×4) + (165×11) + (175×8) + (185×2) = 620 + 1815 + 1400 + 370 = 4205. Estimated mean = 4205 ÷ 25 = **168.2 cm**.

DECIDE — THE VERDICT

How early do you leave for the bus?

Back to the question that opened this section — now with the math to settle it.

Solution. Mornings later than 15 minutes: 6 + 3 = 9 of 40 = **22.5%** — nearly 1 in 4. Mornings 10+ minutes late (missing the 9:00 start on the 8:50 bus): the 10–15 group splits — conservatively, all of 10–15 plus worse: 11 + 6 + 3 = 20 of 40 = **50%.** Estimated mean lateness: 10.75 minutes.

The decision: don't trust the 8:50 bus — take the earlier one. A coin flip's chance of being late, with firing on the second offense, is not a commute; it's a countdown. The average (10.75 min) actually *understates* the danger: what gets you fired isn't the typical morning, it's the red 22.5%.

AI AI Learning Lab 5.5

AI Level 1 — Understand.

Explain discrete vs. continuous data, class intervals, and histograms as if I'm 15, using the bus lateness table. Show me how to estimate a mean from midpoints, then give me one quick check question per idea and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two classifying data as discrete or continuous, one building a grouped frequency table, one drawing/reading a histogram (describe it in words), and one estimating a mean from grouped data. Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The bus decision showed that the estimated mean (10.75 min) hid the real danger — the 22.5% of red mornings. Discuss with me: WHY does grouping change what we can see — think about who chooses the interval widths, and what a different grouping could hide? What are the implications when someone else picks the bins — income brackets, test-score bands, age groups — who benefits from coarse bins, and what would you demand to see? End by asking me for one grouped statistic in MY life whose bins I should question.

Level 4 — Decide. *Should the coffee shop hire a second barista?*

Example decision problem. Sixty morning customers' wait times (minutes): 0–2: 18, 2–4: 22, 4–6: 12, 6–8: 5, 8–10: 3. The owner hires a second barista if more than 10% of customers wait over 6 minutes. Hire or not?

AI Level 4 — Decide.

I manage a coffee shop. Sixty morning wait times: 0–2 min: 18 customers, 2–4: 22, 4–6: 12, 6–8: 5, 8–10: 3. My rule: hire a second barista if over 10% wait more than 6 minutes. Act as my decision coach: make me compute the percentage and the estimated mean wait myself, then ask me whether to hire — don't decide for me.

Check your verdict: Over 6 minutes: 5 + 3 = 8 of 60 ≈ 13.3% > 10% — hire. Estimated mean: (1×18 + 3×22 + 5×12 + 7×5 + 9×3) ÷ 60 = 206 ÷ 60 ≈ 3.43 min. Note the trap: the "fine" 3.4-minute average hides the 8 customers fuming past 6 minutes — the tail, not the mean, loses regulars.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.5.

Ask your AI: "Group these 30 commute times into a histogram and describe the shape: [invent 30 values between 5 and 45]." Then do the detective work: regroup the *same* data with different bin widths (5-minute vs. 15-minute bins) and compare the two histograms. If the shape story changes — one looks normal, the other skewed — write a one-paragraph explanation of how bin choice manufactured the difference. AI chatbots pick bins without thinking; in this chapter, *you* choose the bins.

⚑ PROJECT MILESTONE

M4: Group it

If your data are numerous or continuous, build a grouped frequency table with sensible class intervals and draw a histogram. Deliverable: table + histogram, plus one sentence on what the grouping reveals that the dot plot hid. (Full brief: §5.10.)

Practice Problems 5.5

1. Classify as discrete or continuous: (a) number of pets, (b) marathon finish time, (c) goals in a soccer game, (d) the weight of a backpack.
2. Data: 12, 15, 18, 22, 25, 29, 33. Group into 10–20, 20–30, 30–40 and give the frequency table.
3. From #2, estimate the mean using midpoints.

4. A histogram of commute times has its tallest bar at 20–30 minutes with bars shrinking on both sides. Describe the shape and estimate where the mean sits.
5. Why do histogram bars touch but bar-chart bars don't? Answer in one sentence.

5.6 Measures of Dispersion: Range, IQR, Standard Deviation

THE DECISION

Which fund gets your $1,000?

Your grandmother gives you $1,000 for your future — "invest it, don't spend it." Two index funds both advertise **"8% average annual return"**. Their last 6 yearly returns (%):

- **Fund Steady:** 8, 9, 7, 8, 9, 7
- **Fund Rocket:** 2, 14, −3, 18, 5, 12

Same 8% average. Same $1,000. Very different rides.

Which fund do you pick? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Verify both averages are really 8%. (Add each row, divide by 6.)
- **Hint 2:** For each fund, find the range: (biggest year) − (smallest year). Which fund's range would keep you up at night?
- **Hint 3:** For each year, compute (return − 8), square it, average the squares, and take the square root. That number — the standard deviation — is the typical year's distance from average. Which fund's "typical distance" is bigger, and what does that mean for your $1,000?

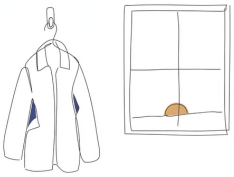
Range: The Quick Glance

The **range** = (maximum) − (minimum): the total stretch of the data. Fund Steady: 9 − 7 = **2**. Fund Rocket: 18 − (−3) = **21**. One number, instantly honest — but it only sees the two extremes and ignores everything between.

STORY

The Weather Report.

"Today's range: 48°F to 72°F" tells you to bring a jacket — one glance, full wardrobe decision. But the range can't tell you whether it was 60° all day with one cold dawn, or swinging wildly all afternoon. *When is the quick glance enough — and when do you need the whole story?*



Variance & Standard Deviation: The Full Story

Variance measures the *average squared distance* from the mean: square each deviation, average them. **Standard deviation (SD)** is the square root of the variance — back in the original units, the "typical distance from average."

Fund Steady: deviations 0, 1, −1, 0, 1, −1 → squares 0, 1, 1, 0, 1, 1 → variance = 4/6 ≈ 0.67 → SD ≈ **0.82**. A typical year lands within a point of 8%.

Fund Rocket: deviations −6, 6, −11, 10, −3, 4 → squares 36, 36, 121, 100, 9, 16 → variance = 318/6 = 53 → SD ≈ **7.28**. A typical year misses 8% by seven points — in either direction.

One convention note: we divide by n — the *population* formula. Some calculators and spreadsheets divide by $n - 1$ instead (the *sample* formula) — for Fund Steady that gives 0.89 instead of 0.82 — so know which one your tool uses.

STORY

The Two Free-Throw Shooters.

Two players both shoot 70% from the line. Player A: 7, 7, 7, 7, 7, 7, 7, 7, 7, 7 every ten attempts (SD = 0). Player B: 10, 4, 10, 4, 10, 4, 10, 4, 10, 4 (SD = 3). Same average — but with one second left and the game on the line, you want Player A at the stripe. *The average promised 70%. What did the standard deviation promise?*

Same Mean, Different Meaning

Two data sets can share a mean and tell opposite stories: (−2, −1, 0, 1, 2) and (−4, −2, 0, 2, 4) both average 0 — but the second spreads twice as wide (range 8 vs. 4; SD ≈ 2.83 vs. 1.41). **Dispersion is information the mean deletes.** Range gives the quick glance, IQR (§5.4) gives the middle 50%, and standard deviation gives the typical distance — three lenses on the same question: *how spread out is this?*

STORY

The Pill Bottles.

A pill-filling machine targets 100 mg per bottle. Machine X: SD 1 mg — nearly every bottle lands between 97 and 103 (about ±3 SD). Machine Y: SD 15 mg — some bottles 70, some 130. Both average 100 mg. One of them is a lawsuit. *The average passed inspection. Which number should the inspector have asked for?*

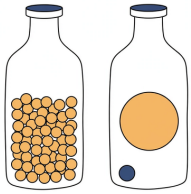
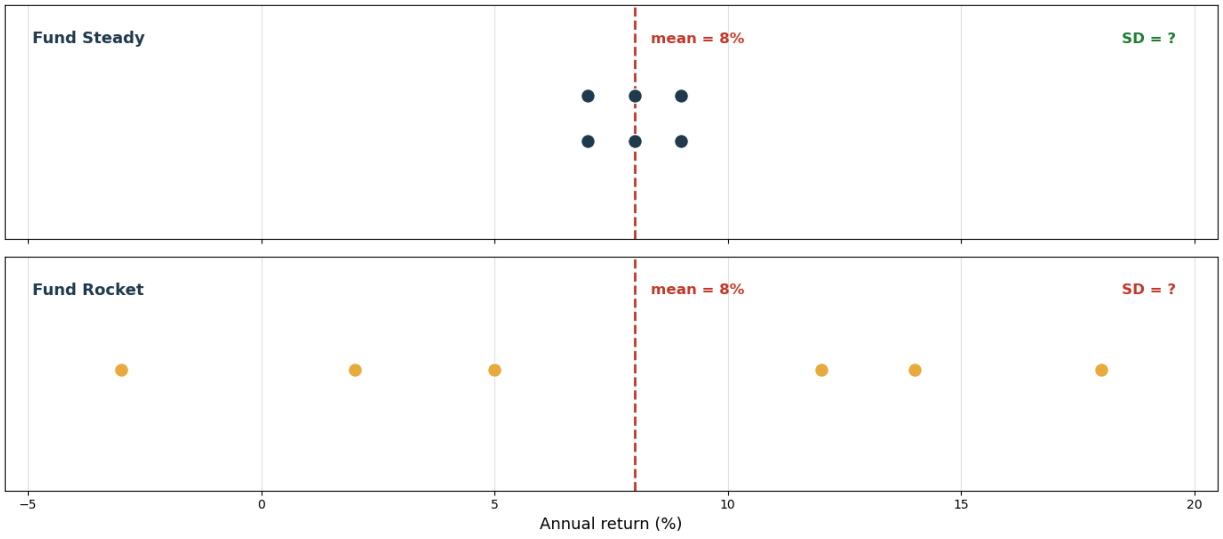


Fig. 5.6

Fig. 5.6 — Same mean, different spread



Identical averages, opposite risks — spread is the half of the story the mean never tells.

Worked Example 1 — Range and what it misses. Data set A: −2, −1, 0, 1, 2. Data set B: −4, −2, 0, 2, 4. Find each mean, range, and standard deviation.

Solution. Both means = **0**. Range A = 2 − (−2) = **4**; range B = 4 − (−4) = **8**. SD A: squared deviations 4, 1, 0, 1, 4 → 10/5 = 2 → SD = √2 ≈ **1.41**. SD B: 16, 4, 0, 4, 16 → 40/5 = 8 → SD = √8 ≈ **2.83**. Same center, double the spread — the mean knew nothing.

Worked Example 2 — Reading an SD. A coffee chain's drink wait times average 3 minutes with SD 0.5 minutes. A rival averages 3 minutes with SD 2.5 minutes. Which is better run?

Solution. Same mean — but the rival's typical wait misses 3 minutes by 2.5 (some 30-second waits, some 8-minute fumes), while the chain's stays near 3. **The chain is better run:** low SD means consistent, and consistency is what customers feel.

DECIDE — THE VERDICT

Which fund gets your $1,000?

Back to the question that opened this section — now with the math to settle it.

Solution. Both average 8% — the ads are identical and the reality isn't. Fund Steady: SD ≈ 0.82 — your $1,000 grows about 8% nearly every year. Fund Rocket: SD ≈ 7.28 — years of −3% and +18%; the average is 8% but you may never *experience* an 8% year.

The decision: it depends on whose $1,000 it is — and that's the lesson. For a 16-year-old with decades ahead, Rocket's wild ride might be worth it. For Grandma's gift that you can't afford to watch halve, take Steady. "8% average" was never an answer; it was a question — *8% with what spread?* Never invest in an average without asking for its standard deviation.

AI AI Learning Lab 5.6

AI Level 1 — Understand.

Explain range, variance, and standard deviation as if I'm 15, using the two funds that both average 8%. Show me the full SD computation for Fund Steady step by step, then give me one quick check question per measure and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two finding range, two computing SD step by step (small data sets), and one comparing two data sets with the same mean but different spreads. Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The fund decision showed that "8% average" describes two opposite investments — SD 0.8 vs. SD 7.3. Discuss with me: WHY does advertising only the mean serve the seller — think about investment ads, "average" salary quotes, or schools reporting average test scores? What are the implications when spread stays hidden — who takes risks they didn't choose, and what does the pill-bottle story say about who should be *required* to report dispersion? End by asking me for one "average" in MY life whose spread I should demand.

Level 4 — Decide. *Two job offers, same $50k average — which do you take?*

> **Example decision problem.** Offer A: $50k salary, guaranteed. Offer B: $30k base + commission; last 6 hires earned 35, 42, 48, 52, 58, 65 (thousands — mean 50). You have $1,800/month in fixed bills. Which offer?

AI Level 4 — Decide.

Two job offers, both "averaging $50k." Offer A: $50k salary guaranteed. Offer B: $30k base + commission; the last 6 hires earned 35, 42, 48, 52, 58, 65 (thousands). My fixed bills are $1,800/month. Act as my decision coach: make me compute Offer B's range and SD myself, then ask me which I'd take given my bills — don't decide for me.

Check your verdict: Offer B: mean = 300 ÷ 6 = $50k ✓; range = 65 − 35 = $30k; deviations −15, −8, −2, 2, 8, 15 → squares 225, 64, 4, 4, 64, 225 = 586 → variance ≈ 97.7 → SD ≈ $9.9k. Bills need $21.6k/year; Offer B's base alone ($30k) covers them — so the risk is survivable and the upside ($65k) is real. Take B if you can stomach a $35k year; take A if you can't. The SD ($9.9k) turned a gamble into a measurable risk.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.6.

Ask your AI: "Invent two data sets of 8 values with the same mean (50) but SDs of 2 and 15, and show the SD computations." Then verify both computations yourself, every squared deviation. If the AI's SDs don't match its own data — the most common failure — write a one-paragraph explanation of where its arithmetic broke (forgot to square? divided by the wrong n?). AI chatbots are fluent at formulas and sloppy at following them — in this chapter, *you* are the auditor.

⚑ PROJECT MILESTONE

Milestone

[PROJECT MILESTONE — M5: How spread out?] Compute the range, IQR, and standard deviation of your data. Deliverable: the three numbers plus one sentence — "The spread of my data means ___." (Full brief: §5.10.)

Practice Problems 5.6

1. Find the range of: 12, 5, 18, 9, 22, 5, 14.
2. Data: 10, 10, 10, 10, 10. What is the SD? Explain in one sentence what it means.
3. Two classes both average 75% on the final. Class A: SD 3. Class B: SD 15. Which class had more consistent scores, and what might Class B's histogram look like?
4. A data set has mean 100 and SD 15. A value of 145 is how many SDs above the mean? (This is its z-score.)
5. Your basketball team scores a mean of 80 points with SD 4; your rival averages 80 with SD 18. The championship is one game. Which team's fans should be more nervous, and why?

5.7 Misleading Statistics: How Graphs Lie

THE DECISION

Do you pay double for the "4× whiter" toothpaste?

A toothpaste ad shows a bar chart: their brand's bar towers **four times taller** than the competitor's. *"4× whiter teeth!"* it shouts. It costs twice as much. The fine print gives the lab scores (out of 100): their brand **98**, competitor **92**.

Do you pay double? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Compute the real difference: 98 − 92 = 6 points. As a percent of the competitor's 92, how big is that?
- **Hint 2:** Look at the chart's vertical axis. Does it start at 0 — or at 90? What happens to the *visual* difference when 90 points of every bar are chopped off?

- **Hint 3:** If the axis started at 0, how tall would the 98-bar look next to the 92-bar? Now ask: is "4× whiter" describing the teeth — or the chart?

The Truncated Axis

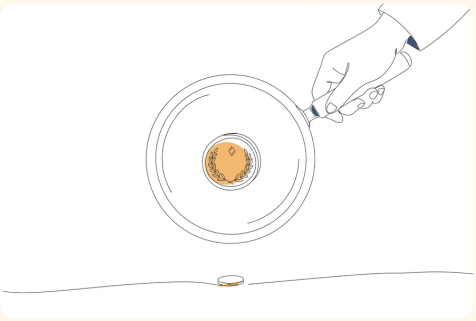
A **truncated axis** starts somewhere above zero, stretching small differences into dramatic ones. The toothpaste chart starts at 90: the 98-bar shows 8 units, the 92-bar shows 2 — visually 4× taller. Start at 0 and the bars are 98 vs. 92 — nearly identical. Same data, opposite impression. The lie isn't in the numbers; it's in the *frame*.

STORY

The "Soaring" Stock.

A company's ad shows its stock "soaring"

from $101 to $103 — the chart starts at $100, so the bar triples. The real gain: under 2%. The chart is technically accurate and completely dishonest — every pixel of drama was manufactured by the axis. *If the axis choice changes your feelings but not the facts, who should be allowed to choose the axis?*



Cherry-Picked Ranges

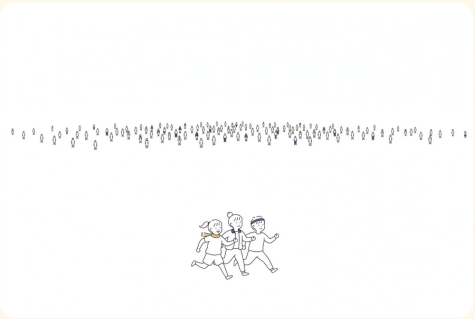
Cherry-picking means choosing the time window (or group) that tells your story. "Crime up 200%!" — from 1 incident to 3, in one cherry-picked month, while the yearly trend falls. "Our graduates earn $80k!" — surveyed only the employed ones, six months after graduation, in a boom year.

STORY

The Diet Ad.

"Lose 10 pounds in a month!" — the fine print:

results from the 3 dieters (of 200) who also ran daily and skipped dinner. The 197 others are the data; the 3 are the advertisement. *When someone shows you a spectacular result, what three questions should you ask about the other 197?*



Pictures That Lie

Even honest numbers can wear dishonest pictures: a **pictograph** where the "twice as big" icon is drawn twice as *tall* — making it four times the *area*. Or a 3D pie chart tilted so the front slice looks biggest. Or two lines on different vertical scales, "proving" they move together.

STORY

The Fundraising Thermometers.

Two charities both raised

$50k of a $100k goal. Charity A's thermometer runs $40k to $55k — nearly full! Charity B's starts at $0 — half full. Same achievement, and the thermometer told two different stories to two donor lists. *Which charity would you donate to — and did the data decide that, or the drawing?*

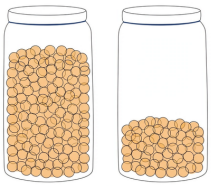
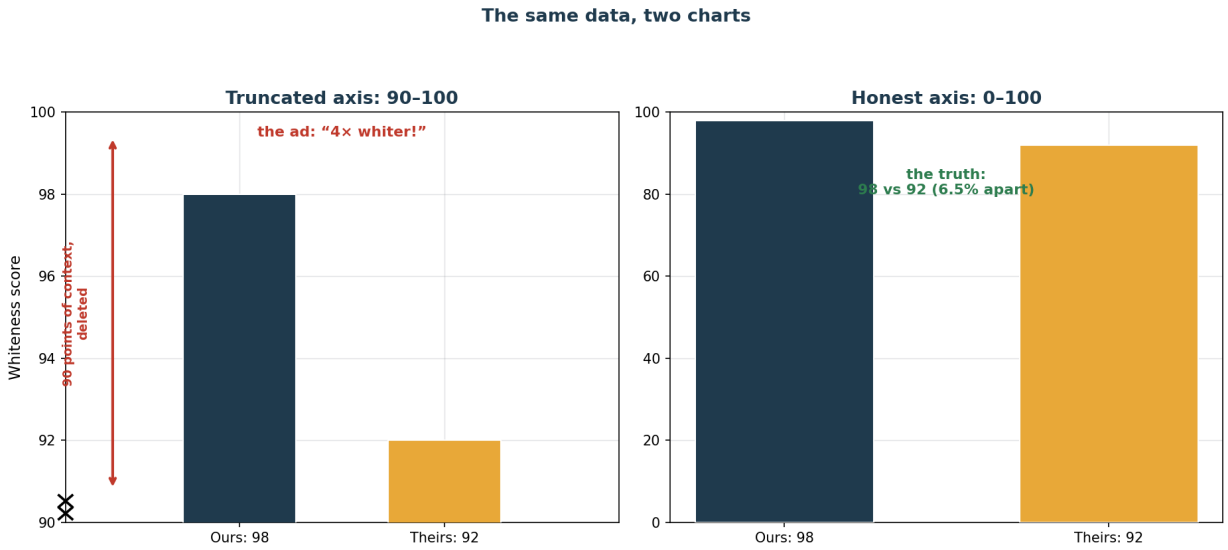


Fig. 5.7



Nothing about the data changed — only the frame. Always check where the axis starts.

Worked Example 1 — Percent vs. percentage points. The ad's claim: 98 vs. 92 is "4× whiter." What is the honest comparison?

Solution. Absolute difference: 98 − 92 = **6 points**. Relative difference: 6 ÷ 92 ≈ **6.5%** whiter — not 400%. (And "4×" would need a score of 368 out of 100 — impossible. The claim fails its own math.)

Worked Example 2 — The 200% crime wave. A flyer warns: "Burglaries up 200% this month!" Last month: 1 burglary. This month: 3. Evaluate.

Solution. Percent change = (3 − 1) ÷ 1 = **200%** — arithmetically true. But the *base* is 1: two extra burglaries in a town of 40,000 is noise, not a wave. Big percentages on tiny bases are the oldest trick in the book — always ask "200% *of what?*"

DECIDE — THE VERDICT

Do you pay double for the "4× whiter" toothpaste?

Back to the question that opened this section — now with the math to settle it.

Solution. Real difference: 6 points, **≈6.5%** — the "4×" lives entirely in the truncated axis (90–100), not in your teeth. Double the price for 6.5% more whiteness is a terrible trade — and a company that charts like this will chart like this about everything.

The decision: don't pay double. Buy the competitor's — or better, buy whichever is on sale and spend the savings on something that actually changes your life. "4× whiter" was never a measurement; it was a drawing of a measurement.

AI AI Learning Lab 5.7

AI Level 1 — Understand.

Explain truncated axes, cherry-picked ranges, and dishonest pictographs as if I'm 15, using the toothpaste's 98-vs-92 chart. Show me how the same data looks with an honest axis, then give me one quick check question per trick and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two spotting truncated axes (described in words), one computing the honest percent difference behind a hyped claim, one evaluating a "200% increase" with a tiny base, and one redescribing a cherry-picked time window honestly. Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The toothpaste decision showed that a true 98-vs-92 became a false "4× whiter" through axis choice alone — nothing about the data changed. Discuss with me: WHY does graphical dishonesty threaten democracy itself — think about election polls, public-health charts, or school funding votes? What are the implications when citizens can't read axes — who gets to manufacture public opinion, and what does that make statistical literacy: a school subject or a civic duty? End by asking me for one chart in MY life (news, ads, school) I should re-draw honestly.

Level 4 — Decide. *The battery ad says "lasts TWICE as long!" — upgrade?*

> **Example decision problem.** Two phone batteries, lab-tested hours: standard 10.0, "Ultra" 10.5. The Ultra ad's chart starts the axis at 9.5 hours, making its bar look 2× taller. The Ultra costs $40 more. Upgrade?

AI Level 4 — Decide.

A battery ad claims "lasts TWICE as long!" Lab tests: standard 10.0 hours, Ultra 10.5 hours. The ad's chart axis starts at 9.5 hours. The Ultra costs $40 more. Act as my decision coach: make me compute the honest percent difference and sketch what the chart hides, then ask me whether to pay $40 — don't decide for me.

Check your verdict: Honest difference = 0.5 ÷ 10 = 5% — "twice as long" would need 20 hours. The axis starts at 9.5 to manufacture drama from half an hour. Don't pay $40 for 5%: that's $80 per extra hour of battery you'll never notice.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.7.

Ask your AI: "Find me a real misleading graph about [a topic you care about] and explain the trick." Then do the detective work: verify the underlying numbers yourself from the cited source. If the AI can't produce a real source — or its "misleading graph" is invented — write a one-paragraph explanation of why you must never trust an AI's examples without sources. Then find a real one yourself (news sites are full of them) and re-draw it honestly. AI chatbots hallucinate evidence with total confidence — in this chapter, *you* are the fact-checker.

⚑ PROJECT MILESTONE

M6 prep: Audit your own poster

Sketch your project poster draft. Now attack it with this section's tools: any truncated axis? Cherry-picked range? Picture exaggerating a small difference? Deliverable: a marked-up draft with every misleading element circled and fixed. An honest graph is part of the grade. (Full brief: §5.10.)

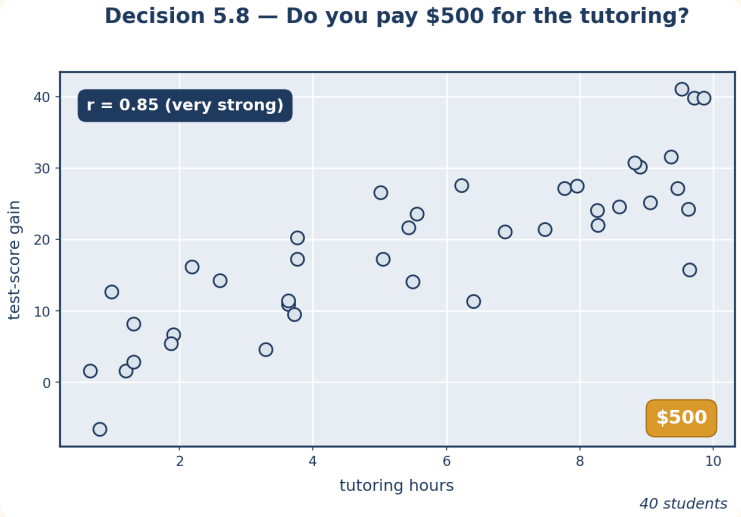
Practice Problems 5.7

1. An ad claims "50% more!" — the old bottle held 200 mL, the new holds 300 mL. Is the claim honest? What if the price also rose 50%?
2. A chart shows two bars, 30 and 33, with the axis starting at 28. How many times taller does the 33-bar look? What is the honest percent difference?
3. "Unemployment fell 2%!" — from 6% to 4%. Is that a 2% fall? Explain the percent vs. percentage-point distinction.
4. A school's "average SAT 1300!" excludes students who didn't report scores. Name the trick and explain who it misleads.
5. Re-draw honestly in words: a pictograph shows one pizza icon vs. two pizza icons (each twice as tall) to mean "twice as much pizza." What's wrong, and how would you fix it?

5.8 Scatter Plots & Correlation

THE DECISION

Do you pay $500 for the tutoring?



A tutoring company shows you a scatter plot: hours of their tutoring vs. test-score gains for 40 students. The dots climb steeply — the **correlation** is 0.85, "very strong." *"Our program works!"* the salesman says. *"Just $500."* Your parents will pay if you say yes.

Do you sign up? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** A correlation of 0.85 says the two variables move together. Write down what "move together" proves — and what it doesn't prove.
- **Hint 2:** Who buys $500 tutoring? Probably students who already study hard. Could *studiousness* raise both the tutoring hours and the scores?
- **Hint 3:** Design the fair test: what second group of students would you need to prove the *tutoring* caused the gains? Now ask: did the company show you that group?

Scatter Plots

A **scatter plot** puts one variable on each axis — one dot per observation. Patterns jump out: dots climbing left-to-right mean "as x grows, y grows" (**positive correlation**); falling means **negative correlation**; a formless cloud means no correlation.

STORY

The Texting Survey, Revisited.

Plot your classmates' two

survey answers — texting frequency (x) vs. self-rated popularity (y). One class found dots drifting down-right: more texting, *less* popularity (a negative correlation!). Nobody proved texting causes unpopularity — but the plot asked a question no frequency table could. *What would you plot about your own life — and what are you afraid the dots would say?*

Correlation Strength

The **correlation coefficient (r)** measures how tightly dots hug a line: r = +1 is a perfect climb, r = −1 a perfect fall, r = 0 no pattern. Use this strength scale:

|r|	Strength
0.81 – 1.00	Very strong
0.61 – 0.80	Strong
0.41 – 0.60	Moderate
0.21 – 0.40	Weak
0.00 – 0.20	Very weak / none

(The sign just gives the direction.) The tutoring company's r = 0.85: very strong — the dots really do climb together. That part is true.

STORY

The Line of Best Fit.

Draw the straight line that best threads

the dots — the **line of best fit** — and you can predict: at x = 170 cm tall, the line says y ≈ 65 kg. But a line is a promise, and r tells you whether to believe it: r = 0.95, trust the prediction; r = 0.25, the line is decoration. *When has someone shown you a confident prediction from a cloud of dots?*

Correlation Is Not Causation

Here is the chapter's most important sentence: **correlation never proves causation.** Ice-cream sales and drownings correlate strongly (r ≈ 0.9) — but banning ice cream won't save swimmers. The **lurking variable** is summer heat, driving both. The tutoring dots climb — but studiousness drives both hours *and* scores. Without a **control group** (studious students *without* tutoring), the company proved nothing except that motivated kids do well.

STORY

The Breakfast Study.

"Students who eat breakfast score higher!"

— so schools pushed breakfast programs, and scores barely moved. The lurking variable: stable home routines, which produce both breakfast and studying. The correlation was real; the *cause* was elsewhere. *What "obvious cause" in your life might be a lurking variable wearing a costume?*

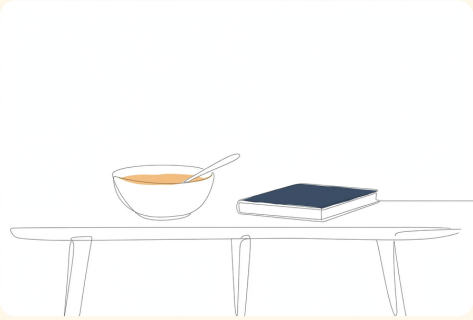
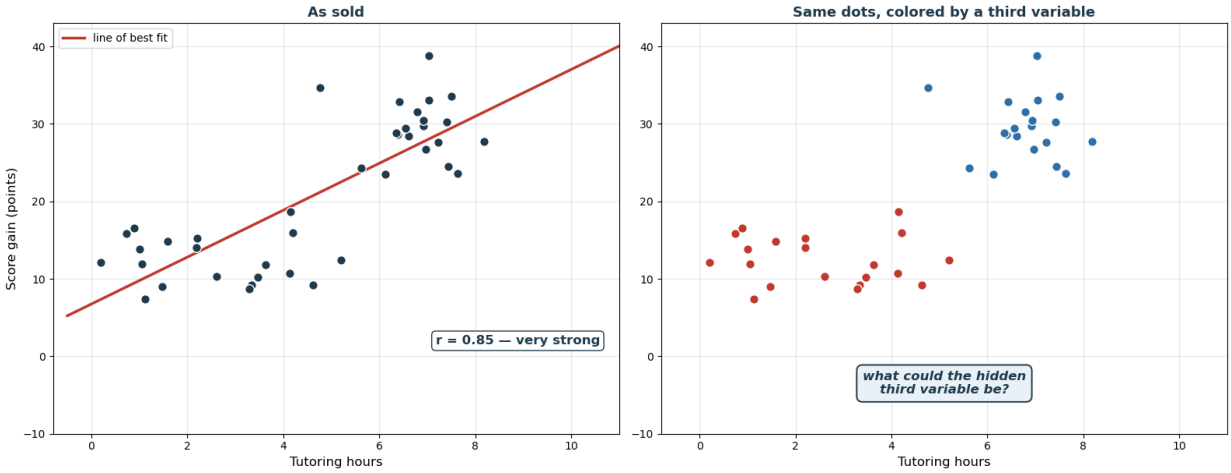


Fig. 5.8

Fig. 5.8 — Correlation is not causation



Strong correlation, unproven causation — always ask what third variable could drive both.

Worked Example 1 — Judging strength. Nine measurements of two variables: (0, 3), (1, 4.47), (2, 5.91), (3, 7.43), (4, 8.95), (5, 10.6), (6, 11.96), (7, 13.46), (8, 15.04). The correlation coefficient is r ≈ 0.9999. Interpret it.

Solution. $|r| \approx 1.00$ falls in 0.81–1.00 → **very strong positive correlation**. The dots essentially sit on a line (roughly $y = 1.5x + 3$: at $x = 8$, $1.5(8) + 3 = 15$ ✓). Prediction is trustworthy *here* — but remember: even $r = 1$ can't say *why*.

Worked Example 2 — A weak correlation. A class plots outfit rating (x) vs. shyness (y). Line of best fit: $y = -0.396x + 8.03$, with $R^2 = 0.063$. Interpret.

Solution. $R^2 = 0.063 \rightarrow |r| = \sqrt{0.063} \approx \mathbf{0.25}$ → weak (borderline negligible) negative correlation. The line *exists*, but the dots are a cloud — predicting someone's shyness from their outfit is numerology, not statistics. **A line of best fit with weak r is a costume, not a model.**

DECIDE — THE VERDICT

Do you pay $500 for the tutoring?

Back to the question that opened this section — now with the math to settle it.

Solution. $r = 0.85$ is genuinely very strong — the dots climb. But the company showed you one group: students who *bought* tutoring. The lurking variable (studiousness) was never measured, and no control group of equally studious non-buyers exists. The correlation is real; the *causation* is unproven — and $500 is a lot to pay for unproven.

The decision: don't sign up — not yet. Tell your parents: "Show me the control group." If the company can't produce one, the 0.85 is marketing wearing a lab coat. Correlation tells you *where to look*; only a fair comparison tells you *what works*.

AI AI Learning Lab 5.8

AI Level 1 — Understand.

Explain scatter plots, correlation direction/strength, and the line of best fit as if I'm 15, using the tutoring example ($r = 0.85$). Show me the strength scale, then give me one quick check question per idea and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two judging correlation strength from r values, one predicting with a line of best fit, one computing r's meaning from R^2, and one spotting the lurking variable in a described correlation. Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The tutoring decision showed that a very strong correlation (r = 0.85) proved nothing about causation — the lurking variable did all the work. Discuss with me: WHY is "correlation ≠ causation" so profitable to forget — think about supplement ads, "studies show" headlines, or education companies? What are the implications when millions act on unproven causes — who profits, who pays $500, and what does the breakfast study say about good intentions? End by asking me for one "X causes Y" claim in MY life whose lurking variable I should hunt.

Level 4 — Decide. *The district wants later start times — $50,000 pilot. Vote?*

> **Example decision problem.** Across districts, later school start times correlate with higher test scores (r = 0.6, moderate). Wealthier districts both start later and score higher. The board proposes a $50,000 later-start pilot in your district. How do you vote?

AI Level 4 — Decide.

My school board cites a moderate correlation (r = 0.6): districts with later start times score higher. But wealthier districts both start later AND score higher. A $50,000 pilot is proposed. Act as my decision coach: make me name the lurking variable and design the fair test myself, then ask me how I'd vote — don't decide for me.

Check your verdict: The lurking variable is district wealth. The honest test: randomize start times within *similar schools, or pilot in your own district and compare against its own past scores. Vote yes on a* small randomized pilot — *the correlation is worth investigating, but $50,000 of unproven causation is a gamble. Correlation buys you a question, never an answer.*

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.8.

Ask your AI: "Give me three real-world examples where a strong correlation turned out NOT to be causation, with the lurking variable identified." Then do the detective work: pick one and verify it from an independent source — did the lurking variable really explain it, or did the AI invent a tidy story? If the AI's example falls apart under checking, write a one-paragraph explanation of how a plausible-sounding causal story fooled even the storyteller. AI chatbots are fluent at causes and sloppy at evidence — in this chapter, *you* demand the control group.

⚑ PROJECT MILESTONE

M6 (part 1): Two variables

Your two questions give paired numerical data (say, gaming hours vs. sleep hours) — draw a scatter plot and judge the correlation's direction and strength. Deliverable: the plot plus one sentence on what the correlation can — and cannot — claim. (Full brief: §5.10.)

Practice Problems 5.8

1. A study finds r = 0.92 between yoga hours and happiness. State the direction and strength. Does this prove yoga causes happiness?
2. Line of best fit: $y = 1.5x + 3$. Predict y when $x = 10$.
3. $R^2 = 0.25$ for nightly sleep hours vs. morning alertness score. Find $|r|$ and judge whether you'd trust predictions from this line.
4. Ice-cream sales and drownings correlate at $r \approx 0.9$. Name the lurking variable and explain why banning ice cream wouldn't help.
5. A headline: "Coffee drinkers live longer!" Suggest one lurking variable and describe the control-group study that would test the claim fairly.

5.9 Line of Best Fit & the Correlation Coefficient

THE DECISION

Is the $12,500 car a fair deal?

A dealer lists a 2019 sedan with 60,000 miles for $12,500. "Priced to sell!" the ad says. You pull the listing prices of eight similar 2019 sedans:

Miles (thousands)	20	30	40	50	60	70	80	90
Price ($)	16,200	14,900	14,100	12,900	12,100	10,900	10,200	9,100

Is $12,500 fair — or are you the one being sold? Hold that question: the math you need comes next.

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Plot price (y) against miles (x). Describe the pattern in one sentence — which direction do the dots move?

- **Hint 2:** Sketch the straight line that best threads the dots. Read off its prediction at x = 60 — roughly what price does the line say?
- **Hint 3:** Subtract: $12,500 minus the line's prediction. The gap has a name — the **residual** — and its sign tells you who gets the better deal.

Scatter Plots: Seeing the Relationship

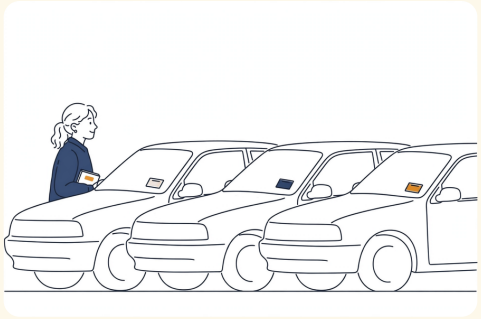
§5.8 taught you to read a scatter plot; now you will *use* one. Put miles on the x-axis, price on the y-axis, one dot per car. The dots fall left-to-right — more miles, lower price — a **negative correlation** you can see before you compute anything.

STORY

The Lemon Lot.

You walk the dealer's lot with a notebook,

plotting each windshield sticker: miles vs. price, dot by dot. By the eighth car the cloud has a clear downhill slope — and the $12,500 car sits *above* the cloud's center, lonely at the top. Nobody told you it was overpriced; the dots did. *What cloud of dots in your life is waiting to be plotted — and what would it accuse?*



The Line of Best Fit

Draw the straight line that best threads the dots — the **line of best fit** (trend line). "Best" has a precise meaning: the line that makes the squared misses as small as possible (software computes it instantly; by eye, aim for roughly as many dots above the line as below). For our eight cars:

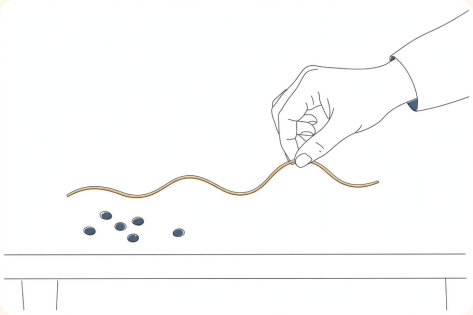
y = −99.5x + 18,024 (x = thousands of miles, y = price in dollars)

Read it like a sentence: the **slope** −99.5 says each extra thousand miles shaves about $100 off the price; the **intercept** 18,024 says a zero-mile car of this model would list near $18,024. Now the line is a prediction machine: at x = 55 (55,000 miles), y = −99.5(55) + 18,024 = **$12,552**.

STORY

The Thread Through the Dots.

Lay a spaghetti noodle across your plotted dots and wiggle it until the misses look balanced — congratulations, you just did regression by hand. Your noodle won't match the software's line exactly, but it will tell the same story: downhill, about $100 per thousand miles. *When is a rough line you drew yourself more trustworthy than a precise one someone handed you?*



Residuals: Measuring the Miss

No dot sits exactly on the line — each one misses by a **residual**:

residual = actual value − predicted value

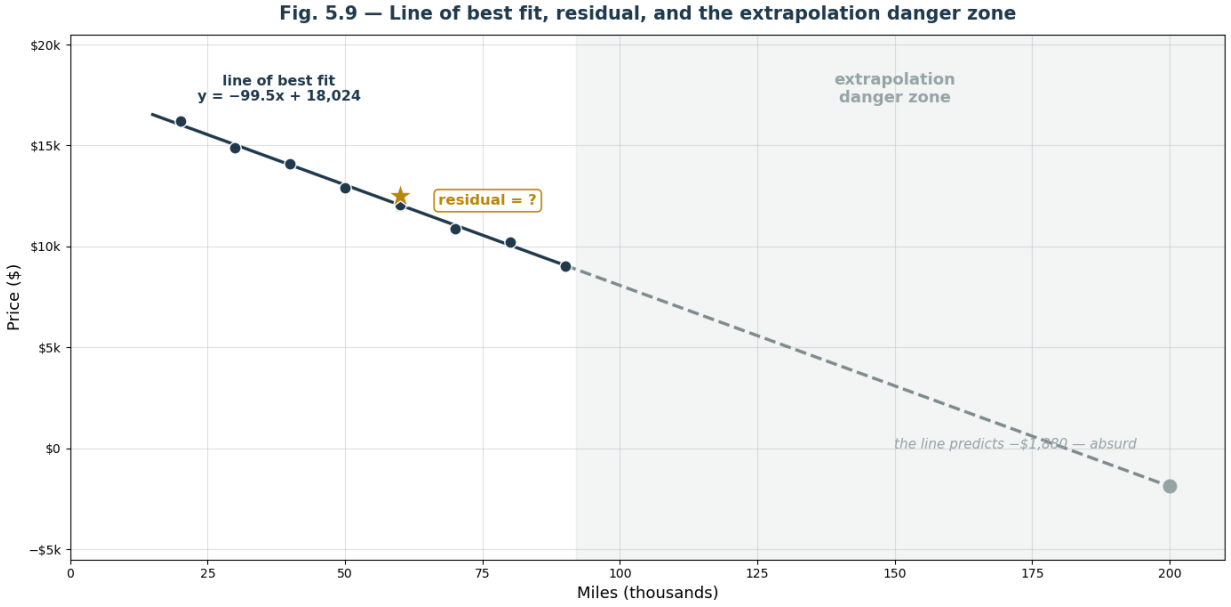
Positive residual: the dot sits *above* the line (pricier than typical). Negative: *below* (cheaper than typical). The $12,500 car at 60,000 miles: the line predicts −99.5(60) + 18,024 = **$12,054**, so its residual is 12,500 − 12,054 = +**$446**. It isn't "priced to sell" — it's priced about $450 *above* the market trend.

STORY

The $446 Question.

Two cars, same model year: one lists $446 above the line, one $552 below it. The window stickers look identical — same mileage, same shine. The residual sees what the shine hides: one dealer is charging you for the wax job. *What "identical" choices in your life would a residual tell apart?*

Fig. 5.9



Inside the data, the line is a tool. Outside it, a rumor.

Worked Example 1 — Predict and judge with residuals. Using y = −99.5x + 18,024, what price does the line predict at 55,000 miles? A second dealer lists a 55,000-mile car at $12,000. Compute its residual and judge the deal.

Solution. Predicted: y = −99.5(55) + 18,024 = −5,472.5 + 18,024 = **$12,552** (about $12,550). Residual: 12,000 − 12,552 = **−$552**. Negative residual — the car lists about $550 *below* the trend. **Good deal** (assuming similar condition — the line knows prices, not accidents).

The Correlation Coefficient r

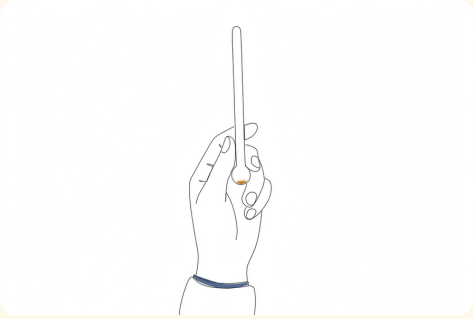
How much should you trust the line? The **correlation coefficient r** — from §5.8's strength scale — is the answer: r = +1 means the dots sit on a perfect climbing line, r = −1 a perfect falling line, r = 0 a formless cloud. For our eight cars, **r ≈ −0.999**: very strong negative. The dots hug the line, so predictions *inside the data's range* are trustworthy. Rule of thumb: |r| above ~0.8, lean on the line; below ~0.4, the line is decoration.

STORY

The Thermometer of Trust.

Think of r as a thermometer you

hold up to any trend line: near ±1, boiling with reliability; near 0, frozen — the line predicts nothing. Salespeople show you the line; they never show you the thermometer. *What trend have you been shown lately — and did anyone mention its temperature?*



Interpolation vs. Extrapolation

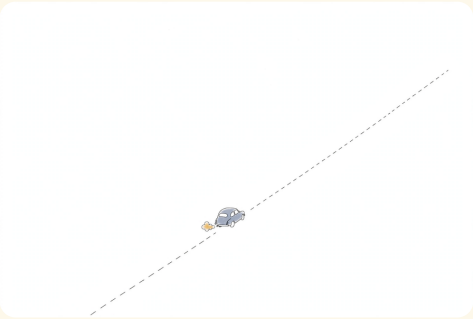
Predicting *inside* the data's x-range is **interpolation** — relatively safe. Predicting *outside* it is **extrapolation** — gambling. Our data spans 20,000–90,000 miles: predicting at 55,000 is interpolation (safe). But push the line to 200,000 miles: $y = -99.5(200) + 18{,}024 \approx$ **−$1,880**. A negative car price — the dealer pays *you*? Absurd. The line describes the data's neighborhood; beyond it, new forces (scrap value, collector interest) take over. Even the $18,024 intercept is a mild extrapolation (no zero-mile car was measured) — harmless here, but always ask how far past the data a prediction reaches.

STORY

The 200,000-Mile Car.

Your little brother runs the line out

to 300,000 miles and announces the dealer will owe him $11,826. He's doing the math *right* — and the thinking wrong. Every formula has a jurisdiction; outside it, arithmetic is just confident nonsense. *Where in your life are you extrapolating — grades, growth, savings — past the data?*



Correlation Still Isn't Causation

§5.8's warning stands: $r \approx -0.999$ does **not** prove miles *cause* the price drop. Age, wear, warranty expiry, and model-year updates all ride along with mileage — **lurking variables** again. The line predicts beautifully and explains nothing. Use it to *price*, never to *blame*.

STORY

The Shoe-Size Study.

In elementary schools, shoe size correlates strongly with reading ability ($r \approx 0.8$). Should we buy kids bigger shoes to make them read better? The lurking variable is age — older kids have bigger feet *and* read better. The correlation is real; the "cause" is a costume. *What "bigger shoes" fix is being sold to you right now?*

Worked Example 2 — Interpolate, don't extrapolate. For the car data (20k–90k miles), classify each prediction and judge it: (a) price at 50,000 miles; (b) price at 200,000 miles.

Solution. (a) 50,000 sits inside 20k–90k → **interpolation**: $y = -99.5(50) + 18{,}024 =$ **$13,049** — trustworthy ($r \approx -0.999$). (b) 200,000 is far outside → **extrapolation**: the line says $\approx -\$1{,}880$ — impossible. **Reject it**; the model has no jurisdiction there.

DECIDE — THE VERDICT

Is the $12,500 car a fair deal?

Back to the question that opened this section — now with the math to settle it.

Solution. The eight-car trend line is $y = -99.5x + 18{,}024$ with $r \approx -0.999$ — a very strong fit, and 60,000 miles sits inside the data (safe interpolation). The line predicts $-99.5(60) + 18{,}024 =$ **$12,054**. The asking price is $12,500: residual = 12,500 − 12,054 = +**$446**. "Priced to sell" is marketing — the data says it's priced about $450 *above* the market.

The decision: don't pay $12,500. Offer $12,000 — a fair price sits on or below the line — or keep looking; the −$552 car from the worked example shows better deals exist. The line can't tell you about that car's accident history (always check!), but it can tell you exactly what the *typical* 60,000-mile sedan costs — and this one isn't it.

AI AI Learning Lab 5.9

AI Level 1 — Understand.

Explain the line of best fit, residuals, the correlation coefficient r, and interpolation vs. extrapolation as if I'm 15, using the used-car example (8 cars, line $y = -99.5x + 18{,}024$, $r \approx -0.999$). Give me one quick check question per idea and wait for my answer before continuing.

AI Level 2 — Apply.

Give me 5 problems: two reading predictions from a given line of best fit, one computing a residual and judging the deal, one classifying predictions as interpolation or extrapolation, and one where a strong r still doesn't prove causation. Hide the answers. I'll solve all five; then check my work step by step.

AI Level 3 — Matter.

The car decision showed that a single residual (+$446) exposed a "priced to sell" lie — the dealer profits exactly when buyers can't read a trend line. Discuss with me: WHY is line-of-best-fit literacy a consumer-protection skill — think about "comparable sales" chosen by the seller, salary bands, or rent listings? What are the implications when only one side can draw the line — who picks the data, who profits from innumeracy, and what does "residual thinking" (am I above or below the line?) change about everyday choices? End by asking me for one price in MY life I should check against a trend line.

Level 4 — Decide. *7 hours of studying — and what about 14?*

> **Example decision problem.** Six students' study hours (1–6) and test scores (62, 68, 70, 78, 82, 88) give a line of best fit y ≈ 5.14x + 56.7 with r ≈ 0.993 — very strong. You're deciding whether to study 7 hours for the next test. What score does the line predict? And what does it predict for 14 hours?

AI Level 4 — Decide.

I have study data: hours 1–6, scores 62, 68, 70, 78, 82, 88. The line of best fit is y ≈ 5.14x + 56.7 with r ≈ 0.993. Act as my decision coach: make me predict the score for 7 hours myself, then make me predict for 14 hours and confront whether that answer is possible. Then ask me: is 7 hours interpolation or extrapolation, and should I reshape my week around this line? Make me state my final study plan with reasons — don't decide for me.

Check your verdict: At 7 hours, y ≈ 5.14(7) + 56.7 ≈ 93 — just past the data with a very strong r, so it's a plausible, cautiously-trusted prediction. At 14 hours the line says ≈ 129 — impossible on a 100-point test: pure extrapolation, reject it. Verdict: add the 7th hour, but don't rebuild your life around the line — it models the middle of the data, not the extremes of your week.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.9.

Ask your AI: "Show me Anscombe's quartet — four data sets with nearly identical correlation coefficients ($r \approx 0.816$) — and describe what each scatter plot actually looks like." Then do the detective work: verify the claim yourself (the four r values really are ≈ 0.816, yet one plot is a clean line, one is a curve, one has a wild outlier, and one is a vertical pile with a single far dot). Write a one-paragraph explanation of what the quartet proves: *r summarizes the dots; it never replaces looking at them.* If the AI can't produce the four plots accurately, explain which one it mangled and why "trust, but plot" is the statistician's motto. AI chatbots love single numbers — in this chapter, *you* demand the picture.

⚑ PROJECT MILESTONE

M6 (part 2): Predict — carefully

Fit a line of best fit to your scatter plot. Make one interpolation and write down one prediction you must NOT make (extrapolation), with one line on why it's forbidden. Deliverable: both predictions labeled, the dangerous one crossed out. Then assemble your final poster or 3-minute talk. (Full brief: §5.10.)

Practice Problems 5.9

1. For another car model, $r = -0.94$ between mileage and price. State the direction and strength. Would you trust this line's predictions?
2. Using $y = -99.5x + 18{,}024$ (x = thousands of miles), predict the price of a 45,000-mile car. A listing at 45,000 miles asks \$14,200 — compute the residual and judge the deal.
3. The car data spans 20,000–90,000 miles. Classify each as interpolation or extrapolation: (a) predicting at 50,000 miles; (b) predicting at 150,000 miles. Which do you trust, and why?
4. Across towns, the number of churches and the number of bars correlate at $r = 0.88$. Name a lurking variable and explain why building more churches wouldn't create more bars.
5. A class finds $y = 5.2x + 57$ for hours studied (x, data from 1–6 hours) vs. test score (y). Predict the score at 4 hours and at 20 hours. Which prediction is legitimate, and why?

Vocabulary

- **Categorical data** — data sorted into buckets (lunch lines, blood types, jersey numbers); summarize with counts, percents, mode — never a mean.
- **Numerical data** — data measured in genuine numbers; splits into discrete and continuous.
- **Discrete data** — counted in whole chunks with real gaps between values (siblings, goals scored, dice rolls).
- **Continuous data** — measured ever-finer, with values between any two values (wait time, height, temperature).
- **Mean** — the average: sum of values ÷ number of values; pulled by extreme values.
- **Median** — the middle value when data are ordered; resistant to outliers.
- **Mode** — the most frequent value; a data set can have none, one, or several.
- **Frequency table** — a table counting how many times each value (or group) occurs.
- **Dot plot** — one dot per observation stacked over a number line; shows shape at a glance.
- **Distribution** — the pattern of how data spread out: shape, center, and spread.
- **Box-and-whisker plot** — a five-number summary (min, Q1, median, Q3, max) drawn as a box with whiskers.
- **Quartiles** — Q1 (25th percentile), Q2 (the median), Q3 (75th percentile); they split ordered data into quarters.
- **Interquartile range (IQR)** — Q3 − Q1; the spread of the middle half.
- **Outlier** — a value far from the rest: below Q1 − 1.5×IQR or above Q3 + 1.5×IQR.
- **Range** — max − min; the crudest measure of spread.
- **Standard deviation** — the typical distance of values from the mean; larger means more spread.
- **Grouped data** — data bundled into class intervals for tables and histograms.
- **Class interval** — one group in grouped data (e.g., 10–19 minutes).
- **Histogram** — bars (touching) showing frequencies of class intervals.
- **Skewed** — a distribution with a longer tail on one side; mean gets dragged toward the tail.
- **Scatter plot** — points (x, y) showing the relationship between two variables.
- **Correlation** — how strongly two variables move together; not causation.
- **Correlation coefficient (r)** — a number from −1 to 1 measuring the strength and direction of a linear relationship.
- **Line of best fit** — the straight line threading closest through scatter-plot points; used for prediction.
- **Residual** — actual value − predicted value; how far a point sits off the line.
- **Interpolation** — predicting inside the data's range (safer).
- **Extrapolation** — predicting outside the data's range (dangerous).
- **Lurking variable** — a hidden third variable driving both x and y, faking a causal story.
- **Sample** — the group you actually collect data from; bigger and more representative beats smaller and cozier.
- **Survey** — a set of questions asked to a sample to gather data; the questions shape every answer before it's given.

- **Bias** — a systematic tilt in how data are collected or reported (loaded questions, coerced answers, cherry-picked samples) that pushes results one way.

5.10 The Personal Data Project

THE DECISION

Which two questions deserve a month of your life?

Naomi has a hunch: her late-night gaming is wrecking her sleep, and her sleep is wrecking her grades. Her teacher offers her a deal — she may ask her class exactly two survey questions, and the answers become her personal data set for every milestone in this chapter: dot plots, box plots, histograms, standard deviation, correlation, all of it, on numbers she collected herself about something she actually cares about. Two questions. One month. Which two does she ask — and which tempting questions must she reject?

Two questions, asked well, can carry a whole chapter. Two questions, asked badly, poison all of it.

HINTS

1. **Start here.** Write the one thing you genuinely want to know, as a single sentence: "I want to know whether ___." If you can't finish that sentence, you don't have a question yet — you have a topic. Curiosity first, questions second.
2. **Find the direction.** Test every candidate question three ways: (a) Will the answers be numbers you can actually analyze — or clean categories? (b) Would a classmate answer it honestly? (c) Could the answers surprise you? A question that fails all three is decoration. Kill it.
3. **Just before you decide.** Run your two finalists through the five ethics rules below and the bad-question gallery in "What Makes a Question Good (or Poison)." If either question needs a warning label, it isn't finished — rewrite it.

The Math You Need

What Makes a Question Good (or Poison)

A good survey question is **specific** (one thing, precisely defined), **neutral** (no thumb on the scale), and **answerable** (the respondent knows the answer and will give it). Poison questions fail at least one:

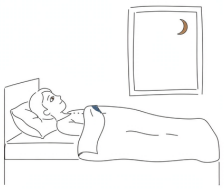
- **Vague:** "Do you sleep well?" — *well* means everything and nothing.
- **Leading:** "Don't you think homework is too much?" — the answer is baked in.
- **Double-barreled:** "How many hours do you sleep and study?" — two questions wearing a trench coat; the answer is uninterpretable.
- **Sensitive:** "What is your exact GPA?" — true answers dry up, and the missing ones aren't random.

STORY

The 2 a.m. Confession.

Derek asked his class "Do you get

enough sleep?" — 90% said "no," and he learned exactly nothing. Aria asked "How many hours did you sleep last night? (nearest half hour)" — and got a distribution: mean 6.4, median 6.5, one glorious outlier at 11. *One word — "enough" — was the difference between data and noise.*



The Ethics Rules — Read Before You Ask

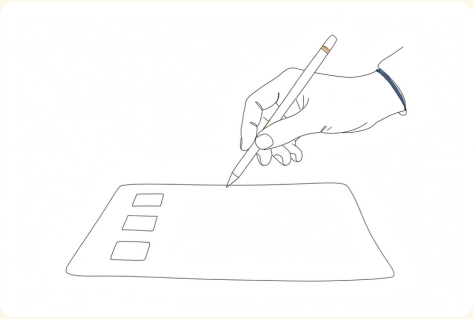
Five rules. No exceptions, no "just this once":

1. **Anonymous by default** — no names on responses, ever.
2. **No coercion** — "no" is a complete answer; never pressure, bribe, or guilt a classmate into responding.
3. **No sensitive questions** — weight, grades/GPA, family income or job loss, health conditions, and immigration status are off-limits.
4. **Right to skip** — every question gets a "prefer not to answer" option, and skips are never punished.
5. **Report honestly** — never invent, round toward, or delete responses to improve the story. The data is the data.

STORY

The Skipped Question.

Marcus put "prefer not to answer" on every question, and 6 of 24 classmates skipped the one about allowance. He almost deleted the question in shame — then realized the skips *were* the finding: the question touched money, and money is personal. *A skipped question is an answer about the question.*



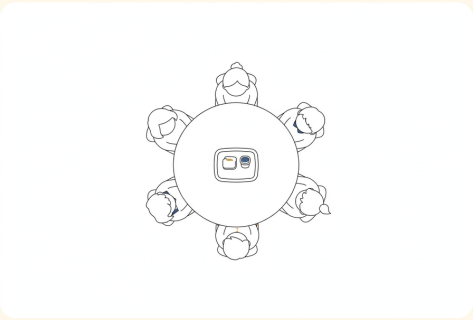
How Much Data Is Enough

Minimum n = 20. The whole class is better. Why 20? Below about twenty responses, a single outlier can drag your mean wherever it wants, your dot plot is mostly empty space, and the Law of Large Numbers — the engine under half this chapter — never gets a crowd to work with. Twenty is the floor, not the goal.

STORY

The Table of Six.

Nadia surveyed her lunch table — six friends — and announced "our class averages 9 hours of sleep." Her teacher surveyed all 28 students: 6.8. *Six friends are not a class; they're a clique with a mean.*



The Milestone Roadmap

You don't build this project at the end. You build it in nine small pieces, each right after you learn the tool it needs:

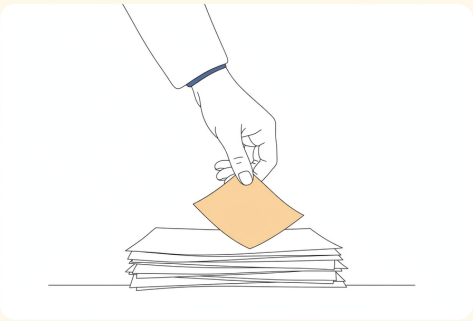
Milestone	After §	You deliver
M1	5.1	Two teacher-approved *numerical* questions that might be related
Collect	5.2	At least n = 20 honest responses
M2	5.3	Frequency table + dot plot + one shape sentence
M3	5.4	Box-and-whisker plot + one outlier paragraph

Milestone	After §	You deliver
M4	5.5	Grouped frequency table + histogram (if it fits your data) + one insight sentence
M5	5.6	Range, IQR, standard deviation + one spread sentence
M6 prep	5.7	Poster draft, self-audited for misleading tricks
M6	5.8–5.9	Scatter plot + r + line of best fit on your paired data
Final	ch. end	1-page poster or 3-minute talk + 3-paragraph essay

STORY

The Slow Build.

Darius thought the project was "due at the end." It wasn't — it was due in nine pieces. By M5 he had already found his outlier, fixed his bins, and caught his own truncated axis in the M6-prep audit. *The project isn't a deadline; it's a staircase.*



The Final Products

1. **A 1-page poster or 3-minute talk:** your two questions, your n, two or three visuals, one honest conclusion, and one limitation — what your data *can't* say.
2. **A 3-paragraph decision essay:** you may use your project data for the chapter's Summative Task. Same rules: decision and claim, mathematical evidence, implications and what would change your mind.

STORY

The Honest Poster.

Aisha's poster concluded: "No clear link between sleep and quiz scores in our class ($r = 0.18$, $n = 26$)." No fireworks — and the best grade in the room. *In statistics, "I found nothing" is a finding, if you can prove you looked.*

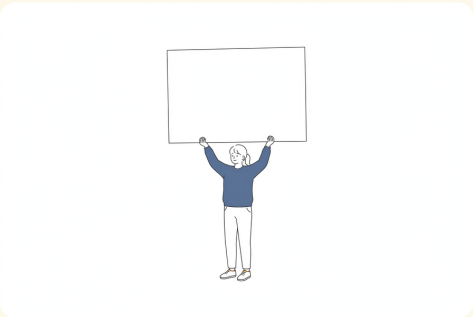
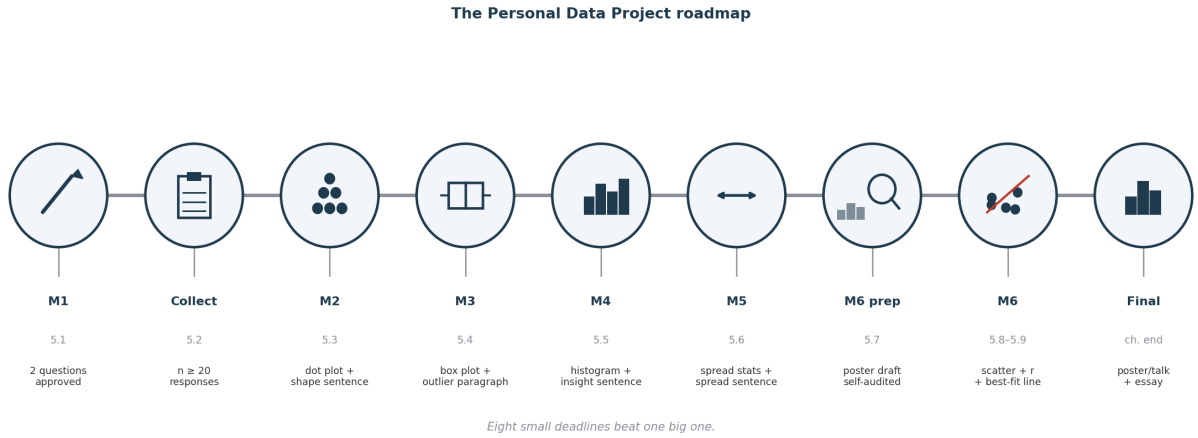


Fig. 5.10



Nine small deadlines beat one big one.

DECIDE — THE VERDICT

Naomi's two questions — final answer.

Back to the question that opened this section — now with the math of good questions to settle it.

Solution. Naomi's keeper pair: **Q1 (numerical)** — "How many hours did you play video games last night? (nearest half hour)"; **Q2 (numerical, continuous)** — "How many hours did you sleep last night? (nearest half hour)". Both pass all three tests (specific, neutral, honest answers) — and because both are numerical *and* plausibly related, every milestone works: dot plots and box plots per question, then in M6 a scatter plot of gaming hours vs. sleep hours with r and a line of best fit. Rejected: "Are you a good student?" (vague *and* leading — poison twice over) and "What is your exact GPA?" (ethics rule 3 — sensitive; it would crater her response rate and bias whatever survived).

The decision: two questions asked well beat ten asked badly. Naomi asks about gaming and sleep — and her whole chapter now runs on data she owns.

AI AI Learning Lab 5.10

AI Level 1 — Understand.

Explain what makes a survey question good or bad as if I'm 15. Then give me 6 candidate questions about MY topic (I'll tell you the topic), one at a time: I'll sort each into good or bad with my reason, and you check each before moving on.

AI Level 2 — Apply.

Here is my draft survey question: [paste it]. Stress-test it like an auditor: flag vagueness, leading words, double barrels, and sensitivity problems. Then rewrite it 3 different ways. I'll pick the best rewrite and defend my pick; check my reasoning.

AI Level 3 — Matter.

We learned that vague or leading questions poison data before the first answer. Discuss with me: WHO benefits when survey questions are rigged — think push polls, customer-satisfaction surveys that never offer "terrible," or "studies" funded by the product's maker? What are the implications for democracy and markets when the questions are fixed before anyone answers? End by asking me to find one real survey question in the wild and dissect it.

Level 4 — Decide. *Are our two questions good enough to bet a month on?*

> **Example decision problem.** Your team's topic is video games and sleep. Candidate questions: (a) "Do you game too much on school nights?" (b) "How many hours did you sleep last night? (nearest half hour)". Naomi pilots (b) on 5 friends: 7, 6.5, 8, 5.5, 7 hours. Does each question pass the three tests and the five ethics rules? Is the pilot data promising?

AI Level 4 — Decide.

My topic is [mine]. My two draft questions are [paste both]. Act as my decision coach: make me test each against the three tests (analyzable? honest answers? could surprise me?) and all five ethics rules, one at a time. Make me rewrite any question that fails until both pass. Then make me state my final two questions with reasons — don't choose for me.

Check your verdict: (a) fails twice — "too much" is leading and vague (poison). (b) passes everything: specific, neutral, numerical. Pilot mean = (7 + 6.5 + 8 + 5.5 + 7) ÷ 5 = 34 ÷ 5 = 6.8 hours, all 5 friends answered, zero skips — the question works. Verdict: keep (b), kill (a), and find a clean

numerical partner like "How many hours did you game last night? (nearest half hour)" — two numerical questions, plausibly related, so M6's scatter plot and r have something to chew on. Never let a poison question into M1 — everything downstream inherits it.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 5.10.

Ask your AI: "Invent a push poll — 5 questions that look neutral but steer respondents toward saying our school lunch is great." Then do the detective work: dissect each question, name the manipulation technique (leading, loaded, false choice, double-barreled…), and rewrite all five honestly. Write one paragraph on who profits when a survey is rigged this way. AI chatbots are fluent at persuasion disguised as research — in this chapter, *you* are the ethics board.

Practice Problems 5.10

1. Sort each into good or bad survey question, with one reason: (a) "How many hours did you sleep last night?" (b) "Don't you agree the cafeteria food is terrible?" (c) "What is your height in inches?" (d) "How many siblings do you have, and do you get along with them?"
2. A student wants to ask "What is your family's income?" to study allowance vs. income. Which ethics rule blocks it, and how would you redesign the project so the question isn't needed?
3. Pilot data for "text messages sent yesterday": 12, 45, 30, 200, 25, 18, 40, 33, 28, 22. Compute the mean. Is n = 10 enough under the project rules? What should the team do next?
4. On the topic "music and mood," write two *numerical* questions that might be related (so you can run M6 on them) and that pass all three tests. State which test each question's wording is designed to satisfy.
5. Your team's two questions are "hours of sleep last night" and "favorite music genre." Which milestone can't you complete with this pair, and how should you fix the pair? Explain using the M6 requirements.

Summative Task: The 3-Paragraph Decision Essay

The prompt. Your friend is choosing between the two café jobs from §5.2: Café A pays $11/hour flat ($220/week guaranteed); Café B pays $9/hour plus tips, advertising "our crew

averages $240 a week!" — but the last 10 paychecks were $200, $200, $205, $210, $210, $220, $225, $230, $240, $460. Advise your friend in a 3-paragraph essay: which job should she take, and why does the math support it?

Planning your essay (do this before you write). 1. **Compute first.** Run the numbers yourself: the 10 paychecks sum to $2,400 → mean = $240 (the flyer's number — true); sorted, the middle two are $210 and $220 → median = $215; $200 appears twice and $210 appears twice → the data is bimodal, modes = $200 and $210. Verify these yourself — don't take them on faith. 2. **Thesis template.** "I recommend ___ because ***, even though*** ." Fill it in for YOUR side — either side is defensible if the numbers back it. 3. **Evidence checklist.** Every claim needs a number behind it: the $240 mean vs. the $215 median vs. the $200 and $210 modes, the single $460 week that drags the mean $25 above the typical week, plus at least one idea from this chapter (how an outlier skews the mean, or why the median describes the typical week) that strengthens the case. 4. **Flip test.** Name one specific condition that would change your recommendation — pooled tips that smooth every week toward $240? A guaranteed $220 minimum under the tip system? If your advice wouldn't flip under any condition, it's a slogan, not a decision.

Paragraph 1 — Your decision and claim. State your advice in one clear sentence, then one or two sentences on why it matters. (Use the thesis template: "I recommend ___ because ***, even though*** ." Both sides are arguable — the "take Café A" side leans on the guaranteed $220; the "take Café B" side leans on the $240 average and the upside of good weeks.)

Paragraph 2 — Mathematical evidence. Show all three measures: mean = 2400 ÷ 10 = $240 (the flyer's number — true), median = $215 (the typical week), modes = $200 and $210 (the most common weeks). Explain how the single $460 week drags the mean $25 above reality. Every claim needs a number behind it.

Paragraph 3 — Implications, limitations, and what would change your mind. What does "typical $215 vs. guaranteed $220" mean for someone with fixed bills? What are this calculation's limits (10 weeks is a small sample — what if summer tips run higher)? Name one specific condition that would flip your advice — for example, pooled tips that smooth every week toward $240, or a guaranteed minimum — and explain why.

Length: 200–300 words. Three paragraphs, no more, no fewer.

Using AI. Write your first draft entirely on your own — no AI. Then you may use AI as a revision coach: ask it to check your logic, flag weak evidence, and suggest clearer sentences. Never let it write paragraphs for you. And remember this book's golden rule: AI makes mistakes — verify everything, especially any numbers it gives you. (Ask it for the median of the 10 paychecks and watch whether it sorts first.)

Rubric and exemplar essays are in the Teacher Guide.

Answer Key — Practice Problems (odd-numbered)

5.1 — 1. (a) categorical (b) discrete numerical (c) continuous numerical (d) categorical — zip codes are labels, not quantities (e) continuous numerical (f) discrete numerical. 3. Mean = 213 ÷ 50 = 4.26. Misleads: it hides the 5 unhappy customers (10% rated ≤2 stars) behind a shiny average. Useful: quick comparison across products or over time, as long as you also check the distribution. 5. Example: categorical — "How do you get to school? (bus / car / walk / bike)"; discrete — "How many buses do you take?"; continuous — "How many minutes does the trip take?"

5.2 — 1. Sorted: 4, 8, 8, 15, 16, 23, 42. Mean = 116 ÷ 7 ≈ 16.57; median = 15; mode = 8. 3. Below — the single low score (30) drags the mean under the median. 5. Median = 6'1" (5th of 9). Misleading: even the true mean is only ≈ 6'3.1", and the typical player is 6'1".

5.3 — 1. 2→3, 3→2, 4→1, 5→4. 3. Route A — the mean is identical, but A's tight cluster means you won't miss the flight; B's spread might. 5. Mode is 3–4 people — the most common household size.

5.4 — 1. Median = (20+22)/2 = 21; Q1 = (15+18)/2 = 16.5; Q3 = (25+30)/2 = 27.5. 3. 50% between Q1 and Q3; ~100% between min and max (excluding outliers). 5. Answers vary — must cite median $58k vs. mean $96k: a few high earners pull the mean up; the typical household earns $58k.

5.5 — 1. (a) discrete (b) continuous (c) discrete (d) continuous. 3. Midpoints 15, 25, 35: (45 + 75 + 35) ÷ 7 = 155 ÷ 7 ≈ 22.14. 5. Because the data is continuous — intervals flow into each other with no gaps.

5.6 — 1. Range = 22 − 5 = 17. 3. Class A — SD 3 means scores hug the mean; Class B's histogram sprawls (SD 15). 5. The rival's fans — SD 18 means any single game could be a blowout either way; your team's SD 4 means ~80 almost every night.

5.7 — 1. "50% more" is honest (300 ÷ 200 = 1.5) — but if the price rose 50% too, you're paying 50% more for 50% more: no real gain. 3. No — that's 2 *percentage points* (a 33% relative fall from 6%). Saying "fell 2%" understates it — the real fall (33%) is about 17 times bigger than 2% suggests. 5. Twice as tall = 4× the area — the picture quadruples what the numbers double. Fix: use two same-sized icons, or an honest bar chart.

5.8 — 1. Very strong positive — but no: no control group, no proven causation (lurking variables like lifestyle could drive both). 3. $|r| = \sqrt{0.25} = 0.5$ — moderate; predictions beat guessing but stay shaky. 5. Answers vary — must name a plausible lurking variable (e.g., income: wealthier people afford both coffee habits and healthcare) and a control-group design (compare similar people who differ only in coffee).

5.9 — 1. Very strong negative — this line's predictions are trustworthy within the data range (r still proves nothing about causation). 3. (a) Interpolation — 50k sits inside the 20k–90k data; trust it. (b) Extrapolation — the model breaks down out there (pushed far enough, it prices cars below $0). 5. At 4 hours ≈ 78 — legitimate interpolation; at 20 hours = 161 — impossible, an extrapolation beyond both the data and the 100-point scale.

5.10 — 1. (a) good — specific, neutral, numerical; (b) bad — leading, the answer is baked in; (c) good — specific and numerical (height isn't sensitive); (d) bad — double-barreled, two questions in one. 3. Mean = 453 ÷ 10 = 45.3. No — n = 10 is below the n = 20 floor; collect at least 10 more responses. Also investigate the 200 (possible outlier) — don't delete it. 5. M6 can't be completed — "favorite music genre" is categorical, so there is no paired

numerical data for a scatter plot, r, or line of best fit. Fix: replace it with a numerical question plausibly related to sleep, e.g. "How many hours did you game last night?" — then M6 runs on gaming hours vs. sleep hours.

Level 4 example checks — **5.1:** Books are numerical discrete — total 65 ÷ 30 ≈ 2.17, legitimate (median 2 confirms). Favorite subject is categorical — the coded "1.87" is a mirage; report the mode: Math, 40%. **5.2:** Mean $1,200 (ad truthful); median $1,100 — budget is safe, but the $2,200 penthouse does the inflating. **5.3:** Both means = 15. P: tight tower (13–17) — hire P; Q: lottery (2–25). **5.4:** Q1 = 12.1, Q3 = 13.65, IQR = 1.55, lower fence ≈ 9.78 ($9,780) — $6,500 is a flagged low outlier; budget ≈ $12,900 (median). **5.5:** Over 6 min: 8/60 ≈ 13.3% > 10% — hire; est. mean ≈ 3.43 min (the tail, not the mean, loses regulars). **5.6:** Offer B: mean $50k, range $30k, SD ≈ $9.9k; $30k base covers $21.6k bills — take B if you can stomach a $35k year, else A. **5.7:** Honest diff = 5% ("twice" would need 20 hrs) — don't pay $40. **5.8:** Lurking variable = district wealth; vote for a small randomized pilot, not $50,000 of unproven causation. **5.9:** Line y ≈ 5.14x + 56.7, r ≈ 0.993 (very strong). 7 hours → ≈ 93: plausible, just past the data — trust cautiously. 14 hours → ≈ 129: absurd — extrapolation past the scale. Verdict: add the 7th hour, but don't rebuild your life around the line. **5.10:** (a) fails — "too much" is leading and vague (poison); (b) passes — specific, neutral, numerical. Pilot mean = 34 ÷ 5 = 6.8 h, 5/5 answered, 0 skips — the question works. Verdict: keep (b), kill (a).

CHAPTER 6

Finance

A Note on AI Learning Labs in This Chapter

Chapter 5 taught you to interrogate averages. This chapter teaches you to interrogate *money* — the place where math meets human nature at full speed. AI chatbots will happily compute a loan payment, project your retirement, or build you a budget, and every number will look precise. Precision is not honesty: a chatbot will not warn you that the 0% offer has deferred interest in paragraph 14, or that "average return" hides the years you lost half. So this chapter's standing rule: **never sign a number you cannot recompute yourself.** Every AI Learning Lab keeps the four levels (understand, apply, matter, decide), and every Level 4 ends the way the chapter begins — with a decision about your money that is yours to make.

About This Chapter

Money runs on mathematics most people never do. The difference between a comfortable retirement and a painful one is often a single decision made at 25 — and it is pure exponential functions. The difference between a fair car loan and a trap is one page of fine print — and it is pure arithmetic. This chapter gives you the tools: compound interest (an old friend from §3.7, back in a suit), depreciation, loan payments and the time value of money, paychecks and deductions, budgeting, and credit scores. Then you will use all of it to decide: when to start saving, new car or used, 36 months or 60, what "$18 an hour" really pays, whether the rent is affordable, where your $2,800 goes, and whether to build credit now or pay for skipping it later.

Running through the whole chapter is **The $2,800 Month** (§6.8): you get a scenario — your first real job, $2,800 a month take-home — and one milestone per section. Price your savings plan, your car, your loan, your paycheck math, your full budget, your credit future. By the final pitch, the budget won't be the book's. It'll be yours.

By the end of this chapter, you will be able to:

- Compute compound interest for any compounding frequency — and explain why starting 10 years earlier beats saving twice as much.
- Model depreciation two ways (declining-balance and straight-line) and use it to price used cars.

- Compute monthly loan payments, total repayment, and interest — and explain the 36-month vs. 60-month trade-off with numbers.
- Convert hourly wages to annual, monthly, and per-paycheck gross across pay periods, including overtime.
- Read a paystub: FICA, withholding, marginal tax brackets — and compute net pay from gross.
- Build a balanced budget on the 50/30/20 frame, and diagnose a budget that balances on paper but fails in real life.
- Explain the five credit-score factors, keep utilization under 30%, and price the lifetime cost of a low score.
- Use AI prompts at four levels (understand, apply, matter, decide) to learn finance — and force every AI money number to show its work before you trust it.

6.1 Compound Interest — Time Is the Money

THE DECISION

Who retires richer — the early saver or the bigger saver?

Two 25-year-olds make a bet:

- **Olivia** invests **$200/month** starting at age **25**, earning 7% a year, and never stops until 65.
- **Grace** waits, then invests **$400/month** — twice as much — starting at age **35**, same 7%, until 65.

Grace contributes $48,000 *more* over her lifetime. But someone's account ends up roughly $37,000 bigger — and it isn't Grace's. Who wins the bet, and why does waiting ten years cost more than doubling your payment? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Olivia's money gets 40 years; Grace's gets 30. At 7%, money roughly doubles every 10 years (you'll prove this in the Deep Dive). How many doublings does each saver get?

- **Hint 2:** Contributions aren't the score — the *final balance* is. Olivia puts in $96,000 total; Grace puts in $144,000. The winner's balance beats both of those numbers. Where does the extra come from?
- **Hint 3:** Compute it: future value = payment × [((1 + r)^n − 1) ÷ r], with r = 0.07 ÷ 12 and n = months. Run both. Then ask: what did the first ten years buy Olivia?

The Compound Interest Formula

You met this function in §3.7 as pure mathematics: A = P(1 + r/n)^(nt). In finance it wears a suit, but the engine is identical — **interest earning interest**. Each period's interest joins the principal, so next period you earn interest on your interest. The exponent is time, which is why time is the most powerful variable in the formula: r and n are chosen by banks, but t is chosen by *you*, by starting early.

For regular monthly investing, the twin formula is the **future value of an annuity**: FV = PMT × [((1 + r)^n − 1) ÷ r], where PMT is the monthly payment, r the monthly rate, n the number of months.

STORY

The Two Savers.

Olivia and Grace from the opening bet are real people now — cousins, actually. At every Thanksgiving, the two compare notes: Grace points to her $400 months, Olivia points to the calendar. The $400 months look bigger — but they miss what the calendar knows: every dollar Olivia invests at 25 gets forty years to double and double again, while Grace's dollars get only thirty. The first ten years don't just add money — they buy *time* for every dollar after them. *If someone offered you $48,000 to wait ten years before saving, would you take the deal?*

Compounding Frequency

Banks slice the year into compounding periods: annually (n = 1), quarterly (n = 4), monthly (n = 12), daily (n = 365). More slices means interest joins the principal sooner — but the effect is small next to the rate and the years. $10,000 at 6% for one year: annual compounding → $10,600.00;

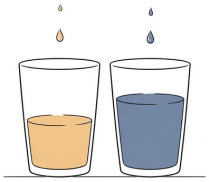
daily → ≈ $10,618.31. The daily drip earns $18.31 extra — real, but not life-changing. **Frequency is a footnote; time and rate are the story.**

STORY

The Daily Drip.

Two savings accounts, both 6%, both

$10,000. One compounds annually, one daily. After a year the daily one leads by $18.31 — enough for a pizza, not a retirement. A banker who brags about "daily compounding!" while offering 0.5% is selling you the footnote and hiding the story. *Which number would you ask about first — the compounding frequency or the rate?*



Simple vs. Compound — The Direction of the Engine

Simple interest pays only on the original principal: A = P(1 + rt). **Compound interest** pays on principal *plus* accumulated interest. The gap starts small and explodes: $2,000 at 8% for 5 years → simple: $2,800; compound (annual): ≈ $2,938.66. A $138.66 difference on a small account — scale it to a mortgage or a retirement and it's a house.

And the engine runs both directions. Compound interest *for* you is a retirement; compound interest *against* you is credit-card debt. Chapter 1's $2,400 balance at 24% was this same formula wearing a mask.

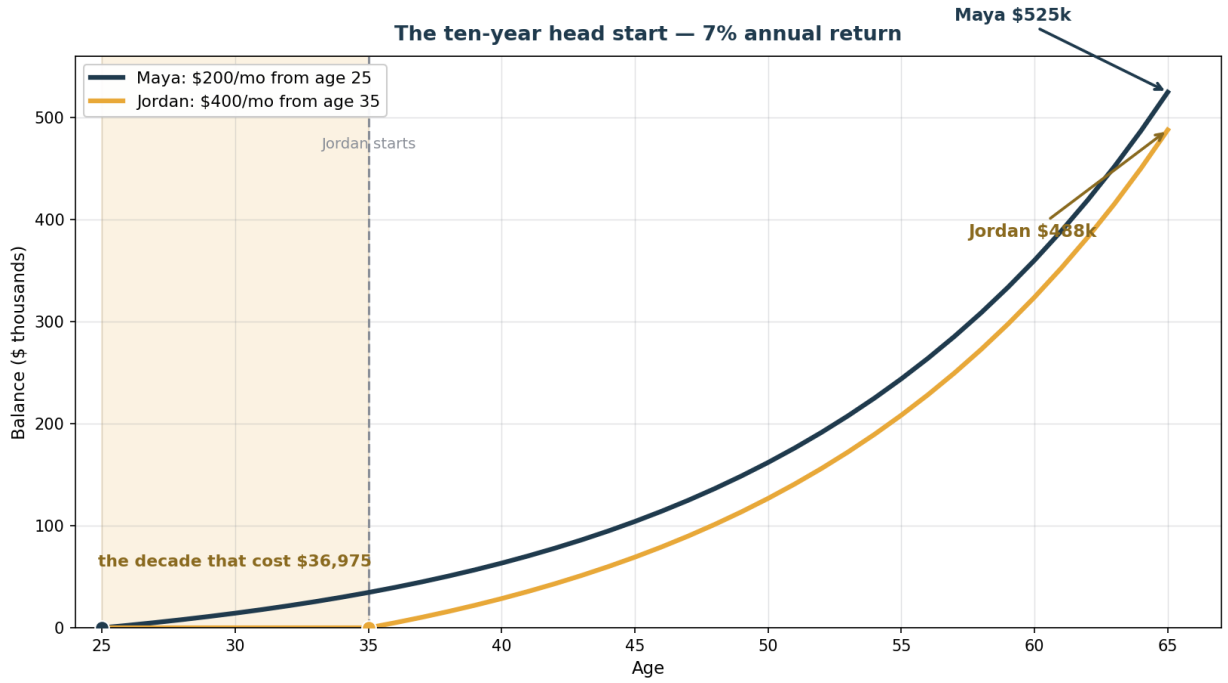
STORY

The Minimum-Payment Ghost.

Remember the $50-a-month trap

from Chapter 1's summative essay? That was compound interest collecting from *you*: every month, 2% interest joined the balance, and next month you paid interest on the interest. The formula doesn't care whose side it's on. *Is there any "good" version of paying interest on interest?*

Fig. 6.1



Time in the market beats money in the market.

DECIDE — THE VERDICT

Who retires richer — the early saver or the bigger saver?

Back to the question that opened this section — now with the math to settle it.

Solution. Monthly rate r = 0.07 ÷ 12 ≈ 0.005833. Olivia: n = 480 months. FV = 200 × [((1.005833)^480 − 1) ÷ 0.005833] ≈ 200 × 2,624.81 ≈ **$524,963**. Contributions: $96,000. Grace: n = 360 months. FV = 400 × [((1.005833)^360 − 1) ÷ 0.005833] ≈ 400 × 1,219.97 ≈ **$487,988**. Contributions: $144,000.

The decision: Olivia wins by ≈ **$36,975** while contributing $48,000 less. The first ten years — money doubling roughly once — did more work than Grace's extra $200 every month for thirty years. The lesson isn't "save $200 at 25" — it's that **every year you wait has a price, and the price is exponential**. Start with whatever you can; the amount matters less than the start date.

AI AI Learning Lab 6.1

AI Level 1 — Understand.

Explain compound interest as if I'm 15, using a snowball rolling downhill. Then explain the difference between the lump-sum formula A = P(1+r/n)^(nt) and the monthly-investing formula, one at a time, with a tiny numeric example for each. Check that I can tell them apart before moving on.

AI Level 2 — Apply.

Give me 5 compound-interest problems: mix lump sums and monthly investing, and mix compounding frequencies. Hide the answers. I'll solve each showing r, n, and the formula; then check my work step by step and point out exactly where I went wrong if I slipped.

AI Level 3 — Matter.

Olivia beat Grace with half the monthly payment because she started ten years earlier — time beat money. Discuss with me: WHY does almost nobody act on this in their twenties, and who profits from that delay — banks selling "start later, pay more" products? employers? the "buy now, pay later" industry? What are the implications for someone whose family can't spare $200 a month at 25 — is "start early" advice that ignores poverty, or does it matter even more? End by asking me what one small, automatic transfer I could start this month.

Level 4 — Decide. *If you were 25 with a $5,000 bonus, what would you do?*

> **Example decision problem.** At 25 you get a $5,000 bonus. Option A: invest all of it at 7% and touch nothing for 30 years. Option B: spend $2,500 now and invest $2,500. Which option, and what does the difference buy you at 55?

AI Level 4 — Decide.

I'm 25 with a $5,000 bonus. Option A: invest all $5,000 at 7% for 30 years. Option B: spend $2,500 now, invest $2,500. Act as my decision coach: make me compute both future values myself (5000 × 1.07^30 and half that), then ask me what the $2,500 of spending today would have to be worth to justify giving up the difference. Make me state my final choice with reasons — don't decide for me.

Check your verdict: A → 5000 × 1.07^30 ≈ 5000 × 7.612 ≈ $38,061. B → half: ≈ $19,031. The $2,500 spent today costs $19,030 at 55 — every dollar spent at 25 costs about $7.60 at 55. The decision isn't "never spend" — it's knowing the price tag has an exponent on it.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.1.

Ask your AI: "Explain the Rule of 72 and give me the doubling time at 4%, 7%, and 12%." Then do the detective work: verify each with the exact formula t = ln(2) ÷ ln(1 + r), compute the rule's error at each rate, and find a rate where the rule breaks badly. Write one paragraph on *why* a rule of thumb works at 7% and fails at 40% — and why bankers love rules you don't check. AI chatbots quote the Rule of 72 fluently and never mention its error bars — in this chapter, *you* are the auditor.

⚑ PROJECT MILESTONE

M1: Price your future

The $2,800 Month starts now (full brief: §6.8). From your $2,800 monthly take-home, commit a savings number — start with 10% ($280). Project it: $280 a month at 7% for 40 years ≈ **$734,948**. Write one sentence on what surprised you, and one on what would make you raise it to 15%.

Practice Problems 6.1

1. (Site warm-up) Calculate the final balance of $10,000 at 6% compounded annually for 12 years.
2. $3,000 at 5% compounded quarterly for 10 years. Find the final balance.
3. $2,000 at 8% for 5 years: compute the simple-interest total and the annually-compounded total. How big is the gap?
4. Use the Rule of 72 to estimate the doubling time at 6%. Then compute the exact doubling time with t = ln(2) ÷ ln(1.06). How far off was the rule?
5. $5,000 at 6% for 10 years: compare annual compounding vs. monthly compounding. Which wins, and by how much?

6.2 Depreciation — The New-Car Smell Costs $10,800

THE DECISION

New car or 3-year-old car?

Same model, same dealership lot:

- **New:** $28,000, zero miles, that smell.
- **3 years old:** 36,000 miles, one previous owner, **$17,200**.

Cars lose value following the same exponential decay you graphed in §3.7 — roughly 15% a year. At that rate, what *should* the 3-year-old cost? Is $17,200 fair, a steal, or a trap — and who already paid the $10,800 difference? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** After one year at 15% depreciation, the $28,000 car is worth 28,000 × 0.85. After two years, multiply by 0.85 again. What does "15% a year" do *three* times?
- **Hint 2:** Compare your result to the $17,200 sticker. If your number is lower, the used car is overpriced; if higher, it's a deal. Which is it?
- **Hint 3:** The $10,800 gap didn't vanish — someone paid it. Who, and what did they buy with it besides transportation?

Declining-Balance Depreciation

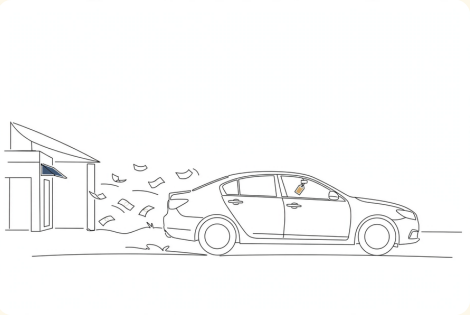
Most cars (and phones, and laptops) depreciate like compound interest in reverse: **V = P(1 − r)^t**, where r is the annual depreciation rate. The dollar losses shrink each year because each year's 15% comes off a smaller value — the cruelest losses happen first. Year 1 on the $28,000 car: −$4,200. Year 5: the same 15% is only −$2,192.

STORY

The Drive-Off-the-Lot Drop.

A buyer pays $28,000 on Monday

and totals the car on Friday. Insurance pays market value: ≈ $26,600. Five days, $1,400 gone — but the 15%-a-year curve explains only about $60 of it. The rest is the market's instant "used-car" haircut: the moment you drive off the lot, dealers reprice the car as used, and no formula governs that first cut. Depreciation models describe the slow slide *after* — don't confuse the two. *What did the buyer actually purchase for that $1,400?*



Straight-Line Depreciation

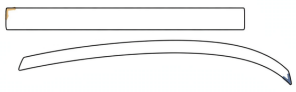
Businesses often use a simpler ruler: **straight-line** depreciation spreads the loss evenly — (cost − salvage value) ÷ useful life. A $20,000 delivery van with a 5-year life and $2,000 salvage value loses ($20,000 − $2,000) ÷ 5 = **$3,600 a year**, every year, on the books. It's less realistic than declining-balance and much easier to audit — which is why accountants love it.

STORY

The Accountant's Two Rulers.

A bakery buys a $12,000 oven

(10-year life, $2,000 salvage). Straight-line: $1,000/year, neat as a ledger. But the resale market says ovens lose 20% a year — declining-balance: $2,400 gone in year 1. The tax return uses one ruler; reality uses the other. *Which ruler would you trust when deciding to sell the oven in year 3?*



When Depreciation Is Your Friend

Depreciation is a loss only if *you* held the asset while it fell. Buy *after* the steep years and the same curve pays you: the 3-year-old car lost its most expensive years on someone else's watch. Businesses get a second gift — depreciation as a tax deduction, a paper loss that lowers taxable income while the machine still works.

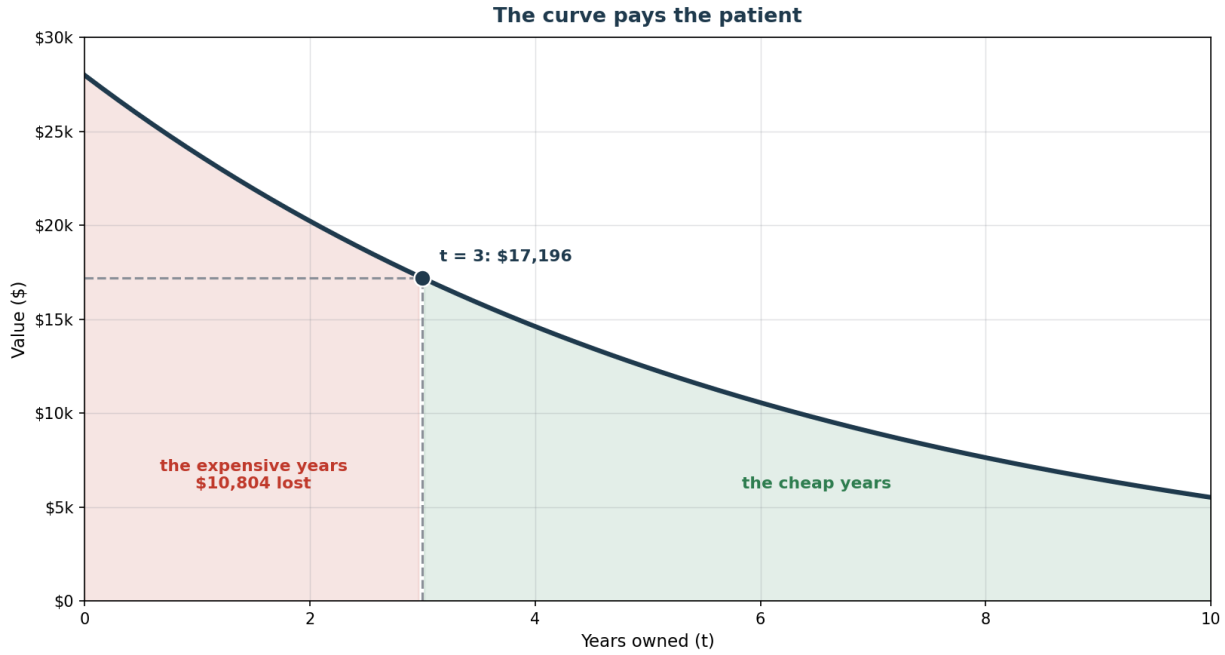
STORY

The Landlord's Refrigerator.

A landlord buys a $1,200 fridge

for a rental, depreciates it over 5 years ($240/year straight-line), and deducts it against rental income — while the fridge keeps humming for 12 years. The math says it "lost value"; the tenants say it keeps the milk cold. *When is a loss on paper a win in the wallet?*

Fig. 6.2



The curve's two halves.

DECIDE — THE VERDICT

New car or 3-year-old car?

Back to the question that opened this section — now with the math to settle it.

Solution. V(3) = 28,000 × 0.85^3 = 28,000 × 0.614125 ≈ **$17,196**. The $17,200 sticker is essentially fair — priced right on the curve, not a steal, not a trap.

The decision: buy the 3-year-old. The first owner paid $10,804 for three years of newness — about $300 a month for the smell. Unless those three newest years matter to you (warranty, safety tech, reliability for a long commute), you're buying the same transportation at a 39% discount. Depreciation is a tax on impatience; let someone else pay it.

AI AI Learning Lab 6.2

AI Level 1 — Understand.

Explain depreciation as if I'm 15, using a melting ice cube. Then explain the difference between declining-balance and straight-line with one tiny example each, one at a time. Check that I can pick the right method for a car vs. a tax return before moving on.

AI Level 2 — Apply.

Give me 5 depreciation problems mixing both methods. Hide the answers. I'll solve each showing the formula and every step; then check my work and flag exactly where I went wrong if I slipped.

AI Level 3 — Matter.

The 3-year-old car is $10,804 cheaper because someone else absorbed the steepest years. Discuss with me: WHY does the market punish the first owner so hard — is it rational (wear, risk) or psychological (the "used" stigma)? Who benefits from our obsession with "new" — dealers? advertisers? lenders? And what are the implications for a family that can barely afford the $17,200 car — is buying used a smart move or a forced one? End by asking me what I'd check on a used car before trusting the curve.

Level 4 — Decide. *If your laptop were worth $650 today, would you sell?*

Example decision problem. Your laptop cost $1,200 new and loses 25% of its value each year (declining-balance). A buyer offers $650 today (end of year 2). If you keep it one more year, what will it be worth? Sell now or keep it?

AI Level 4 — Decide.

My $1,200 laptop depreciates 25% a year. A buyer offers $650 today (end of year 2). Act as my decision coach: make me compute its current curve value and its value one year from now myself, then ask me what "keeping it" is really worth to me (a working laptop vs. cash). Make me state my final choice with reasons — don't decide for me.

Check your verdict: curve value now = 1200 × 0.75^2 = $675 — the $650 offer is fair (a $25 haircut for a fast sale). One more year: 1200 × 0.75^3 ≈ $506. Keeping it "costs" about $144–169 in lost resale. Sell if you don't need it; keep it only if a working laptop is worth more to you than ~$650 cash.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.2.

Ask your AI: "Build a 5-year depreciation schedule for a $25,000 car at 15% declining-balance, and tell me in which year its value drops below half." Then do the detective work: recompute every year yourself, find the half-value year with logarithms (0.85^t < 0.5), and check the AI's answer. If it rounded its way into the wrong year, write one paragraph on how a $500 rounding error compounds. AI chatbots love schedules and hate checking them — in this chapter, *you* are the auditor.

⚑ PROJECT MILESTONE

M2: Choose your car

The $2,800 Month continues. Three options for your transportation: new ($28,000), the 3-year-old ($17,200), or the bus ($80/month pass). Price each over 3 years using depreciation for the cars. Which one survives your budget? Write the decision in two sentences with numbers.

Practice Problems 6.2

1. A $24,000 car depreciates 12% a year (declining-balance). What is it worth after 4 years?

2. A $15,000 machine has a 6-year useful life and zero salvage value. Find the straight-line depreciation per year and its book value after 4 years.
3. Which loses more value in year 1: a $30,000 car at 15% declining-balance, or $30,000 of equipment on 10-year straight-line? By how much?
4. A $900 phone loses 30% of its value each year. What is it worth after 2 years?
5. A new SUV costs $32,000; the 2-year-old version costs $23,120. At 15% annual depreciation, is the used price fair? How much do you save buying used?

6.3 Loans & Monthly Payments — The $188 Question

THE DECISION

36 months or 60 months?

You need to borrow **$18,000** for a car at **8% APR**. The bank offers two terms:

- **36 months:** higher payment, done sooner.
- **60 months:** lower payment, done later.

The monthly payments differ by about $199. The total interest differs by about $1,593. One choice costs you $1,593; the other costs you $199 a month of breathing room. Which is the right call *for you* — and what single number decides it? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Monthly payment = P × r ÷ (1 − (1 + r)^(−n)), with r = 0.08 ÷ 12 and n = months. Compute both payments. (Yes, by hand — that's the point.)
- **Hint 2:** Multiply each payment by its number of months and subtract $18,000. That's the interest — the price of borrowing. Which term's price is higher, and by how much?
- **Hint 3:** Now forget the bank and look at your life: after rent, food, and gas, how many dollars are actually left each month? The right term is the one your budget can survive for its whole life.

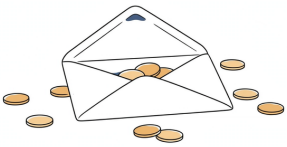
How Monthly Payments Work

Every payment splits into **interest** (the bank's cut, charged on the remaining balance) and **principal** (your cut, shrinking the debt). Early on, the balance is big, so interest eats most of the payment; late in the loan, it's nearly all principal. This split schedule is called **amortization** — and it explains why the first year of a loan feels like running in place.

STORY

The First Payment's Secret.

Borrow $7,500 at 9.5% for 4 years (a real bank offer): the payment is **$188.42** a month. Month one's interest: 7,500 × 0.095 ÷ 12 ≈ **$59.38** — nearly a third of the payment buys nothing but time. Principal: $129.04. *If you could make one extra $129 payment in month one, what would it buy you?*



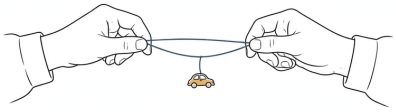
The TVM Idea

Behind every loan calculator sits one idea: the **time value of money** — a dollar today beats a dollar next year, because today's dollar can earn interest. The TVM calculator just enforces fairness: the lender gives you $18,000 *now*, so your stream of future payments must be worth $18,000 *now* at the agreed rate. The payment formula is that fairness, solved for the monthly number. You don't need the calculator's buttons; you need its logic: **every loan is a trade of timing, priced by r.**

STORY

The Finance Office.

At the dealership, the finance manager never asks "what price works for you?" — she asks "what *monthly payment* works for you?" Then she stretches the term until the payment fits, and the price quietly grows. A $24,000 car at 12% for 72 months is $469 a month — sounds fine, costs $33,783. *What question should you ask before you answer hers?*



Term Length: The Real Trade-off

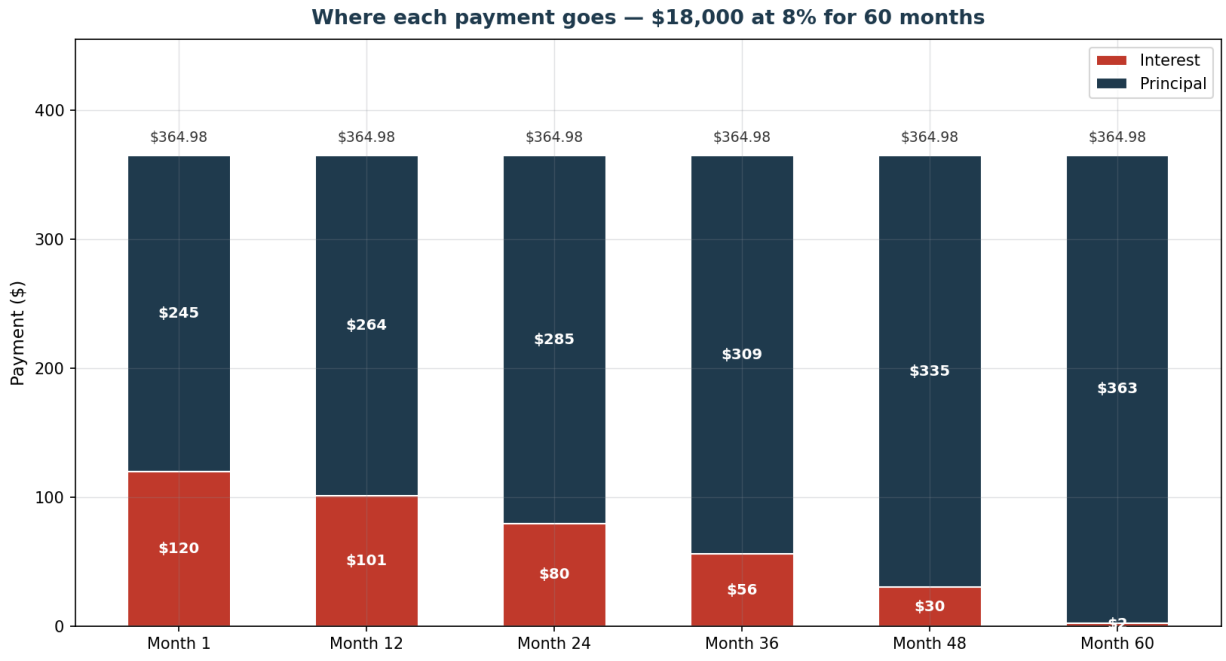
Shorter term → bigger payment, less interest, freedom sooner. Longer term → smaller payment, more interest, more risk (the car may die before the loan does). Neither is "correct" — the math prices both, and your budget picks. The trap is choosing the term by the payment alone and never asking the interest question.

STORY

The Two Neighbors.

Ana takes the 36-month loan: $564.05 a month, $2,306 in total interest, free and clear in 3 years. Ben takes 60 months: $364.98 a month, $3,899 in interest, paying in year 5 on a car worth half. Ben's months were easier; Ana's total was $1,593 cheaper. *Five years from now, who sleeps better — and who drove the nicer-feeling months?*

Fig. 6.3



The bank eats first.

DECIDE — THE VERDICT

36 months or 60 months?

Back to the question that opened this section — now with the math to settle it.

Solution. r = 0.08 ÷ 12 ≈ 0.006667. 36 months: PMT = 18,000 × 0.006667 ÷ (1 − 1.006667^(−36)) ≈ **$564.05**/month. Total = 564.05 × 36 ≈ **$20,306**; interest ≈ **$2,306**. 60 months: PMT = 18,000 × 0.006667 ÷ (1 − 1.006667^(−60)) ≈ **$364.98**/month. Total ≈ **$21,899**; interest ≈ **$3,899**. The 60-month term costs **$1,593** more in interest for $199/month of relief.

The decision: it depends on the number from Hint 3 — your free monthly cash. If $564 fits with room to spare, take 36 months and keep the $1,593. If it doesn't, take 60 months *honestly*: you're buying $199 of monthly breathing room for $1,593, and you should kill the loan early with any windfall. The wrong answer is the one you chose by payment alone.

AI AI Learning Lab 6.3

AI Level 1 — Understand.

Explain loan amortization as if I'm 15, using a melting ice cube that someone keeps pouring water onto (interest) while I scoop some off (payment). Then walk me through one payment split — interest vs. principal — with the $7,500 example, one step at a time. Check my understanding before moving on.

AI Level 2 — Apply.

Give me 5 loan problems: compute the monthly payment, then the total interest, for different principals, rates, and terms. Hide the answers. I'll solve each showing r, n, and the formula; then check my work step by step.

AI Level 3 — Matter.

Ben chose 60 months for easier months and paid $1,593 extra — and the dealership *wants* you to choose by monthly payment. Discuss with me: WHY is "what monthly payment works for you?" such an effective sales question — what does it hide, and who profits when buyers never ask about total interest? What are the implications for someone with shaky income — is the longer term a trap or a lifeline? End by asking me to find one real loan ad and compute the total interest the ad doesn't mention.

Level 4 — Decide. *If you had to borrow $5,000, which term?*

> **Example decision problem.** You must borrow $5,000 at 12% APR for an emergency car repair. The lender offers 24 months or 36 months. Which term do you take, and what does each cost you in total interest?

AI Level 4 — Decide.

I must borrow $5,000 at 12% APR. Options: 24 months or 36 months. Act as my decision coach: make me compute both monthly payments and both total-interest numbers myself, then ask me whether the $69/month savings is worth the extra interest given that this is emergency debt I want gone. Make me state my final choice with reasons — don't decide for me.

Check your verdict: 24 mo → $235.37/month, total ≈ $5,649, interest ≈ $649. 36 mo → $166.07/month, total ≈ $5,979, interest ≈ $979. The longer term saves ≈ $69/month but costs ≈ $330 more in interest. For emergency debt you want dead fast: take 24 months if your budget survives $235 — every extra month of emergency debt is risk.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.3.

Ask your AI: "A store offers 0% APR for 12 months on a $3,000 TV, then 24% after. If I pay $100/month, what's my balance at month 12?" Then do the detective work: compute it yourself *including* the deferred interest most 0% offers charge retroactively (roughly $805 on the full $3,000), and compare with the AI's answer. If the AI forgot the deferred interest, write one paragraph on how "0%" cost $805 — and who designed it that way. AI chatbots read the headline rate and skip paragraph 14 — in this chapter, *you* read paragraph 14.

⚑ PROJECT MILESTONE

M3: Finance your car

Take your M2 car choice into the loan office. Price it at 8% APR for both 36 and 60 months: monthly payment, total interest. Which term fits inside your $2,800 budget with the savings from M1 intact? Decide in two sentences with numbers.

Practice Problems 6.3

1. Compute the monthly payment, total repayment, and interest on $12,000 at 6% APR for 5 years (60 payments).

2. Same $12,000 at 6%, but 3 years (36 payments). How much interest does the shorter term save versus problem 1?
3. (Site warm-up) A used-car loan: borrow $7,500 at 9.5% APR for 4 years, monthly payments. Find the payment, total repayment, and interest.
4. On the $18,000/8%/60-month loan ($364.98/month), you pay $400/month instead. About how many months early do you finish, and roughly how much interest do you save?
5. A store offers 0% for 12 months on $3,000, then 24% APR — with deferred interest on the full $3,000 if any balance remains. (a) If you pay it off in 12 months, what is your total cost? (b) If you pay $100/month, what is your balance at month 12 including deferred interest?

6.4 Income & Paychecks — What "$18 an Hour" Really Pays

THE DECISION

Is $18 an hour enough?

A warehouse offers you **$18/hour, 40 hours a week**, paid biweekly. Before you accept, you need three numbers: the annual salary this *really* is, the gross amount of each biweekly paycheck, and — the one nobody quotes — what actually lands in your account (that's §6.5). For now: what does $18 an hour buy you in a year, and is the offer what it sounds like? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** A full-time year is 40 hours × 52 weeks = 2,080 hours. Multiply. That's the gross annual — before a single deduction.
- **Hint 2:** "Biweekly" means 26 paychecks a year (not 24 — that's semimonthly). Divide the annual by 26.
- **Hint 3:** Twice a year, a biweekly worker gets a month with *three* paychecks. Sketch which months those are — that third check is a savings opportunity most people spend by accident.

Hourly, Salary, and Overtime

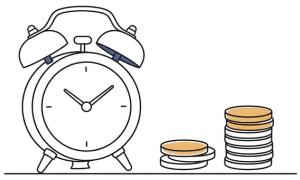
Hourly pay × hours = gross. **Salary** quotes the annual number directly. **Overtime** (usually over 40 hours/week) pays time-and-a-half: 1.5 × the hourly rate. One 42-hour week at $18: 40 × 18 + 2 × 27 = $720 + $54 = **$774**. Two overtime hours bought $54 — a 50% premium for the week's least pleasant hours.

STORY

The Overtime Week.

During holiday rush, Luis works 48 hours

at $18/hr: 40 × 18 + 8 × 27 = $720 + $216 = **$936**. His buddy on salary works the same 48 hours for the same flat pay. *Who got the better deal that week — and who gets the better deal in a quiet week?*



Pay Periods

How often you're paid changes the *size* of each check, not the annual total: weekly (52), biweekly (26), semimonthly (24), monthly (12). On $37,440: biweekly = **$1,440**; semimonthly = **$1,560**. The semimonthly check looks bigger — but there are fewer of them. And the biweekly calendar hides a gift: 26 paydays means two months a year with three checks. Budget on two; bank the third.

STORY

The Third Paycheck.

Priya budgets her $1,440 biweekly checks

as "$2,880 a month" — and in March and September a third check lands: an extra $1,440 she didn't plan for. Year one she spent both on a vacation and a couch. Year two she sent both to her Roth IRA: $2,880 at 7% for 40 years ≈ **$43,100**. *Which March would you rather have?*

Commission and Piecework

Some jobs replace wages with **commission** (a percent of sales) or **piecework** (pay per unit). $2,000/month base + 5% on $40,000 of sales = $2,000 + $2,000 = **$4,000** that month — and $2,000

in a dead month. The math is easy; the *risk* is the lesson: variable pay means variable budgets, which is why §6.6 budgets on the worst month, not the best.

STORY

The Car Salesperson.

Ellis's lot pays $2,500 base + 4% of

sales. In June he sells $180,000 of cars: 2,500 + 7,200 = **$9,700**. In February: $2,500 plus almost nothing. His budget is built on $2,500 — everything above is savings and fun. *Could you live on your worst month?*

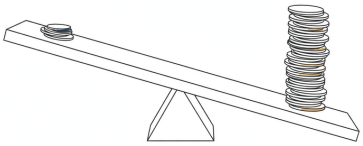


Fig. 6.4

Same year, different slices.

Annual	Weekly (×52)	Biweekly (×26)	Semimonthly (×24)	Monthly (×12)
$37,440	$720	$1,440	$1,560	$3,120

Same year, different slices.

DECIDE — THE VERDICT

Is $18 an hour enough?

Back to the question that opened this section — now with the math to settle it.

Solution. Annual gross = 18 × 2,080 = **$37,440**. Biweekly gross = 37,440 ÷ 26 = **$1,440**. Monthly equivalent = 37,440 ÷ 12 = **$3,120** — but that's *gross*, and §6.5 will take its bite (roughly 23% → ≈ $2,382/month spendable).

The decision: $18/hour is a **$37,440/year** offer wearing a small number's clothes — respectable for a first job, but the biweekly $1,440 is not spendable cash. Accept it with eyes open: the real question isn't "is $18 enough?" but "is ≈ $2,382 a month enough?" — and that question needs the next section's math.

AI AI Learning Lab 6.4

AI Level 1 — Understand.

Explain hourly vs. salary pay as if I'm 15, using a lemonade stand (paid per cup) vs. a weekly allowance. Then explain the four pay periods and why biweekly gives 26 checks but semimonthly gives 24, one at a time. Check my understanding before moving on.

AI Level 2 — Apply.

Give me 5 pay problems: hourly-to-annual, overtime weeks, and per-check amounts across pay periods. Hide the answers. I'll solve each showing every step; then check my work.

AI Level 3 — Matter.

Priya's third paychecks became $43,100 of retirement money because she budgeted on two checks and banked the third. Discuss with me: WHY do most people spend windfalls — is it innumeracy, or does money that arrives "extra" feel psychologically different? Who benefits when workers budget on the rosy number — lenders? landlords? What are the implications for someone paid monthly with no third-check gift — how do *they* manufacture a windfall? End by asking me where my next "extra" check would go.

Level 4 — Decide. *Hourly or salary — which offer?*

Example decision problem. Offer A: $20/hour, 35 hours/week, no overtime. Offer B: $38,000/year salary, occasional late nights unpaid. Which pays more gross per year?

AI Level 4 — Decide.

Offer A: $20/hour × 35 hrs/week. Offer B: $38,000 salary. Act as my decision coach: make me compute A's annual gross myself, compare it to $38,000, then ask me to price the unpaid late nights (how many hours before B's effective hourly rate drops below $20?). Make me state my final choice with reasons — don't decide for me.

Check your verdict: A = 20 × 35 × 52 = $36,400 — B pays $1,600 more gross. But B's edge melts fast: 100 unpaid extra hours a year drops B to 38,000 ÷ 1,920 ≈ $19.79/hour — below A's $20. The decision is a bet on how "occasional" those late nights really are.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.4.

Ask your AI: "At $22/hour, what is my gross per paycheck under weekly, biweekly, semimonthly, and monthly pay?" Then do the detective work: compute all four yourself (annual = 22 × 2,080 = $45,760) and check each. If the AI divided by 24 for "biweekly," write one paragraph on why the 26-vs-24 confusion is the most expensive calendar mistake in payroll — and who it hurts. AI chatbots are fluent at pay periods and sloppy at counting them — in this chapter, *you* count.

⚑ PROJECT MILESTONE

M4: Reverse-engineer your paycheck

The $2,800 Month needs a job behind it. Work backwards: ≈ $2,800/month take-home ≈ $33,600/year net. At ~23% total deductions, that's ≈ **$43,636** gross — about **$21/hour** full-time. Write your project's job description in one sentence: "$21/hour, 40 hrs/week, biweekly."

Practice Problems 6.4

1. At $16.50/hour, 40 hours/week: find the annual gross and the biweekly gross paycheck.
2. A 45-hour week at $20/hour (time-and-a-half over 40). What is the week's gross?
3. On a $39,000 salary: which paycheck is bigger, semimonthly or biweekly? By how much?
4. A $52,000 salary: find the monthly and weekly gross amounts.

5. A job pays $2,000/month base plus 5% commission. In a month with $40,000 in sales, what is the gross? In a month with $8,000?

6.5 Deductions & Net Pay — The $1,099 Paycheck

THE DECISION

Can you afford the $950 apartment?

The warehouse job is yours: **$1,440 gross every two weeks**. You found an apartment for **$950/month** — and the 30% rule says housing should stay under 30% of your pay. 950 ÷ 1,440 ≈ 66%… wait, that's per *paycheck*, and rent is monthly. So what fraction is it really — and of *which* pay? Because $1,440 gross is not $1,440 spendable: FICA, federal withholding, and Georgia state tax all take their cut first. What's your actual take-home — and does the apartment pass the 30% test on *that* number? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Start with FICA: Social Security 6.2% + Medicare 1.45% = 7.65% of gross. Compute 1,440 × 0.0765.
- **Hint 2:** Subtract estimated withholding — say 11% federal and 5% Georgia state. What's left of the $1,440?
- **Hint 3:** Convert take-home to monthly (× 26 ÷ 12, not × 2 — why?). Then compute 950 ÷ monthly take-home. Over 30%? The rule just spoke.

FICA: Social Security & Medicare

FICA takes **7.65%** of every gross dollar: 6.2% Social Security + 1.45% Medicare. No deductions, no exemptions, no negotiation — on $1,440, that's **$110.16** gone before anything else. It funds retirees' checks today and your Medicare at 65; whether it funds *your* retirement is a political question, not a math one.

STORY

The Paystub's First Bite.

Jamal's first paycheck ever:

$720 gross for a half-month of part-time work. FICA: $55.08. "I didn't agree to this!" he tells his manager. She laughs: "Nobody agrees to FICA. It's the cover charge for having a job." *Is there any legal way to earn wages and skip FICA?*

Income Tax Withholding

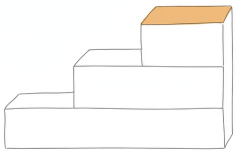
Employers withhold estimated **federal** and **state** income tax from each check — a prepayment, settled at tax time. Federal withholding on $1,440 ≈ 11% ≈ **$158.40**; Georgia ≈ 5% ≈ **$72.00**. And here's the myth to kill: **tax brackets are marginal** — a raise that "pushes you into a higher bracket" taxes only the dollars *above* the line at the higher rate. Nobody's take-home ever *falls* because of a raise.

STORY

The Bracket Myth.

Coworker: "Don't work overtime — it'll

push you into the 22% bracket and you'll take home *less*!" The math: say 12% applies up to $47,000 and 22% above. On $50,000: 47,000 × 0.12



- 3,000 × 0.22 = $5,640 + $660 = **$6,300**. The myth's version (all of it at 22%): $11,000. The myth costs whoever believes it $4,700 of courage. *Why does this myth survive — who repeats it, and why?*

Net Pay

Net pay = gross − all deductions. Our $1,440: −$110.16 (FICA) −$158.40 (federal) −$72.00 (state) = **$1,099.44** take-home. The offer letter said $37,440/year; your life runs on ≈ $28,586/year. Every budget, every rent test, every "can I afford it" runs on the *net* — which is why this section, not §6.4, owns your decisions.

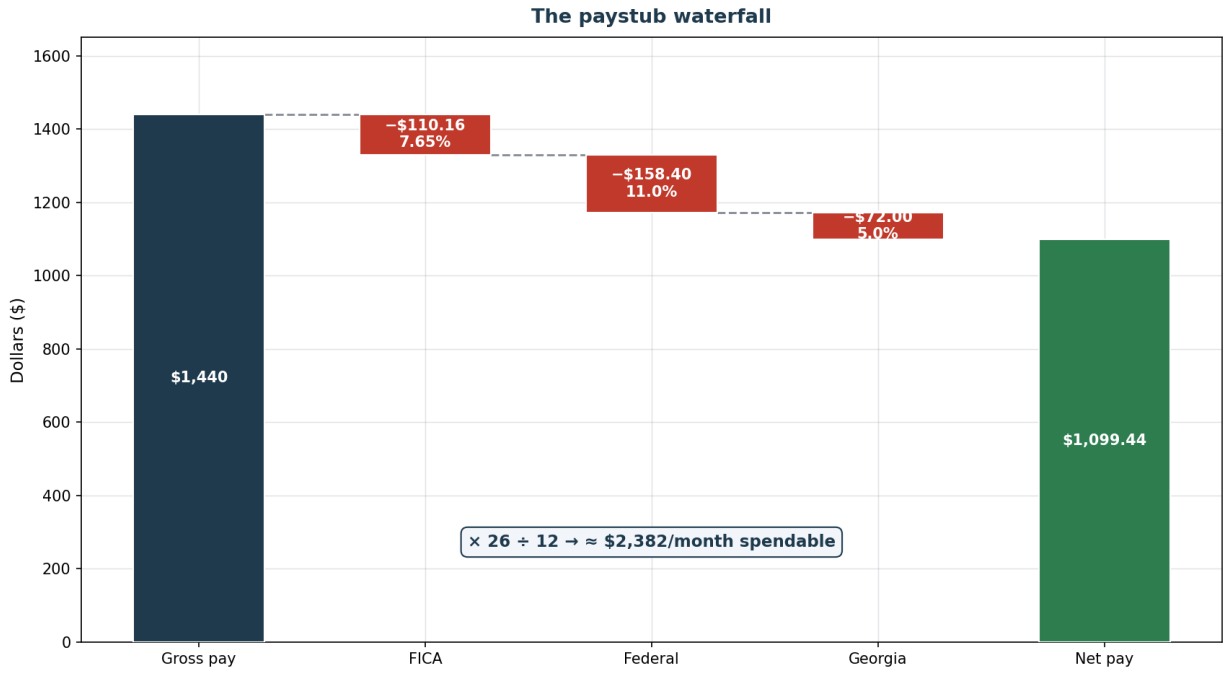
STORY

The Offer Letter.

Two offers: $19/hour with no benefits

($39,520 gross) vs. $17.50/hour ($36,400) with health insurance worth $200/month ($2,400/year). Gross says offer one wins by $3,120. But if you'd buy insurance anyway, offer two's real value is $38,800 — and the gap shrinks to $720. *Which number would you compare — and what does "benefits" have to be worth to flip your choice?*

Fig. 6.5



Your salary is a rumor; your net pay is the news.

DECIDE — THE VERDICT

Can you afford the $950 apartment?

Back to the question that opened this section — now with the math to settle it.

Solution. Net per paycheck = 1,440 − 110.16 − 158.40 − 72.00 = **$1,099.44**. Monthly take-home = 1,099.44 × 26 ÷ 12 ≈ **$2,382**. Rent test: 950 ÷ 2,382 ≈ **39.9%** — well over the 30% line ($715).

The decision: not on this paycheck alone. At 39.9%, one car repair or medical bill turns rent into a crisis — the 30% rule exists because life has variance (§5.6 sends its regards). Options: a roommate (rent → $475, test → 19.9% ✓), a cheaper place, or a higher-paying job. Signing at 39.9% isn't impossible — it's *fragile*, and now you know exactly how fragile.

AI AI Learning Lab 6.5

AI Level 1 — Understand.

Explain deductions as if I'm 15, using a pizza: gross is the whole pizza, and FICA, federal, and state each take slices before I eat. Then walk me through the $1,440 paystub one deduction at a time, and check that I can compute net pay on a new number before moving on.

AI Level 2 — Apply.

Give me 5 paystub problems with different gross amounts and pay periods. Hide the answers. I'll compute FICA, withholding, and net pay for each, showing every step; then check my work.

AI Level 3 — Matter.

The bracket myth scared someone out of $4,700 of overtime courage — and nobody's take-home ever falls from a raise. Discuss with me: WHY does this myth survive for generations — who repeats it, and what does its survival say about how tax is taught (or not taught)? Who benefits when workers fear earning more — and who benefits when they understand marginal rates? End by asking me to explain marginal brackets to a coworker in two sentences.

Level 4 — Decide. *Which offer — $19/hour bare or $17.50 with benefits?*

Example decision problem. Offer A: $19/hour, 40 hrs/week, no benefits. Offer B: $17.50/hour, 40 hrs/week, health insurance worth $200/month. You'd buy insurance on your own if you took A. Which offer is really better?

AI Level 4 — Decide.

Offer A: $19/hr, no benefits. Offer B: $17.50/hr + health insurance worth $200/month. I'd buy insurance anyway under A. Act as my decision coach: make me compute both annual grosses myself, add the insurance value to B, then ask me what could still flip the choice (a sick year? a healthy year? a spouse's plan?). Make me state my final choice with reasons — don't decide for me.

Check your verdict: A = 19 × 2,080 = $39,520. B = 17.50 × 2,080 = $36,400 + $2,400 insurance = $38,800 effective. A leads by $720 in cash — but one bad health year can erase that ten times over. If you're young, healthy, and disciplined, take A and buy a cheap plan; if anyone depends on you, B's certainty is worth more than $720. The decision prices your risk, not just your wage.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.5.

Ask your AI: "My coworker says a raise pushed her into a higher tax bracket so her take-home went down. Is that possible?" Then do the detective work: build the marginal math yourself on a $3,000 raise crossing a bracket line, and prove take-home *rises*. If the AI hedges, agrees with the myth, or invents fake bracket numbers, write one paragraph on exactly where it went wrong — and why a confident wrong answer about taxes is more dangerous than no answer. AI chatbots repeat folk tax wisdom fluently — in this chapter, *you* do the brackets.

⚑ PROJECT MILESTONE

M5: Verify your scenario

The $2,800 Month assumes ≈ $2,800 take-home. Prove it: your project job pays $1,678 biweekly gross. Stack the deductions — FICA, 11% federal, 5% state — and convert to monthly. You should land near **$2,776** (the scenario rounds to $2,800). If your number is far off, find your arithmetic error before touching the budget.

Practice Problems 6.5

1. Biweekly gross $1,600: compute FICA (7.65%), federal withholding (12%), state (5%), and net pay.

2. Weekly gross $950: compute FICA, federal (10%), state (5%), and net pay.
3. Say the 12% bracket covers income up to $47,000 and 22% applies above. Compute the federal tax on $50,000 of taxable income — then compute what the "whole thing at 22%" myth claims. What's the difference?
4. Monthly gross $4,200: compute FICA, federal (12%), state (5%), and net pay.
5. Which costs the employer more per year: a $25/hour wage, or $23/hour plus $300/month health insurance? (Assume 2,080 hours.) By how much?

6.6 Budgeting — Where Does the $2,800 Go?

THE DECISION

Does your month balance?

Your take-home: **$2,800/month**. Your life:

- Needs: rent $950, car payment $365, car insurance $140, phone $45, groceries $320, gas $110 → **$1,930**
- Wants: dining out $120, subscriptions $30 → **$150**
- Savings: **$280** (10% — pay yourself first)

Total out: $2,360. Leftover: **$440**. It balances — with room to spare. But the 50/30/20 rule says needs should stay near 50% ($1,400), and yours are at 68.9%. So: does this budget *work*, or does it just *add up*? And what's the single line item you'd change first? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Check the arithmetic: 1,930 + 150 + 280 = ? And 2,800 − that = ? Confirm the $440 buffer is real.
- **Hint 2:** Convert each bucket to a percent of $2,800. Which bucket breaks the 50/30/20 frame, and by how many dollars?
- **Hint 3:** The car payment is $365 (the 60-month loan from §6.3). What would it be on the cheaper $12,000 car from §6.2's logic — and what does that do to the buffer?

The 50/30/20 Rule

50% needs (housing, food, transport, minimums), **30% wants**, **20% savings and extra debt payments**. It's a diagnostic, not a law — but when needs hit 69%, the diagnosis is clear: *fragile*. One $600 car repair eats the buffer and then some, and the "then some" goes on a credit card at 24% (§6.7 sends its regards). A budget that balances is arithmetic; a budget that survives is engineering.

STORY

The Envelope Grandmother.

Rosa's grandmother ran her household on cash envelopes: RENT, FOOD, LIGHT BILL, CHURCH, SAVINGS. When an envelope was empty, spending stopped — no overdraft, no debate. The 50/30/20 rule is her envelopes with percentages. *What would your grandmother say about a budget with no savings envelope?*

Fixed vs. Variable — and Pay Yourself First

Fixed expenses barely move (rent, loan payments, insurance). **Variable** ones do (groceries, gas, dining). Two disciplines: list *every* annual bill as a monthly twelfth — a $600 registration is $50/month, not a "surprise" — and **pay yourself first**: the $280 savings leaves on payday, like a bill. "Save what's left" saves what's left — usually nothing.

STORY

The Leaky Bucket.

Devon "couldn't save." Audit found: four streaming services ($52), a gym he visited twice ($40), app subscriptions ($23) — $115/month of variable spending he couldn't name. He cut $80, auto-transferred it to savings: $960/year, then ≈ $13,800 in 10 years at 7%. *What's leaking from your bucket right now?*

When It Doesn't Balance

Three levers, in order: **cut** (cheaper car, roommate), **earn** (overtime at 1.5× — §6.4), **reframe** (the goal, not the math, was wrong). Never the fourth lever — borrowing to cover the gap — because that's not a lever, it's a trapdoor into §6.7.

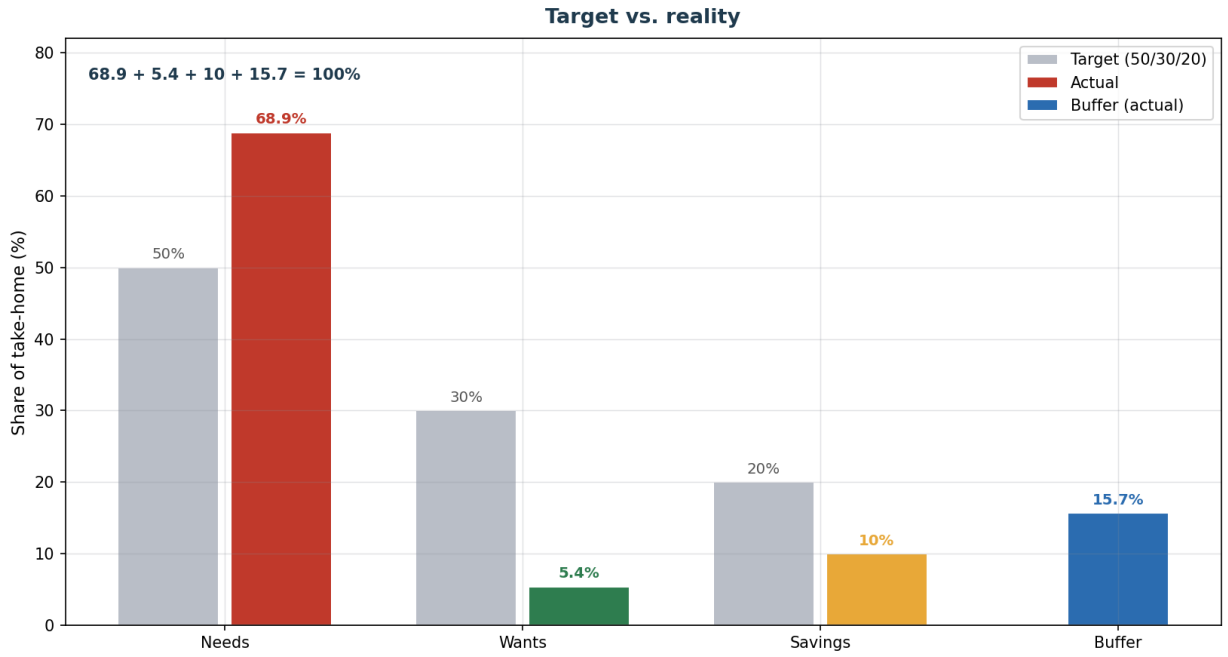
STORY

The $440 Question.

Your budget balances with a $440 buffer.

Your friend's doesn't — she's $120 short every month and covers it with a credit card. In a year she's $1,440 deeper at 24%: ≈ $1,642 owed. Same income, different arithmetic. *At what point does "tight month" become "debt spiral" — and how would you know you're crossing it?*

Fig. 6.6



It adds up — and it's still fragile.

DECIDE — THE VERDICT

Does your month balance?

Back to the question that opened this section — now with the math to settle it.

Solution. Arithmetic: 1,930 + 150 + 280 = $2,360; buffer = 2,800 − 2,360 = **$440** ✓. Percents: needs 1,930 ÷ 2,800 ≈ **68.9%**, wants ≈ **5.4%**, savings **10%**, buffer ≈ **15.7%**. The 50% needs-line is missed by $530. The lever: the car. A $12,000 car at 8% for 60 months → ≈ **$243**/month (§6.2–6.3 working together), saving $122/month. Needs → $1,808 (64.6%), buffer → **$562**.

The decision: yes, sign the lease — the month balances — but know it's fragile, and fix the car first. The $440 buffer is one emergency from zero; the $562 buffer is two. Budgeting isn't about the month that adds up; it's about the month that *doesn't*, and whether you survive it.

AI AI Learning Lab 6.6

AI Level 1 — Understand.

Explain the 50/30/20 rule as if I'm 15, using three jars. Then explain fixed vs. variable expenses and "pay yourself first," one at a time, with a tiny example each. Check my understanding before moving on.

AI Level 2 — Apply.

Give me 5 budgeting problems: build a budget from an income and an expense list, test it against 50/30/20, and find the buffer. Hide the answers. I'll solve each showing every bucket; then check my work.

AI Level 3 — Matter.

Your budget balanced at $2,360 but failed the 50% needs test — arithmetic said yes, engineering said fragile. Discuss with me: WHY do budgets that "add up" still collapse — what does the 50/30/20 rule know that raw arithmetic doesn't? Who gets hurt when budgets are built on best-case months — and who profits (hint: §6.7)? What are the implications for someone whose needs are 80% through no fault of their own — is 50/30/20 realistic advice or a luxury metric? End by asking me which of my expenses is fixed-but- negotiable.

Level 4 — Decide. *If your take-home were $2,400, could you afford $800 rent?*

Example decision problem. Take-home $2,400/month. Apartment: $800. Other needs: groceries $280, gas $100, phone $40, insurance $120. Build the budget: test rent against 30%, needs against 50%, and leave 10% savings. Does it work?

AI Level 4 — Decide.

My take-home is $2,400; rent is $800. Other needs: groceries $280, gas $100, phone $40, insurance $120. Act as my decision coach: make me compute the rent percent and the needs percent myself, set aside 10% savings first, then tell me what's left for wants + buffer. Ask me: what breaks first if groceries rise 20%? Make me state my final verdict with reasons — don't decide for me.

Check your verdict: rent = 800 ÷ 2400 ≈ 33.3% — just over the 30% line ($720), close enough to argue about. Needs = 800 + 280 + 100 + 40 + 120 = $1,340 ≈ 55.8% — over 50% but not alarming. Savings (10%) = $240. Left for wants + buffer: 2400 − 1340 − 240 = $820. Verdict: workable — the rent is defensible, the buffer is real, but groceries +20% ($56) eats the buffer's edge first. That's the number to watch.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.6.

Ask your AI: "Build me a perfect $2,800/month budget." Then do the detective work: audit it line by line for what's *missing* — annual bills divided by 12 (registration? gifts? haircuts?), irregular income, the "miscellaneous" that isn't miscellaneous. Add the missing lines yourself and recompute the buffer. If the AI's "perfect" budget forgot $600 of annual costs, write one paragraph on why a budget with no annuals÷12 is a fantasy — and whose fantasy it serves. AI chatbots build beautiful budgets with no birthdays in them — in this chapter, *you* count the birthdays.

⚑ PROJECT MILESTONE

M6: Build the full budget

Assemble your $2,800 Month: every line item in three buckets (needs / wants / savings), one annual÷12 line, the buffer. Test against 50/30/20 and write the one-paragraph verdict: does it balance *and* survive? If not, pull your M2/M3 levers first.

Practice Problems 6.6

1. On $3,000/month net, what are the 50/30/20 dollar amounts? If rent is $1,100, does it fit the needs bucket with $400 left for other needs?

2. Net $2,500; fixed $1,700; variable $500; savings $200. Does the month balance? What is the buffer?
3. A $720 annual insurance bill: what monthly budget line does it become? Why is forgetting it dangerous?
4. Wants are $450 on $2,800 income; needs are $2,000. Which bucket is the real problem, and what does 50/30/20 say to do?
5. "Save what's left": $2,800 income, $2,700 spending → $100 saved. "Pay yourself first": save $300, then spend $2,500. How much more is saved per year with the second method?

6.7 Credit & Credit Scores — The 3-Digit Number That Prices Your Life

THE DECISION

Secured card or debit-only?

You're 18. Two paths:

- **Path A:** Open a **secured credit card** ($500 deposit as collateral), spend a little, pay in full every month, build a score.
- **Path B:** Debit card only. No debt, no score — "if I can't afford it, I don't buy it."

By 23, Path A can hold a score near 720; Path B holds no score at all. Then both of you finance the same $18,000 car for 60 months — one at 8% APR, one at 14%. How many dollars is the score worth over the life of that loan? And what does "no debt" cost the person who chose it? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Compute both monthly payments with r = APR ÷ 12, n = 60. (You've done this in §6.3 — it's the same formula.)
- **Hint 2:** Multiply each payment by 60 and subtract. The difference is the lifetime price of the score gap.
- **Hint 3:** Now price Path A's risk: what happens if the secured-card user carries a balance one month at 24%? (Try it: a $500 balance × 2% monthly = **$10** of interest for one lazy month —

the card's convenience has a price tag.) The score is a tool — when does the tool become the trap?

What Credit Is

Credit is borrowed trust: someone gives you money now because the system believes you'll return it later. It comes as revolving credit (cards — borrow, repay, repeat), installment loans (car, mortgage — fixed payments), and service credit (phone plans, utilities). Debit spends money you have; credit spends money you're *promised* to repay — and the promise is priced.

STORY

The Library Card.

A library card is pure credit: they hand you $30 of books on nothing but your name in the system. Return them and your "limit" grows; lose them and the card dies. A credit score is the library's memory of you, written in three digits, read by every lender you'll ever meet. *What would your library record say about you?*

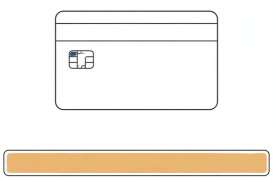
The Score: Five Factors

FICO scores (300–850) weight five things: **payment history 35%** (pay on time — nothing matters more), **amounts owed / utilization 30%** (keep balances under **30%** of limits), **length of history 15%**, **credit mix 10%**, **new inquiries 10%**. Translation for an 18-year-old: get the secured card, put one small recurring bill on it, autopay in full, never exceed $150 of a $500 limit. Time does the rest.

STORY

The 30% Line.

Nina's limit is $500. She charges $480 "for the points" and pays in full — responsible, right? Her score *drops*: utilization hit 96%, and the algorithm read desperation. Next month she charges $120 (24%) — score climbs. Same payer, different signal. *If you needed to spend $400 on a $500 limit, what would you do?*



Good Debt, Bad Debt, and the Minimum-Payment Trap

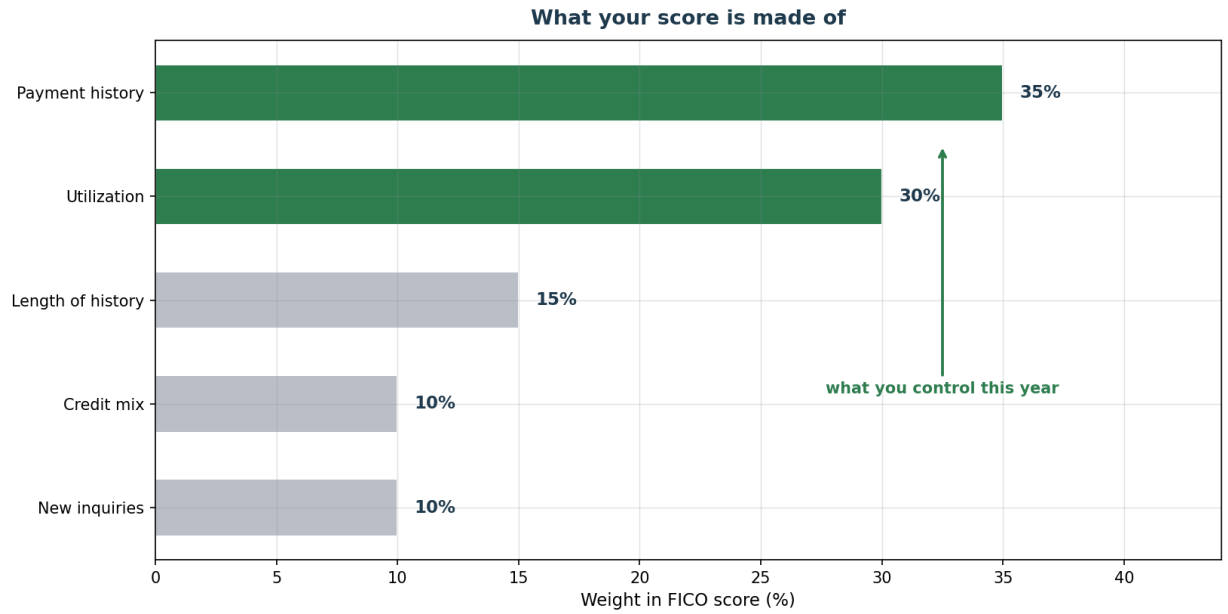
Debt is a tool with a direction. **Good debt** buys something that grows or earns (education, a reasonable mortgage); **bad debt** buys something that shrinks while interest compounds (vacations on a 24% card). And the minimum-payment trap from Chapter 1 is this section's ghost: $1,000 at 24% takes **26 months and ≈ $300 in interest** at $50/month — or about **6 months and ≈ $65** at $200/month. Same debt, same rate; the payment is the decision.

STORY

The $50 Ghost.

Chapter 1's summative essay buried a friend's "$50 is a reasonable plan" under $5,730 of interest. The ghost is back: every minimum payment is a vote to stay in debt longer. *What would you write on a sticky note attached to a minimum-payment bill?*

Fig. 6.7



65% of your score is two habits.

DECIDE — THE VERDICT

Secured card or debit-only?

Back to the question that opened this section — now with the math to settle it.

Solution. 8% path: PMT = 18,000 × (0.08÷12) ÷ (1 − 1.006667^(−60)) ≈ **$364.98**/month. 14% path: r = 0.14÷12 ≈ 0.011667; PMT ≈ **$418.83**/month. Difference: ≈ **$53.85**/month × 60 = ≈ **$3,231** over the loan's life.

The decision: Path A — the secured card — *if* you run it like a debit card with a memory: small charges, autopay in full, utilization under 30%. The 720 score is worth **$3,231** on one car loan alone, and it compounds across every loan you'll ever sign. But the condition is everything: one carried balance at 24% starts the same compounding engine that Chapter 1 warned you about. Build the score; never feed the trap. "No debt" is safe. "Managed credit" is profitable. Choose profitable — carefully.

AI AI Learning Lab 6.7

AI Level 1 — Understand.

Explain a credit score as if I'm 15, using a video-game reputation meter. Then explain the five factors one at a time, with a tiny example for each, and check that I can rank them by weight before moving on.

AI Level 2 — Apply.

Give me 5 credit problems: compute utilization, compare two loan offers by score tier, and price a minimum-payment trap. Hide the answers. I'll solve each showing every step; then check my work.

AI Level 3 — Matter.

The secured card is worth $3,231 on one car loan — but one carried balance at 24% wakes the Chapter 1 ghost. Discuss with me: WHY is credit simultaneously the cheapest money and the most expensive — who designed it that way, and who profits most from the confusion? What are the implications for someone whose family has no credit history at all — is the system a ladder or a gate? Think about who gets a 720 by 23 and who never gets the $500 deposit. End by asking me what my first credit move will be after this chapter.

Level 4 — Decide. *If you owed $1,000 at 24%, how fast would you kill it?*

> **Example decision problem.** You owe $1,000 on a card at 24% APR. You can pay $50/month or $200/month. How long does each take, and what interest does each cost?

AI Level 4 — Decide.

I owe $1,000 at 24% APR. I can pay $50 or $200 a month. Act as my decision coach: make me compute both payoff times myself with n = −ln(1 − r×PV÷PMT) ÷ ln(1+r), then ask me what the $150/month difference in my life would have to fund to justify the slow plan. Make me state my final choice with reasons — don't decide for me.

Check your verdict: r = 0.02/month. $50 → n ≈ 25.8 → 26 months, total ≈ $1,300, interest ≈ $300. $200 → n ≈ 5.3 → ~6 months, total ≈ $1,065, interest ≈ $65. The slow plan costs $225 more and 20 extra months of risk. Kill it fast — every minimum payment is a donation to the bank.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.7.

Ask your AI: "A company promises to raise my credit score 100 points in 30 days for $99. Is it legit?" Then do the detective work: list what *actually* moves a score in 30 days (pay down utilization — the 30% factor; correct errors) vs. what takes years (history length), and check the AI's answer against the five weights. If it endorses the scheme, write one paragraph on the business model of selling hope to people the system already failed — and who reports *them*. AI chatbots will happily summarize a scam's sales pitch — in this chapter, *you* read the weights.

⚑ PROJECT MILESTONE

M7: Write your credit future

You're 18 with no score. Write your 4-year plan to 720: the secured card, the utilization rule, the autopay. Then price it: what does your 720 save you on your M3 car loan versus the 14% no-score rate? One paragraph, with the dollar number.

Practice Problems 6.7

1. Your limit is $2,000 and your balance is $750. What is your utilization? Is it over the 30% line? What balance gets you under it?
2. Which damages a score more: one 30-day-late payment, or one hard inquiry from a card application? Use the factor weights to explain.
3. Compute the monthly payment, total repayment, and interest on $6,000 at 18% APR for 36 months.
4. Classify each as good debt, bad debt, or "it depends" — and explain in one sentence: (a) student loans for a nursing degree, (b) a vacation on a 24% credit card, (c) a mortgage at 7%.
5. On a $500 secured card, you spend $120/month and pay in full. What is your utilization, and what does it build? What if you spent $400?

6.8 The $2,800 Month Project

THE DECISION

Which life does $2,800 buy?

Your scenario: first real job, **$2,800 a month take-home**, first apartment, first everything. Two sketches from last year's class:

- **Sketch A — "The Driver":** the 3-year-old car ($17,200, 60 months at 8% → ≈ $349/month), insurance $140, gas $110. Transport: **$599**.
- **Sketch B — "The Saver":** the $80 bus pass. Transport: **$80**. The $519 difference goes straight to savings.

Same income, same city — $519 a month apart. Sketch A has freedom and a payment book; Sketch B has wealth and a bus schedule. Over the chapter you'll price both, milestone by milestone — savings, car, loan, job, paystub, budget, credit — and at the end you defend your choice against two shocks. Which sketch survives contact with real life? *Hold that question: the build starts now.*

HINTS

Stuck? Work through these in order — do the math yourself.

1. **Start here.** The $519 gap goes to savings every month. Ask what $519 a month becomes in 10 years at 7% (future value of a monthly annuity — §6.1). The bus isn't free; it's *paid in time*.
2. **Find the direction.** List what Sketch A buys besides transport: late-night freedom, job access, dates without a timetable. Now price each in dollars or don't count it. Feelings are real; budgets only read numbers.
3. **Just before you decide.** Run both sketches through the two shocks below (the $600 repair, the $75 rent hike). The sketch that survives both *with savings intact* is the honest answer — even if it isn't the fun one.

The Rules of the Game

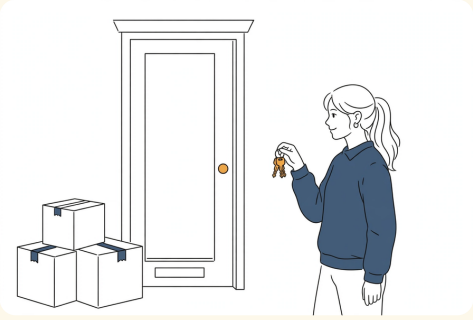
You are 22, employed full-time (≈ $21/hour — you proved that in M4), taking home $2,800 a month. No trust fund, no parental subsidy, no roommate yet (that changes in M6 if the math demands it). Every milestone below is a real decision with your real-looking numbers; the final pitch is you, defending your month against two shocks in three minutes.

STORY

The First Apartment.

Last year, a student ran this project

with Sketch A, signed the lease, bought the car — then the transmission died in month four ($1,900). Her buffer was $210. She finished the semester on a credit card at 24%. "The math was right," she said. "My *shocks* were wrong." *What shock is your budget pretending won't happen?*



The Milestone Roadmap

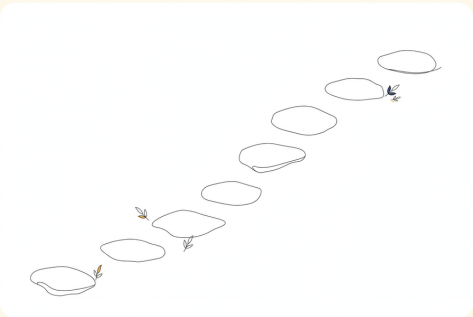
Milestone	After §	Deliverable
M1	6.1	Savings plan: 10% ($280/mo) → 40 yrs @ 7% ≈ **$734,948**
M2	6.2	Car choice: new / 3-yr-old / bus — priced over 3 years
M3	6.3	Loan terms: 36 vs. 60 months on your car, interest priced
M4	6.4	The job behind the money: ≈ $21/hr reverse-engineered
M5	6.5	Paystub proof: $1,678 biweekly → ≈ $2,776/mo net
M6	6.6	Full budget table + 50/30/20 survival verdict
M7	6.7	4-year credit plan + dollars saved vs. the no-score rate
Final	ch. end	One-page budget + 3-minute pitch vs. the two shocks

STORY

The Slow Build.

Nobody builds a financial life in a weekend

— that's why lotteries sell tickets. The roadmap builds yours in eight small decisions, each checkable, each reversible early and expensive late. M1's $280 is easy to raise at 22 and nearly impossible to catch up at 40. *Which milestone are you most likely to skip — and what does skipping it cost?*



The Two Shocks (Final Defense)

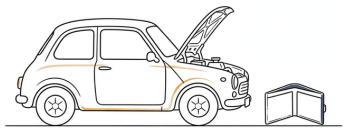
Your pitch must survive: **Shock 1** — a $600 car repair in month 3 (Sketch B: a $600 medical bill instead — nobody escapes shocks); **Shock 2** — rent rises $75/month at renewal ($900/year, permanent). Show the buffer absorbing both *without* touching savings or borrowing. If it can't, your pitch must say what changes: roommate, cheaper car, second job. Honesty outranks optimism.

STORY

The Transmission.

The student's $1,900 transmission (above)

was Shock 1 with real teeth. Her $210 buffer covered 11% of it. The rest went on plastic, and the 24% APR turned $1,690 of bad luck into $2,100 of debt. *What size buffer would have saved her — and what monthly choice builds it?*



What "Done" Looks Like

One page: every line item, three buckets, the buffer, the 50/30/20 test, the two shocks absorbed on paper. Three minutes: you, pitching it like a founder — "here's my month, here's what breaks it, here's why it doesn't." The class votes: *survives* or *fantasy*. The math decides; the room confirms.

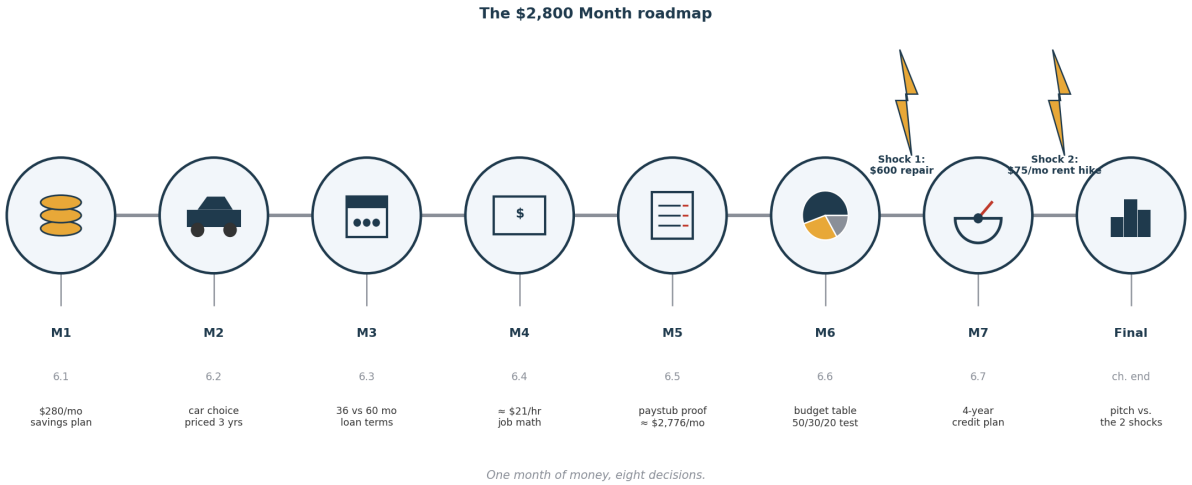
STORY

The Honest Budget.

One of the best pitches last year came

from a bus rider whose budget survived both shocks *and* a third she invented (a $400 flight home for Thanksgiving). "I budgeted the year I actually live," she said, "not the month I wish I lived." *What does the year you actually live cost?*

Fig. 6.8



One month of money, eight decisions.

DECIDE — THE VERDICT

Which life does $2,800 buy?

Back to the question that opened this section — now with the whole chapter's math to settle it.

Solution. Sketch A transport: $349 (loan) + $140 + $110 = **$599**. Sketch B: **$80**. Gap: **$519/month** — about $6,228 a year. Invested monthly at 7% for 10 years, that gap compounds to ≈ **$89,831.02**. Sketch B's saver is nearly $90k richer in a decade — *if* the bus actually serves her life. Shock test: Sketch A's $440 buffer absorbs the $600 repair (painfully) and the $75 rent hike (buffer → $365, permanently thinner). Sketch B's buffer — $440 + $519 = $959 — shrugs at both.

The decision: Sketch B wins the math by a mile — but the honest answer has a condition: the commute. If the bus turns 25 minutes into 90, Sketch B costs 65 minutes a day × 250 workdays ≈ **271 hours a year** — at $21/hour, that's $5,688 of *time*, nearly erasing the $6,228. The winning move: run Sketch B for **one year**, bank the $6,228 as your emergency fund and credit-builder, then revisit with data instead of dreams. The $2,800 Month isn't about the bus or the car — it's about making the trade with your eyes open, in numbers.

AI Learning Lab 6.8

AI Level 1 — Understand.

Explain this project as if I'm 15: I get a pretend $2,800/month and eight milestones. Then explain what each milestone's deliverable is, one at a time, and check that I know what "done" looks like for M1 before moving on.

AI Level 2 — Apply.

Quiz me on the roadmap: for each milestone M1–M7, ask me what decision it answers and what number proves it. Hide nothing — then check my answers and correct me where I'm vague.

AI Level 3 — Matter.

Sketch B beats Sketch A by $89,831.02 in ten years — yet most 22-year- olds choose the car. Discuss with me: WHY — is it freedom, status, or the simple fact that the bus costs *time*, which young people feel they have? Who profits when young workers choose Sketch A — dealers? lenders? insurers? What are the implications for someone whose bus genuinely doesn't run — is "take the bus" advice that ignores their reality? End by asking me what my actual commute would cost in each sketch, in dollars *and* hours.

Level 4 — Decide. *Is your variable budget honest?*

> **Example decision problem.** You budgeted $600/month for variable spending (groceries + gas + dining). You tracked one real week: $185. Is the $600 budget honest?

AI Level 4 — Decide.

I budgeted $600/month variable. One tracked week: $185. Act as my decision coach: make me annualize the week myself ($185 × 52 ÷ 12), compare it to $600, then ask me whether the week was unusual or my budget is fantasy. Make me fix the budget or defend the week with evidence — don't decide for me.

Check your verdict: $185 × 52 ÷ 12 ≈ $802/month — $202 over budget. One week doesn't convict, but it indicts: track three more weeks. If the average holds, the budget is fantasy and the fix is cut $200 of variable, earn $200 more, or admit the savings rate is 3%, not 10%. A budget that ignores its own data is a wish.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 6.8.

Ask your AI: "Build me a complete $2,800/month budget that survives a $600 emergency." Then do the detective work: audit every line — annuals ÷ 12? irregular income? the "miscellaneous" that's actually $200 of DoorDash? Rebuild it honestly yourself, then stress-test both versions against the two shocks. If the AI's version dies at Shock 1, write one paragraph on *which* missing line killed it — and why "miscellaneous" is where budgets go to lie. AI chatbots build budgets for months nobody lives — in this chapter, *you* live the month.

Practice Problems 6.8

1. M1 check: $280/month at 7% for 40 years ≈ $734,948. What if you raise savings to 15% ($420/month)? What is the new 40-year total?
2. Sketch A vs. Sketch B: what is the annual transportation gap? (Use $599 vs. $80.)
3. Your buffer is $440/month. Shock 1 ($600 repair) hits in month 3; Shock 2 (rent +$75) starts month 7. What is your buffer in month 8? How much have you saved by month 12 if savings ($280) was never touched?
4. Your roommate candidate has no income but "will pay you back." Using the chapter's math, list two numbers that make this a good idea and two that make it a trap.
5. Write your 3-minute pitch in three sentences: your sketch, your buffer, and how it survives both shocks. (Answers vary — must cite numbers.)

Vocabulary

- **Compound interest** — interest computed on principal *plus* accumulated interest; the engine of both retirement and debt.
- **Principal** — the original sum invested or borrowed, before interest.
- **APR (annual percentage rate)** — the yearly cost of borrowing (or yield on savings) *before* compounding frequency is applied.
- **APY (annual percentage yield)** — the true yearly rate *after* compounding; always ≥ APR for the same nominal rate.
- **Compounding period** — how often interest joins the principal (annually, monthly, daily…).
- **Rule of 72** — doubling time ≈ 72 ÷ rate percent; a rule of thumb, not a law.

- **Future value of an annuity** — FV = PMT × [((1+r)^n − 1) ÷ r]; what regular payments grow into.
- **Depreciation** — loss of an asset's value over time.
- **Declining-balance depreciation** — V = P(1 − r)^t; biggest losses first, like compound interest in reverse.
- **Straight-line depreciation** — (cost − salvage) ÷ life; the same dollar loss every year.
- **Salvage value** — what an asset is worth at the end of its useful life.
- **Amortization** — the schedule splitting each loan payment into interest and principal.
- **Loan term** — the repayment period; longer term = smaller payment, more interest.
- **Time value of money (TVM)** — a dollar today beats a dollar next year; the logic behind every loan formula.
- **Deferred interest** — "0% for 12 months" fine print: miss the payoff and interest applies retroactively to day one.
- **Gross pay** — earnings before any deductions; the rumor.
- **Net pay (take-home)** — gross minus all deductions; the news.
- **FICA** — 7.65% of gross (6.2% Social Security + 1.45% Medicare), no exceptions.
- **Withholding** — estimated income tax taken per paycheck, settled at filing time.
- **Marginal tax rate** — only income *above* a bracket line is taxed at that bracket's rate; raises never shrink take-home.
- **Overtime** — typically 1.5× hourly pay beyond 40 hours/week.
- **Pay period** — weekly, biweekly (26), semimonthly (24), monthly (12): same year, different slices.
- **Budget** — a plan assigning every dollar a job before the month begins.
- **50/30/20 rule** — ≈50% needs, 30% wants, 20% savings: a diagnostic, not a law.
- **Fixed expense** — barely moves month to month (rent, loan payment).
- **Variable expense** — moves with choices (groceries, dining).
- **Pay yourself first** — savings leaves on payday like a bill; "save what's left" saves nothing.
- **Credit** — borrowed trust: money now for a priced promise later.
- **Credit score (FICO)** — 300–850; payment history 35%, utilization 30%, length 15%, mix 10%, inquiries 10%.
- **Utilization** — balance ÷ limit; keep under 30%.
- **Hard inquiry** — a lender's credit check; small, temporary score dip.
- **Secured credit card** — your deposit is the limit; training wheels for a score.

Summative Task: The 3-Paragraph Decision Essay

The prompt. Your cousin is car shopping and texts you two offers:

- **Offer 1:** $24,000 car at **12% APR for 72 months** — "only $469/month!"

- **Offer 2:** $15,000 car at **8% APR for 48 months** — $366/month.

She loves the bigger car and the "low" $469 payment. Advise her in a 3-paragraph essay: which offer should she take, and why does the math support it?

Planning your essay (do this before you write). 1. **Compute first.** Run the numbers yourself: Offer 1 → $469.20 × 72 = **$33,782.73** total, **$9,782.73** in interest; Offer 2 → $366.19 × 48 = **$17,577.30** total, **$2,577.30** in interest. Verify these yourself — don't take them on faith. 2. **Thesis template.** "I recommend ___ because ***, even though*** ." Fill it in for YOUR side — either side is defensible if the numbers back it. 3. **Evidence checklist.** Every claim needs a number behind it: the $103/month payment gap, the 24 extra months, the $7,205 interest gap, the $9,000 day-one value gap — plus at least one idea from this chapter (depreciation vs. loan term, or the minimum-payment trap) that strengthens the case. 4. **Flip test.** Name one specific condition that would change your recommendation — a 0% manufacturer offer on the bigger car? Contract work that genuinely requires it? If your advice wouldn't flip under any condition, it's a slogan, not a decision.

Paragraph 1 — Your decision and claim. State your advice in one clear sentence, then one or two sentences on why it matters. (Example: "Take Offer 2 — the $469 payment is a 6-year trap dressed as a deal, and the cheaper car saves you over $16,000.")

Paragraph 2 — Mathematical evidence. Show the numbers: Offer 1 → $469.20 × 72 = **$33,782.73** total, **$9,782.73** in interest. Offer 2 → $366.19 × 48 = **$17,577.30** total, **$2,577.30** in interest. The "low" payment costs **$103 more per month for 24 extra months** and **$7,205 more in interest** — for a car worth $9,000 more on day one (and depreciating per §6.2). Every claim needs a number behind it.

Paragraph 3 — Implications, limitations, and what would change your mind. What does 72 months of payments mean for someone whose car may not *last* 72 months (owing more than it's worth — §6.2's curve meets §6.3's term)? What are this calculation's limits (insurance and repairs aren't included — the $24,000 car may cost more to insure)? Name one specific condition that would flip your advice — for example, a 0% manufacturer offer on the bigger car, or contracting work that genuinely requires it — and explain why.

Length: 200–300 words. Three paragraphs, no more, no fewer.

Using AI. Write your first draft entirely on your own — no AI. Then you may use AI as a revision coach: ask it to check your loan math, flag weak evidence, and suggest clearer sentences. Never let it write paragraphs for you. And remember this book's golden rule: AI makes mistakes — verify everything, especially any numbers it gives you. (Ask it for the 72-month payment on $24,000 at 12% and check it against $469.20.)

Rubric and exemplar essays are in the Teacher Guide.

Answer Key — Practice Problems (odd-numbered)

6.1 — 1. 10,000 × 1.06^12 ≈ **$20,121.96**. 3. Simple: 2,000 × 1.40 = $2,800. Compound: 2,000 × 1.08^5 ≈ $2,938.66. Gap ≈ **$138.66** — small here, a house on a mortgage. 5. Annual: 5,000 × 1.06^10 ≈ $8,954.24. Monthly: 5,000 × 1.005^120 ≈ $9,096.98. Monthly wins by ≈ **$142.74** — frequency is a footnote.

6.2 — 1. 24,000 × 0.88^4 ≈ **$14,392.69**. 3. Car: 30,000 × 0.15 = $4,500 lost. Equipment: 30,000 ÷ 10 = $3,000 lost. The car loses **$1,500 more** in year 1 — the declining curve bites early. 5. Fair: 32,000 × 0.85^2 = **$23,120** exactly on the curve. You save $8,880 — about 28% off for two years of use.

6.3 — 1. r = 0.005, n = 60: PMT ≈ **$231.99**; total ≈ $13,919.62; interest ≈ **$1,919.62**. 3. (Site warm-up) PMT = 7,500 × (0.095÷12) ÷ (1 − 1.007917^(−48)) ≈ **$188.42**; total ≈ $9,044.33; interest ≈ **$1,544.33**. 5. (a) **$3,000** — paid off, no interest, the offer works. (b) $1,800 remaining + deferred interest ≈ $804.73 → balance ≈ **$2,604.73**. "0%" cost $804.73 — read paragraph 14.

6.4 — 1. Annual: 16.50 × 2,080 = **$34,320**; biweekly: 34,320 ÷ 26 = **$1,320**. 3. Semimonthly: 39,000 ÷ 24 = **$1,625**; biweekly: 39,000 ÷ 26 = **$1,500**. Semimonthly is bigger by **$125** — but there are only 24 of them. 5. $40,000 month: 2,000 + 0.05 × 40,000 = **$4,000**. $8,000 month: 2,000 + 400 = **$2,400**. Budget on the $2,400.

6.5 — 1. FICA $122.40 + federal $192.00 + state $80.00 = $394.40; net = **$1,205.60**. 3. Real: 47,000 × 0.12 + 3,000 × 0.22 = **$6,300**. Myth: 50,000 × 0.22 = $11,000. Difference: **$4,700** of courage. 5. $25/hr: 25 × 2,080 = **$52,000**. $23/hr + insurance: 23 × 2,080 + 3,600 = **$51,440**. The straight wage costs **$560** more.

6.6 — 1. 50/30/20 → **$1,500** / **$900** / **$600**. Rent $1,100 ≤ $1,500 ✓ with $400 left for other needs. 3. $720 ÷ 12 = **$60/month** — forget it and December ambushes you. 5. "Save what's left": $100/mo. "Pay yourself first": $300/mo. Difference: **$200/mo** = **$2,400/year**.

6.7 — 1. 750 ÷ 2,000 = **37.5%** — over the line. Get under $600 (30%). 3. r = 0.015, n = 36: PMT ≈ **$216.91**; total ≈ $7,808.92; interest ≈ **$1,808.92**. 5. $120: 120 ÷ 500 = **24%** ✓ — builds history, autopay in full. $400: **80%** — the algorithm reads distress.

6.8 — 1. 15% = 1.5 × $734,948 ≈ **$1,102,422**. Five percent more now is $367,474 later. 3. Month 8 buffer: 440 − 75 (the rent hike, in effect since month 7) = **$365**. Savings by month 12: 12 × $280 = **$3,360** — but only if the $160 Shock-1 overrun went on plastic; if it came from savings, **$3,200**. (Month 3: 440 − 600 = **−$160** — the buffer failed, and the $160 came from savings or a card. That's the lesson.) 5. Answers vary — must cite the sketch, the buffer, and both shocks with numbers.

Level 4 example checks — **6.1:** A: 5,000 × 1.07^30 ≈ **$38,061**; B: half ≈ **$19,031**. Every dollar spent at 25 costs ≈ $7.60 at 55. **6.2:** Now: 1,200 × 0.75^2 = **$675** (offer $650 is fair). One more year: ≈ **$506**. **6.3:** 24 mo → $235.37/mo, interest ≈ **$649**; 36 mo → $166.07/mo, interest ≈ **$979**. Take 24 if $235 fits. **6.4:** A = 20 × 35 × 52 = **$36,400** vs. $38,000 — B leads $1,600 until ~100 unpaid hours drag it under $20/hr. **6.5:** A = **$39,520**; B = $36,400 + $2,400 = **$38,800** effective — A leads $720 cash; health risk flips it. **6.6:** Rent ≈ **33.3%** (just over the line); needs ≈ **55.8%**; $820 left for wants + buffer — workable, watch groceries. **6.7:** $50/mo → **26 months**, ≈ $300 interest; $200/mo → **~6 months**, ≈ $65 interest. Kill it fast. **6.8:** $185 × 52 ÷ 12 ≈ **$802**/month vs. $600 budgeted — track 3 more weeks; the budget is indicted.

CHAPTER 7

Networks & Graphs

A Note on AI Learning Labs in This Chapter

Chapter 6 taught you to recompute money. This chapter teaches you to distrust pretty routes. Ask an AI for the "shortest delivery route" or the "cheapest cable layout" and it will answer instantly, confidently — and sometimes wrong, because route problems are exactly where confident guessing looks like reasoning. Your superpower in this chapter is that every graph here is small enough to check by hand: count the degrees, trace the path, add the weights. So the standing rule: **never trust an AI's route until you've walked it yourself.** Every AI Learning Lab keeps the four levels (understand, apply, matter, decide), and every Level 4 ends the way the chapter begins — with a decision about a real route, a real schedule, or real money that is yours to make.

About This Chapter

Some of the most expensive decisions in the world are drawing problems: which streets the plow covers, which order the truck drives, where the cable goes, which exams share a day, which delay sinks the festival. The mathematics of all of them is the **graph** — dots and lines — and a handful of ideas from the 1700s that still route your pizza. This chapter gives you the tools: Euler paths and circuits, the traveling salesman problem, shortest paths, minimum spanning trees, graph coloring, and project scheduling. Then you will use all of it to decide: whether the snowplow can finish without retracing, which delivery order is truly shortest, which detour to take when the walkway closes, whether a $71,000 cable bid is a scam, how many days finals really need, and which $200 fee saves the festival.

Running through the whole chapter is **The Campus Network** (§7.7): a fictional campus with real numbers — walkway minutes, cable costs, exam conflicts, fair-planning tasks — and one milestone per section. Patrol it, deliver across it, wire it, schedule it. By the final report, the campus won't be the book's. It'll be yours.

By the end of this chapter, you will be able to:

- Model streets, conflicts, and schedules as graphs — vertices for the things, edges for the connections.
- Use parity (odd vs. even degrees) to decide whether an Euler path or circuit exists — and repair a broken route with the cheapest fix.

- Contrast brute force with the nearest-neighbor heuristic on the traveling salesman problem — and explain why 25 cities breaks brute force forever.
- Run Dijkstra's algorithm by hand on a small graph — and read exactly what a closed road costs you.
- Build minimum spanning trees with Kruskal's algorithm — and audit a contractor's bid with the cut property.
- Color conflict graphs, find chromatic numbers, and turn them into real schedules.
- Map projects as PERT networks — find the critical path, measure float, and decide which delay to pay to prevent.
- Use AI prompts at four levels (understand, apply, matter, decide) to learn networks — and force every AI route to show its work before you trust it.

7.1 Euler Paths & Circuits — The Snowplow Problem

THE DECISION

Can the snowplow finish without retracing a single street?

A small town's snowplow parks at the garage (G). Tonight it must plow *every* street in the neighborhood below — and the driver wants to drive each street exactly once and roll back into the garage, no repeated streets, no wasted diesel.

The map (miles):

- G–A: 0.4, A–B: 0.5, B–C: 0.6, C–G: 0.5
- A–C: 0.7 (the diagonal), B–D: 0.4, C–D: 0.5

Is the perfect route possible? If not, what is the cheapest fix — and how many extra miles does it cost? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** At each intersection, count how many streets meet. Write the counts: G? A? B? C? D?
- **Hint 2:** Euler proved a route covering every edge exactly once exists only when the odd counts are 0 (circuit) or 2 (path). How many odds do you have — and what does that rule out?

- **Hint 3:** If the circuit is impossible, the cheapest repair pairs up the odd intersections with the shortest path between them. Which pair, and how long is it?

What Is a Graph?

A **graph** is dots and lines: **vertices** (the dots — intersections, buildings, people) and **edges** (the lines — streets, cables, conflicts). That is the whole definition, and it is enough to model half the modern world. Forget geography: in a graph, only *connections* matter. The 0.4-mile street and the 0.7-mile street are both just edges; their lengths become **weights** when the decision needs them.

STORY

The Seven Bridges of Königsberg.

In 1736, the people of

Königsberg wondered: can you walk through town crossing each of its seven bridges exactly once? Leonhard Euler drew the town as four dots (land areas) and seven lines (bridges) — inventing graph theory on the spot — and proved the walk was impossible. Not "hard." *Impossible.* One drawing settled every argument in every tavern. *What argument in your life could one good drawing settle?*

Fig. 7.0

[illustration]

Four land areas, seven bridges — every vertex odd, so no Euler walk exists.

Illustration to come — production art in final layout.

FIGURE SPEC (FOR ILLUSTRATOR): The Königsberg graph: vertices N (north bank), S (south bank), A (big island), B (small island); seven bridge-edges in the real layout — two parallel N–A, two parallel S–A, one N–B, one S–B, one A–B. Degree shown inside each vertex circle (N: 3, S: 3, A: 5, B: 3); all four vertices red-ringed as odd degree. In-image title "Fig. 7.0 — The Seven Bridges." Caption: "Four land areas, seven bridges — every vertex odd, so no Euler walk exists."

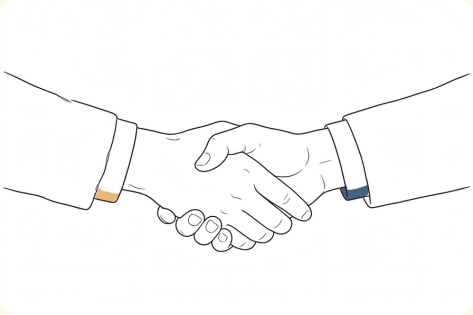
Degrees and Parity

The **degree** of a vertex is the number of edges touching it. Count ours: G has 2, A has 3, B has 3, C has 4, D has 2. Now the fact that runs this section: **every time a route enters a vertex through one edge, it must leave through another** — so every middle vertex of a route uses up its edges in pairs. Consequence: in a route that uses every edge exactly once and returns to its start, *every* vertex must have even degree.

STORY

The Handshake Count.

At a party, each handshake involves two hands — so the total number of "hands shaken" is always even, and the number of people who shook an *odd* number of hands is even too. Same law: every edge contributes 2 to the sum of degrees, so **the number of odd-degree vertices is always even.** If your count of odds comes out odd, you miscounted — recount. *When has a parity check saved you from a silly error?*



Euler's Rule

Here is the whole decision procedure, proved by Euler nearly 300 years ago:

- **0 odd vertices** → an **Euler circuit** exists: cover every edge exactly once and return to start.
- **2 odd vertices** → an **Euler path** exists: cover every edge exactly once, but you must start at one odd vertex and end at the other.
- **Any other number of odds** → impossible. No route. Don't bother searching — prove it instead.

Our snowplow graph: odds are A and B — exactly two. So an Euler *path* exists (start at A, end at B, or the reverse), but no *circuit*. The driver cannot both start and end at the garage on a perfect route. That is not bad news; it is a *priced* problem, which is better than a vague one.

STORY

The Mail Carrier.

Every morning a carrier must walk every block of a route and return to the post office. Carriers don't know Euler's name, but the good ones know his rule: they plan routes that retrace as little as possible, and the retracing always happens between odd-degree corners. The "Chinese postman problem" — finding the shortest closed walk covering every edge — is Euler plus optimization, and it schedules real mail trucks today. *Where in your town would the carrier's trick save the most steps?*

Fixing a Broken Route

Two odds, one missing circuit. The repair: **retrace the shortest path between the two odd vertices**, which makes them even and creates the circuit. A–B has a direct edge of 0.5 miles (shorter than A–C–B at 1.3 or A–G–C–B at 1.5). Drive every street once (3.6 miles of plowing), retrace A–B (0.5 miles of deadheading), done.

STORY

The Extra Pass.

A Zamboni driver at a hockey rink faces the same math: resurface all the ice, return to the gate, minimize dry passes. The pros' secret is identical to the plow's — find the odd corners, pair them with the shortest dry path, and never argue with parity. *What machine in your world makes "extra passes," and does its driver know the cheapest one?*

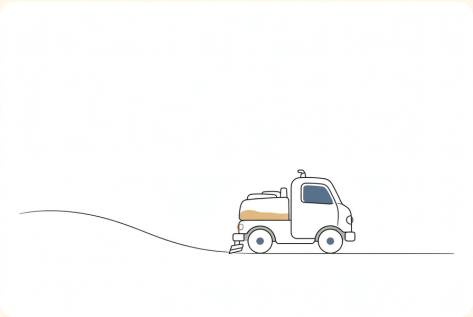
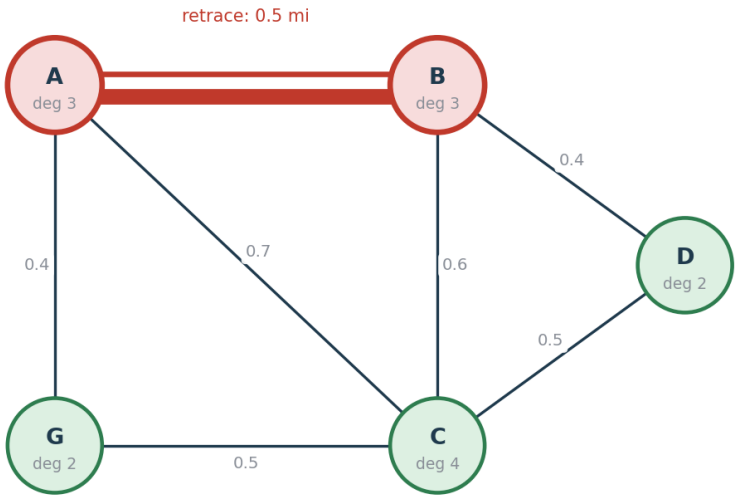


Fig. 7.1

Fig. 7.1 — Parity decides.



odd vertices: A, B (2) → Euler path yes, circuit no

Two odds, one repair.

Two odds, one repair.

DECIDE — THE VERDICT

Can the snowplow finish without retracing?

Back to the question that opened this section — now with the math to settle it.

Solution. Degrees: G:2, A:3, B:3, C:4, D:2. Two odds (A, B) → Euler path yes, Euler circuit no. The perfect no-retrace round trip from the garage is impossible — proven, not guessed.

The decision: run the route and retrace A–B. Plowing total: 0.4 + 0.5 + 0.6 + 0.5 + 0.7 + 0.4 + 0.5 = **3.6 miles**; deadhead: **0.5 miles**; night's total: **4.1 miles**. Don't move the garage, don't build a street — the 0.5-mile retrace is the cheapest possible repair, because any fix must pair the two odd vertices and 0.5 is their shortest connection. Parity doesn't negotiate; it invoices.

AI AI Learning Lab 7.1

AI Level 1 — Understand.

Explain Euler paths and circuits as if I'm 15, using the "draw the envelope without lifting your pen" puzzle. Then explain why odd-degree vertices are the whole story, one idea at a time. Check that I can count degrees correctly before moving on.

AI Level 2 — Apply.

Give me 5 small graphs (describe each as a vertex list with degrees, or as an edge list). Hide the answers. For each I'll state the number of odd vertices and whether an Euler path, circuit, both, or neither exists, showing my degree count; then check my work and point to the exact vertex I miscounted if I slipped.

AI Level 3 — Matter.

The snowplow driver can't argue with parity — the math proved the perfect route impossible before a single tire moved. Discuss with me: WHY is an *impossibility proof* so powerful — what does it save us (time? money? arguments)? Who profits when routes are optimized this way — the town? the driver? the fuel company (less)? And what are the implications of a 300-year-old theorem still scheduling mail trucks — what does that say about "old" math? End by asking me where in MY town I'd look for wasted retracing.

Level 4 — Decide. *Can the park ranger's patrol be a circuit?*

> **Example decision problem.** A park ranger must walk every trail of a small park once and return to the ranger station. Trails (miles): P–Q: 0.3, Q–R: 0.4, R–S: 0.5, S–P: 0.3, P–R: 0.6. The station is at P. Is a no-retrace round trip possible? If not, what's the cheapest retrace, and what's the night's total?

AI Level 4 — Decide.

I need to decide the ranger's patrol route. Trails: P–Q 0.3, Q–R 0.4, R–S 0.5, S–P 0.3, P–R 0.6 miles; station at P. Act as my decision coach: make me count every degree myself, apply Euler's rule myself, and find the cheapest retrace myself. Then ask me: is the retrace worth it, or should the ranger just accept it? Make me state my final route with reasons — don't decide for me.

Check your verdict: degrees P:3, Q:2, R:3, S:2 — two odds (P, R), so no circuit; Euler path yes. Cheapest retrace: P–R direct, 0.6 miles (shorter than P–Q–R at 0.7 or P–S–R at 0.8). Trail total: 2.1 miles; night's total: 2.7 miles. Accept the 0.6 — any circuit needs the odds paired, and 0.6 is their shortest link.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 7.1.

Ask your AI: "Use Fleury's algorithm to find an Euler path on the snowplow graph from §7.1, starting at A." Then do the detective work: trace its path edge by edge on paper, checking at every step that it never crosses a bridge it didn't have to (Fleury's rule: avoid bridges unless forced). If the AI's path strands itself — a classic AI failure on this exact task — finish the path yourself and write one paragraph on *where* it got stuck and why. AI chatbots love announcing routes and hate walking them — in this chapter, *you* are the walker.

⚑ PROJECT MILESTONE

M1: The patrol route

The Campus Network project starts now (full brief: §7.7). The security cart must cover every campus walkway once and return to North Gate. Count the degrees on the campus map, apply Euler's rule, and either produce the circuit or price the cheapest retrace. Deliverable: the verdict plus the total patrol time.

Practice Problems 7.1

1. A graph has vertices X, Y, Z, W with edges X–Y, Y–Z, Z–X, X–W. Count each degree, list the odd vertices, and state whether an Euler path, circuit, both, or neither exists.
2. Königsberg's seven bridges touched four land areas with degrees 3, 3, 3, and 5. Was the famous walk possible? Explain in one sentence.
3. A plow district is a square A–B–C–D–A (1 mile per side) plus a diagonal A–C (1.4 miles). The garage is at A. Find the odd vertices, the cheapest retrace, and the night's total mileage.
4. For the district in problem 3, where would you build one new street (one new edge) so that a perfect Euler circuit becomes possible? Explain why that edge — and not any other — works.
5. A connected graph has an Euler circuit. How many odd-degree vertices does it have? Can it have exactly two? Explain the difference between the two cases in one sentence each.

7.2 The Traveling Salesman — The Pizza Run

THE DECISION

Which delivery order is actually shortest?

A pizza driver starts at the shop (S), must visit four houses (A, B, C, D) exactly once each, and return to S. The distances (miles):

	S	A	B	C	D
S	–	3	5	4	7
A	3	–	2	6	5
B	5	2	–	3	4
C	4	6	3	–	2
D	7	5	4	2	–

The driver's gut says: always drive to the nearest unvisited house — S→A→B→C→D→S, 17 miles. With only 12 possible tours, is the gut right? And what happens when it's 25 houses, not 5? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Fix the start at S. How many orders are there really? (5−1)! ÷ 2 = 12 — few enough to list. Write them.
- **Hint 2:** Price the tour S→A→B→D→C→S by hand: 3 + 2 + 4 + 2 + 4. Compare it to the gut's 17.
- **Hint 3:** The gut rule never reconsiders an early choice. Which early choice in S→A→B→C→D→S quietly costs 2 extra miles at the end?

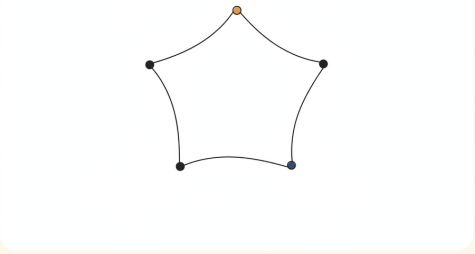
Hamiltonian Cycles

A **Hamiltonian cycle** visits every *vertex* exactly once and returns to the start (Euler visited every *edge*; Hamilton visits every *vertex* — don't mix them up). The traveling salesman problem (TSP): find the shortest Hamiltonian cycle in a complete weighted graph. It sounds like §7.1's cousin. It is not — it is famously, brutally harder, and that difference is the whole section.

STORY

The Salesman's Complaint.

In the 1800s, traveling salesmen rode circuits of towns and grumbled about wasted miles. Mathematicians formalized the grumble: given n cities, find the shortest tour. For 5 cities there are 12 tours; for 10, 181,440; for 25, about 3 × 10^23. The salesman's complaint became computer science's hardest famous problem — and your pizza is its descendant. *What everyday grumble of yours is secretly a famous math problem?*



Brute Force: Twelve Tours

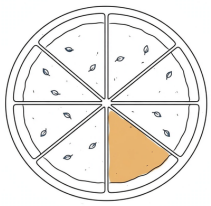
For 5 houses, brute force wins: list all 12 tours, price each, take the minimum. The champion: **S→A→B→D→C→S** = 3 + 2 + 4 + 2 + 4 = **15 miles**. The gut's S→A→B→C→D→S costs 17 — two miles worse, and now we know *exactly* how much the gut tax is. Brute force is dumb, honest, and — at this size — unbeatable.

STORY

The Dispatcher Who Counts.

A dispatcher with 6 stops has 60

tours — too many for one person, perfect for six: split the list, each person prices 10 tours, and the true optimum is found before the pizzas cool. Brute force parallelizes beautifully; it is only *scale* that kills it. *When is the dumb-but-sure method actually the smart one?*



The Nearest-Neighbor Heuristic

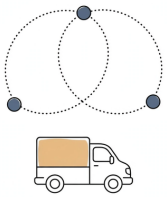
The gut has a name: the **nearest-neighbor heuristic** — always go to the closest unvisited city. It is fast (no counting, no tables) and often decent. It is also *never guaranteed*: here it pays 17 for a 15-mile problem, a 13% tax, because the early bargain S→A→B→C strands the driver at C with only the long hauls left (C→D is 2, but then D→S is 7 — the trap). Heuristics are bets, and this section prices the bet.

STORY

The Hungry Driver.

Watch a real delivery driver and you'll

see nearest-neighbor in the wild — they chase the closest pin on the app, then curse the last two stops across town. The app companies know: for 100+ stops they use fancier heuristics (and still can't promise optimal). The driver feels the 13% in gas and tips. *What heuristic do you run daily without naming it?*



Why TSP Is Hard

Count the tours: n cities → (n−1)! ÷ 2. At a billion tours checked per second, 25 cities need about **9.8 million years**. Not a bigger computer — *physics* says no. This is why TSP sits in the famous class **NP-hard**: no known shortcut, and finding one would break modern cryptography. Some problems yield to cleverness (§7.1's parity, §7.4's greedy); TSP yields only to patience or approximation.

STORY

The 25-City Wall.

Chip designers route millions of

connections; biologists assemble DNA from fragments; airlines schedule crews — all disguised TSPs. Nobody solves them exactly at scale. They solve them *well enough*, and "well enough" is a whole science. *Where in your life is "well enough" the right standard — and where isn't it?*

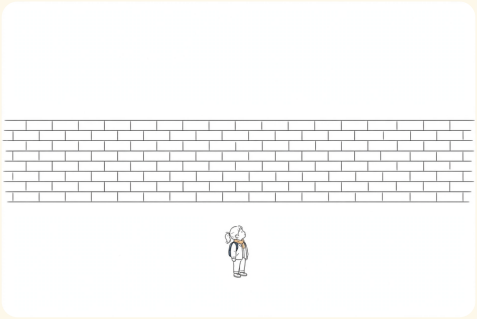
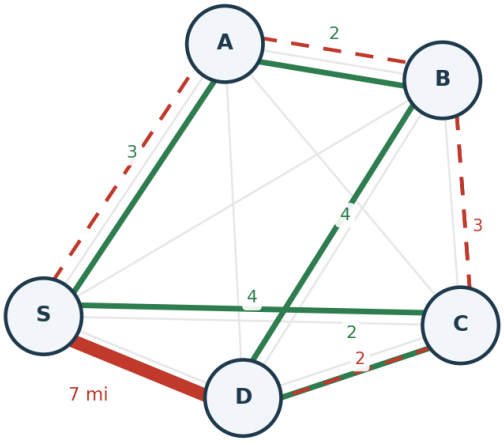


Fig. 7.2

Fig. 7.2 — The gut tax.



red dashed: nearest-neighbor S→A→B→C→D→S = 3+2+3+2+7 = 17 mi

green: optimal S→A→B→D→C→S = 3+2+4+2+4 = 15 mi

Two extra miles for trusting your gut.

Two extra miles for trusting your gut.

DECIDE — THE VERDICT

Which delivery order is actually shortest?

Back to the question that opened this section — now with the math to settle it.

Solution. All 12 tours priced; the minimum is **S→A→B→D→C→S at 15 miles**. Nearest-neighbor's 17 is not optimal — the early choice C→D looks cheap (2 miles) but strands the driver for the 7-mile D→S finale.

The decision: run the 15-mile tour. The 2-mile gut tax, at 30 deliveries a day and $0.70 a mile, is **$42 a day** — over $15,000 a year for one driver. At 5 stops, brute force is free and perfect; save the heuristic for the days with 25 stops, when perfection costs 9.8 million years.

AI AI Learning Lab 7.2

AI Level 1 — Understand.

Explain the traveling salesman problem as if I'm 15, using the pizza driver. Then explain the difference between a Hamiltonian cycle (visits every vertex) and an Euler circuit (covers every edge) with one tiny example each. Quiz me on which is which until I stop mixing them up.

AI Level 2 — Apply.

Give me 3 TSP instances with 4 cities each (distance tables), one at a time. Hide the answers. For each I'll list all 3 tours, price them, and name the optimum; then check my work tour by tour.

AI Level 3 — Matter.

The gut route cost 2 extra miles — $15,000 a year for one driver. Discuss with me: WHY do humans love greedy rules like nearest-neighbor even when counting would win — is it laziness, time pressure, or something deeper? Who profits when delivery companies settle for "good enough" routes — and who pays the 13%? What are the implications of a problem being NP-hard: does "no perfect answer exists" excuse a bad one? End by asking me for one decision in MY life where I use a gut heuristic I should probably price.

Level 4 — Decide. *Which order for four houses?*

Example decision problem. Shop S, houses A, B, C. Distances: S–A 2, S–B 4, S–C 5, A–B 3, A–C 6, B–C 2. Find the shortest tour starting and ending at S. Then run nearest-neighbor from S and compare.

AI Level 4 — Decide.

I need the shortest pizza tour: shop S, houses A, B, C; distances S–A 2, S–B 4, S–C 5, A–B 3, A–C 6, B–C 2. Act as my decision coach: make me list all 3 tours and price each myself, then make me run nearest-neighbor myself. Then ask me: did the gut win this time, and does that prove anything? Make me state my final order with reasons — don't decide for me.

Check your verdict: tours — S→A→B→C→S = 2+3+2+5 = 12; S→A→C→B→S = 14; S→B→A→C→S = 18. Optimum 12 (the reverse S→C→B→A→S ties it — optima aren't always unique). Nearest-neighbor: S→A→B→C→S = 12. The gut wins this time, but one win wouldn't prove the rule — §7.2's lesson is that heuristics are bets, and bets need pricing, not faith.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 7.2.

Ask your AI: "Give me a 6-city TSP distance table and your best tour." Then do the detective work: with 5 students, split the 60 possible tours (12 each), price every one by hand, and find the *true* optimum. Compare it to the AI's answer. If the AI claimed "optimal" for a tour that isn't, write one paragraph on how it fooled itself — and why "sounds optimal" is the most expensive phrase in routing. AI chatbots announce optima the way drivers announce shortcuts — in this chapter, *you* hold the map.

⚑ PROJECT MILESTONE

M2: The mail run

The Campus Network continues. The mail room (North Gate) must hit the Library, Science Hall, Dorm, and Stadium, then return. Using the campus courier times, find the true optimum by brute force (12 tours) AND run nearest-neighbor. Deliverable: both tours with prices, and one sentence on whether the gut won this time.

Practice Problems 7.2

1. Shop S and houses A, B, C with distances S–A 1, S–B 4, S–C 5, A–B 2, A–C 6, B–C 3. List all 3 tours and find the shortest.

2. How many distinct tours exist for 8 cities? (Start fixed, direction doesn't matter.) Show the formula.
3. Four stops with S–A 2, S–B 3, S–C 4, A–B 5, A–C 1, B–C 6. Run nearest-neighbor from S, then find the true optimum. Does the gut win here?
4. True or false: "The nearest-neighbor heuristic always finds the optimal TSP tour." If false, cite the counterexample from §7.2 with numbers.
5. Six houses mean 60 tours. Your group of 5 splits them evenly; each tour takes 30 seconds to price by hand. How many minutes does each student work? Is brute force realistic at this size?

7.3 Shortest Paths — The Closed Walkway

THE DECISION

Which detour — and how early do you leave?

Your 9:00 exam is in the Stadium (S). Your usual walk from the Dorm (D) is D→L→C→G→S — 4 + 3 + 5 + 3 = **15 minutes** via the Library (L), Cafeteria (C), and Gym (G). But this morning the C–G walkway is closed for construction. The campus map (minutes) — H is the Health Center:

- D–L: 4, D–C: 7, L–C: 3, L–H: 8
- C–G: 5 (⚠️ closed today), C–S: 12
- H–G: 4, H–S: 6, G–S: 3

Two detours look plausible: D→L→H→S or D→C→S. Which is faster, by how much — and what time do you leave? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Price both detours by hand first: D→L→H→S = 4 + 8 + 6; D→C→S = 7 + 12. Which wins?
- **Hint 2:** Dijkstra's idea: always lock in the closest *unsettled* vertex next. From D, which neighbor locks first — L (4) or C (7)?
- **Hint 3:** The winner isn't just faster — it's *proven* fastest once every vertex is locked. What does that proof buy you at 8:40 a.m.?

Weighted Graphs

§7.1's streets had lengths; §7.2's cities had distances. A **weighted graph** puts a number on every edge — minutes, miles, dollars — and the question becomes: cheapest *total*, not fewest edges. Your campus map is a weighted graph, and every morning you solve a shortest-path problem with your feet before you've had coffee.

STORY

The Subway Map.

A subway map lies about geography and tells the truth about connections — every rider reads it as a weighted graph (stations = vertices, ride-times = weights) and runs a mental Dijkstra to catch a transfer. The map's designers *want* you thinking in graphs. *What map in your life is secretly a weighted graph?*

Dijkstra's Algorithm

Edsger Dijkstra's 1956 algorithm finds the shortest path from a start to *every* vertex, and it fits on an index card:

1. Set the start's distance to 0, everyone else's to ∞.
2. Repeatedly: among unsettled vertices, lock in the one with the smallest tentative distance.
3. From it, relax every neighbor: if going through it is shorter, update.
4. Stop when all vertices are locked (or your target is).

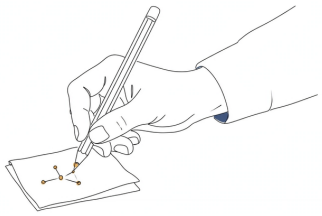
The magic: once a vertex locks, its distance is *final* — no later discovery can beat it. That is why the algorithm is trustworthy and why your GPS uses it (well, its souped-up descendants) a billion times a day.

STORY

The GPS Whisperer.

Dijkstra scribbled the algorithm in about

20 minutes in 1956, without a computer, while shopping with his fiancée — and called it one of his most productive coffee breaks ever. Every "recalculating…" you've ever heard is his idea, scaled to continents. *What 20-minute idea of yours might outlive you?*



Reading the Table

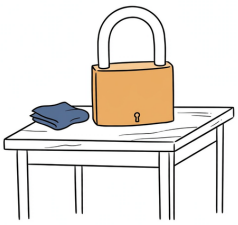
Run it from the Dorm with C–G closed. Lock D (0). Neighbors: L→4, C→7. Lock L (4): via L, C→min(7, 4+3)=7, H→12. Lock C (7): via C, S→19 (C–S=12; C–G is closed). Lock H (12): via H, G→16, S→min(19, 18)=18. Lock G (16): via G, S→min(18, 19)=18. Lock S (18). **Fastest detour: D→L→H→S, 18 minutes** — three minutes worse than your usual 15, and now *proven* optimal, not guessed.

STORY

The 8:57 Decision.

At 8:57 you don't need the philosophy of

shortest paths — you need a number. Dijkstra's table is that number with a certificate: every locked vertex is a promise. The detour cost isn't a guess — it's exact, down to the minute. *When did a "proven" number beat a confident guess for you?*



When Shortcuts Close

Closures are where shortest-path math earns its keep: the algorithm doesn't care *why* C–G closed — it just reprices the world. Note the trap D→C→S (19): it *looks* direct (only two edges!) but loses to the three-edge D→L→H→S (18). Fewest edges ≠ shortest path — the oldest intuition error in routing.

STORY

The Cone Zone.

Road crews close lanes every summer, and every summer drivers "detour" by instinct onto the road that *looks* shortest — usually the one that parallels the closure, usually jammed with everyone else's instinct. The algorithm has no instincts, which is why it beats them. *What "obvious" shortcut of yours deserves a Dijkstra check?*

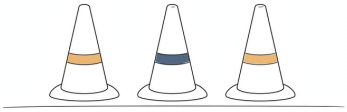
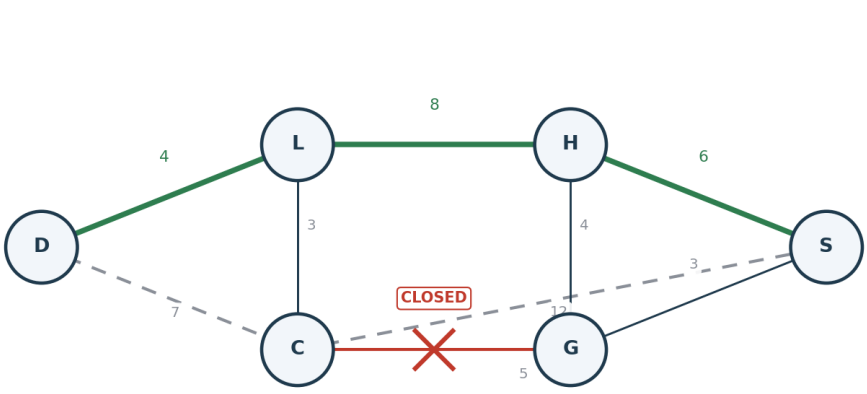


Fig. 7.3

Fig. 7.3 — The closure, priced.



Dijkstra lock order

1. D locks at 0 min
2. L locks at 4 min
3. C locks at 7 min
4. H locks at 12 min
5. G locks at 16 min
6. S locks at 18 min

gray dashed: D→C→S = 7+12 = 19 min (the trap)

green: D→L→H→S = 4+8+6 = 18 min (proven fastest)

Three minutes, proven.

Three minutes, proven.

DECIDE — THE VERDICT

Which detour — and how early do you leave?

Back to the question that opened this section — now with the math to settle it.

Solution. With C–G closed, Dijkstra locks S at **18 minutes** via **D→L→H→S** (4+8+6). The tempting D→C→S costs 19 — one minute slower, and "two edges" fooled nobody who ran the table.

The decision: take D→L→H→S and leave by **8:39** for the 9:00 exam — 18 minutes of walking plus a 3-minute buffer, because construction zones lie about everything except being annoying. The old 15-minute habit (out the door at 8:45, arriving on the dot) would have made you exactly 3 minutes late, which is the difference between Dijkstra and vibes.

AI AI Learning Lab 7.3

AI Level 1 — Understand.

Explain Dijkstra's algorithm as if I'm 15, using the idea of a spreading ink stain that always expands through the cheapest frontier first. Then walk me through one 4-vertex example, one lock at a time, and don't move on until I can say why a locked vertex never changes.

AI Level 2 — Apply.

Give me 4 small weighted graphs, one at a time, and ask for the shortest path between two named vertices. Hide the answers. I'll build the lock-order table myself; then check each table line by line and show me the first line where I went wrong if I slipped.

AI Level 3 — Matter.

The detour cost exactly 3 minutes — proven, not estimated. Discuss with me: WHY does a *guaranteed* shortest path matter — who needs guarantees rather than guesses (ambulances? airlines? exam-takers?)? What are the implications of everyone following the same optimal route — can the "shortest path" get crowded into not being shortest (traffic!)? And who decides what the weights *are* — is "18 minutes" the same for everyone? End by asking me where in MY life I'd want a Dijkstra guarantee.

Level 4 — Decide. *Which way to the gym?*

Example decision problem. From home (H) to the gym (G): H–P 2, H–T 5, P–T 2, P–C 6, T–C 3, T–G 4, C–G 2 (minutes). Find the shortest route. A friend insists H→T→G "looks shorter" — settle it.

AI Level 4 — Decide.

I need the fastest route from home (H) to the gym (G). Paths (min): H–P 2, H–T 5, P–T 2, P–C 6, T–C 3, T–G 4, C–G 2. Act as my decision coach: make me run Dijkstra's lock-order table myself — no skipping to the answer. Then make me price my friend's H→T→G against the winner. Make me state my final route with reasons — don't decide for me.

Check your verdict: lock H(0) → P(2) → T(4, via P) → C(7, via T) → G locked at 8 via H→P→T→G (2+2+4). The friend's H→T→G = 5+4 = 9 — one minute slower, and "looks shorter" is exactly the intuition §7.3 warns about. Take H→P→T→G.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 7.3.

Ask your AI: "Run Dijkstra on the §7.3 campus graph with C–G closed and show the full lock-order table." Then do the detective work: rebuild the table yourself, line by line, and compare. AIs frequently "lock" a vertex too early or relax an edge wrong on this exact exercise — if yours did, pinpoint the first wrong line and write one paragraph on why that single error corrupts everything after it. Then the twist: invent a story where a road has *negative* weight (a downhill toll *refund*?) and explain in two sentences why Dijkstra's lock-in promise breaks. AI chatbots recalculate routes all day and rarely show the table — in this chapter, *you* audit the table.

⚑ PROJECT MILESTONE

M3: The game-day detour

The Campus Network continues. On game day, two of the stadium walkways (C–T and G–T) close — D–T stays open. Find the fastest North Gate → Stadium route on the closed map and price it against the normal 15-minute N→C→T. Deliverable: the detour path, its time, and your "leave X minutes early" recommendation.

Practice Problems 7.3

1. Weighted graph: A–B 2, A–C 5, B–C 1, B–D 4, C–D 2, C–E 6, D–E 3. Find the shortest path from A to E. Show the lock order.

2. In the §7.3 campus story with C–G closed, the winner was D→L→H→S (18). What is the *second*-best D→S route, and are there ties?
3. True or false: "Dijkstra's algorithm settles on the final answer only after every vertex is locked." Explain the precise stopping rule in one sentence.
4. In a graph where every edge weighs 1, what does the shortest path become, and what simpler algorithm finds it? (One word each.)
5. Propose one new walkway for the §7.3 campus graph (with a reasonable walking time) that would beat the 18-minute detour when C–G is closed. What is the new shortest D→S time?

7.4 Minimum Spanning Trees — The Fiber-Optic Bid

THE DECISION

Is the $71,000 cable bid a scam?

Six municipal buildings need fiber-optic cable: Admin (A), Library (L), Science (S), Dorm (D), Gym (G), Cafeteria (C). A contractor bids **$71,000**: connect every building straight to Admin (A–L $8k, A–S $12k, A–D $15k, A–C $16k, A–G $20k). The IT director suspects the "star" layout is lazy — and lazy is expensive. Trenching costs ($ thousands) between buildings:

- A–L 8, A–S 12, A–D 15, A–C 16, A–G 20
- L–S 6, L–C 10, L–D 13, S–D 9, S–G 14, D–G 7, D–C 11, G–C 5

What is the cheapest network connecting all six — and how much does the bid overcharge? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Sort every cable by price, cheapest first: 5, 6, 7, 8, …
- **Hint 2:** Take each cable in order — keep it unless it forms a cycle with cables you already kept. (This is Kruskal's algorithm.)
- **Hint 3:** Six buildings need exactly five cables. Stop at five and add up. Compare with $71,000.

Trees and Spanning Trees

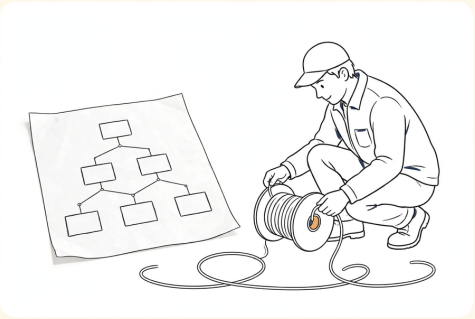
A **tree** is a graph with no cycles — one and only one path between any two vertices. A **spanning tree** of a graph is a tree using *every* vertex (all six buildings) with the fewest edges that still connects them: always exactly (vertices − 1) edges. Six buildings → five cables, no more, no less. Fewer disconnects someone; more wastes a trench.

STORY

The Family Tree.

Your family tree (the bloodline kind) has no

cycles — exactly one path of ancestry between any two relatives (ignoring royal families). A spanning tree is the same discipline applied to cable: connect everyone, no loops, nothing wasted. *What system in your life has expensive loops nobody noticed?*



Kruskal's Algorithm

Joseph Kruskal's 1956 recipe is almost insultingly simple — take the cheapest edge that doesn't form a cycle, repeat:

1. Sort all edges by weight.
2. Walk down the list; keep an edge unless it closes a cycle.
3. Stop when you have (vertices − 1) edges.

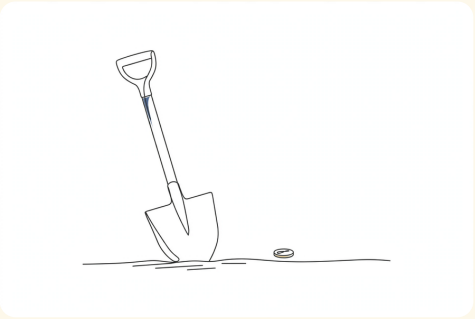
On our bid: G–C ($5k) ✓, L–S ($6k) ✓, D–G ($7k) ✓, A–L ($8k) ✓, S–D ($9k) ✓ — five cables, all six buildings connected, **$35,000**. The next cheapest, L–C ($10k), would close a loop — skip it, and everything after it. Done in one pass down a sorted list.

STORY

The Cheapest-First Contractor.

Imagine a contractor so cheap

she buys cable one trench at a time, always the cheapest trench that isn't redundant. You'd expect a mess. You get the *optimal* network — Kruskal proved the greediest possible strategy is perfect here, which is shocking because greed fails almost everywhere else in this chapter (looking at you, §7.2). *When has "cheapest first" worked for you — and when did it backfire?*



Why Greedy Works Here

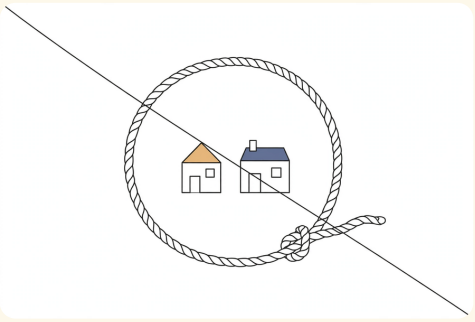
The one-line intuition (the **cut property**): split the buildings into any two groups; the cheapest cable crossing between the groups *must* be in the minimum spanning tree — because any spanning tree has to cross that split somehow, and crossing it cheaper never hurts. Kruskal's cheapest-first rule is just this property applied repeatedly. No proof required in this course — but notice the contrast with TSP: there, an early cheap choice can strand you (§7.2's 7-mile finale); here, an early cheap choice is *provably safe*. Same greed, opposite guarantees.

STORY

The Cut Property.

Draw a loop around just the Gym;

the cheapest cable out of the loop is G–C at $5k. Any connected network must escape that loop *somewhere* — escaping at $5k beats escaping at $7k, always, no exceptions. That tiny local certainty compounds into a global optimum. *What "tiny certainty" in your life compounds?*



The Bid Audit

The contractor's star: $8k + $12k + $15k + $16k + $20k = **$71,000**. Kruskal's tree: **$35,000**. The bid overcharges by **$36,000** — more than double — for a layout that isn't even *wrong*, just lazy: the star ignores every cheap building-to-building trench (G–C at $5k!) because drawing lines to Admin is easier than thinking.

STORY

The $36,000 Question.

The IT director walks into the negotiation with one sorted list and one question: "Show me why your $71,000 beats my $35,000." There is no answer — the cut property ate it. She knows the cheapest possible network down to the dollar, so every dollar above it is negotiable. *What bill of yours deserves a Kruskal audit?*

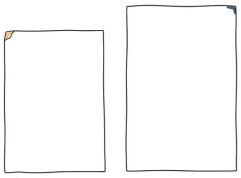
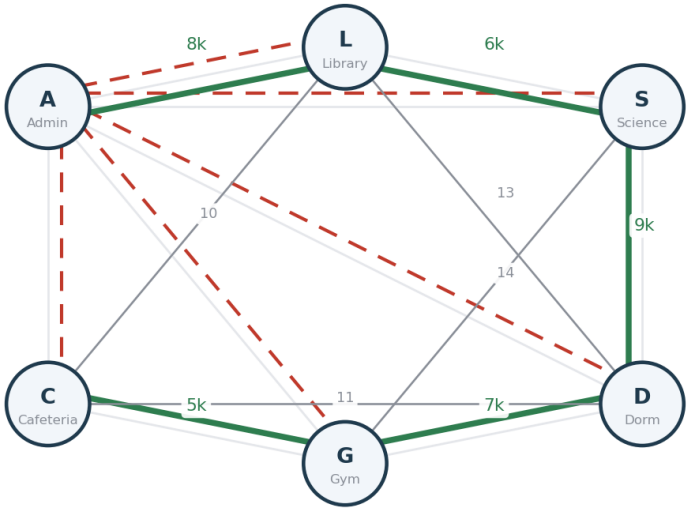


Fig. 7.4

Fig. 7.4 — Laziness, priced.



red dashed: contractor's star = 8+12+15+16+20 = $71,000 bid

green: Kruskal's 5 cables = 5+6+7+8+9 = $35,000

Same six buildings. Half the cost.

Same six buildings. Half the cost.

DECIDE — THE VERDICT

Is the $71,000 cable bid a scam?

Back to the question that opened this section — now with the math to settle it.

Solution. Kruskal: take G–C (5), L–S (6), D–G (7), A–L (8), S–D (9) — five cables, no cycles, all connected: **$35,000**, provably minimal by the cut property. The $71,000 star is $36,000 of pure laziness.

The decision: reject the bid and counter at $38,000 — the optimal $35,000 plus a fair profit of about 8%. "Scam" is a strong word; "we did the math and you didn't" is stronger. If the contractor walks, the town digs Kruskal's five trenches itself and banks the $36,000.

AI AI Learning Lab 7.4

AI Level 1 — Understand.

Explain spanning trees and Kruskal's algorithm as if I'm 15, using the idea of connecting treehouses with the cheapest rope bridges without making any loops. Walk me through one 4-vertex example edge by edge, and make me say out loud why we skip the edge that forms a cycle.

AI Level 2 — Apply.

Give me 4 weighted graphs (5–6 vertices each), one at a time. Hide the answers. For each I'll sort the edges, run Kruskal, and give the MST weight; then check my sorted list and tell me the first edge where I wrongly kept or skipped.

AI Level 3 — Matter.

The same greed that fails in TSP (§7.2) is provably perfect here. Discuss with me: WHY does context change everything — what is it about spanning trees that makes "cheapest first" safe, when the same instinct strands the pizza driver? What are the implications for trusting "rules of thumb" in general — is any rule good or bad on its own? Who benefits when a contractor hopes you *won't* run Kruskal — and what does that say about who should learn this math? End by asking me which bill in MY life I'd audit first.

Level 4 — Decide. *Wire the four cabins.*

Example decision problem. Four cabins A, B, C, D need power cable ($ thousands): A–B 5, A–D 7, B–C 4, B–D 6, C–D 3. A contractor quotes $18,000 (A–B, A–D, B–D — "one per cabin, simple"). Find the MST, price it, and decide: accept or counter?

AI Level 4 — Decide.

I need to wire four cabins A, B, C, D. Cable costs ($k): A–B 5, A–D 7, B–C 4, B–D 6, C–D 3. The quote is $18k for A–B, A–D, B–D. Act as my decision coach: make me sort the edges and run Kruskal myself, then make me check whether the quoted edges even form a valid network (do all four cabins connect?). Make me state accept-or-counter with reasons — don't decide for me.

Check your verdict: sorted — C–D 3, B–C 4, A–B 5, B–D 6, A–D 7. Keep C–D, B–C, A–B → $12,000, all connected, no cycles. The $18,000 quote's edges (A–B, A–D, B–D) don't even reach cabin C — it fails the "spanning" in spanning tree! Counter at $13,000 and ask the contractor whether C was supposed to stay dark.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 7.4.

Ask your AI: "Explain the cut property and use it to justify ONE step of Kruskal's algorithm on the §7.4 bid graph." Then do the detective work: verify its justification against the actual cheapest-crossing edge for a split *you* choose (draw the loop yourself). AIs often "justify" with the wrong crossing edge — if yours did, write one paragraph on why the wrong edge breaks the guarantee. Then the contrast question, in your own words: why is greedy *safe* here and *suicidal* in TSP? AI chatbots love the word "optimal" — in this chapter, *you* check what it covers.

⚑ PROJECT MILESTONE

M4: Wire the campus

The Campus Network continues. Fiber must reach all 7 campus buildings; trenching costs $1,800 per walkway-minute. Find the MST of the campus walkway graph, price it, and compare with digging under *every* walkway. Deliverable: the six cables, the price, and the savings.

Practice Problems 7.4

1. Cabins A, B, C, D with cable costs ($k): A–B 6, A–C 9, A–D 4, B–C 5, B–D 8, C–D 7. Find the MST and its weight.

2. A tree has 8 vertices. How many edges does it have? Explain why in one sentence.
3. True or false: "Every spanning tree of the same graph has the same number of edges." Defend your answer in one sentence.
4. Five buildings H, A, B, C, D. Hub costs: H–A 10, H–B 12, H–C 9, H–D 11; cross-links: A–B 4, B–C 5, C–D 6, D–A 20. Find the MST weight and the savings versus connecting everything to H.
5. A 6-vertex MST must have exactly 5 edges. Explain why 4 edges can't work and 6 edges can't work, in one sentence each.

7.5 Graph Coloring — How Many Exam Days?

THE DECISION

Can finals fit in 2 days?

The principal wants to compress finals from 3 days to 2, saving $8,000 in operating costs. Six subjects — Math (M), English (E), Science (S), History (H), Art (A), Music (U) — but some share students and can't run simultaneously. The conflicts:

- M conflicts with E, S, H
- E conflicts with S, A
- S conflicts with H, U
- H conflicts with U
- A conflicts with U

Can all six exams fit in 2 time slots with no student double-booked? If not, what is the minimum — and is the $8,000 saving real? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Math, English, and Science pairwise share students. How many slots do those three alone force?
- **Hint 2:** Try to 2-color the whole conflict graph. Where does it break — which exam has no legal color left?
- **Hint 3:** Exhibit 3 slots explicitly (which exams share each slot?). Then ask: what does the principal's plan do to the shared students?

Coloring a Graph

Coloring a graph means assigning colors to vertices so that adjacent vertices (conflicting exams) never share a color. The **chromatic number** is the fewest colors that work. Mapmakers knew the game first: color a map so neighbors differ. Exams are the same game with time slots as colors.

STORY

The Mapmaker's Dilemma.

In 1852, a map-coloring student noticed every map he'd ever seen needed at most four colors — and asked whether four *always* suffice. The Four Color Theorem took 124 years and, in 1976, a computer checking 1,936 cases to prove. A child can ask it; a machine had to answer it. *What simple question of yours might take a century?*

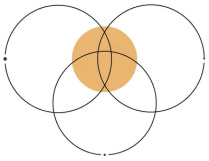
The Chromatic Number

Look for a **clique** — a set of pairwise-adjacent vertices. Math, English, Science: M–E, E–S, M–S all conflict — a triangle, a 3-clique. Three mutually conflicting exams need three different slots, so the chromatic number is *at least* 3. Cliques are lower bounds you can read off the graph; the principal's 2-day plan dies right here, on one triangle.

STORY

The Clique That Decides.

In every scheduling fight there's a clique — three exams sharing the same honors students, four teams sharing one field — that single-handedly sets the minimum. Find the clique first and the argument is half over before it starts. *Where's the clique in your schedule?*



Greedy Coloring

The **greedy algorithm**: order the vertices, give each the first color that doesn't conflict. Order matters — color the troublemakers (highest degree) first. Science (degree 4) first, then Math, English, History, Music (degree 3 each), then Art (degree 2):

- S → Slot 1; M → Slot 2 (conflicts S); E → Slot 3 (conflicts M, S); H → Slot 3 (M is in Slot 2 and S is in Slot 1, so Slot 3 is H's first free slot); U → Slot 2 (conflicts S and H, so Slot 1 is out); A → Slot 1 (conflicts E in Slot 3 and U in Slot 2, so Slot 1 is free).

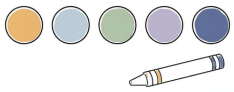
Result: Slot 1: Science, Art. Slot 2: Math, Music. Slot 3: English, History. Three slots — matching the clique's lower bound, so this is *optimal*, not just greedy luck.

STORY

The Registrar's Algorithm.

Real registrars color graphs with

thousands of exams — they can't try everything, so they run greedy with clever orderings and then haggle. Your school's final schedule is a graph coloring someone argued about in April. *Who colored your schedule, and what would you change?*



What Coloring Can't Do

Coloring minimizes *slots*, not happiness: it won't balance slot sizes, won't keep mornings free, won't notice that Slot 3 has the two hardest exams back-to-back. And greedy isn't always optimal — a bad ordering can waste colors. The math gives the *minimum*; humans still do the *mercy*.

STORY

The Two-Day Dream.

The principal's 2-day plan saves $8,000

on paper. Graph coloring has a stubborn habit: the biggest group of mutually conflicting exams sets a floor no clever schedule can go under. The question isn't how badly the school wants two days — it's what the conflict graph allows. *When has a plan of yours hit a fact it couldn't negotiate with?*

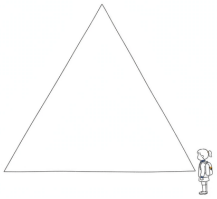
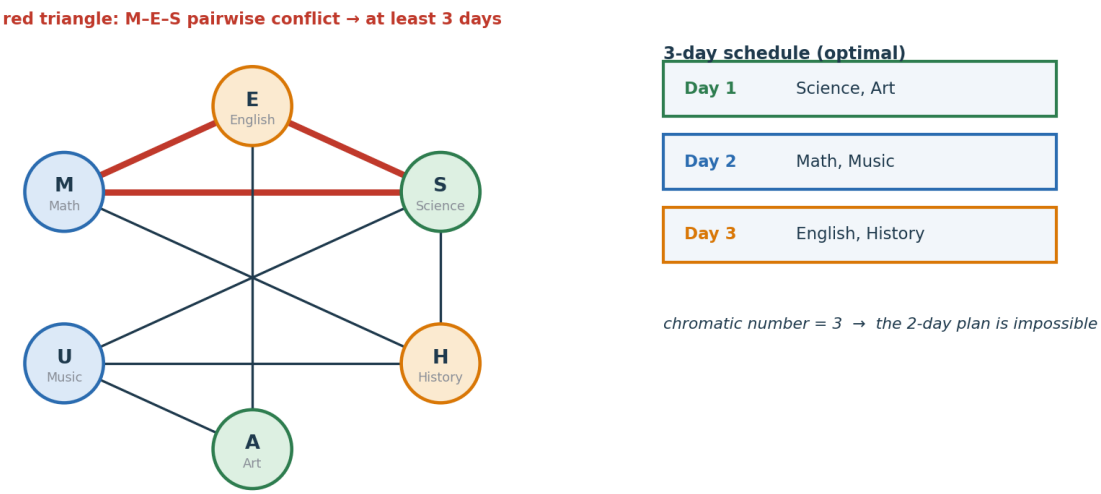


Fig. 7.5

Fig. 7.5 — The triangle rules.



One triangle, three days. The $8,000 was never real.

One triangle, three days. The $8,000 was never real.

DECIDE — THE VERDICT

Can finals fit in 2 days?

Back to the question that opened this section — now with the math to settle it.

Solution. M–E–S form a triangle (pairwise conflicts) → chromatic number ≥ 3. Greedy coloring achieves 3: **Day 1: Science, Art; Day 2: Math, Music; Day 3: English, History.** Two days is *impossible*, not difficult — no 2-coloring exists, proven by the triangle.

The decision: keep 3 days. The $8,000 "saving" would force shared students to choose between finals — the cost doesn't vanish, it lands on the honors kids taking all three triangle subjects. If the board wants savings, color *smarter* (balance the hard exams across days), not *fewer*.

AI AI Learning Lab 7.5

AI Level 1 — Understand.

Explain graph coloring as if I'm 15, using exam scheduling: same color = same time slot, and conflicting exams can't share. Then explain the chromatic number with the triangle example. Make me color two tiny graphs myself before moving on.

AI Level 2 — Apply.

Give me 4 conflict graphs (5–6 vertices each), one at a time. Hide the answers. For each I'll find a clique (lower bound), run greedy coloring, and state the chromatic number with my coloring as proof; then check my clique and my coloring separately.

AI Level 3 — Matter.

The triangle killed the principal's $8,000 saving with pure logic. Discuss with me: WHY is a *lower-bound proof* ("at least 3 days") politically powerful — who can't argue with it, and who will try anyway? What are the implications of scheduling being "solved" by math — does the registrar become unnecessary, or more necessary (mercy vs. minimum)? Who benefits when the minimum is *hidden* — and what does the Four Color Theorem's computer proof say about trusting answers no human fully checked? End by asking me what in MY schedule I'd recolor.

Level 4 — Decide. *How many days for five courses?*

> **Example decision problem.** Five courses with conflicts: Math–English, Math–Science, English–Science (a triangle), Science–History, English–Art. What's the minimum number of exam days? Propose the actual schedule.

AI Level 4 — Decide.

I need to schedule five courses: conflicts Math–English, Math–Science, English–Science, Science–History, English–Art. Act as my decision coach: make me find the clique first, then greedy-color the rest myself, then make me write the day-by-day schedule and check every conflict pair against it. Make me state the minimum days with reasons — don't decide for me.

Check your verdict: the M–E–S triangle forces ≥ 3 days. Greedy: M → Day 1, E → Day 2, S → Day 3, H → Day 1 (conflicts S only), A → Day 1 (conflicts E only). Schedule — Day 1: Math, History, Art; Day 2: English; Day 3: Science. Three days, optimal, every conflict pair checked.

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 7.5.

Ask your AI: "Explain the controversy around the Four Color Theorem's 1976 computer proof." Then do the detective work: verify its two or three key claims (the 1,936 cases, Appel and Haken, the 'no human can check it all' objection) against a second source. If the AI invented a detail — a wrong year, a wrong name — document it. Then write one paragraph: *does a proof no human can fully check still count as proof?* AI chatbots state controversies smoothly and source them rarely — in this chapter, *you* check the footnotes.

⚑ PROJECT MILESTONE

M5: Exam week

The Campus Network continues. Five campus courses — Calculus, Physics, Chemistry, English, History — with conflicts: Calc–Phys, Calc–Chem, Phys–Chem, Chem–Eng, Phys–Hist. Find the chromatic number and publish the exam-week schedule. Deliverable: the number, the day-by-day schedule, and one sentence on which course pair forced it.

Practice Problems 7.5

1. A graph is a triangle (3 mutually adjacent vertices) plus one isolated vertex. What is its chromatic number? Explain the isolated vertex's role in one sentence.
2. A square (4-cycle: A–B–C–D–A) needs how many colors? Exhibit the coloring.
3. Four subjects: Math–English, English–Science, Science–Math (a triangle), plus History conflicting only with Math. Find the chromatic number and give a valid schedule.
4. True or false: "If the greedy algorithm uses 4 colors on a graph, the graph's chromatic number is 4." If false, explain the flaw in one sentence.
5. Four regions on a map are pairwise adjacent (each touches the other three). How many colors does the map need? What famous theorem says you'll never need more than 4 for *any* map?

7.6 Project Scheduling — The Festival Countdown

THE DECISION

Pay the $500? Pay the $200? Pay neither?

The school festival is in **21 days**. Eight tasks remain (days):

- A: Book the venue — 5 (no prerequisite)
- B: Design posters — 4 (no prerequisite)
- C: Print posters — 3 (after B)
- D: Sell tickets — 10 (after C)
- E: Book bands — 7 (after A)
- F: Sound check — 2 (after E)
- G: Book food vendors — 6 (after A)
- H: Final setup — 3 (after D, F, G)

Two people want your money: the band demands **$500** to confirm *now* instead of whenever, and the printer wants **$200** to rush the posters. Which fee protects the festival — if either? *Hold that question: the math you need comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** Find the longest chain of dependent tasks (the critical path). B→C→D→H = 4 + 3 + 10 + 3. How long? Any other chain longer?
- **Hint 2:** For each task, compute its *float*: how many days it can slip without moving the finish. Which tasks have zero float?
- **Hint 3:** Money should buy *float where it matters*. Which fee protects a zero-float task — and which one buys peace of mind you already own?

Tasks and Dependencies

A project is a **directed acyclic graph (DAG)**: vertices are tasks (labeled with durations), arrows are dependencies ("C can't start until B finishes"). "Acyclic" means no loops — you can't need the posters to print the posters. Draw the festival: A and B start immediately; C waits on B; D waits on C; E and G wait on A; F waits on E; H waits on D, F, and G. The drawing *is* the plan.

STORY

The Recipe Order.

You can't frost the cake before baking it,

can't bake before preheating. Every recipe is a tiny PERT chart, and every ruined dinner is a dependency violated. Projects are recipes with budgets. *What "recipe" in your life has a hidden dependency?*

The Critical Path

The **critical path** is the longest path through the DAG — it sets the project's finish date, because everything else can (in principle) fit inside it. Festival chains:

- B→C→D→H: 4 + 3 + 10 + 3 = **20 days** ← critical
- A→E→F→H: 5 + 7 + 2 + 3 = 17
- A→G→H: 5 + 6 + 3 = 14

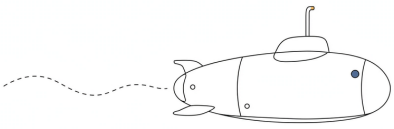
Twenty days against a 21-day deadline: feasible, with exactly **one day of buffer**. Every task on B→C→D→H is *critical* — slip any of them by two days and the festival slips.

STORY

The Polaris Missile.

PERT was born in 1957, when the U.S.

Navy needed the Polaris submarine missile *fast* during the Cold War. Managers mapped thousands of dependent tasks, found the critical path, and aimed all urgency at it — finishing two years early. The math of your school festival once aimed nuclear submarines. *What in your life deserves a critical path?*



Float: The Schedule's Shock Absorber

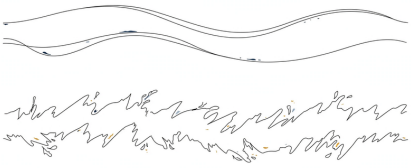
Float (or slack) is how long a task can slip without delaying the project. E (bands) sits on the 17-day chain with a 20-day project: **3 days of float**. F: 3 days. G: 6 days. A: 3 days. Float is the schedule's shock absorber — and the most misunderstood number in management, because float *looks* like laziness and *is* insurance.

STORY

The Lazy River.

Non-critical tasks are a lazy river next to

the critical path's rapids — they arrive when they arrive, as long as it's before the merge. Managers who "optimize" the lazy river accomplish nothing; managers who guard the rapids accomplish everything. *Where's your lazy river — and are you guarding the rapids?*



Crashing

Crashing means paying to shorten a task — but only *critical* tasks repay it. Paying to crash E (3 days of float) buys literally nothing: the project still ends at 20 days. Paying $200 to rush C (critical) from 3 days to 1 buys 2 days of buffer for $100/day. Crash the rapids, never the river.

STORY

The $200 Question.

The printer's rush fee isn't "$200 for

faster posters" — it's "$200 for 2 days of *something*." Reframe every fee as *what it buys on the critical path* and half of all business decisions get easy: days bought where there are none are insurance; days bought where you already own them are a donation. *What fee in your life is really insurance?*

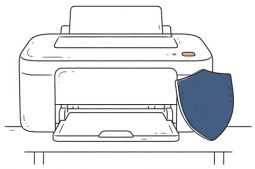
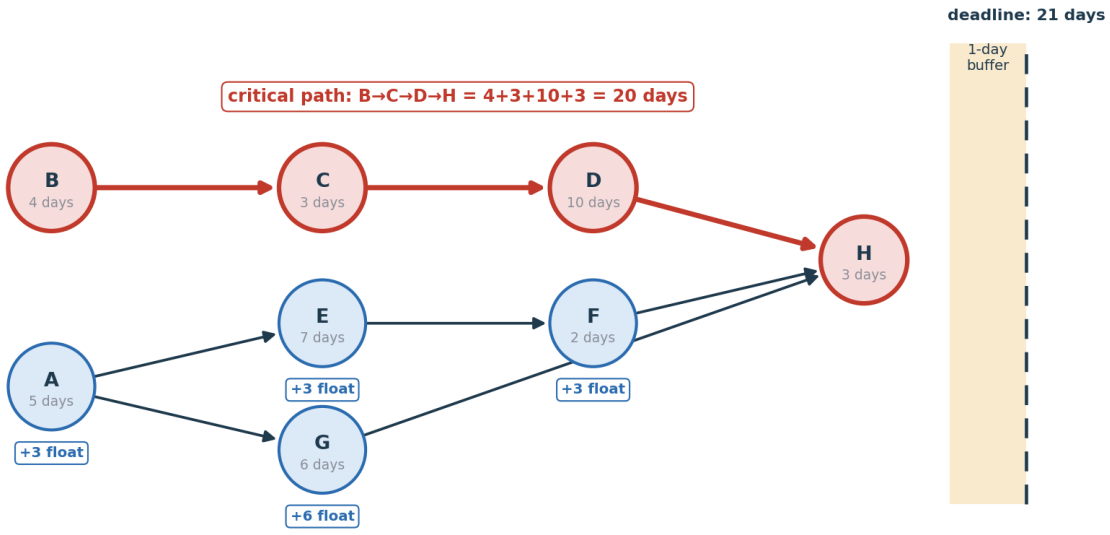


Fig. 7.6

Fig. 7.6 — The rapids and the river.



Crash the rapids, never the river.

Crash the rapids, never the river.

DECIDE — THE VERDICT

Pay the $500? Pay the $200? Pay neither?

Back to the question that opened this section — now with the math to settle it.

Solution. Critical path B→C→D→H = **20 days**; deadline 21; buffer 1 day. The band's task E has **3 days of float** — paying $500 to hurry it buys zero days. The printer's task C is **critical** — a 2-day print delay pushes the finish to 22 > 21, killing the festival.

The decision: pay the $200, refuse the $500. The $200 buys 2 days of buffer on the critical path ($100/day of insurance); the $500 buys nothing but an earlier confirmation email. Tell the band they'll be confirmed on the float — day 8, same stage, no surcharge.

AI AI Learning Lab 7.6

AI Level 1 — Understand.

Explain PERT, the critical path, and float as if I'm 15, using planning a birthday party (book room, send invites, buy cake…). Draw the tiny DAG with me, one dependency at a time, and don't move on until I can point to the critical path myself.

AI Level 2 — Apply.

Give me 4 small projects (5–8 tasks with durations and dependencies), one at a time. Hide the answers. For each I'll find the critical path, the project duration, and each task's float; then check my chains against yours and show me where my longest path went wrong.

AI Level 3 — Matter.

The $500 bought nothing and the $200 bought the festival. Discuss with me: WHY do people pay to hurry non-critical tasks — is it anxiety, ignorance, or pressure to "look busy"? What are the implications for how organizations waste money — how much of every budget is $500s paid for float nobody needed? Who benefits from float being invisible — the band? the printer? And what does it mean that Cold War missile math now plans school festivals — is that beautiful or unsettling? End by asking me what MY current project's critical path is.

Level 4 — Decide. *Will the garage open on time?*

> **Example decision problem.** You're building a garage: A: design 3 days (–), B: permits 5 days (–), C: foundation 4 days (after A), D: framing 2 days (after B, C), E: roofing 1 day (after D). The inspector comes in 11 days. The concrete supplier offers a $150 rush cutting C to 2 days. Take it?

AI Level 4 — Decide.

I'm building a garage: A design 3d (–), B permits 5d (–), C foundation 4d (after A), D framing 2d (after B,C), E roofing 1d (after D). Inspector in 11 days; $150 rush cuts C to 2 days. Act as my decision coach: make me find every chain and the critical path myself, compute every float myself, then make me price the $150 against the deadline myself. Make me state take-it-or-leave-it with reasons — don't decide for me.

Check your verdict: chains — A→C→D→E = 3+4+2+1 = 10 days (critical); B→D→E = 5+2+1 = 8. Deadline 11: buffer 1 day. The $150 rushes C (critical) 4→2 days, buying 2 days of buffer for $75/day.

Take it — cheap insurance on the rapids. (If the rush were on B instead, with its 2 days of float, it would buy nothing — same $500-band lesson.)

DEEP DIVE CHALLENGE

AI Deep-Dive Challenge 7.6.

Ask your AI: "Invent a 7-task project where TWO paths tie for the critical path, and tell me the project duration." Then do the detective work: find both critical paths yourself and check the AI's duration — AIs often miss the tie and report one path's length as if the other didn't exist. If it missed it, write one paragraph on why a hidden tie is dangerous (crash one path and the *other* still sinks you). Then answer in your own words: with two critical paths, which tasks have zero float? AI chatbots love a single answer — in this chapter, *you* count the paths.

⚑ PROJECT MILESTONE

M6: The spring fair

The Campus Network continues. Plan the spring fair: A reserve quad 4d (–), B design posters 3d (–), C print 2d (after B), D ticket sales 8d (after C), E food trucks 5d (after A), F stage/sound 3d (after A), G setup 2d (after D, E, F). Find the critical path and duration. Deliverable: the path, the duration, and which task you'd crash if the date moved up 2 days.

Practice Problems 7.6

1. Tasks: A 2d (–), B 6d (–), C 3d (after A), D 2d (after B, C), E 1d (after D). Find the critical path and project duration.
2. In problem 1, which tasks have float, and how much each?
3. True or false: "Shortening a non-critical task shortens the project." Explain in one sentence.
4. In the §7.6 festival, poster printing (C) slips by 2 days (3 → 5). What is the new completion day? Does the festival survive the 21-day deadline?
5. The printer offers: $200 to cut printing (C) from 3 days to 1 day. What is the new project duration, and what is the cost per day of buffer gained?

7.7 The Campus Network Project — One Map, Six Decisions

THE DECISION

The board funds only three. Which three?

Six milestones, one campus map, one budget meeting. The board will fund exactly three of the six — and your proposal has to say which three, with a number behind every claim. *Hold that question: the brief comes next.*

HINTS

Stuck? Work through these in order — do the math yourself.

- **Hint 1:** For each milestone, write one line: the decision, the key number, the tool that produced it.
- **Hint 2:** Rank the six by dollars saved (or earned) per dollar spent. Which three float to the top?
- **Hint 3:** Now argue against your own ranking with one non-financial value — safety, fairness, student stress. Does the ranking survive?

You have spent this chapter collecting tools. This section is the workshop where they all fit together. Read the brief, then complete any milestones you haven't finished — M1 through M6 are posted in §§7.1–7.6. Your final deliverable: the board proposal.

The Brief

The school board is planning a five-year campus upgrade. One map, six decisions — every tool from this chapter applies to the same ground. Here is the campus (walkway times in minutes; fiber trenching costs $1,800 per minute of walkway):

- North Gate (N): N–L 3, N–C 8
- Library (L): L–S 4, L–C 5, L–D 6
- Science Hall (S): S–D 7, S–G 8
- Dorm (D): D–G 3, D–T 9
- Gym (G): G–T 4, G–C 6
- Cafeteria (C): C–T 7
- Stadium (T)

Milestones M1–M6 (see §§7.1–7.6 for full prompts):

- **M1 (§7.1):** The security cart must cover every walkway once and return to North Gate. Euler verdict + total patrol time.
- **M2 (§7.2):** The mail run: North Gate → Library, Science, Dorm, Stadium → return. Optimum (brute force) vs. nearest-neighbor.
- **M3 (§7.3):** Game-day closures (C–T, G–T): fastest North Gate → Stadium detour, priced against the normal 15-minute N→C→T.
- **M4 (§7.4):** Fiber to all 7 buildings: the MST, its price, and the savings versus digging under every walkway.
- **M5 (§7.5):** Five courses, conflicts Calc–Phys, Calc–Chem, Phys–Chem, Chem–Eng, Phys–Hist: chromatic number + exam-week schedule.
- **M6 (§7.6):** Plan the spring fair: critical path, project duration, and which task to crash if the date moves up 2 days.

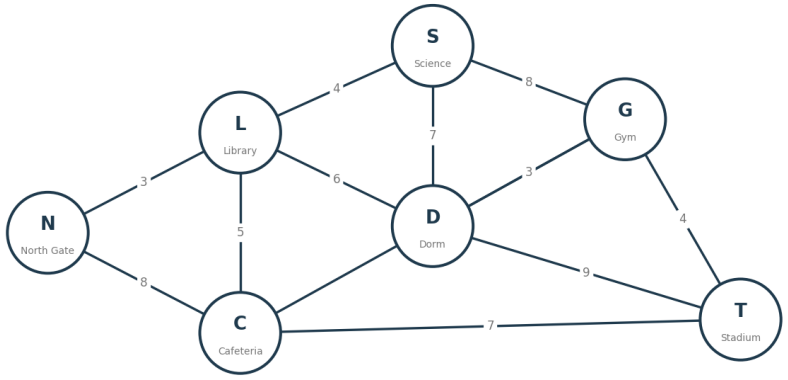
The Board Proposal

Assemble M1–M6 into a single proposal document for the board:

1. **One page per milestone:** the decision, the math, the number.
2. **The budget table:** fiber cost (M4), exam-week days (M5), patrol cost (M1), detour policy (M3), mail-run route (M2), fair date (M6).
3. **The recommendation paragraph:** if the board funds only *three* of the six, which three — and why? (There is no answer key for judgment. There is a rubric: every claim needs its number.)

Fig. 7.7

Fig. 7.7 — The whole campus, one map.



campus walkways (minutes) — 12 walkways, 70 min total

M1 · Patrol (Euler)

odd vertices: S, T (2)

→ Euler path, no circuit

retrace S→G→T: 12 min

patrol total:

70 + 12 = 82 min

M2 · Mail run (TSP)

N → L → S → T → D → N

brute-force optimum: 35 min

N → L → S → D → T → N

nearest-neighbor: 36 min

M3 · Detour (Dijkstra)

closed: C–T, G–T

normal N→C→T: 15 min

detour N→L→D→T: 18 min

leave 3 min early

M4 · Fiber (Kruskal)

MST: N–L, L–S, L–C,

L–D, D–G, G–T = 25 min

fiber cost: $45,000

dig everything: $126,000

M5 · Exams (coloring)

triangle Calc-Phys-Chem

→ chromatic number 3

Day 1: Calc, Eng, Hist

Day 2: Phys · Day 3: Chem

M6 · Fair (PERT)

critical: B→C→D→G

= 3+2+8+2 = 15 days

crash D if date moves up

crash the rapids, not the river

Six decisions, one map.

Six decisions, one map.

DECIDE — THE VERDICT

The board funds only three. Which three?

Back to the question that opened this section — now with six milestones of math behind it.

There is no answer key for judgment — but there is a wrong way to argue: without numbers. A strong proposal might fund **M4** (the fiber MST: $81,000 saved, guaranteed optimal by the cut property), **M1** (the patrol: every walkway covered, cheapest retrace priced), and **M6** (the spring fair: a 15-day critical path with the crash decision worked out) — because those three carry the hardest guarantees. Or safety first: fund the patrol, the detour policy, and the exam schedule, and tell the board the fiber savings will still be there next year. Either way, the rubric is the same: every claim needs its number, every number needs its tool, and "the math says so" is never the whole argument.

🤖 Decision Project Lab 7.7

AI Assemble.

I'm building my Campus Network board proposal (Chapter 7 project). Act as my assembly coach: for each milestone M1–M6, make me state the decision, the key number, and the one-sentence justification — from MY work, not yours. Flag any milestone where my number disagrees with my own earlier work and make me reconcile it before moving on.

AI Challenge.

Now challenge my proposal like a skeptical board member: attack my three funded milestones — ask what assumption each one rests on, what breaks if the assumption is wrong, and who pays. Make me defend or revise each choice. Don't let me hide behind "the math says so."

AI Matter.

Discuss with me: after six milestones on one map, WHY do all these different tools (Euler, TSP, Dijkstra, Kruskal, coloring, PERT) live in one chapter — what's the deep unity? What are the implications of one map holding all six decisions — does optimizing them separately risk breaking something together? End by asking me which tool I'd reach for first in MY school, and why.

AI Decide.

I must recommend only three of the six milestones for funding. Act as my decision coach: make me rank all six by (dollars saved ÷ dollars spent), then make me argue the ranking against one non-financial value (safety? fairness? student stress?). Make me write the final recommendation paragraph myself — my judgment, my numbers, my signature.

Note: the AI assembles, challenges, and coaches — the proposal is yours. Rubric and exemplar proposals are in the Teacher Guide.

Chapter 7 Vocabulary

- **Graph** — dots and lines modeling connections: vertices joined by edges.
- **Vertex / vertices** — a dot in a graph: an intersection, building, exam, or task.
- **Edge** — a line joining two vertices: a street, cable, conflict, or dependency.

- **Degree** — the number of edges touching a vertex.
- **Weighted graph** — a graph with a number (cost, time, distance) on each edge.
- **Euler path** — a route using every edge exactly once (start and end differ).
- **Euler circuit** — a route using every edge exactly once and returning to the start.
- **Hamiltonian cycle** — a tour visiting every *vertex* exactly once and returning to the start.
- **Traveling salesman problem (TSP)** — find the shortest Hamiltonian cycle in a weighted graph.
- **Brute force** — solving by checking every possibility.
- **Heuristic** — a fast rule of thumb that is not guaranteed optimal.
- **Nearest-neighbor heuristic** — for TSP: always go to the closest unvisited city.
- **NP-hard** — problems with no known efficient exact solution; TSP is the famous example.
- **Shortest path** — the minimum-weight route between two vertices.
- **Dijkstra's algorithm** — locks in vertices in order of increasing distance to find shortest paths.
- **Tree** — a graph with no cycles: exactly one path between any two vertices.
- **Spanning tree** — a tree reaching every vertex of a graph.
- **Minimum spanning tree (MST)** — the cheapest spanning tree.
- **Kruskal's algorithm** — build the MST by taking the cheapest edge that doesn't form a cycle.
- **Cut property** — the cheapest edge crossing any split of the vertices must be in the MST.
- **Coloring** — assigning labels (colors) to vertices so adjacent vertices differ.
- **Chromatic number** — the fewest colors that can color a graph.
- **Clique** — a set of pairwise-adjacent vertices; forces a lower bound on colors.
- **Greedy coloring** — color vertices in some order, each with the first legal color.
- **DAG (directed acyclic graph)** — tasks as vertices, dependencies as arrows, no loops.
- **Critical path** — the longest path through a project DAG; sets the finish date.
- **Float (slack)** — how long a task can slip without delaying the project.
- **Crashing** — paying to shorten a task; only pays off on the critical path.
- **PERT** — Program Evaluation and Review Technique: scheduling with task networks.

Chapter 7 Summative Task

Write exactly 3 paragraphs (200–300 words total). Draft it yourself; use AI only as a revision coach — it may question your numbers and your tone, but the judgment must be yours.

The board's $45,000 question. The school board will fund exactly one campus-network milestone this year: the fiber MST (M4), the security patrol route (M1), the exam-week schedule (M5), or the spring fair plan (M6). Write a 3-paragraph decision essay recommending ONE.

Planning your essay (do this before you write). 1. **Compute first.** Pull the key numbers from your own milestone work: the MST price or saving, the patrol route total, the exam-week days, the fair's critical path. Verify each against your own work — don't take them on faith. 2. **Thesis template.** "I recommend ___ because ***, even though*** ." Fill it in for YOUR milestone — any of the four is defensible if the numbers back it. 3. **Evidence checklist.** Every claim needs a number behind it: your milestone's price or saving, one comparison figure from a rejected milestone, and the tool (Euler, Kruskal, coloring, PERT) with the guarantee it does or doesn't carry. 4. **Flip test.** Name one specific number that would change your recommendation — a cheaper MST bid? A patrol route 2 miles shorter? If your advice wouldn't flip under any number, it's a slogan, not a decision.

- **Paragraph 1 — Decide.** State your recommendation and your central claim in the first two sentences. Which milestone gets funded, and what is the one reason it beats the others?
- **Paragraph 2 — Prove.** Give the numbers: the key figure from your milestone work, the price or saving, and one comparison with a rejected milestone. Cite the tool (Euler, Kruskal, coloring, PERT) and the guarantee it does or doesn't carry.
- **Paragraph 3 — Qualify.** Name the strongest limitation of your choice and the strongest argument for a rival milestone. State exactly what would change your mind — a number, not a feeling.

"Rubric and exemplar essays are in the Teacher Guide."

Answer Key — Chapter 7 (Odd-Numbered Problems)

7.1. 1. Degrees: X:3, Y:2, Z:2, W:1. Odd vertices: X, W — two odds, so an Euler path exists (X to W) but no circuit. 3. Degrees: A:3, B:2, C:3, D:2. Odds: A, C — Euler path yes, circuit no. Cheapest retrace: the A–C diagonal, 1.4 mi (vs. 2.0 either way around). Plowing: 5.4 mi; night's total: **6.8 mi**. 5. Zero odd-degree vertices. No — exactly two odds gives an Euler *path* (different start and end), not a circuit.

7.2. 1. Tours: S→A→B→C→S = 1+2+3+5 = 11; S→A→C→B→S = 14; S→B→A→C→S = 17. Shortest: **S→A→B→C→S, 11** (reverse ties). 3. Nearest-neighbor: S→A→C→B→S = 2+1+6+3 = 12. All tours: 17, 12, 13 — optimum **12**. The gut ties

the optimum here: heuristics can win, but never promise. 5. 12 tours each × 30 s = 360 s = **6 minutes** per student — brute force is realistic at this size.

7.3. 1. Lock order: A(0) → B(2) → C(3, via B) → D(5, via C) → E(8, via D). Shortest path: **A→B→C→D→E, 8**. 3. False — stop as soon as the *destination* locks; its distance is final then. 5. Example: a direct D–S walkway at 10 min makes the new shortest D→S **10 min** (any walkway under 18 min improves the detour; under 15 beats the original).

7.4. 1. Sorted: A–D 4, B–C 5, A–B 6, C–D 7, B–D 8, A–C 9. Keep A–D, B–C, A–B (the rest close cycles). MST weight: **15**. 3. True — every spanning tree of an n-vertex graph has exactly $n - 1$ edges. 5. Four edges on six vertices leave at least two disconnected components (a forest); six edges on six vertices must contain a cycle — so neither can be a spanning tree.

7.5. 1. Chromatic number **3** — the triangle forces it; the isolated vertex reuses any color. 3. The M–E–S triangle forces ≥ 3; chromatic number **3**. Schedule: Day 1: Math; Day 2: English, History; Day 3: Science. 5. Four pairwise-adjacent regions need **4 colors**; the **Four Color Theorem** says no map ever needs more.

7.6. 1. Chains: A→C→D→E = 8; B→D→E = 9. Critical path: **B→D→E**, duration **9 days**. 3. False — only shortening a *critical* task (and not creating a new critical path) shortens the project. 5. C: 3 → 1 day; new duration **18 days**; 2 days of buffer for \$200 = **\$100 per day**.